

Hidden Region Finder

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ABSTRACT: The Method of Regions is a systematic way to derive asymptotic expansions of Feynman integrals. For Euclidean integrals there is a well-established algorithm to determine the complete set of regions as facets of a Newton polytope defined by the graph polynomials. In many physically relevant Minkowski limits, additional regions known as *hidden regions* are needed to obtain the correct asymptotic expansion. By making a precise connection with pinch singularities in parameter space and the associated Landau conditions, we identify the general mechanism giving rise to these regions and devise an algorithm, the *Hidden Region Finder*, to determine them. We show that hidden regions arise through a delicate interplay between asymptotic scaling of the edge parameters, which enhances certain monomials of the graph polynomial, and cancellations amongst these monomials, which reduce their collective contribution to the same order as other terms in the polynomial. We explore massless four-, five- and six-point integrals in a variety of wide-angle mass-shell, planar, collinear, double-collinear and Regge limits. Across these different expansions, hidden regions recur in the same small set of seed topologies. In all cases studied, they can be traced to wide-angle configurations with multiple hard-scattering subdiagrams. Distinct kinematic limits of a given seed inherit a common singular locus, while each limit fixes its own scaling vector and cancellation depth. This exploratory study showcases the potential of the proposed algorithm both to uncover missing contributions to asymptotic expansions and to organise them through their underlying singular geometry.

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1 Introduction

Identifying the leading regions of Feynman integrals is fundamental whenever a scattering problem contains widely separated scales. Their most immediate use is to construct asymptotic expansions of individual Feynman integrals. Such expansions have supplied essential ingredients in a wide range of state-of-the-art computations [1–7]. The same region analysis determines the momentum configurations that underlie factorisation, evolution and resummation of large logarithms in scattering amplitudes and cross sections. In the classic analyses of Collins, Soper and Sterman, for example, hard, collinear, soft and Glauber configurations enter the proof of parton-distribution-function (PDF) factorisation, with the contour deformation or cancellation of the Glauber contribution providing a central subtlety [8–11]. More generally, factorisation into hard, jet and soft functions, followed by their evolution, underlies perturbative resummations of threshold and transverse-momentum logarithms [12–14]. Effective field theories take this separation of momentum modes a step further by encoding it directly in their degrees of freedom and Lagrangian. In heavy-quark effective theory and soft-collinear effective theory, the modes relevant at a given scale are represented by explicit fields, while harder fluctuations are integrated out [15–17]. A systematic procedure for identifying regions is therefore a starting point both for asymptotic expansion at the level of individual integrals and for establishing the factorisation and resummation properties of amplitudes and cross sections.

The Method of Regions (MoR) implements this separation of scales through an integrand-level expansion. Once an integration chart (a chosen local coordinate system for the integration variables) and its overlap prescription have been fixed, a region is characterised

by the scaling law of the integration variables as the limit is approached, which determines the corresponding Taylor expansion of the integrand. Remarkably, the correct asymptotic expansion of the original integral is obtained by summing the contributions of all regions, with each expanded integrand integrated over the full original domain. Extending all regional approximations in this way makes them overlap. The corresponding overlap terms are obtained by applying the expansions associated with two or more regions to the same integrand. When all ultraviolet and infrared singularities are appropriately regulated, the sum of these overlap contributions, with the corresponding inclusion–exclusion signs, normally reduces to scaleless integrals and therefore vanishes. Jantzen established this relation in momentum space under explicit conditions and showed that overlap contributions must instead be retained when the chosen regularisation makes them non-scaleless [18]; see also refs. [19–21]. The construction assumes that one is already given a set of regions together with non-overlapping domains that cover the full momentum integration space and satisfy the required admissibility conditions. It therefore validates the MoR once a complete admissible set has been supplied, but does not determine that set; for a generic expansion, the corresponding separation must still be constructed case by case. Determining a complete region set for non-Euclidean expansions is the central problem addressed in this paper.

The MoR has its origins in the expansion-by-subgraphs programme for Euclidean large-momentum and large-mass limits [22–27]. Smirnov placed these expansions on a systematic footing and related their terms to asymptotically irreducible subgraphs [28, 29]. The more general strategy of regions was established by Beneke and Smirnov in their treatment of threshold expansion [30], and was subsequently developed for other non-Euclidean regimes, notably Sudakov limits [31, 32]. In such limits the hard region is supplemented by soft and collinear modes and, where appropriate, potential or Glauber modes. For general non-Euclidean expansions, however, no procedure is known that determines the complete set of regions: a plausible list of momentum modes may be suggested by physical insight, but omitting even one region gives an incorrect asymptotic expansion.

Parameter space provides an important route towards making this problem algorithmic.¹ A key advance was the geometric formulation of Pak and Smirnov [34]: it maps every monomial of a graph polynomial to its exponent vector and identifies region vectors with inward normals to lower facets of the resulting Newton polytope. The Lee–Pomeransky representation [35], in which the polynomial $\mathcal{P} = \mathcal{U} + \mathcal{F}$ is integrated over the positive orthant ($x_e > 0 \ \forall e$), provides a particularly economical setting for this construction. For Euclidean integrals, and more generally when the relevant polynomial admits a same-sign representation, the lower facets determine the regions directly from the polynomial support. The reason is that the asymptotics in parameter space are then controlled by endpoints: a region that is fully characterised by a homogeneous scaling of the parameters towards a boundary is encoded by a lower facet. We refer to such contributions as *facet regions*. In

¹Early developments along this parametric route include the alpha-parameter reformulation of the strategy of regions in ref. [32] and the sector-decomposition and Mellin–Barnes algorithm of ref. [33], which provides an independent power expansion, supplies the ansatz for a differential-equation analysis, and offers a numerical check on a regions analysis.

this setting the geometric construction turns the endpoint analysis into a convex-geometric problem. It underlies implementations such as ASY2, ASPIRE and the expansion-by-regions module of PYSECDEC [36–38].

The limitation is that not every region is characterised by endpoint scaling in the original parameters. Unlike in the fixed-sign case, in Minkowski kinematics $\mathcal{F}$ generally contains terms of opposite sign, which can cancel at finite positive values of the parameters. Such cancellation surfaces may pinch the integration contour and need not correspond to facets of the original Newton polytope. In momentum space, contour pinches are intrinsic to the identification of contributing modes; once their analogues are relevant in parameter space, the polynomial support alone is therefore insufficient. Potential and Glauber regions provided important early examples of this difficulty. For particular classes of such examples, the relevant cancellations can be exposed before the Newton-polytope construction by suitable linear changes of parameters, as implemented in ASY2 [36]. This is a valuable but intrinsically restricted mechanism: it requires a cancellation surface that can be resolved by the available linear transformations. It does not provide a general determination of regions in Minkowski kinematics, where cancellation surfaces may be polynomial and may only become relevant after a nontrivial asymptotic scaling.

The missing information is supplied by the singularity structure of the integral. The Landau equations give necessary conditions for a singularity: propagator singularities must trap the integration contour, producing a pinch surface [39–42]. Ref. [43] made the connection between such pinches and *facet* regions precise for the wide-angle on-shell expansion: facets of the Newton polytope were related to infrared solutions of the Landau equations and to hard, collinear and soft momentum configurations. This led to a graph-theoretical construction of the facet regions, subsequently proved and developed further in ref. [44].

A Newton polytope retains the exponents of the monomials but not the signs and kinematic information carried by their coefficients. It therefore cannot decide whether a cancellation locus is pinched or whether an asymptotic scaling promotes it to a genuine region. Supplying precisely this missing information is the task required to complete the MoR beyond facet regions.

Ref. [45] subsequently showed that genuinely nonlinear pinches, and hence non-facet regions, do occur in the same wide-angle on-shell expansion. In massless $2 \rightarrow 2$ scattering, the first such example appears for a special nonplanar three-loop topology known as the Crown. A dissection of parameter space at the singular locus converts the interior pinch into endpoint singularities, after which ordinary geometric sector-decomposition and MoR algorithms can be applied in the local coordinates. The resulting additional contributions, absent from the facets of the original Newton polytope, were termed *hidden regions*. For the Crown, the hidden region describes Landshoff scattering in the wide-angle on-shell expansion and a Glauber configuration in the Regge limit [45]. The underlying fixed-angle momentum configuration predates the HR terminology: related perturbative and independent-scattering analyses include Refs. [46–48], with Botts and Sterman providing the momentum-space power counting used in Ref. [45]. Identifying the wide-angle and Regge configurations as HRs of the same seed was the first indication that a special topology can generate hidden

regions in more than one kinematic expansion.

Dissection provides a decisive certificate for a hidden region and a cover by useful local coordinates for evaluating its contribution, but it presupposes knowledge of the pinch locus. The next step was taken in the five-point spacelike-collinear problem of ref. [49]. In that example the leading graph polynomial separates into a factorised, individually superleading sector and an obstruction. The pinch conditions apply to the former only after a non-uniform LP-parameter scaling has made the obstruction subleading. This example made explicit the essential interplay between cancellation and scaling and motivated a general procedure for discovering the region directly in the original parameters.

The purpose of the present paper is to formulate that procedure as the *Hidden Region Finder* (HRF), thereby addressing a central obstacle to the practical use of the MoR in non-Euclidean limits. The core construction has two stages. First, HRF extracts candidate polynomial cancellation factors and uses them to decompose the relevant leading polynomial into a superleading cancellation sector $\mathcal{F}_{\text{SL}}$ and an obstruction $\mathcal{F}_{\text{Obs}}$. It then determines a consistent scaling and measures separately the first resolved weights of $\mathcal{F}_{\text{SL}}$ and of the rest of the Lee–Pomeransky polynomial. Their equality is observed in all examples below rather than imposed by the algorithm. Positivity of the LP parameters and a source-resolved dissection, culminating in an exact local lower facet, certify the result. Boundary hidden regions are included by applying the same construction on contraction strata. Composite kinematic limits are handled by an outer step, termed *asymptotic-order alignment*, which first exposes monomials that occur at different orders in the native expansion parameter but must participate in the same cancellation. The central HRF logic itself is unchanged.

We formulate HRF at the level of scalar Feynman integrals with unit numerator. This entails no loss of generality for identifying candidate regions: their existence and location are fixed by the propagator denominators and do not depend on the numerator. The numerator can, however, change the order at which a region contributes, or cause its contribution to vanish at a given order. This distinction is especially important at leading power in gauge theories, where numerator structure can produce behaviour different from that of the corresponding scalar integral. Accounting for such numerator-dependent effects in the various asymptotic expansions lies beyond the scope of this paper.

We develop the construction for massless internal propagators. In this case both Symanzik polynomials are multiaffine, or equivalently linear in every individual edge parameter. Internal masses add $\mathcal{U} \sum_e m_e^2 x_e$ to $\mathcal{F}$, generating quadratic dependence on individual parameters and new possible cancellation structures. Extending HRF to this case is non-trivial and lies outside the scope of this paper.

The applications are exploratory rather than an exhaustive classification. They are ordered to expose progressively the structures that a general region-finding procedure must accommodate. The four-point examples of Sec. 3.1 begin with a direct interior cancellation with no obstruction; their contraction descendants then motivate the search on boundary strata and already show that the derivative harvest must retain general polynomial cancellation factors, including factors with more than two monomials. The five-point Fish family of Sec. 3.2 has a non-vanishing obstruction already at the level of the seed topology, and this obstruction fixes a non-uniform parameter scaling, making

the decomposition into $\mathcal{F}_{\text{SL}}$ and $\mathcal{F}_{\text{Obs}}$ essential. Further examples in that section test the inheritance of this cancellation geometry under a local vertex correction and introduce multi-generator cancellation ideals. The six-point examples of Sec. 3.3 involve composite kinematic limits and show why asymptotic orders must sometimes be aligned before the core construction is applied and why more than one cancellation layer may be active. At the same time, the examples show both sides of the organising pattern: special graph topologies recur across physically different expansions, while the existence and form of a hidden region remain sensitive to the chosen limit and to the permutation of external legs. A separate momentum-space analysis reconstructs the corresponding loop modes and distinguishes propagator virtualities, values at the pinch and local integration widths.

The broader survey suggests a topological organisation of Landshoff hidden regions by complete-bipartite incidence cores, while also showing that the external-leg configuration and the kinematic limit remain essential. It compares several complete-bipartite cores and their four-, five- and six-point external-leg configurations, distinguishing certified positive Landshoff regions from algebraic stationary loci excluded by positivity. These results motivate, but do not establish, conjectures about minimal Landshoff seeds, inheritance under label-preserving graph contraction and the transverse order of positive hidden-region pinches.

Section 2 presents the parameter-space construction independently of any particular graph. Section 3 develops it through a sequence of massless examples, each introducing an additional feature of the algorithm. Section 4 addresses the logically separate translation from certified parameter regions to momentum modes, pole pinches and Glauber routings. Section 5 examines the complete-bipartite organisation of Landshoff hidden regions and states the conjectures suggested by the survey. Appendix A gives the local ideal-theoretic relation between the Landau conditions and the squared cancellation ideal. A notation summary is provided in the final appendix.

2 Hidden Regions Finder: Principles

In this section we set out the principles underlying the Hidden Region Finder. After introducing the parameter-space setup and definitions in Sec. 2.1, we present the core construction in Sec. 2.2: the Landau conditions identify cancellation factors whose common vanishing defines a singular locus, and an appropriate scaling makes the associated cancellation sector uniformly superleading. Section 2.3 extends this construction to nontrivial starting faces and boundary strata, making it applicable in a wide range of circumstances. Section 2.4 then develops the search-level strategy needed to identify equivalent presentations and to certify either the detection or the absence of hidden regions. Finally, Sec. 2.5 summarises the complete algorithmic workflow.

2.1 Parameter-space setup and definitions

Consider a connected scalar graph $\mathcal{G}$ with N internal edges and L loops. Edge e has propagator power ν_e , with total propagator power $\nu \equiv \sum_{e=1}^N \nu_e$. We use dimensional regularisation in $D = 4 - 2\epsilon$ space-time dimensions, where ϵ is the dimensional regulator.

We denote by $\mathbf{s}$ a set of real kinematic invariants formed from the external momenta. Dependence on the graph is normally suppressed. When several graphs or inequivalent regions are compared, a graph-dependent quantity is written as $X[\mathcal{G}; \alpha]$, where the optional second entry labels the region or branch. Thus graph identity is always an argument, whereas a parenthesised superscript, as in $\mathcal{P}^{(R)}[\mathcal{G}]$, denotes a rescaled region expansion. After choosing orientations and a basis of loop momenta $\mathbf{k} = (k_1, \dots, k_L)$, the momentum through edge e can be written as

$$q_e^\mu(\mathbf{k}, \mathbf{p}) = \sum_{a=1}^L \eta_{ea} k_a^\mu + P_e^\mu(\mathbf{p}), \quad \eta_{ea} \in \{0, \pm 1\}, \quad (2.1)$$

where P_e is a linear combination of external momenta. For the massless internal propagators considered here, define $\mathcal{D}_e = q_e^2 + i0$. The momentum-space integral is

$$I_{\mathcal{G}}(\mathbf{s}) = \int [\mathrm{d}\mathbf{k}] \prod_{e=1}^N \mathcal{D}_e^{-\nu_e}, \quad [\mathrm{d}\mathbf{k}] \equiv \prod_{a=1}^L \frac{\mathrm{d}^D k_a}{i\pi^{D/2}}. \quad (2.2)$$

Introducing one proper-time parameter $\tilde{x}_e$ for each propagator while retaining the loop integrations gives² [50]

$$I_{\mathcal{G}}(\mathbf{s}) = \frac{e^{-i\pi\nu/2}}{\prod_{e=1}^N \Gamma(\nu_e)} \int_0^\infty \left(\prod_{e=1}^N \mathrm{d}\tilde{x}_e \tilde{x}_e^{\nu_e-1} \right) \int [\mathrm{d}\mathbf{k}] \exp[i\mathcal{S}(\mathbf{k}, \mathbf{p}; \tilde{\mathbf{x}})], \quad (2.3)$$

$$\mathcal{S}(\mathbf{k}, \mathbf{p}; \tilde{\mathbf{x}}) \equiv \sum_{e=1}^N \tilde{x}_e \mathcal{D}_e = \sum_{e=1}^N \tilde{x}_e [q_e^2(\mathbf{k}, \mathbf{p}) + i0].$$

The sign of the exponent and the prefactor $e^{-i\pi\nu/2}$ follow from the Feynman causal prescription $\mathcal{D}_e = q_e^2 + i0$: for $\tilde{x}_e > 0$, the factor $e^{i\tilde{x}_e \mathcal{D}_e}$ is damped.

For our purposes it is convenient to work in the Lee–Pomeransky (LP) representation [35]. It is obtained from the Schwinger representation by carrying out the Gaussian loop integrations followed by an auxiliary-scale transformation, as described in Ref. [43, App. B]:

$$I_{\mathcal{G}}(\mathbf{s}) = \mathcal{N}_{\text{LP}} (\mu_{\text{LP}}^2)^{LD/2-\nu} \int_0^\infty \prod_{e=1}^N \frac{\mathrm{d}x_e}{x_e} x_e^{\nu_e} \mathcal{P}_{\mu_{\text{LP}}}(\mathbf{x}, \mathbf{s})^{-D/2}, \quad (2.4)$$

$$\mathcal{N}_{\text{LP}} \equiv \frac{\Gamma(D/2)}{\Gamma((L+1)D/2-\nu) \prod_{e=1}^N \Gamma(\nu_e)}, \quad \mathcal{P}_{\mu_{\text{LP}}}(\mathbf{x}, \mathbf{s}) = \mathcal{U}(\mathbf{x}) + \frac{\mathcal{F}(\mathbf{x}; \mathbf{s})}{\mu_{\text{LP}}^2}.$$

Here μ_{LP} is an arbitrary reference mass and the LP parameters x_e are dimensionless. The power of μ_{LP}^2 outside the integral restores the mass dimension of the original momentum integral. In both the Schwinger and LP representations the N edge parameters are independent, with each parameter integrated over $(0, \infty)$.

²For the Fourier-form Schwinger parametrisation with arbitrary propagator powers, see Eqs. (25) and (26) of Ref. [51]; the original proper-time construction is given in Eq. (2.24) of Ref. [52].

Before the division by μ_{LP}^2 in Eq. (2.4), the first and second Symanzik graph polynomials for massless internal propagators are [50]

$$\begin{aligned}\mathcal{U}(\mathbf{x}) &= \sum_{T^1} \prod_{e \notin T^1} x_e, \\ \mathcal{F}(\mathbf{x}; \mathbf{s}) &= \sum_{T^2} (-s_{T^2}) \prod_{e \notin T^2} x_e - i0.\end{aligned}\tag{2.5}$$

Here T^1 runs over spanning trees and T^2 over spanning 2-forests of $\mathcal{G}$; s_{T^2} is the squared total external momentum entering either connected component of T^2 . Writing the coefficients as $-s_{T^2}$ makes manifest that every monomial of the real part of $\mathcal{F}$ is positive in Euclidean kinematics, where all channel invariants s_{T^2} are spacelike.

For an L -loop graph, $\mathcal{U}$ is homogeneous of degree L , while $\mathcal{F}$ is homogeneous of degree $L + 1$ in the edge parameters. Each edge occurs at most once in every complementary tree or 2-forest product. Hence both graph polynomials are linear in each individual x_e . This multiaffinity is a standing assumption below.

In Eq. (2.5), the causal Feynman $i0$ prescription for the propagators in Eq. (2.3) translates into the overall $-i0$ prescription for the second Symanzik graph polynomial $\mathcal{F}$, thereby fixing the physical-sheet boundary value. The term $-i0$ is a limiting prescription, not an additional constant monomial. A necessary condition for a singularity of the parametric integral is given by the Landau conditions [39, 42, 50]

$$\mathcal{F} = 0, \quad x_e \frac{\partial \mathcal{F}}{\partial x_e} = 0 \quad (e = 1, \dots, N),\tag{2.6}$$

where the parameters x_e are not all zero, and $\mathcal{F}$ in these algebraic equations denotes its real polynomial part. By Euler's identity, homogeneity of $\mathcal{F}$ makes the first condition a consequence of the logarithmic derivative conditions; we display it separately to make the singular hypersurface explicit. The factors x_e allow solutions on boundary strata. For an interior first-sheet pinch all $x_e > 0$, and Eq. (2.6) reduces to the stationary conditions $\partial \mathcal{F} / \partial x_e = 0$ for every edge. These equations give the necessary condition for the positive integration contour to be pinched, while the displayed $-i0$ fixes the causal approach to the singular locus.

Geometric expansion by regions. Having introduced the parametric representation, we now turn to the asymptotic expansion of the Feynman integral in a specified kinematic limit, using the geometric parametric formulation of the Method of Regions [34, 36–38]. Let $\mathcal{K}$ denote the physical domain in the real kinematic coordinates $\mathbf{s}$, specified by the appropriate inequalities among them, and let $\delta \rightarrow 0^+$ parametrise the chosen asymptotic limit.³ No universal sign convention is imposed on the individual components of $\mathbf{s}$; the defining inequalities of $\mathcal{K}$ fix the physical domain in each application. This allows the same variables to be retained across different physical regions, making changes of sign under crossing or analytic continuation explicit.

³The natural kinematic deformation need not itself be dimensionless. If a chart uses a dimensionful variable δ_{phys} , its powers are understood as powers of δ_{phys}/Q for a fixed nonzero reference scale Q . Retaining the chart's natural symbol when recording these powers does not affect region vectors or weight comparisons.

After expanding the kinematics, we write each monomial in the LP polynomial as

$$m_i \equiv c_i(\mathbf{s}) \delta^{a_i} \mathbf{x}^{\mathbf{r}_i}, \quad \mathbf{x}^{\mathbf{r}_i} \equiv \prod_{e=1}^N x_e^{r_{i,e}}. \quad (2.7)$$

To retain its origin in the two graph polynomials, we partition the monomial index set as $\mathcal{M} \equiv \mathcal{M}_{\mathcal{U}} \sqcup \mathcal{M}_{\mathcal{F}}$ and write

$$\begin{aligned} \mathcal{U}(\mathbf{x}) &= \sum_{i \in \mathcal{M}_{\mathcal{U}}} m_i, & c_i(\mathbf{s}) &\equiv 1, \quad a_i \equiv 0 \quad (i \in \mathcal{M}_{\mathcal{U}}), \\ \frac{\mathcal{F}(\mathbf{x}; \delta, \mathbf{s})}{\mu_{\text{LP}}^2} &= \sum_{i \in \mathcal{M}_{\mathcal{F}}} m_i, \\ \mathcal{P}_{\mu_{\text{LP}}}(\mathbf{x}; \delta, \mathbf{s}) &\equiv \mathcal{U}(\mathbf{x}) + \frac{\mathcal{F}(\mathbf{x}; \delta, \mathbf{s})}{\mu_{\text{LP}}^2} = \sum_{i \in \mathcal{M}} m_i. \end{aligned} \quad (2.8)$$

Unless stated otherwise, in the region analysis we choose μ_{LP} fixed and nonzero as $\delta \rightarrow 0$. We then express the kinematic coefficients in units of μ_{LP}^2 , suppress the scale label, and use the customary shorthand $\mathcal{P} = \mathcal{U} + \mathcal{F}$. Such a fixed choice changes neither the exponent geometry nor the cancellation locus. If a δ -dependent reference scale is useful, we state that choice explicitly and include its scaling in the monomial weights.

In this and subsequent monomial decompositions, $\mathcal{F}$ denotes its real polynomial part; the $-i0$ remains a boundary-value prescription and is not included in $\mathcal{M}_{\mathcal{F}}$. Thus $\mathbf{r}_i = (r_{i,1}, \dots, r_{i,N})$ is the length- N LP-parameter exponent vector of the monomial m_i . We define the corresponding augmented exponent point by

$$\vec{r}_i \equiv (\mathbf{r}_i; a_i) \in \mathbb{R}^{N+1}. \quad (2.9)$$

The convex hull of these points is the Newton polytope of $\mathcal{P}$; the extremal exponent points are its vertices. An LP-parameter scaling is represented by an augmented vector whose last component is normalised to one,

$$\vec{v}_R = (\mathbf{v}_R; 1), \quad x_e \longmapsto \delta^{v_{R,e}} x_e. \quad (2.10)$$

Under this scaling, the weight of a monomial is defined by

$$w_R(m_i) \equiv \vec{v}_R \cdot \vec{r}_i = a_i + \mathbf{v}_R \cdot \mathbf{r}_i. \quad (2.11)$$

For a given $\vec{v}_R$, the monomials of minimum weight are the dominant terms as $\delta \rightarrow 0^+$. When their exponent points span a codimension-one lower face of the Newton polytope, this face is a lower facet. Its inward-pointing normal is the augmented vector $\vec{v}_R$, normalised as in Eq. (2.10); it gives the corresponding facet-region scaling.

Sign cancellations and hidden regions. The geometric construction above depends only on the augmented exponent points $\vec{r}_i$, and not on the coefficients $c_i(\mathbf{s})$. Once the monomial support is fixed, it is therefore insensitive to the coefficient signs and produces the same Newton polytope and facet regions in every kinematic domain. In particular, it cannot detect cancellations that depend on relative signs between dominant monomials.

Sign cancellations missed by a direct Newton-polytope analysis were considered in Ref. [36]; their role in hidden regions and their resolution by dissecting the positive integration domain were demonstrated in Ref. [45]. To determine whether such a cancellation is possible, we must restore the coefficient information by specifying a physical kinematic domain $\mathcal{K}$. Its defining inequalities fix the signs of the relevant coefficients $c_i(\mathbf{s})$, away from any boundary locus explicitly approached in the expansion. Since $\delta^{a_i} \mathbf{x}^{r_i} > 0$ for $\delta > 0$ and $\mathbf{x} > 0$, the sign of m_i is the sign of $c_i(\mathbf{s})$. These signs may differ between physical domains or crossed channels.

For the scaling vector under consideration, let $\mathcal{P}_{\text{dom}}$ denote the sum of the monomials of minimum weight in Eq. (2.11). If these dominant monomials cannot mutually cancel, as for a same-sign polynomial, the facet normal completely specifies the conventional region. If instead $\mathcal{P}_{\text{dom}}$ contains terms of opposite sign, it may vanish in the positive integration domain. A hidden region can arise when this zero forms a stationary first-sheet pinch. The scaling vector determines $\mathcal{P}_{\text{dom}}$, but not its cancellation locus; the geometric data must therefore be supplemented by the cancellation locus and its resolution.

Although all monomials enter the Newton polytope of $\mathcal{P}$, only the $\mathcal{F}$ -monomials can participate in such a cancellation. Suppose that $\mathcal{P}_{\text{dom}}$ contains both kinds of monomial and decompose it as $\mathcal{P}_{\text{dom}} = \mathcal{U}_{\text{dom}} + \mathcal{F}_{\text{dom}}$. At an interior first-sheet stationary pinch,

$$\mathcal{P}_{\text{dom}} = \mathcal{U}_{\text{dom}} + \mathcal{F}_{\text{dom}} = 0, \quad 0 = \sum_e x_e \frac{\partial \mathcal{P}_{\text{dom}}}{\partial x_e} = L\mathcal{U}_{\text{dom}} + (L+1)\mathcal{F}_{\text{dom}}, \quad (2.12)$$

where the second equality follows from the respective LP-parameter degrees L and $L+1$. These equations require $\mathcal{U}_{\text{dom}} = \mathcal{F}_{\text{dom}} = 0$, but any nonempty sum $\mathcal{U}_{\text{dom}}$ is strictly positive for $\mathbf{x} > 0$. Thus $\mathcal{U}$ -monomials cannot form part of an interior first-sheet pinch.⁴ They must nevertheless be retained because they can affect which terms are dominant in the complete LP polynomial.

For later use, we record how an LP-parameter region vector translates into propagator virtualities. In the Schwinger representation the relation is immediate: $\tilde{x}_e$ multiplies $\mathcal{D}_e$ in Eq. (2.3), so their regional scalings are inverse. The auxiliary-scale transformation relating the Schwinger and LP representations ensures that x_e and $\tilde{x}_e$ have identical regional scaling vectors [43, App. B]. Hence

$$x_e \sim \delta^{v_{R,e}} \iff Q^2 \tilde{x}_e \sim \delta^{v_{R,e}} \implies \frac{\mathcal{D}_e}{Q^2} \sim \delta^{-v_{R,e}}, \quad (2.13)$$

where Q^2 is a fixed hard scale. Thus a parametrically large edge parameter corresponds to a parametrically small propagator virtuality, and conversely. Equation (2.13) determines the propagator virtualities associated with a parameter-space region; recovering a consistent set of loop-momentum modes requires the additional momentum-routing analysis developed in Sec. 4.

⁴The same conclusion is immediate in the homogenised LP representation $\hat{\mathcal{P}}(\mathbf{x}, x_0) = \mathcal{F}(\mathbf{x}) + x_0\mathcal{U}(\mathbf{x})$: stationarity with respect to x_0 requires $\mathcal{U} = 0$, which has no solution in the positive interior for a connected graph [53].

2.2 The core HRF construction

The core construction takes as external input a specified starting polynomial $\mathcal{F}_\star$ and a specified subvector $\mathbf{x}_{\text{active}}$ of nonzero LP parameters, with N_{active} components. All other LP parameters, if any, have already been set to zero. This selects the stratum on which the core construction is applied: a sector of the closure of the positive LP domain with a specified subset of parameters set to zero and all remaining active parameters positive. The choice of these inputs is not part of the core construction.

The mechanism sought by HRF is the following. A subset of the monomials of $\mathcal{F}_\star$ may sum to a polynomial $\mathcal{F}_{\text{SL}}$ that admits a first-sheet Landau pinch in the positive LP domain, even though neither the complete polynomial $\mathcal{F}$ nor the starting polynomial $\mathcal{F}_\star$ does. The remaining monomials obstruct this cancellation. A hidden-region scaling makes the individual monomials of $\mathcal{F}_{\text{SL}}$ superleading relative to the obstruction, while their mutual cancellation counters this individual enhancement, so that their sum contributes at the physical leading order. We therefore call $\mathcal{F}_{\text{SL}}$ the superleading cancellation sector and the complementary part of $\mathcal{F}_\star$ the obstruction. Concrete instances of this mechanism are worked out in Sec. 3.

The construction has two parts. Part I uses cancellation factors exposed by derivatives of $\mathcal{F}_\star$, and products of these factors, to construct a candidate $\mathcal{F}_{\text{SL}}$ and validate its common positive Landau locus. Part II determines the scaling from the complete LP polynomial, imposes the required homogeneity and separation in weight, and checks that resolving the cancellation produces a scaleful lower-facet region.

To state the ordinary core construction, expand

$$\mathcal{F}(\mathbf{x}; \delta, \mathbf{s}) = \mathcal{F}_0(\mathbf{x}; \mathbf{s}) + \sum_{n>0} \delta^n \mathcal{F}_n(\mathbf{x}; \mathbf{s}). \quad (2.14)$$

In the simplest interior case, $\mathbf{x}_{\text{active}} = \mathbf{x}$ and the cancellation sector lies in $\mathcal{F}_0$, so we take $\mathcal{F}_\star = \mathcal{F}_0$. More generally, boundary hidden regions are treated by first restricting the graph polynomial to a contraction stratum, while cancellations involving monomials from different native orders in δ require preliminary asymptotic-order alignment. These two independent extensions are described in Sec. 2.3.

Writing the starting polynomial as $\mathcal{F}_\star = \sum_{i \in \mathcal{M}_\star} m_i$, with $\mathcal{M}_\star \subseteq \mathcal{M}_{\mathcal{F}}$, let $\mathcal{M}_{\text{SL}} \subseteq \mathcal{M}_\star$ denote the subset of monomials selected for a candidate cancellation sector, and define

$$\mathcal{F}_{\text{SL}} \equiv \sum_{i \in \mathcal{M}_{\text{SL}}} m_i. \quad (2.15)$$

Here $\mathcal{M}_{\text{SL}}$ denotes a subset of the monomial support of $\mathcal{F}_\star$; Eq. (2.15) merely fixes the notation. Section 2.2.1 explains how HRF proposes such a subset and tests whether the corresponding $\mathcal{F}_{\text{SL}}$ defines an admissible cancellation sector. In the special case where $\mathcal{M}_{\text{SL}} = \mathcal{M}_\star$, one has $\mathcal{F}_{\text{SL}} = \mathcal{F}_\star$ and the obstruction vanishes.

2.2.1 Part I: constructing and validating the decomposition

The Landau target. The purpose of the decomposition is to isolate a polynomial $\mathcal{F}_{\text{SL}}$ that can admit a first-sheet pinch in the relative interior of the chosen active stratum.

Applying the interior form of Eq. (2.6) to the active LP variables gives the defining target

$$\frac{\partial \mathcal{F}_{\text{SL}}}{\partial x_e} = 0 \quad \text{for every active } x_e, \quad \mathbf{x}_{\text{active}} \in \mathbb{R}_{>0}^{N_{\text{active}}}, \quad \mathbf{s} \in \mathcal{K}. \quad (2.16)$$

Here “interior” refers to the positive domain of the active parameters. When $\mathbf{x}_{\text{active}} = \mathbf{x}$, this is an interior pinch of the full LP domain; when some parameters have been fixed to zero, the same condition describes a boundary pinch in the full domain. Since $\mathcal{F}_{\text{SL}}$ is homogeneous in the active LP parameters, Euler’s identity implies $\mathcal{F}_{\text{SL}} = 0$ whenever the derivative equations hold. Thus Eq. (2.16) is the complete Landau condition on the chosen active stratum; a separate equation $\mathcal{F}_{\text{SL}} = 0$ would be redundant.

Formulating Landau loci directly through derivatives of an LP-type graph polynomial also underlies the principal Landau determinant framework [54]. That construction studies complex critical loci of the Lee–Pomeransky polynomial $\mathcal{P} = \mathcal{U} + \mathcal{F}$ and its facial restrictions. Here we instead require a first-sheet pinch in the positive orthant of the selected asymptotic polynomial $\mathcal{F}_{\text{SL}}$.

Let C be a smooth component of this common pinch locus. Smoothness allows us to choose a nonredundant set of local defining polynomials $f_1, \dots, f_r$, whose gradients are independent along C , so that C is locally given by $f_1 = \dots = f_r = 0$ and the f_i provide coordinates transverse to C . Their radical local ideal in the active LP-parameter space is $\mathcal{I}_C = \langle f_1, \dots, f_r \rangle$.⁵ The Landau equations imply

$$\mathcal{F}_{\text{SL}} \in \mathcal{I}_C^2. \quad (2.17)$$

To see this, complete the f_i by coordinates along C . Homogeneity and Eq. (2.16) imply that $\mathcal{F}_{\text{SL}}$ vanishes on C , eliminating the constant term in its local expansion, while the derivative equations eliminate every term linear in the f_i . The expansion therefore begins with products $f_i f_j$. The two-factor structure used below is consequently a direct implication of the Landau conditions, not an additional HRF assumption. Appendix A provides an intuitive derivation, followed by the equivalent ideal-theoretic formulation in terms of the conormal exact sequence. The implication in Eq. (2.17) does not require $\mathcal{F}_{\text{SL}}$ to be multiaffine; rather, multiaffinity of the second Symanzik polynomial in the massless case is what makes the cancellation factors accessible through derivatives of $\mathcal{F}_\star$.

Derivative harvesting. Because $\mathcal{F}_{\text{SL}}$ is not known in advance, HRF searches for traces of its cancellation factors in derivatives of $\mathcal{F}_\star$. The harvested objects are candidate non-monomial cancellation polynomials $f_j(\mathbf{x}, \mathbf{s})$. Each f_j must be capable of vanishing for $\mathbf{x} > 0$ and $\mathbf{s} \in \mathcal{K}$; at fixed $\mathbf{s}$, a necessary condition is that its nonzero LP-monomial coefficients are not all of the same sign. In choosing a given f_j , any multiplicative factor that is nonzero throughout the selected domain (such as a monomial in the active LP parameters or a kinematic factor known to be nonzero throughout $\mathcal{K}$) is omitted, since this does not change the zero locus of that f_j , and hence does not change their common zero locus.

⁵Locally, $\langle f_1, \dots, f_r \rangle$ consists of combinations $\sum_i h_i f_i$, with coefficients h_i nonsingular near C . The qualifier “radical” means that, if $g^n \in \mathcal{I}_C$ for some $n \geq 1$, then $g \in \mathcal{I}_C$, so the ideal records the common zero locus without algebraic multiplicity.

The direct harvest has two complementary forms. One may factorise the complete derivatives. When the kinematic dependence admits a useful decomposition into independent structures, one may instead, or additionally, collect each derivative in those structures and factorise the resulting LP-parameter coefficient polynomials separately. This coefficientwise harvest is neither a universal step nor by itself sufficient. In particular, when an intended generator depends irreducibly on several kinematic variables, such a decomposition may obscure rather than expose it. Factors that remain coupled in derivatives of an already selected $\mathcal{F}_{\text{SL}}$ may instead be recovered from the saturated selected-sector gradient ideal, with branches supported on factors required to be nonzero removed, in Eq. (2.28). The factors and kinematic sectors exposed by the applicable direct harvest provide the input for constructing the candidate generators below.

At this stage the derivatives of $\mathcal{F}_\star$ are used only to identify candidate factors, not imposed as equations. When an obstruction is present, the derivatives of $\mathcal{F}_\star$ need not all vanish on the true HR locus. The derivatives of the subsequently selected $\mathcal{F}_{\text{SL}}$ do vanish there; those derivatives of $\mathcal{F}_\star$ that receive no contribution from the obstruction therefore vanish as well, while the remaining ones may not. Thus $\mathcal{F}_\star$ itself need not have a stationary point in the positive orthant, and generically does not.

Generators for massless graph polynomials and the obstruction decomposition. Equation (2.17) identifies the square of the local defining ideal,

$$\mathcal{I}_C^2 = \langle f_i f_j \mid 1 \leq i \leq j \leq r \rangle. \quad (2.18)$$

The diagonal products make the full square ideal independent of the chosen local defining polynomials. Equation (2.18) is therefore the presentation-independent algebraic envelope implied by the Landau conditions; it is not the list of generators admitted by a particular massless graph polynomial.

In the original edge-variable basis, define the LP-parameter support of a cancellation factor by

$$\text{supp}_x(f) \equiv \{e \mid \deg_{x_e} f > 0\}. \quad (2.19)$$

For a massless graph, multiaffinity of the second Symanzik polynomial restricts a candidate generator built from the cancellation factors to the form

$$g_k = \prod_{i \in S_k} f_i, \quad |S_k| \geq 2, \quad \text{supp}_x(f_i) \cap \text{supp}_x(f_j) = \emptyset \quad (i \neq j, i, j \in S_k). \quad (2.20)$$

The factors in S_k must be distinct and have pairwise-disjoint LP-parameter support: a repeated factor or overlapping dependence on any x_e would produce degree two in that parameter and could not reproduce a monomial of the massless graph polynomial. The construction permits $|S_k| > 2$; in all examples considered here, however, every generator contains precisely two cancellation factors.

Multiaffinity is only the first support test. The expanded monomials of g_k must also be embeddable in monomials already present in the starting polynomial $\mathcal{F}_\star$. Writing $\text{Mon}(g_k)$

for the set of monomials m in the expanded generator, and $\mathbf{r}(m)$ for the LP exponent vector of m , this necessary condition is

$$\text{for every } m \in \text{Mon}(g_k), \quad \text{there exists } m_\ell \in \mathcal{M}_\star \text{ such that } \mathbf{r}(m) \leq \mathbf{r}_\ell \text{ componentwise.} \quad (2.21)$$

This test concerns only the LP-parameter support. Equivalently, after disregarding kinematic coefficients, each generator monomial must divide the LP-parameter part of an existing monomial of $\mathcal{F}_\star$. This permits a multiplier to supply the remaining LP-parameter content, but it excludes products which belong abstractly to $\mathcal{I}_C^2$ and cannot occur in the graph polynomial.

For a candidate generator set $G = \{g_1, \dots, g_{n_{\text{gen}}}\}$, with $n_{\text{gen}} \equiv |G|$, define the candidate generator ideal

$$\mathcal{I}_{\text{gen}}(G) \equiv \langle g_1, \dots, g_{n_{\text{gen}}} \rangle \subseteq \mathcal{I}_C^2. \quad (2.22)$$

At this stage G and $\mathcal{I}_{\text{gen}}(G)$ remain tentative. They are retained only if the obstruction search finds a nonzero complement of $\mathcal{F}_\star$ that lies in $\mathcal{I}_{\text{gen}}(G)$; otherwise G is discarded and another candidate generator set is tested.

To perform this search, HRF considers proper subsets of monomials $\mathcal{M}_{\text{Obs}} \subsetneq \mathcal{M}_\star$ to remove from $\mathcal{F}_\star$. For each choice, with $\mathcal{M}_{\text{SL}}$ its complement, define

$$\begin{aligned} \mathcal{M}_{\text{Obs}} &\subsetneq \mathcal{M}_\star, & \mathcal{M}_{\text{SL}} &\equiv \mathcal{M}_\star \setminus \mathcal{M}_{\text{Obs}}, \\ \mathcal{F}_{\text{Obs}} &\equiv \sum_{i \in \mathcal{M}_{\text{Obs}}} m_i, & \mathcal{F}_{\text{SL}} &\equiv \sum_{i \in \mathcal{M}_{\text{SL}}} m_i = \mathcal{F}_\star - \mathcal{F}_{\text{Obs}}. \end{aligned} \quad (2.23)$$

The obstruction support $\mathcal{M}_{\text{Obs}}$ may be empty, but $\mathcal{M}_{\text{SL}}$ must be nonempty. The choice gives an algebraically admissible candidate decomposition only if $\mathcal{F}_{\text{SL}}$ is nonzero and belongs to the ideal $\mathcal{I}_{\text{gen}}(G)$ generated by G in Eq. (2.22):

$$\mathcal{F}_\star = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{Obs}}, \quad \mathcal{F}_{\text{SL}} \neq 0, \quad \mathcal{F}_{\text{SL}} \in \mathcal{I}_{\text{gen}}(G). \quad (2.24)$$

The ideal-membership condition is equivalently expressed as

$$\mathcal{F}_{\text{SL}} = \sum_{k=1}^{n_{\text{gen}}} M_k(\mathbf{x}, \mathbf{s}) g_k. \quad (2.25)$$

The multipliers supply the remaining monomial content. By default, Eq. (2.25) is tested as exact membership in the polynomial ideal $\mathcal{I}_{\text{gen}}(G)$, requiring the polynomial-reduction remainder to vanish. The support test in Eq. (2.21) is necessary, whereas the exact identity in Eq. (2.25) is what ensures that the selected generators reproduce the actual terms of $\mathcal{F}_{\text{SL}}$. Only this complete identity tests kinematic compatibility: no term-by-term or generator-by-generator match is required.⁶ Every such presentation belongs to the ambient square ideal and hence has the quadratic vanishing required by Eq. (2.16). Conversely, Appendix A proves membership in $\mathcal{I}_C^2$ at a smooth pinch component; it does not select the graph-dependent set G .

⁶In the five-point near-planar example, the individual generator contributions contain the rational stationary ratios $\boldsymbol{\rho} = -K^{-1}\mathbf{b}$, whose nonlinear kinematic dependence cancels only in the complete sum; see Sec. 3.2.4, especially Eq. (3.61) and the discussion following Eq. (3.75).

Validation of the common locus. A necessary preliminary condition for every distinct cancellation factor f_i appearing in the generators G is that its nonzero terms include both signs in the kinematic domain $\mathcal{K}$. This is not sufficient: the harvested factors and the resulting decomposition of $\mathcal{F}_\star$ become physically admissible only if all these factors have a common positive zero at one and the same kinematic point:

$$\exists (\mathbf{x}_{\text{active}}, \mathbf{s}) \in \mathbb{R}_{>0}^{N_{\text{active}}} \times \mathcal{K} : \quad f_i(\mathbf{x}, \mathbf{s}) = 0 \quad \text{for every } i \in \bigcup_{k=1}^{n_{\text{gen}}} S_k. \quad (2.26)$$

On a smooth component C of this solution, after removing redundant relations, the selected factors are taken to generate the radical local ideal $\mathcal{I}_C$ used in Eqs. (2.17) and (2.18). Equivalently one solves Eq. (2.16) directly. The common locus, rather than any one factor in isolation, encodes the pinch.

The derivatives of an already selected $\mathcal{F}_{\text{SL}}$ need not display the cancellation factors f_i as literal factors; they may contain only coupled polynomial linear combinations of them.⁷ To recover the complete common locus in this situation, package the stationarity equations for $\mathcal{F}_{\text{SL}}$ in Eq. (2.16) into a gradient ideal and define $\mathcal{I}_{\nabla}^{\text{SL}}$ and the localisation element S by

$$\mathcal{I}_{\nabla}^{\text{SL}} \equiv \left\langle \frac{\partial \mathcal{F}_{\text{SL}}}{\partial x_e} \right\rangle_e, \quad \text{and} \quad S \equiv \prod_{e \in \text{active}} x_e \prod_a h_a(\mathbf{s}), \quad (2.27)$$

where the $h_a(\mathbf{s})$ are kinematic factors known to be nonzero in the domain under consideration. When the coefficient matrix relating these combinations to the f_i is invertible on the physical component, multiplication by its adjugate shows that its determinant times each f_i belongs to the gradient ideal once the kinematic constraints are imposed. Here algebraic localisation means allowing division only by factors known to be nonzero on the selected component. Isolating the individual f_i therefore requires localisation by this nonvanishing determinant; the factors needed for this inversion are included in S . The localisation is implemented by the saturation

$$(\mathcal{I}_{\nabla}^{\text{SL}} + \mathcal{I}_{\text{kin}}) : S^\infty, \quad (2.28)$$

with $\mathcal{I}_{\text{kin}}$ imposing the algebraic constraints of the chosen expansion. Equivalently, this saturation discards solution components supported on $S = 0$. It is an optional completion or validation of an already selected cancellation sector, not a general pre-decomposition harvest of $\partial \mathcal{F}_\star$.

The dissection method of Ref. [45] provides the final geometric certificate used by HRF. Starting from a common positive solution, choose coordinates u_α tangent to the pinch locus and independent transverse coordinates $y_j = f_j$. A neighbourhood is covered by sign sectors and by radial-ratio charts⁸ for the normal directions, retaining the Jacobian of every change of variables and the portion for which every $x_e > 0$. In such coordinates the common locus

⁷An explicit example is the five-point near-planar Fish in Sec. 3.2.4, where saturation of the selected gradient ideal recovers the local cancellation ideal $\langle f_A, f_B, f_C \rangle$, although literal factorisation of an individual raw derivative does not expose these factors; see Eqs. (3.78) and (3.79).

⁸Here a chart is a local coordinate patch; in a radial-ratio chart one variable measures the distance from the common locus while ratios specify the normal direction.

is a boundary or corner, and the transformed LP polynomial is treated by the geometric Method of Regions.

This local chart is not unique. Different charts may assign the cancellation depth differently among the y_j , and consequently expose different lower facets, or a lower-dimensional face common to neighbouring facets. They nevertheless certify the same HR when their normals pull back to the same scaling in the original LP parameters. Endpoint charts, in which one normal coordinate carries the entire suppression, can be especially convenient for integration; they do not replace the simultaneous locus by a codimension-one pinch. Thus dissection tests the local cancellation geometry and the existence of a scaleful region without imposing the full gradient of an obstructed starting polynomial. The chart cover and its relation to a direct original-coordinate test are described in Sec. 2.2.2 and illustrated explicitly by the spacelike-collinear Fish in Sec. 3.2.1.

2.2.2 Part II: determining and certifying the scaling

Hidden-region conditions. Given the selected polynomial $\mathcal{F}_{\text{SL}}$ and its common positive pinch component C , validated in Part I, Part II determines a final scaling

$$\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1). \quad (2.29)$$

Using the monomial-weight definition in Eq. (2.11), we denote by W_{SL} the common individual weight of the monomials in $\mathcal{M}_{\text{SL}}$, and by W_{HR} the weight of the first nonvanishing layer of the complete LP polynomial under this scaling, after all cancellations on C have been taken into account. These weights are determined together with $\mathbf{v}_{\text{HR}}$. A HR is characterised by these data when the following four conditions hold:

- (a) **Homogeneity:** every monomial $m_i \in \mathcal{M}_{\text{SL}}$ has the same individual weight W_{SL} ;
- (b) **Common locus:** the validated $\mathcal{F}_{\text{SL}}$ obeys the simultaneous first-sheet equations (2.16) in the positive LP orthant and in $\mathcal{K}$;
- (c) **Hierarchy:** the cancellation removes the individually superleading behaviour, and the first nonvanishing layer of the complete LP polynomial has weight W_{HR} , with

$$W_{\text{HR}} > W_{\text{SL}}; \quad (2.30)$$

- (d) **Scalefulness:** in local cancellation-resolving coordinates, the transformed leading LP polynomial defines a scaleful lower-facet region.

The gap $W_{\text{HR}} - W_{\text{SL}}$ measures the depth of the cancellation.

The rescaled LP polynomial in a hidden region. The structure to be produced by the HR scaling is most transparent at the level of the rescaled complete LP polynomial. For any region vector $\mathbf{v}_R$, define

$$\mathcal{P}^{(R)}(\mathbf{x}; \delta, \mathbf{s}) \equiv \mathcal{P}(\delta^{\mathbf{v}_R} \mathbf{x}; \delta, \mathbf{s}), \quad (2.31)$$

so every displayed power of δ includes both the explicit kinematic dependence and the scaling of the LP parameters x_e . We refer to the monomials in the lower-weight cancelling sector as individually superleading (red), the monomials at the first resolved nonzero order as resolved leading (blue), and all other monomials as remaining (black). In the generic single-layer case, the hidden-region expansion and the corresponding facet expansion for comparison are

$$\mathcal{P}^{(\text{HR})} \equiv \mathcal{P}(\delta^{v_{\text{HR}}} \mathbf{x}; \delta, \mathbf{s}) = \underbrace{\delta^{W_{\text{SL}}} \mathcal{F}_{\text{SL}}}_{\mathcal{F}_{\text{SL}}^{(\text{HR})}} + \delta^{W_{\text{HR}}} \mathcal{P}_{W_{\text{HR}}} + \mathcal{O}(\delta^{>W_{\text{HR}}}), \quad \text{hidden region.} \quad (2.32a)$$

$$\mathcal{P}^{(R)} = \delta^{W_R} \mathcal{P}_{W_R} + \mathcal{O}(\delta^{>W_R}), \quad \text{facet region.} \quad (2.32b)$$

In a hidden region the individually superleading monomials (red) have a lower individual weight than the resolved leading monomials (blue), but their sum vanishes on the positive pinch component C . By comparison, a facet region has no individually superleading sector. When condition (a) holds, the rescaled superleading polynomial is

$$\mathcal{F}_{\text{SL}}^{(\text{HR})} \equiv \mathcal{F}_{\text{SL}}|_{\mathbf{x} \mapsto \delta^{v_{\text{HR}}} \mathbf{x}} = \delta^{W_{\text{SL}}} \mathcal{F}_{\text{SL}}. \quad (2.33)$$

Thus $\mathcal{F}_{\text{SL}}^{(\text{HR})}$ includes the δ -dependence induced by the LP-parameter scaling. It cancels on C , while the first resolved nonzero LP layer appears at $W_{\text{HR}} > W_{\text{SL}}$.⁹

Determining the HR vector: homogeneity and hierarchy. In the generic case, v_{HR} , W_{SL} and W_{HR} are determined by the homogeneity and hierarchy conditions. The first requirement is homogeneity of the selected cancellation sector,

$$w_{\text{HR}}(m_i) = W_{\text{SL}}, \quad \forall m_i \in \mathcal{M}_{\text{SL}}. \quad (2.34)$$

This condition is necessary because a cancellation can take place only among monomials of the same individual weight. If their weights differ, the lowest-weight subset dominates by itself and the remaining monomials cannot cancel it. In the generic case, no additional occupied layer with weight between W_{SL} and W_{HR} vanishes on the same locus. The second (hierarchy) requirement is then

$$w_{\text{HR}}(m_i) \geq W_{\text{HR}} > W_{\text{SL}}, \quad \forall m_i \in \mathcal{M} \setminus \mathcal{M}_{\text{SL}}. \quad (2.35)$$

This condition constrains only the monomials outside $\mathcal{M}_{\text{SL}}$. Each must have weight at least W_{HR} . Those saturating the inequality are the dominant monomials at that weight outside the superleading sector $\mathcal{F}_{\text{SL}}$. Because $\mathcal{M}$ is the monomial support of the complete LP polynomial, Eq. (2.35) also includes the required balance with $\mathcal{U}$. The strict gap is essential: if the conditions are inconsistent, or if they admit only $W_{\text{HR}} = W_{\text{SL}}$, the decomposition does not define a HR and must be rejected. A factorised cancellation sector and a positive pinch are not sufficient without this hierarchy.

⁹In the five-point spacelike-collinear example of Sec. 3.2.1, the monomials in the individually superleading $\mathcal{F}_{\text{SL}}$ sector (red) of Eq. (3.26) all have $W_{\text{SL}} = -5$, while the resolved leading obstruction and kinematically suppressed monomials (blue) in the same equation, together with the $\mathcal{U}$ layer in Eq. (3.27), enter at $W_{\text{HR}} = -4$; see Eq. (3.23). The resulting unit gap and the obstruction fix the non-uniform HR scaling.

The weight definition (2.11) translates these conditions into a finite linear problem. For the selected monomials, and for the remaining monomials m_j that saturate the hierarchy at weight W_{HR} , one has

$$a_i + \mathbf{v}_{\text{HR}} \cdot \mathbf{r}_i = W_{\text{SL}}, \quad m_i \in \mathcal{M}_{\text{SL}}, \quad (2.36a)$$

$$a_j + \mathbf{v}_{\text{HR}} \cdot \mathbf{r}_j = W_{\text{HR}}, \quad m_j \in \mathcal{M} \setminus \mathcal{M}_{\text{SL}}. \quad (2.36b)$$

These active equalities form a linear system for the $N + 2$ unknowns $(\mathbf{v}_{\text{HR}}, W_{\text{SL}}, W_{\text{HR}})$; the unsaturated instances of Eq. (2.35) test the resulting solution as inequalities. If the active system has full rank, the normalisation of the last component of $\vec{v}_{\text{HR}}$ to one gives a unique candidate vector. If it is rank deficient, Eqs. (2.34) and (2.35) define a family rather than a unique vector, and the candidate is not certified until the dissection analysis below fixes or rejects it. Thus these equations are the practical determining conditions, but uniqueness is a property to be checked, not assumed.

Before applying the hierarchy, all monomials suppressed at fixed LP parameters must be restored. If a monomial from $\mathcal{F}_n$, $n > 0$, is promoted to the SL level or below by the candidate scaling, the proposed $\mathcal{F}_{\text{SL}}$ is incomplete and the analysis must be repeated with that monomial included. For each surviving candidate, it is useful to display the occupied weight layers of the complete rescaled polynomial,

$$\mathcal{P}^{(\text{HR})} \equiv \mathcal{P}(\delta^{\mathbf{v}_{\text{HR}}} \mathbf{x}; \delta, \mathbf{s}) = \sum_k \delta^{W_k} \mathcal{P}_{W_k}(\mathbf{x}, \mathbf{s}), \quad W_k < W_{k+1}. \quad (2.37)$$

The sum runs only over the weights which actually occur. One must not insert fictitious intermediate layers: for example, if a parametrisation produces only even powers of the expansion parameter, the absence of an odd power is not an additional cancellation. In the generic case, $\mathcal{P}_{W_{\text{SL}}} = \mathcal{F}_{\text{SL}}$, and the next resolved occupied weight is W_{HR} .

The natural expectation is that the cancellation suppresses the individually superleading monomials in $\mathcal{F}_{\text{SL}}$ just enough for their sum to enter at the resolved leading weight W_{HR} . Otherwise a sector selected because its monomials are superleading would, after cancellation, become subleading to the resolved leading LP polynomial. The expected saturation occurs in every certified example considered here, and we conjecture that it is a general property of hidden regions satisfying the conditions above. The hierarchy conditions alone do not prove it, however. Logically, the first nonzero term in the expansion of $\mathcal{F}_{\text{SL}}$ in the cancellation-resolving coordinates could lie above W_{HR} , with the resolved leading layer supplied entirely by the rest of the LP polynomial. HRF therefore keeps the transformed $\mathcal{F}_{\text{SL}}$, $\mathcal{F}_{\text{Obs}}$, the remaining $\mathcal{F}$ layers and $\mathcal{U}$ as separate sources in the dissection, measures their first nonzero weights independently, and does not impose the expected equality.

An analogous refinement is needed only if the restored polynomial contains an intervening layer $\mathcal{P}_{W_k}$, with $W_{\text{SL}} < W_k < W_{\text{HR}}$, whose sum itself vanishes on the same pinch component C . Here W_k is the common weight of the monomials in $\mathcal{P}_{W_k}$ before this additional cancellation is resolved; we call it the nominal weight of the layer. Rather than rejecting the candidate by Eq. (2.35), one first tests whether the complete layer $\mathcal{P}_{W_k}$ belongs to the local pinch ideal $\mathcal{I}_C = \langle f_1, \dots, f_r \rangle$. If no intervening layer does so, the generic test above is complete.

If such a layer is present, choose local transverse coordinates $y_j = f_j$ and coordinates $\mathbf{u}$ along C , and expand only as far as its first nonzero transverse term,

$$\mathcal{P}_{W_k}(\mathbf{u}, \mathbf{y}) = \sum_{|\beta|=d_k} \mathbf{y}^\beta h_{k,\beta}(\mathbf{u}) + \mathcal{O}(|\mathbf{y}|^{d_k+1}), \quad y_j \sim \delta^{\tau_j}, \quad \tau_j > 0. \quad (2.38)$$

Each nonzero term at that first transverse order has resolved weight $W_k + \beta \cdot \boldsymbol{\tau}$, which must satisfy

$$W_k + \beta \cdot \boldsymbol{\tau} \geq W_{\text{HR}} > W_{\text{SL}}, \quad (2.39)$$

Thus an intervening layer with $W_k < W_{\text{HR}}$ is admissible only if its vanishing on C suppresses every first nonzero transverse contribution to weight W_{HR} or above. In all examples requiring this refinement, at least one contribution saturates Eq. (2.39), so the layer reaches precisely W_{HR} , just as $\mathcal{F}_{\text{SL}}$ does. HRF tests the more general inequality and does not impose this saturation. This ideal-layer test is a fallback for an explicitly encountered intervening cancellation, not part of the generic determination of $\mathbf{v}_{\text{HR}}$. It becomes particularly relevant when the full δ -dependent polynomial is restored after asymptotic-order alignment, as discussed in Sec. 2.3.

In a chart centred on the full common pinch C , every independent transverse coordinate $y_j = f_j$ tends to zero, so every τ_j is positive. This does not require the hierarchy to determine each τ_j separately. Pullback here means translating the local scaling through the coordinate change to the original LP parameters x_e . A continuous family represents only local ratio freedom if its pullback to the original LP parameters is fixed. If the pullback varies, the HR scaling remains undetermined; the local lower-facet analysis must fix it, otherwise the candidate is rejected.

Scalefulness and dissection. Whether the region integral is scaleful is a property of the resolved monomial support of the LP polynomial. It does not depend on the space-time dimension D . A mere count of variables appearing in selected monomials is a useful diagnostic but is not a general certificate. For N local variables, choose one exponent point $\vec{r}_0$ as a reference. The resolved support has full affine rank when the differences $\vec{r}_i - \vec{r}_0$ contain N linearly independent vectors. This is the dimensionality required for a facet; a smaller rank leaves an additional scaling freedom.

After the cancellations are resolved, HRF always performs the dissection of Ref. [45], briefly described at the end of Sec. 2.2.1. The resulting sign and radial-ratio charts cover the relevant normal directions. In an endpoint chart the leading support of the transformed complete LP polynomial must define an exact lower facet with a positive inward normal. A centred chart may instead expose the lower-dimensional common face of adjacent endpoint facets.¹⁰ These are facets or faces in cancellation-resolving coordinates; in general they need not be faces of the original Newton polytope.

The final certificate is therefore attached to the chart cover, not to one preferred local facet. Local normals are pulled back to the original LP parameters and normalised with last component one. Equivalent charts must give the same total vector there, while their

¹⁰The spacelike-collinear Fish in Sec. 3.2.1 provides the explicit example: its centred face and two bounding endpoint facets all pull back to the same HR vector in the original LP parameters.

ratio coordinates account for any remaining freedom in how the fixed cancellation depth is distributed among the independent transverse coordinates f_j . An endpoint chart can overlap an ordinary facet region, and the corresponding overlap — including any rapidity regulator required by that chart — belongs to the regional decomposition rather than to the definition of the HR.

Before dissection, the active weight system (2.36) determines a candidate total vector. In every example considered below, it already fixes this vector in the original LP parameters; any remaining freedom concerns only ratios among cancellation-resolving coordinates and leaves the pulled-back vector unchanged. Agreement with every relevant pulled-back local normal therefore provides a strong independent check. Thus, in the examples studied here, dissection neither modifies nor rejects an HR that passes the preceding algebraic determination: it supplies the final geometric certificate and exposes its local chart structure. If the homogeneous weight equations were instead to retain a residual rescaling of the total vector or leave a cancellation layer unresolved, their output would remain underdetermined and would not yet constitute a certified HR. The original-coordinate support may likewise display the anticipated facet, but this does not replace the final local certificate.

In the examples below we display adapted local coordinates when they illuminate a nontrivial feature of the cancellation geometry or its certificate — for example, freedom among local charts, a moving or boundary pinch locus, or several independent transverse directions. Throughout the certified examples, dissection confirms the vector determined by the active weight system. Where the local analysis reveals no additional structure, we state this agreement without reproducing the complete coordinate map and local facet analysis.

2.3 Wrappers around the core construction

Two extensions fix the inputs supplied to the core without changing its internal logic. Asymptotic-order alignment selects a nontrivial starting face of the full δ -dependent $\mathcal{F}$ and thereby specifies $\mathcal{F}_*$. A boundary search specifies $\mathbf{x}_{\text{active}}$ by setting the complementary subset of LP parameters to zero, thereby restricting the graph polynomial to a contraction stratum; graphically, this contracts the corresponding edges and merges their incident vertices. Either operation, or both together, is followed by the same decomposition and scaling construction of Sec. 2.2.

2.3.1 Asymptotic-order alignment

When the relevant cancellation structure is not visible in the naive fixed- $\mathbf{x}$ leading polynomial $\mathcal{F}_0$, a preliminary LP-parameter rescaling can bring monomials occurring at different native powers of δ to a common effective leading order before the cancellation decomposition is applied. This need for cross-order alignment is characteristic of the composite limits encountered in the examples below, although the construction does not assume that the limit is composite. This operation is called *asymptotic-order alignment*.

The polynomial relevant to this preliminary operation is the full δ -dependent $\mathcal{F}$, rather than the complete LP polynomial $\mathcal{P} = \mathcal{U} + \mathcal{F}$. As shown in Eq. (2.12), $\mathcal{U}$ -monomials cannot participate in a cancellation producing an interior first-sheet stationary pinch. They

nevertheless remain part of the complete LP problem and are restored when the candidate scaling is tested.

Let $\phi \in \mathbb{Q}^N$ be the preliminary alignment vector and introduce new parameters $\mathbf{y}$ by

$$x_e = \delta^{\phi_e} y_e. \quad (2.40)$$

The corresponding augmented alignment vector is $\vec{\phi} \equiv (\phi; 1)$. For a monomial $m_i = c_i(\mathbf{s})\delta^{a_i}\mathbf{x}^{r_i}$, the rescaling changes its effective expansion order from a_i to

$$\omega_\phi(m_i) \equiv \vec{\phi} \cdot \vec{r}_i = a_i + \phi \cdot \mathbf{r}_i. \quad (2.41)$$

Monomials that belong to different fixed- $\mathbf{x}$ orders in δ can thereby acquire a common effective weight. The relevant value is the minimum of ω_ϕ : the monomials attaining it form the exposed face and provide the support from which the cancellation sector is selected. Before alignment these monomials can occur at different native powers of δ , so they need not all belong to $\mathcal{F}_0$. After alignment they form a common leading polynomial, allowing derivative harvesting to expose their joint cancellation structure.

Selecting the monomials of minimum ω_ϕ turns asymptotic-order alignment into a geometric problem analogous, but not identical, to the conventional geometric Method of Regions. Here the relevant Newton polytope is the convex hull of the augmented exponent points of the full δ -dependent $\mathcal{F}$, and the relevant objects are its exposed lower faces, which need not be facets. The conventional construction instead uses the Newton polytope of the complete LP polynomial $\mathcal{P}$ and associates regions with its lower facets.

HRF enumerates these faces without presupposing the cancellation structure. For asymptotic-order alignment, the relevant faces contain monomials with different native powers of δ . For each such face, one may choose any rational ϕ for which $\vec{\phi}$ lies in the relative interior of its normal cone. All such choices select the same lowest-weight monomials: the alignment is specified by the exposed face, while $\vec{\phi}$ is a convenient normal that exposes it.

For an order-aligned search, let $\mathcal{M}_\star \subseteq \mathcal{M}_\mathcal{F}$ consist of these lowest-weight monomials and define

$$\mathcal{F}^{[\phi]} \equiv \sum_{m_i \in \mathcal{M}_\star} m_i. \quad (2.42)$$

The general HRF starting object is therefore

$$\mathcal{F}_\star = \begin{cases} \mathcal{F}_0, & \text{ordinary search,} \\ \mathcal{F}^{[\phi]}, & \text{order-aligned search.} \end{cases} \quad (2.43)$$

Alignment thus brings these monomials into the modified starting polynomial $\mathcal{F}_\star$, to which the same cancellation-factor and obstruction analysis, followed by the pinch test in the positive orthant, is applied.

Asymptotic-order alignment is not itself the hidden-region scaling. It is the first of two coordinate transformations; HRF then determines the scaling about the cancellation locus in the aligned coordinates.

The core vector and the composition law. Let $\mathbf{v}_{\text{core}} \in \mathbb{Q}^N$ denote the HRF scaling in the aligned variables,

$$y_e = \delta^{(v_{\text{core}})_e} z_e. \quad (2.44)$$

Combining Eqs. (2.40) and (2.44) gives

$$x_e = \delta^{\phi_e + (v_{\text{core}})_e} z_e, \quad \mathbf{v}_{\text{HR}} = \boldsymbol{\phi} + \mathbf{v}_{\text{core}}, \quad \vec{v}_{\text{HR}} = (\boldsymbol{\phi} + \mathbf{v}_{\text{core}}; 1). \quad (2.45)$$

Correspondingly, the alignment and the subsequent HR rescaling assign the same original monomial the successive weights

$$\omega_{\boldsymbol{\phi}}(m_i) = a_i + \boldsymbol{\phi} \cdot \mathbf{r}_i, \quad w_{\text{HR}}(m_i) = a_i + (\boldsymbol{\phi} + \mathbf{v}_{\text{core}}) \cdot \mathbf{r}_i. \quad (2.46)$$

The second is its absolute weight after the composed rescaling.

Only the N edge-parameter exponents are added. The last component is the normalisation of the single expansion parameter and is appended once after the two transformations have been composed. Thus one must never add $\vec{\phi}$ and $(\mathbf{v}_{\text{core}}; 1)$, which would spuriously produce a last component equal to two.

The notation of Sec. 2.2.2 is retained for the composed scaling. Thus, in Eqs. (2.37) and (2.32a),

$$\mathcal{P}^{(\text{HR})} = \mathcal{P}(\delta^{\boldsymbol{\phi} + \mathbf{v}_{\text{core}}} \mathbf{z}; \delta, \mathbf{s}), \quad \mathbf{v}_{\text{HR}} = \boldsymbol{\phi} + \mathbf{v}_{\text{core}}. \quad (2.47)$$

and W_{SL} , W_{HR} and W_k denote absolute weights in the complete native polynomial. Likewise, $\mathcal{F}_{\text{SL}}^{(\text{HR})}$ denotes the selected $\mathcal{F}_{\text{SL}} \subset \mathcal{F}^{[\boldsymbol{\phi}]}$ after the complete rescaling of the original LP parameters, $\mathbf{x} = \delta^{\boldsymbol{\phi} + \mathbf{v}_{\text{core}}} \mathbf{z}$. The simplified expression $\mathcal{F}_{\text{SL}}^{(\text{HR})} = \delta^{W_{\text{SL}}} \mathcal{F}_{\text{SL}}$ in Eq. (2.32a) applies directly to the unaligned core case.

Normally the aligned face alone determines the core vector. Let $\mathbf{v}_{\text{core}}^{(0)}$ denote the vector obtained at this stage. Because the aligned search initially retains only this face of $\mathcal{F}$, $\mathbf{v}_{\text{core}}^{(0)}$ remains provisional until the omitted layers of the complete LP polynomial are restored. If a restored layer contributes at a lower weight and does not vanish on the same pinch locus, the candidate is rejected. In the exceptional case where the entire layer also cancels there, it must instead be included in the simultaneous hierarchy analysis of Eq. (2.39), which may modify the provisional vector. In the generic case $\mathbf{v}_{\text{core}} = \mathbf{v}_{\text{core}}^{(0)}$; in either case the resulting $\mathbf{v}_{\text{core}}$ is then combined with $\boldsymbol{\phi}$ in Eq. (2.45).¹¹

Uniform-shift ambiguity. The second Symanzik polynomial is homogeneous of fixed parameter degree. Consequently,

$$\boldsymbol{\phi} \mapsto \boldsymbol{\phi} + c(1, \dots, 1) \quad (2.48)$$

adds the same amount to the weight of every monomial of $\mathcal{F}$ and does not change the face $\mathcal{F}^{[\boldsymbol{\phi}]}$. The preliminary alignment vector therefore has this universal equivalence in addition to the general freedom to choose a representative within the normal cone.

¹¹This exceptional multi-layer refinement occurs in the double spacelike-collinear twisted-hexagon example of Sec. 3.3: the middle layer in Eq. (3.200) vanishes to first order on the common pinch and changes the face-only vector to the final result in Eq. (3.202).

The coordinate transformation itself shows how this ambiguity is removed. For fixed original $\mathbf{x}$, replacing ϕ by $\phi + c\mathbf{1}$ replaces the aligned coordinates by $\mathbf{y}' = \delta^{-c}\mathbf{y}$. The same second transformation is therefore represented by

$$\mathbf{v}'_{\text{core}} = \mathbf{v}_{\text{core}} - c\mathbf{1}, \quad (\phi + c\mathbf{1}) + \mathbf{v}'_{\text{core}} = \phi + \mathbf{v}_{\text{core}}. \quad (2.49)$$

Hence the physical total vector is independent of the chosen alignment representative, provided the core vector is transformed consistently.

In practice this compensation is fixed by the complete LP polynomial. The polynomials $\mathcal{U}$ and $\mathcal{F}$ have degrees L and $L + 1$, respectively, so a uniform shift changes their relative weights. Requiring the resolved cancellation layer to balance the appropriate $\mathcal{U}$ layer, while satisfying the hierarchy conditions of Sec. 2.2.2, fixes the representative and the physical gap $W_{\text{HR}} - W_{\text{SL}}$. This gap is a property of the region and is not an adjustable normalisation.

The consistent two-stage procedure is therefore:

- (i) enumerate the relevant exposed faces of the augmented support and choose a convenient representative ϕ for each;
- (ii) run HRF in the aligned variables to obtain the cancellation ideal and a provisional $\mathbf{v}_{\text{core}}^{(0)}$;
- (iii) restore $\mathcal{U}$ and every occupied layer of the original δ -dependent $\mathcal{F}$, test the generic monomial hierarchy, and invoke the ideal-layer refinement only if an intervening layer vanishes on the same pinch component;
- (iv) compose the edge exponents according to $\mathbf{v}_{\text{HR}} = \phi + \mathbf{v}_{\text{core}}$, append the final component one, then dissect the common pinch, certify the resolved local lower facet and scalefulness, and pull its normal back.

2.3.2 Boundary hidden regions

An interior HR has all $x_e > 0$. A boundary HR occurs on a contraction stratum

$$x_{e_1} = \cdots = x_{e_k} = 0. \quad (2.50)$$

The graph polynomial is first restricted to that stratum, the vanished variables are removed, and the complete analysis is repeated with every remaining active parameter positive. In graph language this is the corresponding contraction minor; in Landau language it is a subleading Landau singularity. The contraction is performed on the externally labelled graph, so the momentum label and orientation of every external half-edge are retained in the contracted graph.

A complete search must include the relevant contraction strata. Failure to find an interior HR is not a proof that the graph has no HR.

If a contraction stratum also requires asymptotic-order alignment, the restriction is made first and the alignment faces are then constructed from the surviving support. The resulting $\mathcal{F}_\star$ is passed to the core HRF construction. Thus boundary restriction and order alignment are independent wrappers which may be nested without modifying the core logic.

2.4 Equivalence and completeness

The HRF construction may produce several algebraic presentations of the same physical hidden region. Conversely, establishing that no hidden region exists requires a search that is independent of any preferred presentation of the cancellation factors. We formulate the corresponding equivalence and completeness criteria here.

2.4.1 Equivalence of presentations

A physical HR is specified by a positive component of the common pinch locus and the chart-independent scaling data obtained by resolving its normal directions, in particular the resulting total vector in the original LP parameters. A lower facet belongs to a chosen dissection chart and is not by itself an invariant label. Facets in different charts, or a common face in a centred chart and its bounding endpoint facets, represent the same HR when they resolve the same positive component and their normals pull back to the same original-coordinate scaling. In particular, neither a change of local generators of the cancellation ideal nor a redistribution of the fixed cancellation depth among normal coordinates produces a new region.

Distinct HRs may instead arise from:

- different cancellation ideals, corresponding to different singular hypersurface systems; or
- inequivalent resolved normal directions, giving different pulled-back total scaling vectors.

If a cancellation locus has several disconnected positive branches, each must be tested separately; no such case occurs in the examples below. The accepted constructions must therefore be classified only after identifying local facets related by regular chart changes and comparing their pullbacks to the original LP parameters.

2.4.2 A scaling-first formulation of a hidden-region absence statement

The constructive HRF strategy described above is cancellation-first: it uses derivatives of $\mathcal{F}_\star$ to identify candidate cancellation factors, constructs the generators and $\mathcal{F}_{\text{SL}}$, and only then determines the scaling. A conceptually interesting alternative is to reverse this order and begin with the supports exposed by possible scalings. This scaling-first route is presentation-independent at the level of candidate supports, but it has not been implemented or tested.

The homogeneity condition (2.34), together with strict separation from the remaining monomials, implies that the exponent support of $\mathcal{F}_{\text{SL}}$ is an exposed face of the relevant starting support. This observation suggests the following formally exhaustive scaling-first search:

1. enumerate the relevant exposed faces of $\mathcal{F}_0$, or of the augmented support of the full δ -dependent $\mathcal{F}$ when asymptotic-order alignment is required;

2. for each face polynomial, solve the simultaneous positive Landau equations in the physical domain $\mathcal{K}$;
3. test every positive solution against $\mathcal{U}$, all restored δ -dependent layers and the hierarchy conditions, then apply source-resolved dissection using a cover of the relevant sign sectors and projectivised normal directions — normal directions considered modulo an overall rescaling — and test the resulting local lower-facet or common-face criteria;
4. repeat the analysis on all required contraction strata.

This construction should be regarded as a theoretical alternative, not as the standard used for negative audits in this work. In particular, the four-loop No-Crown absence results of Secs. 3.1.3 and 3.1.4 were obtained by exhaustive application of the derivative-first HRF search, rather than by the scaling-first procedure above. More generally, an absence statement requires exhaustive coverage of the relevant possibilities within whichever search strategy is used. A non-exhaustive search may discover HRs, but cannot establish their absence.

2.5 Algorithm summary

The complete conceptual workflow is: steps 1–3 configure the physical problem and its optional wrappers, steps 4–10 apply the two parts of the core HRF construction, and steps 11–12 manage equivalence and completeness at search level.

1. **Fix the kinematic chart.** Specify the expansion parameter $\delta \rightarrow 0^+$, a common set of real kinematic coordinates $\mathbf{s}$, and the inequalities defining the physical domain $\mathcal{K}$. Introduce positive coordinates only locally if they simplify a particular positivity test. Otherwise, retaining common signed coordinates explicitly reveals how cancellation loci appear or disappear under crossing or analytic continuation.
2. **Choose a stratum.** Start in the interior and repeat on the required contraction strata.
3. **Choose the starting face $\mathcal{F}_\star$.** Use $\mathcal{F}_0$ in the ordinary case. If the relevant cancellation may involve monomials at different native powers of δ , consider the order-aligned exposed faces $\mathcal{F}^{[\phi]}$ of the full δ -dependent $\mathcal{F}$.
4. **Generate candidate cancellation structures.** Form the derivatives of $\mathcal{F}_\star$ and apply the mixed-sign test only as a necessary prefilter. Harvest non-monomial cancellation factors directly by literal factorisation and, when compatible with the kinematic organisation, by collecting the derivatives into independent kinematic structures and factorising their LP-parameter coefficients. Do not require this coefficientwise decomposition or treat it as sufficient. Do not solve the simultaneous derivative equations of $\mathcal{F}_\star$, nor saturate their full ideal: they generically have no positive solution at this stage when an obstruction is present.
5. **Build the massless generator layer.** Form candidate generators as products of pairs of distinct harvested factors, $g_k = f_i f_j$, with $i \neq j$. Retain only pairs whose

LP-parameter supports are disjoint and whose expanded products pass the LP-support test against $\mathcal{F}_\star$. Form the candidate ideal $\mathcal{I}_{\text{gen}}(G)$ from the surviving generators.

6. **Find the obstruction and construct the decomposition.** For each candidate ideal $\mathcal{I}_{\text{gen}}(G)$, search for subsets $\mathcal{M}_{\text{Obs}} \subseteq \mathcal{M}_\star$ whose removal leaves a nonzero $\mathcal{F}_{\text{SL}} \in \mathcal{I}_{\text{gen}}(G)$, with $\mathcal{M}_{\text{SL}} = \mathcal{M}_\star \setminus \mathcal{M}_{\text{Obs}}$. Discard G if no such subset exists.
7. **Apply the first-sheet pinch test to $\mathcal{F}_{\text{SL}}$.** Only after $\mathcal{M}_{\text{SL}}$ has been isolated, solve $\partial \mathcal{F}_{\text{SL}} / \partial x_e = 0$ simultaneously for all active parameters with $\mathbf{x} > 0$ and $\mathbf{s} \in \mathcal{K}$. The resulting common locus in the positive orthant defines the physical cancellation ideal; reject the candidate if this system has no solution. When useful, complete these selected-sector equations, or explicitly selected derivative-sector relations, by the saturation (2.28).
8. **Determine a candidate scaling.** Restore the complete $\mathcal{P} = \mathcal{U} + \mathcal{F}$ and solve the homogeneity and single-layer hierarchy conditions (2.34) and (2.35) for $\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1)$, W_{SL} and W_{HR} . Reject an inconsistent system or a solution with zero gap. A full-rank system fixes a unique candidate vector; a rank-deficient system remains provisional until dissection.
9. **Inspect the scaled layers and resolve exceptional cases.** For each candidate vector, form the rescaled LP polynomial $\mathcal{P}^{(\text{HR})} = \mathcal{P}(\delta^{\mathbf{v}_{\text{HR}}} \mathbf{x}; \delta, \mathbf{s})$ and decompose it into its occupied weight layers as in Eq. (2.37). If the first occupied layer above W_{SL} is nonzero on the common pinch component, the case is ordinary and the single-layer hierarchy is sufficient. If one or more occupied layers above W_{SL} vanish there before the first resolved nonzero weight, determine the first nonzero order of each such layer in the local pinch ideal $\mathcal{I}_C = \langle f_1, \dots, f_r \rangle$ and impose the inequalities (2.39). Apply this ideal-layer refinement to every intervening vanishing layer and reject an inconsistent system or a solution with zero gap. This determines a candidate scaling, whose local lower-facet and scalefulness certificate is supplied by the dissection in the next step.
10. **Dissect the common pinch and certify the total vector.** For every candidate surviving either hierarchy, apply the source-resolved dissection while keeping $\mathcal{F}_{\text{SL}}$, $\mathcal{F}_{\text{Obs}}$, the remaining $\mathcal{F}$ layers and $\mathcal{U}$ separate. Cover the relevant sign and normal-ratio sectors, measure the first nonzero local weight of each source, and certify the exact local lower facets or common faces. Pull their normals back to the original LP parameters. The resulting total vector and positive gap, together with the required treatment of endpoint overlaps, provide the final HRF certificate; agreement with an original-coordinate determination is an independent check.
11. **Identify physical equivalence classes.** Merge generator presentations that describe the same pinch component in the positive orthant and local chart facets whose normals have the same original-coordinate pullback.

- 12. State completeness honestly.** A positive HR requires all certification stages. A no-HR conclusion requires exhaustive face and boundary coverage; incomplete searches remain unresolved.

3 Hidden Region Finder: Applications

This section demonstrates the Hidden Region Finder through examples chosen to isolate distinct algorithmic requirements. The point is not merely to catalogue hidden regions, but to make clear which stages of the search are needed as the simplifying features of the most elementary case are removed. We begin with the four-point three-loop Crown. This is the simplest HR from the viewpoint of discovery: the complete leading polynomial participates in a simultaneous positive cancellation, so that $\mathcal{F}_\star = \mathcal{F}_{\text{SL}}$, no obstruction is present and the HR has a uniform scaling. Its nontrivial discovery step is the separation of two generator sectors carrying the independent kinematic coefficients s_{12} and s_{23} ; only after this separation does factorisation expose the individual binomial cancellation factors. Its higher-loop descendants then illustrate hidden regions on contraction strata and show that the corresponding factors can instead contain three or more monomials. The No-Crown analysis provides a converse test, while the Regge limits test the correspondence across kinematic channels. We next turn to the five-point two-loop Fish family, based on the topology commonly called the non-planar double box. Its seed already contains a nonzero obstruction and therefore requires a non-uniform parameter scaling. A vertex-corrected descendant shows how its cancellation locus survives a local loop correction through a polynomial lift. The near-planar wide-angle limit provides the closest five-point parallel to the Crown and is the only example here with a three-generator cancellation ideal. The subsequent rapidity-ordered and correlated MRK limits show the same unlabelled Fish seed recurring in distinct kinematic expansions, while their different region vectors demonstrate that the scaling is fixed by the details of the expansion, not by the topology alone. They separately expose the need for asymptotic-order alignment and, in the MRK-planar case, multiple cancellation layers. Finally, the six-point twisted-box seed and its twisted-hexagon realisation combine several of these ingredients. The NMRK and double-collinear regions inherit their cancellation geometry from a boundary HR on the physical wide-angle self-crossing surface, while their different total vectors again reflect the kinematic expansions used to reach that surface.

3.1 Four-point integrals: the Crown seed

The four-point applications are organised by the three-loop Crown, $\mathcal{G}_C$. We first identify its interior wide-angle HR, then follow the same seed onto contraction strata of higher-loop graphs, complete the converse No-Crown audit, and finally compare the resulting topology classification with the three Regge channels.

3.1.1 Wide-angle kinematics and the three-loop seed

The Crown hidden region in the on-shell wide-angle expansion was first identified and analysed in Ref. [45]. From the viewpoint of the finding problem, it is the simplest example

considered here. The full leading polynomial is the cancellation sector, with no obstructing terms, and the resulting HR scaling is uniform. It is nevertheless not a single-generator example: its derivatives mix two products of cancellation factors, weighted by independent kinematic invariants. We first show how the derivative harvest separates these generator sectors and then certify the region vector. This provides the baseline against which the additional search steps required by the later examples can be distinguished, as well as the seed for the subsequent boundary and topology audits.

Take four external momenta approaching the physical $2 \rightarrow 2$ domain, with limiting invariants satisfying

$$s_{12} > 0, \quad s_{23} < 0, \quad s_{12} > -s_{23}, \quad (3.1)$$

and regulate the on-shell limit by $p_i^2 = \delta P_i^2$, with fixed nonzero P_i^2 and $\delta \rightarrow 0^+$. Momentum conservation gives the exact off-shell relation $s_{12} + s_{23} + s_{13} = \sum_{i=1}^4 p_i^2 = \delta \sum_{i=1}^4 P_i^2$, which reduces to the usual massless relation only in the limit. A common normalisation of the P_i^2 is immaterial for the region vector. The graph and its LP-parameter assignment are shown in Fig. 1.

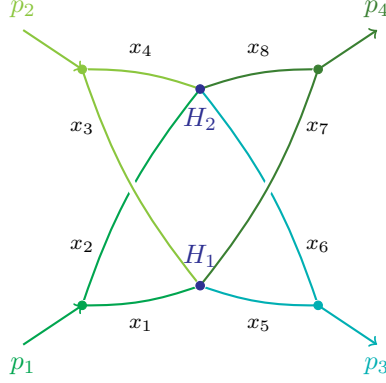


Figure 1. The three-loop Crown $\mathcal{G}_C$ with its LP-parameter assignment and wide-angle momentum configuration. The four shades identify the four collinear directions: each external momentum and the corresponding two-edge path have the same colour. The blue four-point vertices H_1 and H_2 are the two disconnected hard-scattering components.

Each coloured two-edge path carries one of the four collinear directions between the blue vertices H_1 and H_2 . These vertices represent two separate hard scatterings connected by the four jet-like paths. This is the essential Landshoff feature of the Crown HR [45]. It cannot occur in a facet region of the wide-angle on-shell expansion, for which the hard subgraph is necessarily connected [43, 44].

Here and below, displayed LP polynomials use the weight-layer colour convention of Sec. 2.2.2. At fixed parameters the leading polynomial is

$$\begin{aligned} \mathcal{F}_0[\mathcal{G}_C] &= s_{12}g_1[\mathcal{G}_C] - s_{23}g_2[\mathcal{G}_C], \\ g_1[\mathcal{G}_C] &= f_1f_2, & g_2[\mathcal{G}_C] &= f_3f_4, \\ f_1 &= x_4x_5 - x_3x_6, & f_2 &= x_2x_7 - x_1x_8, \\ f_3 &= x_2x_3 - x_1x_4, & f_4 &= x_6x_7 - x_5x_8. \end{aligned} \quad (3.2)$$

Here the four f_i provide the first explicit example of the cancellation factors introduced in Sec. 2.2.1. The full derivatives do not factorise directly into them because they mix the two generator sectors. For example,

$$\frac{\partial \mathcal{F}_0[\mathcal{G}_C]}{\partial x_1} = -s_{12}x_8f_1 + s_{23}x_4f_4, \quad \frac{\partial \mathcal{F}_0[\mathcal{G}_C]}{\partial x_5} = s_{12}x_4f_2 + s_{23}x_8f_3.$$

Separating the coefficients of the independent kinematic structures and omitting the monomial prefactors, which are nonzero in the positive orthant, therefore harvests all four f_i . Pairing factors with disjoint LP-parameter support gives the two generators $g_1[\mathcal{G}_C] = f_1f_2$ and $g_2[\mathcal{G}_C] = f_3f_4$. The corresponding candidate generator ideal is $\mathcal{I}_{\text{gen}}(\mathcal{G}_C) = \langle g_1[\mathcal{G}_C], g_2[\mathcal{G}_C] \rangle$. The positive pinch test then selects the component of its zero set on which all four f_i vanish simultaneously. Once the two generator sectors multiplying s_{12} and $-s_{23}$ have been isolated, the candidate common cancellation locus $f_1 = f_2 = f_3 = f_4 = 0$ is visible directly in the full leading polynomial, without first removing an obstruction.

To display the structure of this locus in its simplest form, we now introduce a ratio parametrisation. This is an analytic description of the result, not the parametrisation used in the HRF search. Regard the graph as four two-edge paths between the two four-point vertices H_1 and H_2 . Factoring one positive overall LP-parameter scale from each path leaves the four ratios

$$r_A = \frac{x_1}{x_2}, \quad r_B = \frac{x_3}{x_4}, \quad r_C = \frac{x_5}{x_6}, \quad r_D = \frac{x_7}{x_8}. \quad (3.3)$$

We collect them in the column vector $\mathbf{r} \equiv (r_A, r_B, r_C, r_D)^T$. Then

$$\begin{aligned} \mathcal{F}_0[\mathcal{G}_C] &= x_2x_4x_6x_8 \Psi_C(\mathbf{r}), \\ \Psi_C(\mathbf{r}) &= s_{12}(r_C - r_B)(r_D - r_A) - s_{23}(r_B - r_A)(r_D - r_C) = \frac{1}{2}\mathbf{r}^T K_C \mathbf{r}, \end{aligned} \quad (3.4)$$

$$K_C = \begin{pmatrix} 0 & s_{12} & -s_{12} - s_{23} & s_{23} \\ s_{12} & 0 & s_{23} & -s_{12} - s_{23} \\ -s_{12} - s_{23} & s_{23} & 0 & s_{12} \\ s_{23} & -s_{12} - s_{23} & s_{12} & 0 \end{pmatrix}.$$

The ratio Landau equations are $K_C \mathbf{r} = 0$. Every row of K_C sums to zero, so $(1, 1, 1, 1)^T$ is a null vector and $\det K_C = 0$. On the other hand, in the domain (3.1) a principal three-by-three minor is nonzero:

$$\det(K_C)_{1\dots 3, 1\dots 3} = -2s_{12}s_{23}(s_{12} + s_{23}) \neq 0. \quad (3.5)$$

Consequently K_C has rank three and its complete stationary locus is

$$\mathbf{r} = \rho(1, 1, 1, 1), \quad \rho > 0, \quad \Psi_C(\mathbf{r}) = 0. \quad (3.6)$$

Here $(1, 1, 1, 1)^T$ specifies the null direction and ρ is the single positive scalar coordinate along it, namely the common value of the four path ratios. Physically, ρ sets the common division of the momentum from each jet between the hard scatterings H_1 and H_2 . The last

equality follows directly from $\Psi_C = \mathbf{r}^T K_C \mathbf{r} / 2$. Thus a positive stationary family exists at every generic point of the physical wide-angle domain; no additional kinematic condition is required. In the analytic treatment of Ref. [45], this ratio change is the rescaling step that precedes the local dissection of the Crown pinch. It turns each binomial factor in (3.2) into a positive path scale times a linear difference of two ratio points. Their common zero is $r_A = r_B = r_C = r_D$, which is fixed by only three relative distances, for example $r_B - r_A$, $r_C - r_A$ and $r_D - r_A$. The four displayed binomial constraints therefore define a codimension-three common locus. This establishes the structure of the cancellation locus, but the decomposition follows separately from Eq. (3.2): $\mathcal{F}_0[\mathcal{G}_C] = s_{12}g_1[\mathcal{G}_C] - s_{23}g_2[\mathcal{G}_C]$ belongs entirely to $\mathcal{I}_{\text{gen}}(\mathcal{G}_C)$, with no remaining monomials to form an obstruction. Therefore

$$\mathcal{F}_\star[\mathcal{G}_C] = \mathcal{F}_0[\mathcal{G}_C] = \mathcal{F}_{\text{SL}}[\mathcal{G}_C], \quad \mathcal{F}_{\text{Obs}}[\mathcal{G}_C] = 0. \quad (3.7)$$

At fixed s_{12} and s_{23} , the external-virtuality expansion is exactly $\mathcal{F}[\mathcal{G}_C] = \mathcal{F}_0[\mathcal{G}_C] + \delta\mathcal{F}_1[\mathcal{G}_C]$, where

$$\mathcal{F}_1[\mathcal{G}_C] = \sum_{i=1}^4 P_i^2 \frac{\partial \mathcal{F}[\mathcal{G}_C]}{\partial p_i^2} \Big|_{p_1^2=\dots=p_4^2=0}.$$

Thus $\mathcal{F}_1$ is linear in the four fixed virtuality coefficients, with the corresponding two-forest polynomials as coefficients. To determine the HR vector, the monomials of $\mathcal{F}_{\text{SL}}[\mathcal{G}_C] = \mathcal{F}_0[\mathcal{G}_C]$ are assigned a common weight according to Eq. (2.34), while the $\delta\mathcal{F}_1[\mathcal{G}_C]$ and $\mathcal{U}[\mathcal{G}_C]$ monomials at resolved-leading weight saturate the hierarchy in Eq. (2.35). The resulting linear system forces the edge part of the region vector to take the form $\mathbf{v}_{\text{HR}} = v(1, \dots, 1)$. Since $\mathcal{F}_1$ has degree four and $\mathcal{U}$ degree three, their equality at resolved-leading weight gives $1 + 4v = 3v$, and hence $v = -1$. Using the augmented-vector convention of Sec. 2.2.2, $\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1)$, the resulting vector and weights are

$$\begin{aligned} \vec{v}_{\text{HR}}[\mathcal{G}_C] &= (-1, -1, -1, -1, -1, -1, -1, -1; 1), \\ (W_{\text{SL}}, W_{\text{HR}}) &= (-4, -3). \end{aligned} \quad (3.8)$$

Consequently, applying the definition in Eq. (2.31), the complete LP polynomial after the HR rescaling is

$$\begin{aligned} \mathcal{P}^{(\text{HR})}[\mathcal{G}_C](\mathbf{x}; \delta, \mathbf{s}) &\equiv \mathcal{P}[\mathcal{G}_C](\delta^{-1}\mathbf{x}; \delta, \mathbf{s}) \\ &= \delta^{-4} \mathcal{F}_0[\mathcal{G}_C] + \delta^{-3} (\mathcal{F}_1[\mathcal{G}_C] + \mathcal{U}[\mathcal{G}_C]). \end{aligned} \quad (3.9)$$

Because the second Symanzik polynomial is linear in the external virtualities and every p_i^2 is proportional to δ , this expansion terminates at $\mathcal{F}_1$. Thus, in the occupied-layer terminology of Eq. (2.37), the complete rescaled Crown polynomial has exactly two nonempty layers. The weight-4 layer $\mathcal{P}_{-4}[\mathcal{G}_C] = \mathcal{F}_0[\mathcal{G}_C]$ is the superleading cancellation sector, while the weight-3 layer $\mathcal{P}_{-3}[\mathcal{G}_C] = \mathcal{F}_1[\mathcal{G}_C] + \mathcal{U}[\mathcal{G}_C]$ is the resolved-leading layer. There is no remaining layer in this example. Neither $\mathcal{F}_0[\mathcal{G}_C]$, the factors f_i , nor the generator ideal depends on the P_i^2 . For the fixed native scaling $p_i^2 \sim \delta$, any proper subset of the external legs may therefore be kept strictly on shell, provided at least one nonzero virtuality remains. In the Crown this changes $\mathcal{F}_1[\mathcal{G}_C]$, but not the cancellation locus or the vector in Eq. (3.8). The nonzero virtualities remain useful as regulators of the ordinary infrared singularities.

The role of the virtuality layer in Part II, despite its absence from the Part-I cancellation locus, becomes clearer upon generalising its native order. Suppose the nonzero virtualities scale as $p_i^2 = \delta^{a_i} P_i^2$, and let $a_{\min} = \min_i a_i$. The first virtuality layer contains the terms from the legs with $a_i = a_{\min}$. For the Crown, the two-forest support supplied by any nonempty set of such legs is sufficient to complete the hierarchy system; their number does not affect its solution. The resolved-leading balance is $a_{\min} + 4v = 3v$, and therefore

$$\begin{aligned} \mathbf{v}_{\text{HR}} &= -a_{\min}(1, \dots, 1), & \vec{v}_{\text{HR}} &= (\mathbf{v}_{\text{HR}}; 1) = (-a_{\min}, \dots, -a_{\min}; 1), \\ (W_{\text{SL}}, W_{\text{HR}}) &= (-4a_{\min}, -3a_{\min}). \end{aligned}$$

Terms with $a_i > a_{\min}$ occur at higher weights and do not alter this augmented vector. Equivalently, using $\lambda = \delta^{a_{\min}}$ as the expansion parameter gives the normalised augmented vector $(-1, \dots, -1; 1)$. A trial branch that omits the first virtuality layer is instead rank deficient: after dissection its resolved support contains no monomial of positive native δ -degree and spans only a lower-dimensional face, not a scaleful lower facet. Restoring the $a_{\min}$ layer fixes the scaling and completes the facet. This conclusion uses the support of the Crown virtuality polynomials and is not asserted here as a topology-independent rule.

The Crown HR corresponds in momentum space to the Landshoff independent-scattering configuration. The central collinear directions can be read directly from the on-shell Landau conditions. In the Coleman–Norton picture of a physical-region pinch, the pinched propagators describe on-shell classical propagation [42, 55]. For each massless two-edge path of the Crown this picture becomes degenerate: the two propagator momenta are null at the pinch and differ by the null external momentum attached to that path. Their scalar product with that external momentum therefore vanishes, which on the physical real branch places both momenta on its collinear ray. These four rays are indicated by the four shades in Fig. 1.

To obtain a momentum-space measure one must additionally determine the local widths about these rays and their correlations. The analysis of Ref. [45]¹² adopted the corresponding jet-like fluctuation envelopes and then used momentum conservation and the pinch constraints to fix the correlated longitudinal support. Its momentum-space and dissected parameter-space counts give the same leading degree, providing a strong consistency check of the mode realisation.

This completes the baseline HRF case: after separating two kinematic coefficient sectors, the derivative harvest exposes the two generators and their simultaneous positive zero; the selected cancellation sector retains all of $\mathcal{F}_0[\mathcal{G}_C]$, and the hierarchy admits the uniform scaling certified by the dissection. No obstruction removal or preliminary asymptotic-order alignment is needed. The later examples are designed to show what must be added when one or more of these simplifications do not occur.

3.1.2 Boundary descendants: SuperCrown and HyperCrown

Ref. [45] had already observed that the Crown organises the three- and four-loop graphs that can support wide-angle HRs. At three loops, the complete class identified there comprises

¹²The momentum-space analysis there builds on the earlier independent-scattering treatment of Botts and Sterman [48].

the Crown itself and nine graphs obtained by opening either or both of its four-point hard vertices. At four loops, the mixed-sign screen retained 1097 topologies: 1081 contain the Crown as a contraction minor, whereas 16 do not. Because this sign pattern is only a necessary prefilter, the census did not by itself certify an HR in every Crown-containing graph. Ref. [45] therefore conjectured that, through four loops in massless $2 \rightarrow 2$ scattering, HRs occur precisely for the topologies in which the Crown structure can be exposed by contractions.

Contractions are performed within the externally labelled parent graph, as in Sec. 2.3.2. For the Crown, however, the external labels impose no additional restriction: its symmetry relates the external-leg permutations, and the corresponding crossed channels all support the HR. In the Crown family below, a Crown contraction minor may therefore be identified at the level of the unlabelled topology. The stronger externally labelled condition becomes essential for less symmetric seeds, such as the Fish, for which the incident seed vertex and momentum assignment must also be preserved.

Here we sharpen both sides of this picture with exact HRF statements. On the positive side, the examples below certify boundary HRs in representative four- and five-loop Crown descendants. Conversely, Sec. 3.1.3 gives complete absence certificates for the 16 exceptional four-loop topologies. In the positive examples below, setting selected additional LP parameters to zero contracts edges without reducing the loop order of the parent graph: it exposes the disconnected Crown hard structure while the remaining propagators accommodate the additional loop momenta.

We denote the SuperCrown and HyperCrown by $\mathcal{G}_{\text{SC}}$ and $\mathcal{G}_{\text{HC}}$, respectively. The two graphs and their parameter assignments are shown in Fig. 2.

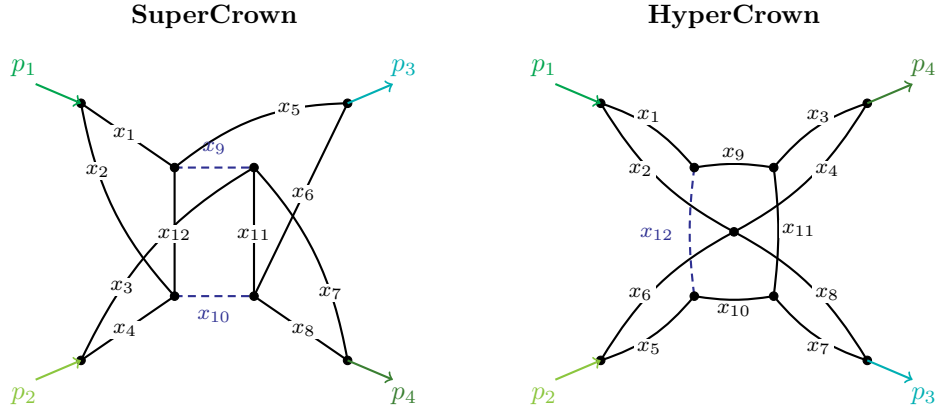


Figure 2. Higher-loop Crown descendants: the five-loop SuperCrown $\mathcal{G}_{\text{SC}}$ (left) and the four-loop HyperCrown $\mathcal{G}_{\text{HC}}$ (right). Blue dashed edges are set to zero on the representative contraction strata used below: $x_9 = x_{10} = 0$ for the SuperCrown and $x_{12} = 0$ for the HyperCrown.

The boundary is essential to this inheritance mechanism, rather than merely a convenient simplification of the parent graph. The Crown cancellation geometry is tied to four jet-like flows meeting at two disconnected hard components. Contracting the indicated edges merges their incident vertices and exposes precisely this hard structure. The propagators that

remain beyond the Crown minor must then carry momenta compatible with the inherited flow, without joining the two hard components into a single connected hard subdiagram. This momentum-space picture motivates the boundary strata shown in Fig. 2, which we test below. Their HR status is established first in parameter space, where this compatibility is reflected in the way the extra active parameters enter the cancellation and scaling equations.

The inheritance has two distinct algebraic signatures. Additional active parameters can occur as common positive monomial factors of the superleading polynomial, thereby altering its weight but not its cancellation locus. They can instead enter longer cancellation factors that remain multihomogeneous in the relevant parameter groups and reduce, upon further contraction, to monomial multiples of the Crown factors. The parameter-space analysis below exhibits both patterns; their momentum-space interpretation is given subsequently.

SuperCrown: a codimension-two inherited Crown. Applying the boundary wrapper of Sec. 2.3.2 at $x_9 = x_{10} = 0$, followed by the Part-I decomposition test of Sec. 2.2.1, HRF finds

$$\mathcal{F}_{\text{SL}}[\mathcal{G}_{\text{SC}}; x_9 = x_{10} = 0] = x_{11}x_{12} \mathcal{F}_{\text{SL}}[\mathcal{G}_{\text{C}}](x_1, \dots, x_8), \quad \mathcal{F}_{\text{Obs}}[\mathcal{G}_{\text{SC}}; x_9 = x_{10} = 0] = 0. \quad (3.10)$$

Thus the surviving superleading polynomial is nothing but the Crown polynomial, multiplied by the two additional active edge parameters. For the cancellation locus, the common positive monomial factor $x_{11}x_{12}$ is immaterial. The two generator polynomials are therefore precisely $g_1[\mathcal{G}_{\text{C}}]$ and $g_2[\mathcal{G}_{\text{C}}]$ defined in Eq. (3.2), with the same simultaneous positive component as the Crown. Solving the Part-II hierarchy and certifying the resulting lower facet gives, in the active order $(x_1, \dots, x_8, x_{11}, x_{12})$,

$$\begin{aligned} \vec{v}_{\text{HR}}[\mathcal{G}_{\text{SC}}; x_9 = x_{10} = 0] &= (-1, -1, -1, -1, -1, -1, -1, -1, -1, -1; 1), \\ (W_{\text{SL}}, W_{\text{HR}}) &= (-6, -5). \end{aligned} \quad (3.11)$$

The weights include the monomial factor $x_{11}x_{12}$. HRF finds a single HR on this stratum. This is the simplest form of parametric inheritance: the boundary exposes the Crown cancellation ideal, while the parameters x_{11} and x_{12} factor out as a positive monomial. They change the overall weights but not the positive cancellation locus.

HyperCrown: two boundary-HR types. The central square of the HyperCrown has dihedral symmetry D_4 . External, edge and kinematic labels are permuted together. Consequently one direct positive witness establishes its entire orbit, including crossing-related representatives. Applying the boundary wrapper and core HRF construction to one representative of each D_4 orbit gives the following wide-angle results through codimension two:

Type	Representative boundary	Certified $\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1)$	Full orbit and conclusion
I	$x_{12} = 0$	$v_{11} = -2, v_e = -1$ otherwise; $(W_{\text{SL}}, W_{\text{HR}}) = (-6, -5)$	four single-edge boundaries $x_e = 0, e = 9, \dots, 12$; one HR on each
II	$x_9 = x_{11} = 0$	ten active entries $v_e = -1$; $(W_{\text{SL}}, W_{\text{HR}}) = (-5, -4)$	$\{x_9, x_{11}\}, \{x_{10}, x_{11}\}, \{x_{10}, x_{12}\}, \{x_9, x_{12}\}$; one HR on each boundary
Excluded	$x_9 = x_{10} = 0$	no admissible HR scaling	$\{x_9, x_{10}\}, \{x_{11}, x_{12}\}$; no HR on either boundary

Table 1. Certified HyperCrown boundary-HR orbits through codimension two.

Here Types I and II label boundary-stratum orbits; Table 1 specifies one representative of each orbit. Graphically, each displayed boundary condition $x_e = 0$ contracts edge e , whose parameter is therefore absent from the active variables in the Landau-locus search.

The two types have different layer structures because their certified vectors are different. For Type I, $v_{11} = -2$, whereas every other active parameter has weight -1 . Since $\mathcal{F}_0[\mathcal{G}_{\text{HC}}]$ is homogeneous of degree five, its terms proportional to x_{11} have weight -6 , while the surviving terms independent of x_{11} have weight -5 . In the notation of Sec. 2.2.2, the Type-I rescaled LP polynomial is

$$\begin{aligned} \mathcal{P}^{(\text{HR})}[\mathcal{G}_{\text{HC}}; \mathbf{I}](\mathbf{x}; \delta, \mathbf{s}) &\equiv \mathcal{P}[\mathcal{G}_{\text{HC}}](\delta^{\mathbf{v}_{\text{HR}}[\mathcal{G}_{\text{HC}}; \mathbf{I}]} \mathbf{x}; \delta, \mathbf{s}) \Big|_{x_{12}=0} \\ &= \delta^{-6} x_{11} Q_{\mathbf{I}} + \delta^{-5} \mathcal{P}_{-5}[\mathcal{G}_{\text{HC}}; \mathbf{I}] + \mathcal{O}(\delta^{>-5}). \end{aligned} \quad (3.12)$$

Here $\mathcal{F}_{\text{SL}}[\mathcal{G}_{\text{HC}}; \mathbf{I}] = x_{11} Q_{\mathbf{I}}$, where the cancellation polynomial is

$$\begin{aligned} Q_{\mathbf{I}} &= s_{12} G_1 + s_{23} G_2, \\ G_1 &= (x_2 x_3 - x_1 x_4 + x_2 x_9)(x_6 x_7 - x_5 x_8 + x_6 x_{10}), \\ G_2 &= (x_2 x_5 - x_1 x_6)(x_4 x_7 - x_3 x_8 - x_8 x_9 + x_4 x_{10}). \end{aligned} \quad (3.13)$$

Thus multi-term polynomial cancellation factors occur already in this Crown descendant: three of the four primitive factors in Eq. (3.13) contain three or four monomials. This is the mixed algebraic form of Crown inheritance. The parameter x_{11} factors from $\mathcal{F}_{\text{SL}} = x_{11} Q_{\mathbf{I}}$, whereas x_9 and x_{10} enter the primitive factors themselves. Setting $x_9 = x_{10} = 0$ reduces them to Crown-form binomials; with these parameters active, multihomogeneity in the corresponding parameter groups gives factor-length profiles 3×3 and 2×4 . Applying the coefficientwise derivative harvest illustrated for the Crown, followed by disjoint-support pairing, gives G_1 and G_2 . The accepted ideal $\langle G_1, G_2 \rangle$ has a component in the positive orthant. This component, together with the derivative conditions, is the Landshoff cancellation locus.

The blue resolved-leading polynomial $\mathcal{P}_{-5}[\mathcal{G}_{\text{HC}}; \mathbf{I}]$ includes every contribution of the complete LP polynomial at weight -5 . In particular, it contains the leading monomials of the restricted $\mathcal{U}[\mathcal{G}_{\text{HC}}]$, as well as the contribution from $\mathcal{F}_0[\mathcal{G}_{\text{HC}}]$, which is the nonzero

polynomial

$$\mathcal{F}_{\text{Obs}}[\mathcal{G}_{\text{HC}}; \text{I}] \equiv \left(\mathcal{F}_0[\mathcal{G}_{\text{HC}}] \Big|_{x_{12}=0} \right)_{-5} \neq 0.$$

This is the first example in which HRF retains a nontrivial obstruction after extracting the superleading cancellation sector. Its role begins in the decomposition stage of the core algorithm: the pinch conditions must be applied to $x_{11}Q_{\text{I}}$, rather than to the complete restricted $\mathcal{F}_0[\mathcal{G}_{\text{HC}}]$, whose obstruction need not be stationary on the selected positive component. In the subsequent scaling stage, placing this obstruction and the leading $\mathcal{U}$ monomials at the same resolved-leading weight fixes the extra unit of scaling on the opposite square edge, $v_{11} = -2$.

For Type II, by contrast, all ten active parameters have weight -1 . Consequently every surviving term of $\mathcal{F}_0[\mathcal{G}_{\text{HC}}]$ has weight -5 : the complete restricted $\mathcal{F}_0$ is the superleading sector and $\mathcal{F}_{\text{Obs}}[\mathcal{G}_{\text{HC}}; \text{II}] = 0$. The HRF search returns the following generator presentation for this superleading sector:

$$\begin{aligned} \mathcal{F}_{\text{SL}}[\mathcal{G}_{\text{HC}}; \text{II}] &= s_{12}(x_2x_3 - x_1x_4) \left[x_{12}(x_6x_7 - x_5x_8 - x_8x_{10}) + x_{10}(x_6x_7 - x_5x_8) \right] \\ &\quad + s_{23}(x_4x_7 - x_3x_8) \left[x_{12}(x_2x_5 - x_1x_6 + x_2x_{10}) + x_{10}(x_2x_5 - x_1x_6) \right] \quad (3.14) \\ &= \mathcal{F}_0[\mathcal{G}_{\text{HC}}] \Big|_{x_9=x_{11}=0}. \end{aligned}$$

Here the Crown inheritance is realised entirely through the second algebraic pattern. Setting $x_{10} = 0$ reduces the two five-term partners to x_{12} times their respective Crown binomials, whereas setting $x_{12} = 0$ gives the same binomials multiplied by x_{10} . With both parameters active, their multihomogeneous lift gives the complete five-term polynomials displayed above.

This factorisation also makes the reciprocal character of the derivative harvest particularly transparent. The variables in each outer binomial are disjoint from those in its five-term partner. Thus, for example, the s_{12} and s_{23} coefficients of $\partial \mathcal{F}_{\text{SL}} / \partial x_3$ are respectively x_2 times the first five-term polynomial and $-x_8$ times the second. After monomial factors are removed, differentiation in the variables of either factor therefore discovers its partner, as expected from the stationary conditions. Differentiation within the five-term factors also produces further candidates, such as $x_2x_5 - x_1x_6 + x_2x_{10}$. The derivative harvest is consequently larger than the factor set entering the final decomposition. The subsequent positive-locus, disjoint-support pairing, decomposition and exact-scaling conditions are essential: here they select the two complete channel products displayed above and give zero obstruction.

With the uniform scaling of the ten active parameters, the complete rescaled LP polynomial is therefore

$$\begin{aligned} \mathcal{P}^{(\text{HR})}[\mathcal{G}_{\text{HC}}; \text{II}](\mathbf{x}; \delta, \mathbf{s}) &\equiv \mathcal{P}[\mathcal{G}_{\text{HC}}](\delta^{\mathbf{v}_{\text{HR}}[\mathcal{G}_{\text{HC}}; \text{II}]} \mathbf{x}; \delta, \mathbf{s}) \Big|_{x_9=x_{11}=0} \\ &= \delta^{-5} \mathcal{F}_{\text{SL}}[\mathcal{G}_{\text{HC}}; \text{II}] + \delta^{-4} \mathcal{P}_{-4}[\mathcal{G}_{\text{HC}}; \text{II}] + \mathcal{O}(\delta^{>-4}). \end{aligned} \quad (3.15)$$

The blue polynomial $\mathcal{P}_{-4}[\mathcal{G}_{\text{HC}}; \text{II}]$ is the resolved-leading layer and contains the leading $\mathcal{U}$ and off-shell contributions.

The opposite-pair boundaries do not support an HR. On $x_9 = x_{10} = 0$, factorisation of the relevant derivatives and kinematic-channel polynomials produces an s_{12} -supported Crown-like generator but no s_{23} -supported companion. The resulting single-generator form admits no scaling satisfying the homogeneous-weight, strict-separation and active-variable coverage conditions. The D_4 -related boundary $x_{10} = x_{11} = 0$ is excluded in the same way. These are algebraic absence statements for the two boundary strata. The HyperCrown interior has not been investigated here.

For comparison across the Crown family, we define the *generator-factor profile* of a pair generator $g_k = f_{i_k} f_{j_k}$ to be $m \times n$ when the two primitive factors f_{i_k} and f_{j_k} contain m and n monomials, respectively; we order the pair so that $m \leq n$. Table 2 lists the profiles of the two generators in each topology, separated by a comma, together with the corresponding parametric inheritance pattern.

Case	Generator-factor profiles	Parametric inheritance pattern	$\mathcal{F}_{\text{Obs}}$
Crown	$2 \times 2, 2 \times 2$	seed binomial pairing	zero
SuperCrown	inherited $2 \times 2, 2 \times 2$	common positive monomial factor	zero
HyperCrown I	$3 \times 3, 2 \times 4$	monomial factor and polynomial lift	nonzero
HyperCrown II	$2 \times 5, 2 \times 5$	polynomial lift	zero

Table 2. Generator structure and parametric realisation of Crown inheritance in the two-generator examples.

The family therefore exposes three indispensable features of HRF. First, the search must descend to contraction strata: an interior-only analysis misses regions present in the parent integral because the contraction can expose the disconnected hard subdiagram required by the inherited Crown mechanism. Second, the cancellation-factor harvest must admit general polynomials: this requirement appears already in the HyperCrown boundary regions, before the five-point examples below. Third, the single-edge HyperCrown boundary $x_{12} = 0$ displays the dual role of a nonzero obstruction. During decomposition it is essential for isolating the polynomial to which the pinch conditions apply; during scaling determination its balance with the leading $\mathcal{U}$ monomials fixes the non-uniform component $v_{11} = -2$. Both roles of the obstruction recur beyond the Crown family.

Having completed the parameter-space certification, we now record a compatible momentum-space realisation. The construction combines momentum conservation at each vertex with the inverse relation between Schwinger-parameter and propagator-virtuality scalings, derived in Ref. [43, App. B] and reviewed in Eq. (2.13). Figure 3 displays the resulting edge-wise momentum modes.

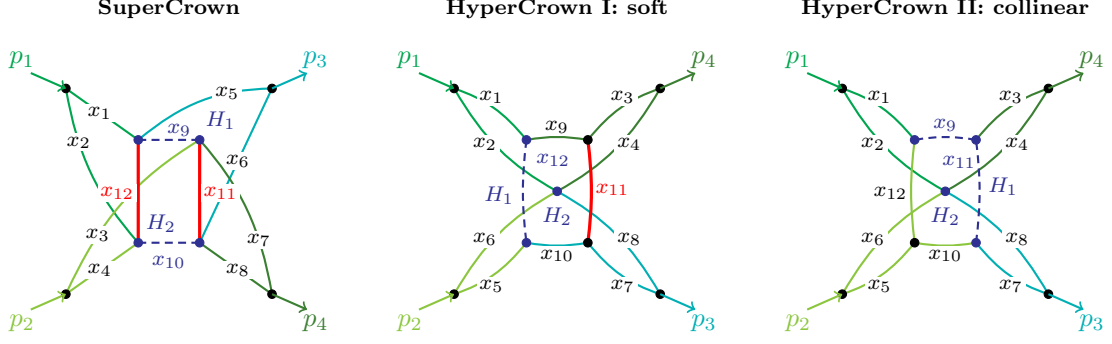


Figure 3. Edge-wise momentum-mode realisations of the Crown descendants. The four green shades denote the external-direction collinear modes, as in Fig. 1; blue dashed edges are contracted into the blue hard components H_1 and H_2 , and red lines are soft.

At the loop-momentum level, choose a basis adapted to the embedded three-loop Crown flow of Fig. 1, whose leading momentum modes remain unchanged. On the SuperCrown stratum, the two additional independent loop momenta can be chosen soft and routed through x_{11} and x_{12} . For HyperCrown Type I, the single additional loop momentum can likewise be chosen soft and routed through x_{11} after x_{12} is contracted. For Type II, after contracting x_9 and x_{11} , the additional loop momentum can be chosen p_2 -collinear and circulates through the two parallel lines x_{10} and x_{12} . In all three cases, the added loops preserve, rather than join, the two hard components: their soft or collinear momentum scalings remain compatible with the inherited Crown flow. This compatibility is universal and explains why any topology containing a compatible Crown contraction minor must contain an HR on the corresponding contraction stratum.

With these loop-momentum assignments established, Fig. 3 gives the physical interpretation of the parametric inheritance patterns summarised in Table 2. For the SuperCrown, x_{11} and x_{12} are the red soft lines; their appearance as a common positive monomial factor is therefore the parametric signature of the factor $x_{11}x_{12}$ in Eq. (3.10). HyperCrown Type I combines the two patterns: the red soft line x_{11} factors out, while the green p_4 - and p_3 -collinear lines x_9 and x_{10} enter the longer multihomogeneous cancellation factors in Eq. (3.13). In Type II, x_{10} and x_{12} are both p_2 -collinear, accounting for the purely polynomial lift in Eq. (3.14).

3.1.3 The four-loop No-Crown converse

The converse wide-angle audit revisits the four-loop graph sample isolated in Ref. [45] and sharpens the picture using the HRF absence test. Sixteen mixed-sign topologies without a Crown contraction minor have no interior or boundary HR. The search is complete on the stated graph sample, so these are no-HR certificates rather than failures to find a candidate. Together with the certified positive HyperCrown cases above, these results provide exact evidence on both sides of the Crown-minor conjecture. They do not, however, constitute a topology-by-topology HR certificate for all 1081 Crown-containing four-loop graphs.

3.1.4 The Crown family in Regge limits: an audited correspondence

The relation between the wide-angle Landshoff region and the Regge Glauber region was first observed in Ref. [45]. The comparison below turns this observation into a topology-by-topology audit through four loops. The audited Crown-containing set includes the graph obtained by opening both four-point hard vertices of the Crown in the t channel. This is Mandelstam's graph G_{tt} , the simplest of a class of diagrams introduced as examples whose Regge asymptotic behaviour is governed by Regge cuts [56]. From this point on, we work strictly on shell. Momentum conservation gives $s_{13} = -s_{12} - s_{23}$, and the three channel charts used in the audit are

$$\begin{aligned} T_{23} : \quad s_{23} &= -\delta s_{12}, & s_{12} &> 0, \\ T_{12} : \quad s_{12} &= -\delta s_{23}, & s_{23} &< 0, \\ T_{13} : \quad s_{23} &= -s_{12} + \delta s_{12}, & s_{12} &> 0, \end{aligned} \tag{3.16}$$

with $\delta \rightarrow 0^+$. In the ratio coordinates of Eq. (3.3), the leading Crown polynomial in each chart is, up to a nonzero kinematic and positive monomial factor,

$$\begin{aligned} \Psi_C^{T_{23}} &\propto (r_C - r_B)(r_D - r_A), \\ \Psi_C^{T_{12}} &\propto (r_B - r_A)(r_D - r_C), \\ \Psi_C^{T_{13}} &\propto (r_C - r_A)(r_D - r_B). \end{aligned} \tag{3.17}$$

The corresponding positive cancellation loci are therefore

$$\begin{aligned} T_{23} : \quad r_B &= r_C, \quad r_A = r_D, \\ T_{12} : \quad r_A &= r_B, \quad r_C = r_D, \\ T_{13} : \quad r_A &= r_C, \quad r_B = r_D. \end{aligned} \tag{3.18}$$

This exposes an important change in codimension. At generic wide angle the Crown locus identifies all four ratio points and hence has three independent relative-distance equations. In each Regge chart the leading polynomial reduces to one product of two difference factors, leaving only two independent equations. Each Regge locus is therefore larger and contains the wide-angle diagonal $r_A = r_B = r_C = r_D$.

Nevertheless, the reduction from three equations to two does not produce new seed topologies in the completed low-loop survey. The positive side is inherited by every audited graph that exposes a Crown contraction minor: the Crown itself has an HR in all three channels, and its higher-loop descendants carry the corresponding region on the contraction stratum. On the negative side, the same sixteen four-loop No-Crown topologies were checked in T_{23} , T_{12} and T_{13} . All 48 interiors and all 343356 nonempty contraction strata with between one and $E - 2$ edges contracted were resolved, with no HR. Deeper contractions leave at most one active parameter, so no nontrivial cancellation locus remains.

Thus, after the channel relabellings implied by crossing, the set of topologies with a Regge HR agrees exactly with the set having a wide-angle Landshoff HR at three and four loops. Equivalently, in this complete audited sample both are precisely the topologies

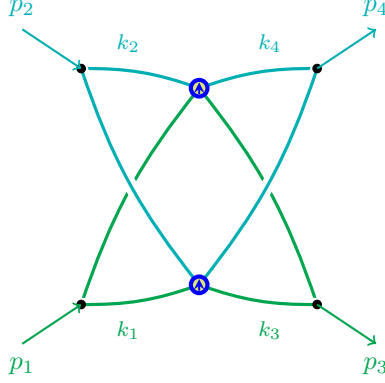


Figure 4. Momentum modes of the Crown HR in the T_{23} Regge channel. All external momenta are exactly lightlike. Of the eight propagator momenta, those shown in green are approximately collinear to both p_1 and p_3 , while those in teal are approximately collinear to both p_2 and p_4 ; the Glauber momentum is instead the loop combination $\ell_G = k_1 - k_3 \sim Q(\delta, \delta, \delta^{1/2})$, with $\delta = -s_{23}/s_{12}$ in this chart. This is the independent Glauber loop of the HR. The markers at the hard vertices identify its local t -channel segments. In the present embedding both local arrows point upward; placing the vertices side by side and opening them into propagators makes the closed-loop orientation explicit, with the Glauber momentum flowing upward on one side and downward on the other. This is the same momentum configuration as Fig. 11(b) of Ref. [45], where the direction of the Glauber exchange was identified; the different vertex marker used here is purely notational. The drawing convention used throughout the paper is as follows. A GreenYellow disc marks a vertex through which a Glauber loop flows; its arrow follows the local t -channel flow, while its rim colour records whether the momentum in the complementary local channel is hard, soft or collinear to the indicated jet direction. Here both arrows follow the upward local t -channel flow and the blue rims record hard complementary channels. Reversing every arrow gives the equivalent opposite loop-momentum convention. The other Regge channels follow by crossing.

containing the corresponding Crown contraction minor. We call this the wide-angle-to-Regge conjecture when extrapolated beyond four loops. Its validity is not a trivial consequence of the Landau equations: algebraically the Regge pinch requires fewer conditions and could have admitted additional topologies, but none occurs in the completed survey.

So far we have focussed on $2 \rightarrow 2$ scattering, where the three-loop Crown emerged as the unique seed topology for HRs in both the wide-angle and Regge limits. We now turn to massless $2 \rightarrow 3$ scattering, where a similar phenomenon occurs already at two loops: again, a special seed topology supports HRs in a variety of kinematic limits.

3.2 Five-point integrals: the Fish seed

The examples below are organised around a single unlabelled two-loop six-propagator topology commonly called the non-planar double box, which we refer to as the Fish seed, but involve different permutations of the external legs and kinematic expansions. The seed was first identified in the spacelike-collinear limit studied in the recent Letter [49], where it was shown to be the unique two-loop topology supporting an HR. The second setting is an exactly on-shell near-planar expansion of five-particle wide-angle scattering.

The cancellation ideals and total vectors are different in these two settings, but in both cases the same sparse topology supports an HR. Imposing standard fixed-transverse MRK and near-planarity simultaneously then yields a continuous one-parameter family of HR scalings that retains the three-generator wide-angle locus. We subsequently consider a central-soft rapidity-ordered limit, which reaches the normalised planar configuration by a more degenerate route and requires order alignment before the core HRF construction.

3.2.1 The spacelike-collinear limit

The spacelike-collinear application of HRF was presented in the recent Letter [49]. Here we apply HRF in parameter space to identify the seed cancellation locus and certify its scaling. These data then provide the baseline for tracking inheritance under graph enlargement. We take p_1 and p_2 incoming and p_3, p_4, p_5 outgoing, with $p_3 \parallel p_2$ in the collinear limit. For this analysis we use the kinematic parametrisation introduced in Ref. [57] and also employed in the recent Letter [49]. It originates from a momentum-twistor construction. In our notation the spacelike-collinear chart is

$$\begin{aligned} s_{12} &= sz, & s_{23} &= -4\delta^2, & s_{34} &= -s\chi(z-1), \\ s_{45} &= s, & s_{15} &= s\chi + c\delta, & \delta &\rightarrow 0^+, \end{aligned} \tag{3.19}$$

where δ tends to zero through positive real values, and

$$s > 0, \quad z > 1, \quad -1 < \chi < 0, \tag{3.20}$$

with fixed nonzero c . In particular, $s_{23} = -4\delta^2 < 0$ away from the strict limit, so Eq. (3.19) parametrises the spacelike-collinear approach. This form makes the dependence on the transverse collinear variable δ especially transparent: the deformation away from the strict collinear surface enters linearly through $c\delta$, while the vanishing collinear invariant starts at $s_{23} = -4\delta^2$. Thus the Mandelstam invariants are polynomial, and at most quadratic, in the expansion variable. The effect of crossing to the timelike channel on the cancellation factors is described below.

The two-loop seed, one representative permutation of the external legs and its LP-parameter assignment are shown in Fig. 5. Its $p_4 \leftrightarrow p_5$ image is an equally relevant permutation in the spacelike-collinear limit. Both permutations, together with the additional graphs whose contractions expose the seed with the required external labelling, were included in the amplitude-level integral analysis of Ref. [49]. To maintain direct correspondence with that analysis, throughout this subsection we retain the one-based LP-parameter labelling $x_1, \dots, x_6$ used in the Letter.

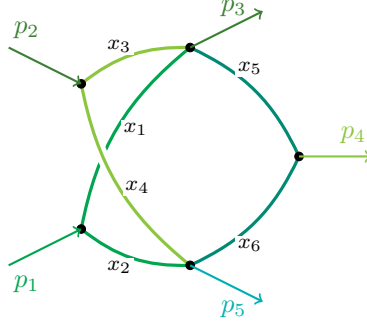


Figure 5. The two-loop non-planar double box, referred to here as the Fish seed topology, with a representative spacelike-collinear permutation of the external legs and the LP parameters used in Eqs. (3.25)–(3.26). The same unlabelled six-edge topology underlies the other Fish-seed limits considered below; each kinematic limit specifies its permutation of external legs separately.

The Fish HRF construction. We first apply the two parts of the HRF construction: Part I identifies the hidden-region decomposition, and Part II determines its scaling. At fixed LP parameters, the Part I decomposition of the strict leading polynomial is

$$\mathcal{F}_0 = \underbrace{s(x_2x_5 + \chi x_1x_6)}_{sf_1f_2} \left[(z-1)x_3 - x_4 \right] + \underbrace{sz(\chi+1)x_2x_3x_6}_{\mathcal{F}_{\text{Obs}}}, \quad (3.21)$$

$$f_1 = x_2x_5 + \chi x_1x_6, \quad f_2 = (z-1)x_3 - x_4.$$

In the HR decomposition, the second term is identified as $\mathcal{F}_{\text{Obs}}$. It is itself part of $\mathcal{F}_0$, and plays its defining role because the HR scaling suppresses it relative to the monomials in $\mathcal{F}_{\text{SL}} = sf_1f_2$, which are individually superleading.

The derivative harvest supplies f_1 and f_2 , which HRF pairs into the single generator

$$g_1 = f_1f_2, \quad G = \{g_1\}, \quad \mathcal{I}_{\text{gen}}(G) = \langle g_1 \rangle, \quad \mathcal{F}_{\text{SL}} = s g_1 \in \mathcal{I}_{\text{gen}}(G). \quad (3.22)$$

Part II then returns the hidden-region vector

$$\vec{v}_{\text{HR}} = (-2, -1, -2, -2, -2, -1; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-5, -4). \quad (3.23)$$

Comparison between the HR and the ordinary facet region. As already discussed in Ref. [49], it is insightful to compare the layer structure of the HR with that of the ordinary facet region. For comparison, the ordinary facet region has scaling vector

$$\vec{v}_C = (-2, 0, -2, -2, -2, 0; 1). \quad (3.24)$$

Its momentum-space interpretation as a collinear region is displayed in Table 3 below. To compare their layer structures directly, we explicitly rescale each graph polynomial as in Eq. (2.31), writing $\mathcal{F}^{(R)} = \mathcal{F}(\delta^{v_R} \mathbf{x}; \delta, \mathbf{s})$ and analogously for $\mathcal{U}^{(R)}$. For the ordinary facet this gives

$$\begin{aligned} \mathcal{F}^{(C)} = & \delta^{-4} \left[s(x_2x_5 + \chi x_1x_6) [(z-1)x_3 - x_4] + 4x_1x_4x_5 \right] \\ & + \left[-c\delta^{-3}x_1x_4x_6 + \delta^{-2}(sz(\chi+1)x_2x_3x_6 - 4x_2x_3x_5) + c\delta^{-1}x_2x_3x_6 \right]. \end{aligned} \quad (3.25)$$

There is no superleading sector. By contrast, the HR scaling gives

$$\begin{aligned}\mathcal{F}^{(\text{HR})} = & \delta^{-5} \underbrace{s(x_2x_5 + \chi x_1x_6)}_{\mathcal{F}_{\text{SL}}} [(z-1)x_3 - x_4] \\ & + \delta^{-4} \underbrace{[sz(\chi+1)x_2x_3x_6 - cx_1x_4x_6 + 4x_1x_4x_5]}_{\mathcal{F}_{\text{Obs}}} + \delta^{-3} [cx_2x_3x_6 - 4x_2x_3x_5].\end{aligned}\tag{3.26}$$

The complete LP comparison also requires the first Symanzik polynomial, which supplies the same leading layer in the two regions,

$$\begin{aligned}\mathcal{U}^{(C)} &= \delta^{-4} \mathcal{U}_{-4} + \mathcal{O}(\delta^{-2}), & \mathcal{U}^{(\text{HR})} &= \delta^{-4} \mathcal{U}_{-4} + \mathcal{O}(\delta^{-3}), \\ \mathcal{U}_{-4} &= x_1x_3 + x_1x_4 + x_1x_5 + x_3x_5 + x_4x_5.\end{aligned}\tag{3.27}$$

In the facet region, the blue terms of Eqs. (3.25) and (3.27) together form the ordinary lower-facet polynomial at weight -4 . The monomial $sz(\chi+1)x_2x_3x_6$ appears only at black weight -2 , while the first c -dependent term is black at weight -3 ; hence the leading power in the asymptotic expansion is independent of c . This is in line with the fact that this region appears in both the timelike and spacelike collinear limits, while the former, which admits strict collinear factorisation, cannot feature c dependence [49]. In the HR, the monomials constituting the factorised part of $\mathcal{F}_0$ are individually superleading at weight -5 . The remaining monomial $sz(\chi+1)x_2x_3x_6$, identified as $\mathcal{F}_{\text{Obs}}$ in the HR decomposition, is suppressed by one power relative to them: it has weight -4 and joins the other terms in the blue physical layer. The obstruction is also essential to the HRF hierarchy conditions selecting the certified non-uniform vector in Eq. (3.23). The c -dependent term $-cx_1x_4x_6$ is promoted to that same layer. The HR therefore depends on c at leading power and exhibits factorisation-violating dependence.

The channel dependence of this positive cancellation locus is already visible in these two factors. To obtain the corresponding physical timelike-collinear limit while retaining a $2 \rightarrow 3$ process, cross particle 2 to the final state and particle 4 to the initial state, so that the process becomes $1 + 4 \rightarrow 2 + 3 + 5$. On the strict collinear surface the physical ranges in the same variables are

$$s < 0, \quad 0 < z < 1, \quad \chi > 0.\tag{3.28}$$

For positive LP parameters these ranges give $f_1 > 0$ and $f_2 < 0$, so neither factor can vanish. Thus the positive common zero in the spacelike domain is lost under the simultaneous crossing, in agreement with the absence of this HR in the timelike-collinear limit [49].

Dissection charts and the invariant region. This example also displays explicitly why a local facet is not an invariant label for the HR. Under the original scaling Eq. (3.23),

$$f_1 = \delta^{-3} \hat{f}_1, \quad f_2 = \delta^{-2} \hat{f}_2, \quad \mathcal{F}_{\text{SL}} = \delta^{-5} s \hat{f}_1 \hat{f}_2.\tag{3.29}$$

The gap $W_{\text{HR}} - W_{\text{SL}} = 1$ fixes only the total transverse suppression. A chart centred on the simultaneous locus may distribute it as

$$\hat{f}_1 \sim \delta^\alpha, \quad \hat{f}_2 \sim \delta^{1-\alpha}, \quad 0 < \alpha < 1.\tag{3.30}$$

Using $y_i = f_i$ and eliminating x_2, x_4 , respectively, the local weights are

$$(v_{x_1}, v_{x_3}, v_{x_5}, v_{x_6}, v_{y_1}, v_{y_2}; 1) = (-2, -2, -2, -1, -3 + \alpha, -1 - \alpha; 1), \quad (3.31)$$

with Jacobian $1/x_5$. Every $0 < \alpha < 1$ approaches the full locus $f_1 = f_2 = 0$, and every such scaling pulls back to the same original vector (3.23). Throughout this open interval, the leading support is the same and has affine dimension five, hence codimension two in the seven-dimensional transformed Newton polytope. It is the common lower face of the two endpoint facets, rather than a facet by itself.

The two endpoint facets provide alternative dissections of the same HR. At the $\alpha \rightarrow 0$ endpoint one may set

$$\begin{aligned} y_2 &= f_2, & x_4 &= (z - 1)x_3 - y_2, \\ \xi_{(2)} &= (x_1, x_2, x_3, x_5, x_6, y_2), & \vec{v}_{(2)} &= (-2, -1, -2, -2, -1, -1; 1), \end{aligned} \quad (3.32)$$

whereas at $\alpha \rightarrow 1$ one may use

$$\begin{aligned} y_1 &= f_1, & x_2 &= \frac{y_1 - \chi x_1 x_6}{x_5}, \\ \xi_{(1)} &= (x_1, x_3, x_4, x_5, x_6, y_1), & \vec{v}_{(1)} &= (-2, -2, -2, -2, -1, -2; 1). \end{aligned} \quad (3.33)$$

In both cases the transformed leading support has affine dimension six and defines an exact lower facet. The first chart has unit Jacobian. In the second, the coordinate measure has weight -11 and the Jacobian $1/x_5$ has weight $+2$; both therefore give the measure weight -9 . Pulling either facet normal back gives Eq. (3.23).

Thus either Eq. (3.32) or Eq. (3.33) may be used as an integration chart for the same invariant cancellation component $f_1 = f_2 = 0$; they are not distinct physical pinches or contributions to be combined. The calculation of Ref. [49] used the $y_2 = f_2$ dissection in Eq. (3.32) alone.

It is useful for comparison with later examples to separate the overall approach to the common zero from the relative approach to its two factors. Here the natural kinematic deformation δ is dimensionful, so let $Q = \sqrt{s}$ and $\lambda = \delta/Q$. The LP parameters, and hence the cancellation factors f_1 and f_2 , are already dimensionless in the convention of Eq. (2.4). Their natural regional weights are removed by defining the dimensionless combinations

$$t_1 = \lambda^3 f_1, \quad t_2 = \lambda^2 f_2.$$

Within a fixed sign sector, a centred radial-ratio parameterisation is

$$|t_1| = \rho e^\eta, \quad |t_2| = \rho e^{-\eta}, \quad \rho \sim \lambda^{1/2}. \quad (3.34)$$

Here ρ measures the approach to the product locus, while η records how its suppression is distributed between the two factors. The scaling $\eta = (\alpha - \frac{1}{2}) \log \lambda + O(1)$ represents the freedom in Eq. (3.30); fixed η gives the symmetric centred chart. This form will be used only for comparison below; the calculation of Ref. [49] used the endpoint chart Eq. (3.32).

Parameter-space power counting. For unit propagator powers, the scalar LP representation contains $\mathcal{P}^{-D/2}$. In the original LP variables, the six edge components of $\vec{v}_{\text{HR}}$ in Eq. (3.23) sum to $\sum_{e=1}^6 v_{\text{HR},e} = -10$. Equations (3.31)–(3.33) show in three equivalent ways that resolving the total cancellation depth contributes one additional power. The parameter measure is therefore

$$\delta^{-10} \delta^{+1} = \delta^{-9}. \quad (3.35)$$

Since the resolved LP polynomial has weight -4 , $\mathcal{P}^{-D/2} \sim \delta^{2D}$, and hence

$$I_{\text{SC}}^{\text{scalar}} \sim \delta^{-9+2D} = \delta^{-1-4\epsilon}, \quad D = 4 - 2\epsilon. \quad (3.36)$$

The power enhancement is a property of the unit-numerator scalar integral. For the gauge-theory numerator relevant to the amplitude, one additional power of δ gives the leading-power behaviour $\delta^{-4\epsilon}$.

Momentum-space interpretation. To connect the parameter-space result with its momentum-space realisation, and to contrast it with the ordinary facet region, we first fix conventions common to both regions. Whenever a component triple is displayed, the subscript on $(+, -, \perp)_{(n_+, n_-)}$ names the null reference directions that define its plus and minus components, in that order. Here we use the global frame $(+, -, \perp)_{(p_2, p_1)}$ defined by the two incoming momenta, $p_1 = (0, p_1^-, \mathbf{0}_\perp)$ and $p_2 = (p_2^+, 0, \mathbf{0}_\perp)$, so that the plus direction is aligned with p_2 , and hence with the collinear momentum p_3 , while the minus direction is aligned with p_1 . We orient q_1, q_2 away from the p_1 vertex, q_3, q_4 away from the p_2 vertex, q_5 from the p_3 vertex towards the p_4 vertex, and q_6 from the p_5 vertex towards the p_4 vertex. These orientation conventions are common to the ordinary facet region and the HR; their virtuality and component scalings distinguish the two cases. The momentum q_e flows through the edge carrying x_e . With the hard scale $Q = \sqrt{s}$ held fixed, we write

$$\frac{q_e^2}{Q^2} \sim \left(\frac{\delta}{Q}\right)^{-v_e}, \quad \frac{1}{Q}(q_e^+, q_e^-, |q_{e\perp}|)_{(p_2, p_1)} \sim \left(\left(\frac{\delta}{Q}\right)^{a_e}, \left(\frac{\delta}{Q}\right)^{b_e}, \left(\frac{\delta}{Q}\right)^{c_e}\right).$$

The first relation follows directly from the parameter-space vector through the inverse Schwinger scaling relation in Eq. (2.13). Determining the separate component exponents (a_e, b_e, c_e) requires additional momentum-space input, notably momentum conservation and the pinch conditions. Below we abbreviate the component scaling as $q_e \sim (\delta^{a_e}, \delta^{b_e}, \delta^{c_e})_{(p_2, p_1)}$; this shorthand records only powers of δ , with the fixed factors of Q understood.

The ordinary facet region admits a homogeneous collinear loop basis and no transverse-dominated loop rerouting. Its component and virtuality scalings are recorded in the first two columns of Table 3.

For the HR, q_1 and q_5 are separately soft at the pinch. With the orientations fixed above, conservation at the vertex carrying p_3 reads $q_1 + q_3 = p_3 + q_5$. A compatible mode-adapted routing is obtained by choosing the independent loop momenta

$$\ell_s = q_5 \sim (\delta, \delta, \delta), \quad \ell_G = q_1 - q_5 = p_3 - q_3 \sim (\delta, \delta^2, \delta), \quad (3.37)$$

at the pinch. The extra suppression of ℓ_G^- is a consequence of the pinch conditions: they correlate the leading $O(\delta)$ minus components of q_1 and q_5 so that these cancel in the

difference. A generic difference of two soft momenta would remain soft; here $\ell_G^+ \ell_G^- \sim \delta^3$ is subleading to $\ell_{G\perp}^2 \sim \delta^2$, making the Glauber loop transverse dominated. A complete oriented routing is then

$$\begin{aligned} q_1 &= \ell_G + \ell_s, & q_2 &= p_1 - \ell_G - \ell_s, & q_3 &= p_3 - \ell_G, \\ q_4 &= p_2 - p_3 + \ell_G, & q_5 &= \ell_s, & q_6 &= p_4 - \ell_s. \end{aligned}$$

This routing gives the compatible HR component scalings in the last two columns of Table 3; the corresponding virtualities follow from the inverse Schwinger relation. Here and below, a momentum valuation is the leading exponent of δ in the indicated momentum component or virtuality.

edge	facet $\vec{v}_C$		hidden $\vec{v}_{\text{HR}}$	
	$(a_e, b_e, c_e)_{(p_2, p_1)}$	$-v_e$	$(a_e, b_e, c_e)_{(p_2, p_1)}$	$-v_e$
q_1	(0, 2, 1)	2	(1, 1, 1)	2
q_2	(0, 0, 1)	0	(1, 0, 1)	1
q_3	(0, 2, 1)	2	(0, 2, 1)	2
q_4	(0, 2, 1)	2	(0, 2, 1)	2
q_5	(0, 2, 1)	2	(1, 1, 1)	2
q_6	(0, 0, 0)	0	(0, 0, 0)	1

Table 3. Edge-momentum valuations of a compatible routing for the five-point spacelike-collinear facet and HR. The component exponents refer to the $(+, -, \perp)_{(p_2, p_1)}$ lightcone frame defined in the text; local loop-integration widths are determined below.

The virtuality exponent $-v_e$ is fixed by the parameter-space region in both columns of Table 3. If the leading terms in $q_e^2 = 2q_e^+ q_e^- - q_{e\perp}^2$ do not cancel, the displayed component exponents reproduce it as $-v_e = \min(a_e + b_e, 2c_e)$. This holds for every facet entry in the table and for $q_1, \dots, q_5$ in the HR column, but is not a defining property of facet regions: leading component cancellations may occur in either type of region, and whether a collinear direction is manifest also depends on the chosen lightcone frame. The HR entry for q_6 illustrates the latter point. Indeed, the explicit routing gives $q_6 = p_4 - \ell_s$. Since p_4 is wide-angle relative to the two reference directions, all three components displayed in the table are of order Q , yielding $(a_6, b_6, c_6) = (0, 0, 0)$. Nevertheless its leading momentum is the lightlike vector p_4 , and hence

$$q_6^2 = (p_4 - \ell_s)^2 = -2p_4 \cdot \ell_s + \ell_s^2 \sim Q\delta,$$

so that $-v_6 = 1$. In a local lightcone basis with p_4 as the plus reference direction and $\bar{p}_4$ a conjugate null direction, the abbreviated component scaling is instead $q_6 \sim (1, \delta, \delta)_{(p_4, \bar{p}_4)}$, making its p_4 -collinear character manifest. This is the scaling shown by the lime edge in Fig. 6. The HR therefore has one soft and one Glauber loop, and the reconstruction is mode complete.

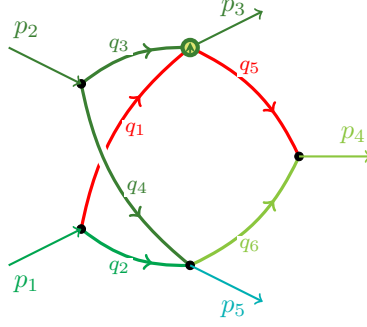


Figure 6. Momentum-space interpretation of the spacelike-collinear Fish HR. Arrowheads define the edge-momentum orientations used in the text. Red denotes the soft edges q_1 and q_5 , olive the edges collinear to the p_2, p_3 direction, and green and lime the remaining jet directions. The oriented GreenYellow disc at the vertex carrying p_3 marks the Glauber loop $\ell_G = q_1 - q_5$ at the collinear-to-soft vertex. It follows the convention defined in Fig. 4; here its olive rim records the collinear complementary local channel.

For the momentum-space power count we need the local widths, not only the routing valuations in Table 3. In the loop basis of Eq. (3.37), the opposite-side poles of the (q_3, q_4) pair restrict $\Delta\ell_G^-$ to $O(\delta^2)$. The (q_1, q_2) pair restricts the corresponding plus transfer to $O(\delta)$, while the (q_5, q_6) and (q_1, q_2) pairs fix the two soft longitudinal widths at $O(\delta)$. Finally, preserving the virtualities in Table 3 fixes both transverse widths at $O(\delta)$, by the homogeneity criterion of Sec. 4.3. Thus this mode-adapted basis has

$$\ell_s|_{\text{pinch}} \sim \Delta\ell_s \sim (\delta, \delta, \delta), \quad \ell_G|_{\text{pinch}} \sim \Delta\ell_G \sim (\delta, \delta^2, \delta), \quad (3.38)$$

where the entries denote $(q^+, q^-, |q_\perp|)$. Thus the compatible routing and the independently determined fluctuation widths have the same scaling in this suitable independent loop basis. The two measures scale as δ^D and δ^{D+1} , respectively. Using the virtuality valuations $-v_e$ already listed in Table 3, the six edge virtualities obey

$$\left(\frac{q_1^2}{Q^2}, \dots, \frac{q_6^2}{Q^2}\right) \sim (\delta^2, \delta, \delta^2, \delta^2, \delta^2, \delta), \quad -\mathbf{v}_{\text{HR}} = (2, 1, 2, 2, 2, 1). \quad (3.39)$$

The six virtuality exponents therefore sum to ten. Hence

$$I_{\text{SC}}^{\text{scalar}} \sim \delta^{D+(D+1)-10} = \delta^{-1-4\epsilon}, \quad (3.40)$$

in agreement with the parameter-space count in Eq. (3.36). In the original edge routing the Glauber momentum is the difference of two soft edge momenta, which is why inspection of individual propagator modes alone does not reveal it.

3.2.2 A vertex correction: polynomial lifting of the seed locus

The three-loop vertex-corrected graph in the same spacelike-collinear kinematics is shown in Fig. 7.

Here the seed cancellation geometry is inherited in a polynomially lifted form. The polynomial-factor harvest returns two cancellation factors. The first is the seed factor

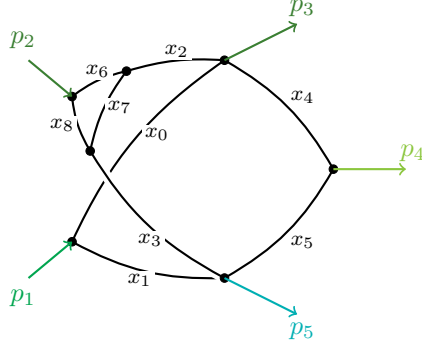


Figure 7. The three-loop vertex-corrected five-point graph, drawn as the two-loop seed of Fig. 5 with a local vertex correction adjacent to p_2 . In particular, x_7 lies outside the other loops. The labels $x_0, \dots, x_8$ are those used in Eqs. (3.41)–(3.42).

f_1 defined in Eq. (3.21), expressed here in the edge labelling of Fig. 7; the second is a polynomial lift of the seed factor f_2 , which we denote by $\tilde{f}_2$. Collecting all terms proportional to $z - 1$, including the $x_6 x_7$ term, gives

$$\begin{aligned} f_1 &= x_1 x_4 + \chi x_0 x_5, \\ \tilde{f}_2 &= (z - 1)A - B, \end{aligned} \quad \begin{aligned} A &= x_2(x_6 + x_7 + x_8) + x_6 x_7, \\ B &= x_3(x_6 + x_7 + x_8) + x_7 x_8, \end{aligned} \quad (3.41)$$

Both A and B are positive in the LP domain. Thus $\tilde{f}_2$ has the same $(z - 1)A - B$ sign structure as the seed factor $f_2 = (z - 1)x_3 - x_4$, while f_1 retains the two-term seed structure on the four edges unaffected by the vertex correction. HRF pairs these factors into the single generator $\tilde{g}_1 = f_1 \tilde{f}_2$. The ensuing Part-I decomposition and Part-II hierarchy solution are

$$\begin{aligned} \mathcal{F}_\star &= \underbrace{-s\tilde{g}_1}_{\mathcal{F}_{\text{SL}}} + \underbrace{-s(1 + \chi)z x_1 x_5 (x_2 x_6 + x_2 x_7 + x_6 x_7 + x_2 x_8)}_{\mathcal{F}_{\text{Obs}}}, \\ \vec{v}_{\text{HR}} &= (-2, -1, -2, -2, -2, -1, -2, -2, -2; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-7, -6). \end{aligned} \quad (3.42)$$

As in the seed, the unit gap fixes the total suppression of the product but need not assign it uniquely to f_1 and $\tilde{f}_2$. A centred normal chart and its endpoint facets are equivalent only after their normals are pulled back to the vector in Eq. (3.42); the discussion in Sec. 3.2.1 applies unchanged to these general polynomial factors. The simultaneous crossing described above acts in precisely the same way on the lifted factors: in the timelike domain of Eq. (3.28), $\tilde{f}_2 < 0$ and $f_1 > 0$ for positive LP parameters. Hence the vertex-corrected graph has no positive cancellation locus there. Its channel dependence is therefore inherited from the Fish seed rather than qualitatively new. The additional loop provides a nontrivial check that HRF retains this cancellation geometry when a seed factor is lifted to a general polynomial and occurs with polynomial multipliers in $\mathcal{F}_{\text{SL}}$.

Momentum-flow inheritance. The vertex-corrected graph has an externally labelled Fish contraction minor and preserves the HR flow of Fig. 6: the two soft edges and their

Glauber difference remain in place. The local correction replaces the two p_2 -collinear seed edges by the larger homogeneous collinear subgraph shown in Fig. 8. The additional loop is therefore collinear and introduces no second Glauber loop. This is also the momentum-space counterpart of the polynomial lift: the longer factor $\tilde{f}_2$ reflects the enlarged collinear subgraph, whereas f_1 retains the two-term structure of the seed.

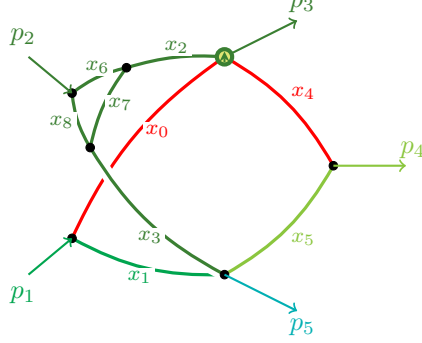


Figure 8. Momentum-space interpretation of the vertex-corrected spacelike-collinear HR. Red denotes the two soft edges, olive the enlarged p_2, p_3 -collinear subgraph, and green and lime the remaining jet directions. At the p_3 vertex the olive-rimmed marker shows the inherited Glauber loop $\ell_G = q_0 - q_4$ at the collinear-to-soft vertex and follows the convention defined in Fig. 4.

3.2.3 Massless five-point kinematics and the near-planar expansion

Let all five external momenta be exactly massless and keep all adjacent Mandelstam invariants at wide angle, with the physical signs

$$s_{12}, s_{34}, s_{45} > 0, \quad s_{23}, s_{15} < 0. \quad (3.43)$$

Define

$$\Gamma_5 = \det(2p_i \cdot p_j)_{i,j=1}^4 = \epsilon_5^2, \quad \epsilon_5 = 4i \varepsilon(p_1, p_2, p_3, p_4). \quad (3.44)$$

Geometrically, Γ_5 is the signed squared four-volume of the parallelotope spanned by $p_1, \dots, p_4$, up to the conventional factor 2^4 ; equivalently it is $2^4(4!)^2$ times the signed squared volume of the corresponding four-simplex. Its vanishing is therefore the condition that the external momenta become coplanar. For real momenta on the physical non-coplanar sheet, ϵ_5 is purely imaginary and $\Gamma_5 < 0$.

To define the approach to this boundary, use an exact parametrisation of general wide-angle five-point kinematics. These variables were introduced in the MRK analysis of Ref. [58].¹³ In the global $(+, -, \perp)_{(p_2, p_1)}$ lightcone frame, with p_2 defining the plus direction and p_1 the minus direction, write the complex transverse momentum as $\mathbf{p} = p^x + ip^y$ and its conjugate as $\bar{\mathbf{p}} = p^x - ip^y$, and define

$$w = -\frac{\mathbf{p}_3}{\mathbf{p}_4}, \quad \bar{w} = -\frac{\bar{\mathbf{p}}_3}{\bar{\mathbf{p}}_4}, \quad X_{34} = \frac{p_3^+}{p_4^+} = \frac{s_{13}}{s_{14}}, \quad X_{45} = \frac{p_4^+}{p_5^+} = \frac{s_{14}}{s_{15}}. \quad (3.45)$$

¹³In the five-point MRK parametrisation of Ref. [59], the transverse variables denoted there by $z, \bar{z}$ coincide, with the present momentum labels, with $w, \bar{w}$. Related multi-Regge-inspired parametrisations beyond five points are discussed in Ref. [60].

The final equalities express the longitudinal variables directly as Lorentz-invariant ratios of Mandelstam invariants. For physical real momenta away from coplanarity, $\bar{w} = w^*$. For comparing the two expansions below it is useful to replace the inverse transverse ratios by

$$\zeta = -\frac{1}{w} = \frac{\mathbf{p}_4}{\mathbf{p}_3}, \quad \bar{\zeta} = -\frac{1}{\bar{w}} = \frac{\bar{\mathbf{p}}_4}{\bar{\mathbf{p}}_3}. \quad (3.46)$$

These four dimensionless ratios and the transverse scale $|\mathbf{p}_4|^2$ form an exact chart, up to an irrelevant longitudinal boost, and therefore determine all five independent Mandelstam invariants. Transverse momentum conservation gives $\mathbf{p}_3 = -w\mathbf{p}_4$ and $\mathbf{p}_5 = -(1-w)\mathbf{p}_4$, together with their barred counterparts; on-shellness then fixes the minus components from the plus components. In this chart the parity-odd invariant factorises exactly as

$$\epsilon_5 = s_{12}|\mathbf{p}_4|^2(w - \bar{w}) = s_{12}|\mathbf{p}_4|^2 \frac{\zeta - \bar{\zeta}}{\zeta \bar{\zeta}}. \quad (3.47)$$

Equations (3.45) and (3.46) describe generic five-point kinematics before any expansion is taken. We shall select three distinct paths from this same generic domain. The near-planar path makes the angular factor $\zeta - \bar{\zeta}$ vanish at fixed $|\mathbf{p}_4|^2$. The central-soft rapidity-ordered path defined later instead sends $\zeta, \bar{\zeta} \rightarrow 0$ with the phase ratio $\zeta/\bar{\zeta}$ held at a generic value, while holding

$$Q_\perp^2 \equiv |\mathbf{p}_3|^2 = \frac{|\mathbf{p}_4|^2}{\zeta \bar{\zeta}}$$

fixed. Equation (3.47) then becomes $\epsilon_5 = s_{12}Q_\perp^2(\zeta - \bar{\zeta}) \rightarrow 0$. This path therefore provides a second approach to the normalised planar surface, through transverse softening of p_4 rather than angular alignment at fixed $|\mathbf{p}_4|^2$. It is rapidity ordered but differs from standard MRK because the transverse ratio $|\mathbf{p}_4|^2/|\mathbf{p}_3|^2 = \zeta \bar{\zeta}$ also tends to zero. The third path combines standard fixed-transverse MRK with $w - \bar{w} \rightarrow 0$; unlike the central-soft path, neither transverse scale is sent to zero.

The near-planar wide-angle expansion is defined by

$$\begin{aligned} w &= -\frac{1}{\xi} + \frac{i}{2}\lambda, & \bar{w} &= -\frac{1}{\xi} - \frac{i}{2}\lambda, \\ \zeta &= \frac{\xi}{1 - i\xi\lambda/2}, & \bar{\zeta} &= \frac{\xi}{1 + i\xi\lambda/2}, \\ \lambda &\rightarrow 0^+, & \xi, X_{34}, X_{45}, |\mathbf{p}_4|^2 &\text{ fixed,} \end{aligned} \quad (3.48)$$

where $\xi > 0$. Thus exact planarity is simply $\zeta = \bar{\zeta} = \xi$, or equivalently $w = \bar{w} = -1/\xi$. The minus sign makes the first-sheet branch with positive LP parameters manifest in the formulae below. There is no rapidity hierarchy or soft external momentum. Thus

$$\epsilon_5 = ic_\epsilon\lambda, \quad \Gamma_5 = -c_\epsilon^2\lambda^2, \quad c_\epsilon = s_{12}|\mathbf{p}_4|^2 > 0. \quad (3.49)$$

This kinematic construction is independent of the loop order and of the integral topology considered below.

3.2.4 The Fish seed in near-planar scattering

Symmetric coordinates for the Fish topology.

The same six-edge topology analysed in Sec. 3.2.1 has a second certified HR in the near-planar expansion defined above. For convenience we use the $p_4 \leftrightarrow p_5$ -interchanged labelling relative to the representative in Fig. 5,

$$(p_1, p_2, p_3, p_5, p_4) \longrightarrow (1, 2, 3, 4, 5), \quad (3.50)$$

as shown in Fig. 9. This interchange is immaterial both here and in the spacelike-collinear limit; it becomes consequential only in the later soft and rapidity-ordered limits that distinguish p_4 from p_5 .

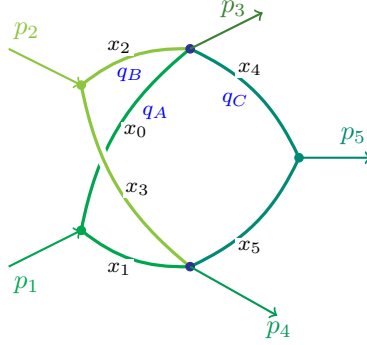


Figure 9. A representative permutation of the external legs and LP-parameter assignment for the near-planar wide-angle analysis. Relative to Fig. 5, p_4 and p_5 are interchanged. The five shades of green distinguish the five wide-angle directions. The pairs of propagators forming paths A, B, C have respectively the colours of p_1, p_2, p_5 , making their collinear assignments explicit. The vertices carrying p_3 and p_4 are blue to identify the two hard vertices, while the other three vertices match their path colours. The blue labels associate the path momenta q_A, q_B, q_C used below with the x_0, x_2, x_4 edges, respectively. All three are oriented outward from the vertex carrying p_3 , so momentum conservation there gives $p_3 + q_A + q_B + q_C = 0$.

For the remainder of the Fish-seed analysis we use the zero-based edge labels $(x_0, \dots, x_5)$, corresponding respectively to $(x_1, \dots, x_6)$ in Fig. 5. As indicated by the matching propagator colours in Fig. 9, group them as the three two-edge paths (x_0, x_1) , (x_2, x_3) and (x_4, x_5) between the two trivalent hard vertices, and put

$$r_A = \frac{x_0}{x_1}, \quad r_B = \frac{x_2}{x_3}, \quad r_C = \frac{x_4}{x_5}, \quad \mathcal{F} = x_1 x_3 x_5 \Phi(r_A, r_B, r_C). \quad (3.51)$$

Here $x_1, x_3, x_5 > 0$ are coordinates for the magnitudes on the three two-edge paths, while r_A, r_B, r_C specify the ratios within them; they are together six independent coordinates on the positive LP-parameter orthant. Write

$$\Phi(\mathbf{r}) = q_{AB} r_A r_B + q_{BC} r_B r_C + q_{CA} r_C r_A + b_A r_A + b_B r_B + b_C r_C = \frac{1}{2} \mathbf{r}^T K \mathbf{r} + \mathbf{b}^T \mathbf{r},$$

$$K = \begin{pmatrix} 0 & q_{AB} & q_{CA} \\ q_{AB} & 0 & q_{BC} \\ q_{CA} & q_{BC} & 0 \end{pmatrix}, \quad \mathbf{b} = (b_A, b_B, b_C)^T. \quad (3.52)$$

For the permutation in Eq. (3.50), the entries are Mandelstam invariants, which may be expressed in terms of the adjacent ones using five-point momentum conservation:

$$\begin{aligned} q_{AB} &= s_{35} = s_{12} - s_{34} - s_{45}, & b_A &= s_{14} = -s_{15} + s_{23} - s_{45}, \\ q_{BC} &= s_{13} = -s_{12} - s_{23} + s_{45}, & b_B &= s_{24} = s_{15} - s_{23} - s_{34}, \\ q_{CA} &= s_{23}, & b_C &= s_{45}. \end{aligned} \quad (3.53)$$

Landau analysis and the origin of the expansion.

We first use the Landau equations at generic five-point kinematics to determine which kinematic limit can support the pinch. Once that limit has been identified, we will fix the corresponding expansion and return to the standard HRF workflow.

Direct differentiation with respect to the ratios gives

$$\nabla_{\mathbf{r}} \Phi = K \mathbf{r} + \mathbf{b}. \quad (3.54)$$

The determinant of the coefficient matrix is

$$\det K = 2q_{AB}q_{BC}q_{CA}. \quad (3.55)$$

Hence, whenever $q_{AB}q_{BC}q_{CA} \neq 0$, ratio stationarity at $\mathbf{r} = \boldsymbol{\rho}$ determines a unique ratio vector and its stationary value:

$$\begin{aligned} K \boldsymbol{\rho} &= -\mathbf{b}, & \boldsymbol{\rho} &= (\rho_A, \rho_B, \rho_C)^T = -K^{-1} \mathbf{b}, \\ \Phi(\boldsymbol{\rho}) &= \frac{1}{2} \mathbf{b}^T \boldsymbol{\rho} = -\frac{1}{2} \mathbf{b}^T K^{-1} \mathbf{b}. \end{aligned} \quad (3.56)$$

Already at this generic kinematic point, define the three polynomials normal to the ratio-stationary locus,

$$f_A = x_0 - \rho_A x_1, \quad f_B = x_2 - \rho_B x_3, \quad f_C = x_4 - \rho_C x_5. \quad (3.57)$$

Here the components ρ_I retain their full kinematic dependence; no kinematic limit has yet been taken.

Ratio stationarity is necessary but is not yet the complete Landau condition. To see the missing equation, use the independent path-magnitude coordinates $u_A = x_1$, $u_B = x_3$, $u_C = x_5$. At fixed ratios,

$$\mathcal{F} = u_A u_B u_C \Phi, \quad \frac{\partial \mathcal{F}}{\partial u_A} = u_B u_C \Phi, \quad \text{and cyclically.} \quad (3.58)$$

Since the path magnitudes are positive, stationarity in these directions also requires

$$\Phi(\boldsymbol{\rho}) = 0. \quad (3.59)$$

In the original LP parameters this condition follows automatically from homogeneity once all derivative equations are imposed. In the ratio chart, however, Φ contains both quadratic and linear terms and is not homogeneous, so Eq. (3.59) does not follow from Eq. (3.56). To

relate the stationary value in Eq. (3.56) to the external kinematics, assemble the coefficients in K and $\mathbf{b}$ into the symmetric matrix

$$\mathcal{B} = \begin{pmatrix} K & \mathbf{b} \\ \mathbf{b}^T & 0 \end{pmatrix}.$$

Using the Mandelstam identifications in Eq. (3.53), five-point momentum conservation gives $\det \mathcal{B} = \det(2p_i \cdot p_j)_{i,j=1}^4 = \Gamma_5$. The relation of this determinant to the stationary value follows from Schur's formula [61],

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B).$$

Taking $A = K$, $B = \mathbf{b}$, $C = \mathbf{b}^T$, and $D = 0$, and using Eq. (3.56), gives

$$\det \mathcal{B} = -\det K \mathbf{b}^T K^{-1} \mathbf{b} = 2 \det K \Phi(\boldsymbol{\rho}) = 4q_{AB}q_{BC}q_{CA} \Phi(\boldsymbol{\rho}).$$

Consequently,

$$\Phi(\boldsymbol{\rho}) = \frac{\Gamma_5}{4q_{AB}q_{BC}q_{CA}}. \quad (3.60)$$

Combining the ratio-stationarity conditions in Eq. (3.56) with the remaining path-magnitude condition in Eq. (3.59), the full Landau system for wide-angle kinematics (here $\det K \neq 0$) fixes $\boldsymbol{\rho}$ uniquely and requires $\Gamma_5 = 0$. Thus vanishing Γ_5 is the only possible kinematic solution in this domain. A positive Landau pinch additionally requires $\rho_A, \rho_B, \rho_C > 0$; the corresponding chamber is identified below.

The wide-angle near-planar Landau structure.

This structure contrasts directly with the Crown. The Crown matrix K_C in Eq. (3.4) has rank three and the kinematics-independent zero mode $(1, 1, 1, 1)$. Its stationary system therefore leaves one common ratio free, while its stationary value vanishes throughout the generic wide-angle domain. For the Fish, by contrast, K is generically invertible and no common-ratio zero mode remains.

To expose the cancellation factors, complete the quadratic form in Eq. (3.52) about its stationary point:

$$\Phi(\mathbf{r}) = \Phi(\boldsymbol{\rho}) + q_{AB}(r_A - \rho_A)(r_B - \rho_B) + q_{BC}(r_B - \rho_B)(r_C - \rho_C) + q_{CA}(r_C - \rho_C)(r_A - \rho_A),$$

where the term linear in $\mathbf{r} - \boldsymbol{\rho}$ vanishes by Eq. (3.56). Using $r_A - \rho_A = f_A/x_1$, $r_B - \rho_B = f_B/x_3$, $r_C - \rho_C = f_C/x_5$, and Eq. (3.60), this gives the exact identity

$$\mathcal{F} = \frac{\Gamma_5}{4q_{AB}q_{BC}q_{CA}} x_1 x_3 x_5 + q_{AB} x_5 f_A f_B + q_{BC} x_1 f_B f_C + q_{CA} x_3 f_C f_A. \quad (3.61)$$

This is an identity for the full massless $\mathcal{F}$ at generic five-point kinematics, not merely its restriction to the Gram surface.

We now express this generic Landau solution in the original x_e variables. The purpose is twofold: to show how the derivatives available to HRF recover the cancellation factors

f_A, f_B, f_C , and to identify the remaining condition they impose on the external kinematics. HRF starts from the usual Mandelstam representation of $\mathcal{F}$, rather than the completed form in Eq. (3.61). Direct differentiation therefore gives, for example,

$$D_0 \equiv \frac{\partial \mathcal{F}}{\partial x_0} = q_{AB}x_2x_5 + q_{CA}x_3x_4 + b_Ax_3x_5. \quad (3.62)$$

Literal collection in q_{AB} , q_{CA} and b_A exposes only monomials. The first component of $K\rho = -\mathbf{b}$,

$$b_A = -q_{AB}\rho_B - q_{CA}\rho_C, \quad (3.63)$$

instead reorganises the same derivative as

$$D_0 = q_{AB}x_5(x_2 - \rho_Bx_3) + q_{CA}x_3(x_4 - \rho_Cx_5) = q_{AB}x_5f_B + q_{CA}x_3f_C. \quad (3.64)$$

Equation (3.64) shows that the f_I emerge from a nontrivial stationary reorganisation of the raw derivatives. Since the final term of Eq. (3.61) is independent of x_0, x_2, x_4 , the three stationary-adapted derivative identities are

$$\begin{aligned} \frac{\partial \mathcal{F}}{\partial x_0} &= q_{AB}x_5f_B + q_{CA}x_3f_C, \\ \frac{\partial \mathcal{F}}{\partial x_2} &= q_{AB}x_5f_A + q_{BC}x_1f_C, \\ \frac{\partial \mathcal{F}}{\partial x_4} &= q_{BC}x_1f_B + q_{CA}x_3f_A. \end{aligned} \quad (3.65)$$

Let

$$D_u = \text{diag}(u_A, u_B, u_C) = \text{diag}(x_1, x_3, x_5), \quad U = u_A u_B u_C = x_1 x_3 x_5.$$

The coefficient matrix multiplying $(f_A, f_B, f_C)^T$ on the right-hand side of the derivative identities in Eq. (3.65) is the original-variable representation of the ratio-space matrix K :

$$\begin{aligned} M_f &= U D_u^{-1} K D_u^{-1} = \begin{pmatrix} 0 & q_{AB}x_5 & q_{CA}x_3 \\ q_{AB}x_5 & 0 & q_{BC}x_1 \\ q_{CA}x_3 & q_{BC}x_1 & 0 \end{pmatrix}, \\ \det M_f &= U \det K = 2q_{AB}q_{BC}q_{CA}x_1x_3x_5. \end{aligned} \quad (3.66)$$

With $\mathbf{f} = (f_A, f_B, f_C)^T$, imposing stationarity on the three derivatives in Eq. (3.65) gives the linear system

$$M_f \mathbf{f} = 0.$$

The relation in Eq. (3.66) incorporates both $f_I = u_I(r_I - \rho_I)$ and the derivative Jacobian. In the interior positive orthant D_u is invertible, and at wide angle K is invertible; hence M_f is invertible and the stationary system has the unique solution $\mathbf{f} = 0$, or $f_A = f_B = f_C = 0$. Thus M_f encodes the original-coordinate derivative system, whereas the bordered matrix $\mathcal{B}$ above relates the stationary value $\Phi(\rho)$ to Γ_5 .

The remaining three derivatives then reduce to

$$\left(\frac{\partial \mathcal{F}}{\partial x_1}, \frac{\partial \mathcal{F}}{\partial x_3}, \frac{\partial \mathcal{F}}{\partial x_5} \right) \Big|_{f_A=f_B=f_C=0} = \frac{\Gamma_5}{4q_{AB}q_{BC}q_{CA}}(x_3x_5, x_1x_5, x_1x_3). \quad (3.67)$$

Likewise, Eq. (3.61) restricted to this locus equals $\Gamma_5 x_1 x_3 x_5 / (4q_{AB}q_{BC}q_{CA})$. The original-variable Landau system therefore reproduces the ratio-space result in two stages: the first three derivatives identify the local defining ideal $\langle f_A, f_B, f_C \rangle$, while the remaining equations require $\Gamma_5 = 0$. This separation explains why the near-planar surface must be selected before the HRF expansion begins.

Positivity of the resulting ρ_I is precisely the usual interior first-sheet condition $x_e > 0$, because the ρ_I are the stationary values of the positive ratios in Eq. (3.51). This condition can be stated directly in Mandelstam variables. Solving Eq. (3.56) gives

$$\begin{aligned}\rho_A &= \frac{q_{BC}b_A - q_{CA}b_B - q_{AB}b_C}{2q_{AB}q_{CA}}, \\ \rho_B &= \frac{-q_{BC}b_A + q_{CA}b_B - q_{AB}b_C}{2q_{AB}q_{BC}}, \\ \rho_C &= \frac{-q_{BC}b_A - q_{CA}b_B + q_{AB}b_C}{2q_{BC}q_{CA}},\end{aligned}\tag{3.68}$$

where every q_{IJ} and b_I is the Mandelstam combination in Eq. (3.53). Where these three q_{IJ} are nonzero, the nondegenerate positive planar Fish chamber is therefore

$$\mathfrak{C}_{\text{Fish}}^+ = \{\mathbf{s} \in \mathcal{K} : \Gamma_5 = 0, \quad \rho_A(\mathbf{s}) > 0, \quad \rho_B(\mathbf{s}) > 0, \quad \rho_C(\mathbf{s}) > 0\}.\tag{3.69}$$

The Gram equation supplies planarity, while the three inequalities select the first-sheet component of the singular locus. This is the generic kinematic statement; it does not assign an asymptotic scaling to any particular approach to the chamber or to its closure.

The spacelike-collinear example above gives the simplest boundary test. Both the wide-angle and $p_2 \parallel p_3$ collinear problems are invariant under the permutation $p_4 \leftrightarrow p_5$, which does not affect the relevant two-edge path. Translating the collinear factor f_2 in Eq. (3.21) to the zero-based edge labels used here gives

$$f_2^{\text{coll}} = (z - 1)x_2 - x_3. \quad f_2^{\text{coll}} = 0 \quad \implies \quad \frac{x_3}{x_2} = z - 1 > 0, \quad r_B = \frac{x_2}{x_3} = \frac{1}{z - 1} > 0.\tag{3.70}$$

Thus the spacelike range $z > 1$ approaches the positive closure of Eq. (3.69). In the timelike-collinear range $0 < z < 1$, the same ratio is negative, so planarity alone does not preserve the pinch. At this boundary $q_{CA} = s_{23} \rightarrow 0$, the matrix K becomes singular and the three-ratio locus degenerates to the two-factor collinear locus. The positive chamber, rather than the Gram surface alone, is therefore the appropriate general organising object.

Returning to HRF: the near-planar leading polynomial.

The generic Landau analysis has now selected $\Gamma_5 = 0$. We therefore fix the near-planar expansion before invoking HRF. This is the point at which the standard HRF workflow begins: the expansion is specified and its leading polynomial can be constructed. Specialising the generic construction, the normals become especially transparent in the exact chart (3.45). Taking $\lambda \rightarrow 0$ in Eq. (3.48) gives the exact-planarity relation $\zeta = \bar{\zeta} = \xi$. To write the resulting solution of $K\boldsymbol{\rho} = -\mathbf{b}$ compactly, we introduce the kinematic ratio

$$R \equiv \frac{\xi + X_{45}(\xi + 1)}{1 + X_{34}X_{45}(\xi + 1)}.\tag{3.71}$$

The solution of $K\boldsymbol{\rho} = -\mathbf{b}$ is then

$$\rho_A = X_{34}\xi R, \quad \rho_B = R, \quad \rho_C = \xi, \quad (3.72)$$

and therefore

$$f_A = x_0 - X_{34}\xi R x_1, \quad f_B = x_2 - R x_3, \quad f_C = x_4 - \xi x_5. \quad (3.73)$$

On the physical positive-Landau branch $X_{34}, X_{45}, \xi > 0$, so $R > 0$ and all three stationary ratios are positive. Beyond the near-planar expansion, no further kinematic restrictions are imposed (in particular, no MRK hierarchy).

To define the near-planar expansion, let $\boldsymbol{\kappa}$ denote four coordinates tangent to the Gram surface and use Γ_5 as a local normal coordinate. Then

$$\mathcal{F}(\Gamma_5, \boldsymbol{\kappa}) = \mathcal{F}_0(\boldsymbol{\kappa}) + \Gamma_5 \mathcal{F}_{\Gamma_5}(\boldsymbol{\kappa}) + O(\Gamma_5^2). \quad (3.74)$$

There is no conflict between the apparently nonlinear kinematic dependence in Eq. (3.61) and the linearity of $\mathcal{F}$ in Mandelstam invariants. The Gram determinant Γ_5 is itself a nonlinear polynomial in those invariants. Replacing one Mandelstam variable by Γ_5 is therefore a nonlinear change of kinematic coordinates, and at fixed $\boldsymbol{\kappa}$

$$\mathcal{F}_{\Gamma_5} = \left. \frac{\partial \mathcal{F}}{\partial \Gamma_5} \right|_{\boldsymbol{\kappa}} = \sum_a \left. \frac{\partial \mathcal{F}}{\partial s_a} \frac{\partial s_a}{\partial \Gamma_5} \right|_{\boldsymbol{\kappa}}. \quad (3.75)$$

The Jacobian factors on the right need not be linear functions of the remaining kinematics. Equivalently, the separate terms in Eq. (3.61) contain the rational ratios $\boldsymbol{\rho} = -K^{-1}\mathbf{b}$; their nonlinear dependence cancels in the sum, restoring the original Mandelstam-linear polynomial.

The complete coefficient $\mathcal{F}_{\Gamma_5}$ away from the stationary locus depends on the choice of the tangent coordinates $\boldsymbol{\kappa}$. Its restriction to $f_A = f_B = f_C = 0$, however, is fixed by Eq. (3.61):

$$\mathcal{F}_{\Gamma_5}|_{f_A=f_B=f_C=0} = \frac{x_1 x_3 x_5}{4q_{AB}q_{BC}q_{CA}}. \quad (3.76)$$

This stationary contribution is not an “obstruction” in the technical HRF sense: it is absent from the leading polynomial by the definition of the chosen kinematic expansion, rather than being made subleading by the HR scaling. Since $\Gamma_5 = O(\lambda^2)$, this contribution first appears two powers beyond $\mathcal{F}_0$. Its precise weight relative to the cancelled leading sector and to $\mathcal{U}$ will follow only after the region vector has been determined below. The leading polynomial itself has the symmetric factorised form

$$\mathcal{F}_0 \equiv \mathcal{F}|_{\Gamma_5=0} = q_{AB}x_5 f_A f_B + q_{BC}x_1 f_B f_C + q_{CA}x_3 f_C f_A. \quad (3.77)$$

Thus the HR generators are the factorised products $f_A f_B, f_B f_C, f_C f_A$, not the individual normal equations. This is a three-generator ideal and is more general than the pair-sector search used in the earlier examples.

HRF factor harvesting, generators and scaling.

With the expansion fixed and $\mathcal{F}_0$ displayed, we now apply the core HRF construction to recover the cancellation locus from its derivatives and determine the region scaling. HRF starts from $\mathcal{F}_0$; in this example its decomposition finds no obstruction, so that $\mathcal{F}_\star = \mathcal{F}_0 = \mathcal{F}_{\text{SL}}$ and the complete leading polynomial is the cancellation sector. The selected-sector saturation of Eq. (2.28) is therefore applicable directly to $\mathcal{F}_0$.

In this example the abstract saturation in Eq. (2.28) becomes completely explicit. Let $\mathcal{R}$ denote the polynomial ring in the LP parameters and the chosen kinematic variables, and define

$$\mathcal{I}_{\nabla}^{(0)} = \left\langle \frac{\partial \mathcal{F}_0}{\partial x_e} \right\rangle_{e=0}^5, \quad \mathcal{I}_{\text{kin}} = \langle \Gamma_5 \rangle, \quad S_{\text{np}} = \left(\prod_{e=0}^5 x_e \right) q_{AB} q_{BC} q_{CA}. \quad (3.78)$$

Here and below, the subscript “np” denotes the near-planar sector. The selected-sector ideal used by HRF is therefore

$$\mathcal{J}_{\text{np}} = (\mathcal{I}_{\nabla}^{(0)} + \mathcal{I}_{\text{kin}}) : S_{\text{np}}^\infty. \quad (3.79)$$

The algebraic recovery of the cancellation-factor vector $\mathbf{f} = (f_A, f_B, f_C)^T$ follows directly from Eq. (3.66). Let $\mathbf{D}$ denote the vector of the first three derivatives in Eq. (3.65). Then $\mathbf{D} = M_f \mathbf{f}$, and the adjugate identity gives

$$\det(M_f) \mathbf{f} = \text{adj}(M_f) \mathbf{D}.$$

Every component on the right belongs to $\mathcal{I}_{\nabla}^{(0)}$. Moreover, $\det M_f = 2q_{AB}q_{BC}q_{CA}x_1x_3x_5$ is, up to a nonzero constant, a factor of S_{np} . Saturation therefore makes $\det M_f$ invertible, so the adjugate identity implies $f_A, f_B, f_C \in \mathcal{J}_{\text{np}}$. This is a genuine recovery step: the q_{IJ} entering the matrix elements of M_f in Eq. (3.66) are known kinematic coefficients, whose nonvanishing specifies the generic wide-angle sector, whereas $\mathbf{f}$ is recovered from the derivatives rather than supplied to HRF. The Gröbner-basis elimination described in the footnote below implements the polynomial combinations and permitted divisions made explicit by the adjugate identity.

In physical terms, this saturation says that the active LP parameters and the kinematic factors q_{AB}, q_{BC}, q_{CA} are nonzero, so division by their products is allowed. It performs all such divisions systematically and removes solution components supported entirely on the excluded surface $S_{\text{np}} = 0$.¹⁴ Conversely, on $\Gamma_5 = 0$, all derivatives of $\mathcal{F}_0$ vanish on that locus. Thus the LP-parameter content of $\mathcal{J}_{\text{np}}$ is precisely the local defining ideal $\langle f_A, f_B, f_C \rangle$.

To connect this ideal calculation back to the HRF harvesting step, return to the three derivative identities in Eq. (3.65). Each identity contains two terms. Across the three identities these six terms fall into the three kinematically independent sectors $q_{AB} = s_{35}$, $q_{BC} = s_{13}$ and $q_{CA} = s_{23}$, using Eq. (3.53). Collecting the first identity by q_{AB} and q_{CA} ,

¹⁴A standard computational implementation introduces an auxiliary variable τ and defines $\tilde{\mathcal{J}}_{\text{np}} = \langle \partial \mathcal{F}_0 / \partial x_e \ (e = 0, \dots, 5), \Gamma_5, 1 - \tau S_{\text{np}} \rangle \subset \mathcal{R}[\tau]$. A Gröbner basis in an elimination order for τ then gives $\mathcal{J}_{\text{np}} = \tilde{\mathcal{J}}_{\text{np}} \cap \mathcal{R}$. The added equation sets $\tau = S_{\text{np}}^{-1}$, so elimination retains precisely the polynomial consequences obtainable after division by powers of S_{np} . This is the Rabinowitsch trick; see, for example, Ref. [62].

for example, exposes $x_5 f_B$ and $x_3 f_C$; the other two identities similarly expose all three local defining polynomials.

For the Crown, direct invariant-sector harvesting already produces this adapted form. Here the f_I contain the kinematic stationary ratios ρ , so literal factorisation or collection of an individual raw derivative does not reveal them. The selected-sector saturation in Eq. (3.79) supplies precisely this algebraic reorganisation and recovers the three normals without assuming them in advance. HRF then combines the harvested normals into candidate factorised generators; their occurrence in $\mathcal{F}_0$ identifies the generator ideal $\langle f_A f_B, f_B f_C, f_C f_A \rangle$.

For the expansion defined in Eqs. (3.48) and (3.49), the original-coordinate vector determined by HRF and its weights are

$$\vec{v}_{\text{HR}} = (-2, -2, -2, -2, -2, -2; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-6, -4). \quad (3.80)$$

The comparison announced above is now fixed. Every cubic monomial in $\mathcal{F}_0$ has raw weight -6 . The Gram-normal factor $\Gamma_5 = O(\lambda^2)$ raises the weight of $\Gamma_5 \mathcal{F}_{\Gamma_5}$ to -4 , while the quadratic polynomial $\mathcal{U}$ also has weight -4 . Cancellation on the singular locus promotes $\mathcal{F}_0$ from $W_{\text{SL}} = -6$ to the same resolved weight $W_{\text{HR}} = -4$. Thus the cancelled leading sector, the first Gram-normal correction and $\mathcal{U}$ all enter the resolved LP polynomial at W_{HR} .

Parameter- and momentum-space power counting.

The HRF construction has now produced the generators and determined the scaling vector in the original coordinates. We first count directly in the original LP-parameter domain. This count does not itself require dissection; the final HR certificate is supplied below. The vector in Eq. (3.80) gives $x_e \sim \lambda^{-2}$ for all six parameters. HRF also computes the cancellation depth as the gap

$$\Delta W = W_{\text{HR}} - W_{\text{SL}} = (-4) - (-6) = 2. \quad (3.81)$$

Every term of $\mathcal{F}_{\text{SL}}$ in Eq. (3.61) has the form $x_e f_I f_J$. If d_I denotes the reduction of the natural normal width by powers of λ , equal resolved weights require $d_A + d_B = d_B + d_C = d_C + d_A = \Delta W = 2$, and hence $d_A = d_B = d_C = 1$. The cancellation domain therefore restricts the three independent normal combinations to

$$|x_0 - \rho_A x_1| = O(\lambda^{-1}), \quad |x_2 - \rho_B x_3| = O(\lambda^{-1}), \quad |x_4 - \rho_C x_5| = O(\lambda^{-1}). \quad (3.82)$$

Unlike the single-product Fish decomposition, the three pair products form a full-rank system for the normal depths, so no continuous allocation freedom remains. Without these restrictions the six measures would give λ^{-12} . Because of the cancellation, each normal has width $O(\lambda^{-1})$ rather than the unrestricted width $O(\lambda^{-2})$, restoring one power of λ . Since the resolved LP polynomial has weight -4 , the original-variable parameter-space count is therefore

$$I_{\text{par}}^{\text{scalar}} \sim \underbrace{\lambda^{-12}}_{\text{unrestricted}} \underbrace{\lambda^3}_{\text{restricted widths}} \underbrace{(\lambda^{-4})^{-D/2}}_{p^{-D/2}} = \lambda^{2D-9} = \lambda^{-1-4\epsilon}, \quad D = 4 - 2\epsilon. \quad (3.83)$$

The dissection supplying that certificate independently reproduces this count, although it is not used here to evaluate the integral. Introduce the signed local coordinates in a first step,

$$\begin{aligned} u_A &= x_1, & y_A &= f_A, & x_0 &= \rho_A u_A + y_A, \\ u_B &= x_3, & y_B &= f_B, & x_2 &= \rho_B u_B + y_B, \\ u_C &= x_5, & y_C &= f_C, & x_4 &= \rho_C u_C + y_C. \end{aligned} \quad (3.84)$$

Thus the six new coordinates are $(u_A, u_B, u_C; y_A, y_B, y_C)$: three tangential path magnitudes and three signed normal displacements from the cancellation locus. At fixed kinematics the transformation from the six x_e to these six coordinates has unit absolute Jacobian. Only in a second step is the domain divided into sign sectors,

$$y_I = \sigma_I t_I, \quad t_I \geq 0, \quad \sigma_I = \pm 1. \quad (3.85)$$

The cancellation locus is now the boundary $t_A = t_B = t_C = 0$. In every sector the corresponding ordinary lower facet has $u_I \sim \lambda^{-2}$ and $t_I \sim \lambda^{-1}$, whose pullback is Eq. (3.80). The dissected parameter-space count is consequently

$$I_{\text{dis}}^{\text{scalar}} \sim \underbrace{(\lambda^{-2})^3 (\lambda^{-1})^3}_{\prod_I du_I dt_I} (\lambda^{-4})^{-D/2} = \lambda^{2D-9} = \lambda^{-1-4\epsilon}, \quad (3.86)$$

in agreement with the direct count in Eq. (3.83).

The paired sign sectors also expose a symmetry of the subleading HR expansion. Reversing the orientation transverse to the planar configuration sends $\epsilon_5 \rightarrow -\epsilon_5$, equivalently $\lambda \rightarrow -\lambda$, while leaving all scalar products and $\Gamma_5 \propto \epsilon_5^2$ unchanged. In the local coordinates this transformation pairs sectors through $(y_A, y_B, y_C) \rightarrow (-y_A, -y_B, -y_C)$. The local expansion is invariant under the simultaneous reversal of λ and the three normal coordinates, so contributions at odd relative powers of λ cancel after the paired sectors are summed. For scalar integrals and parity-even numerators, the HR contribution therefore has the all-order structure

$$I_{\text{HR}} \sim \lambda^{-1-4\epsilon} [c_0(\epsilon) + c_1(\epsilon)\lambda^2 + c_2(\epsilon)\lambda^4 + \dots]. \quad (3.87)$$

Parity-odd numerators are not constrained in this way.

Since Eq. (3.48) gives $w - \bar{w} = i\lambda$ before any MRK hierarchy is imposed, Eq. (3.87) also describes the local two-sided behaviour about the planar locus at general five-point kinematics. This two-sided structure is therefore inherited by its MRK specialisation, in which $w, \bar{w}$ become the customary variables $z, \bar{z}$. Earlier MRK studies encountered the same non-trivial continuation of parity-odd non-planar integrals across $z = \bar{z}$, together with related cancellations in the $\mathcal{N} = 4$ sYM hard function and in four-dimensional QCD finite remainders; the complete two-loop Lipatov vertex has no physical singularity on this locus [58, 59, 63].

Finally, we give a momentum-space consistency count for a physical realisation of this HR. Throughout this momentum-space count, let

$$P_i \equiv p_i|_{\lambda=0}$$

denote the complete external four-momenta at the exactly planar point. The parameter-space vector fixes the six propagator virtualities through Eq. (2.13), but a compatible assignment of momentum components requires additional physical input. Let us use the routing shown in Fig. 9. The two independent loop momenta are q_A and q_B , while conservation at the upper common hard vertex fixes

$$q_C = -p_3 - q_A - q_B. \quad (3.88)$$

The matching edge colours in Fig. 9 display the three path assignments: the two edges on path A carry q_A and $q_A + p_1$, those on path B carry q_B and $q_B + p_2$, and those on path C carry q_C and $q_C - p_5$. The parameter-space vector makes both propagators on every path null at $\lambda = 0$. For example, $(q_A^*)^2 = (q_A^* + P_1)^2 = P_1^2 = 0$ implies $P_1 \cdot q_A^* = 0$, and the analogous relations hold on paths B and C . On the physical real branch, each pair of null momenta with vanishing scalar product is collinear. Hence the central path momenta lie on the P_1 , P_2 and P_5 rays, respectively; this conclusion follows from the virtualities and the on-shell Landau branch rather than being an independent mode assumption.

Expanding each two-propagator pair about its central ray supplies the usual collinear fluctuation envelope. For each path $I = A, B, C$, let $\bar{P}_I$ denote a conjugate null reference direction and use the local basis $(+, -, \perp)_{(P_I, \bar{P}_I)}$, with $P_I = P_1, P_2, P_5$, respectively. The component widths are

$$(\Delta q_I^+, \Delta q_I^-, |\Delta q_I^\perp|)_{(P_I, \bar{P}_I)} \sim (1, \lambda^2, \lambda)_{(P_I, \bar{P}_I)}, \quad I = A, B, C, \quad (3.89)$$

up to interchanging the two lightcone components according to orientation. The companion edge on the same path has the same envelope. Consequently each marginal independent collinear loop measure scales as

$$d^D q_I \sim dq_I^+ dq_I^- d^{D-2} q_I^\perp \sim \lambda^D, \quad I = A, B. \quad (3.90)$$

These two marginal envelopes are not yet the complete joint support: the dependent path C correlates them. Before imposing that correlation, the allowed ranges of the longitudinal fraction, the conjugate lightcone component and each transverse component are $O(1)$, $O(\lambda^2)$ and $O(\lambda)$, respectively. The additional restriction on the two independent longitudinal fractions can be formulated covariantly. For the two independent loop momenta write

$$q_A = \alpha_A P_1 + k_A, \quad q_B = \alpha_B P_2 + k_B, \quad (3.91)$$

where the large components are absorbed into α_A, α_B , while in the respective collinear bases the residual momenta have conjugate lightcone components of order λ^2 and transverse components of order λ . The third path momentum is not independent: as defined in Eq. (3.88), $q_C = -p_3 - q_A - q_B$. At the exact pinch the residual momenta vanish and the three path momenta obey

$$q_A^* = \alpha_A^* P_1, \quad q_B^* = \alpha_B^* P_2, \quad q_C^* = \alpha_C^* P_5.$$

Consequently momentum conservation at $\lambda = 0$ imposes the constraint

$$P_5 = -\frac{P_3 + \alpha_A^* P_1 + \alpha_B^* P_2}{\alpha_C^*}. \quad (3.92)$$

To express the allowed departure of q_C from this ray invariantly, choose a null vector $\bar{P}_5$ along the conjugate lightcone direction, with $P_5 \cdot \bar{P}_5 \neq 0$, and decompose

$$q_C = \alpha_C P_5 + k_{C\perp} + \beta_C \bar{P}_5, \quad k_{C\perp} \cdot P_5 = k_{C\perp} \cdot \bar{P}_5 = 0. \quad (3.93)$$

The collinear envelope of the p_5 path means $k_{C\perp} = O(Q\lambda)$ and $\beta_C = O(\lambda^2)$, where $\bar{P}_5$ is taken to be hard. Two dimensionless Lorentz scalars that vanish on the P_5 ray are

$$\begin{aligned} \Delta_5(q_C) &= \frac{P_1 \cdot q_C}{P_1 \cdot P_5} - \frac{P_2 \cdot q_C}{P_2 \cdot P_5}, \\ \Sigma_5(q_C) &= \frac{(P_1 \cdot P_2)(P_5 \cdot q_C)}{(P_1 \cdot P_5)(P_2 \cdot P_5)}. \end{aligned} \quad (3.94)$$

For $q_C = \alpha_C P_5$, the two ratios in $\Delta_5(q_C)$ both equal α_C , so they cancel in the difference and $\Delta_5(P_5) = 0$. Likewise $\Sigma_5(P_5) = 0$ because $P_5^2 = 0$.

All denominator factors are fixed by Eq. (3.48) at $\lambda = 0$, and are hard and nonzero at generic wide angle. Substitution of Eq. (3.93) gives

$$\begin{aligned} \Delta_5(q_C) &= \frac{P_1 \cdot k_{C\perp}}{P_1 \cdot P_5} - \frac{P_2 \cdot k_{C\perp}}{P_2 \cdot P_5} + \beta_C \left(\frac{P_1 \cdot \bar{P}_5}{P_1 \cdot P_5} - \frac{P_2 \cdot \bar{P}_5}{P_2 \cdot P_5} \right) = O(\lambda), \\ \Sigma_5(q_C) &= \beta_C \frac{(P_1 \cdot P_2)(P_5 \cdot \bar{P}_5)}{(P_1 \cdot P_5)(P_2 \cdot P_5)} = O(\lambda^2). \end{aligned} \quad (3.95)$$

Thus these invariant scalings express the local collinear envelope of the dependent path. Their effect on the joint support of the two independent loop variables follows from the routing constraint.

Set $\delta\alpha_I = \alpha_I - \alpha_I^*$ for $I = A, B$. Inserting the decompositions in Eq. (3.91) into the routing equation (3.88), and subtracting its $\lambda = 0$ form in Eq. (3.92), gives

$$q_C - \alpha_C^* P_5 = -\delta\alpha_A P_1 - \delta\alpha_B P_2 + \ell_{\text{res}}, \quad \ell_{\text{res}} = -(p_3 - P_3) - k_A - k_B. \quad (3.96)$$

Applying the two invariant projections gives

$$\begin{pmatrix} \Delta_5(q_C) \\ \Sigma_5(q_C) \end{pmatrix} - \begin{pmatrix} \Delta_5(\ell_{\text{res}}) \\ \Sigma_5(\ell_{\text{res}}) \end{pmatrix} = (P_1 \cdot P_2) \begin{pmatrix} \frac{1}{P_2 \cdot P_5} & -\frac{1}{P_1 \cdot P_5} \\ -\frac{1}{P_2 \cdot P_5} & -\frac{1}{P_1 \cdot P_5} \end{pmatrix} \begin{pmatrix} \delta\alpha_A \\ \delta\alpha_B \end{pmatrix}. \quad (3.97)$$

The two vectors in Eq. (3.97) have opposite hierarchies. The P_5 -collinear path pair constrains the complete dependent momentum q_C , and Eq. (3.95) gives $\Delta_5(q_C) = O(\lambda)$ and $\Sigma_5(q_C) = O(\lambda^2)$. For the residual vector, the contractions with P_1 and P_2 entering Δ_5 see the conjugate lightcone components of k_A, k_B , and the corresponding components of $p_3 - P_3$, all of which are $O(\lambda^2)$. By contrast, $P_5 \cdot \ell_{\text{res}}$ entering Σ_5 generically sees the $O(\lambda)$ in-plane transverse components of k_A, k_B . Hence

$$\begin{pmatrix} \Delta_5(q_C) \\ \Sigma_5(q_C) \end{pmatrix} \sim \begin{pmatrix} \lambda \\ \lambda^2 \end{pmatrix}, \quad \begin{pmatrix} \Delta_5(\ell_{\text{res}}) \\ \Sigma_5(\ell_{\text{res}}) \end{pmatrix} \sim \begin{pmatrix} \lambda^2 \\ \lambda \end{pmatrix}.$$

Thus the upper row of Eq. (3.97) is dominated by $\Delta_5(q_C)$. In the lower row the $O(\lambda)$ contribution from $\Sigma_5(\ell_{\text{res}})$ must instead be cancelled by the longitudinal fractions, leaving the required $\Sigma_5(q_C) = O(\lambda^2)$. At fixed k_A, k_B , this cancellation fixes the centre of the allowed $(\delta\alpha_A, \delta\alpha_B)$ domain. The determinant of the coefficient matrix is

$$-\frac{2(P_1 \cdot P_2)^2}{(P_1 \cdot P_5)(P_2 \cdot P_5)} \neq 0$$

at generic wide angle. Since a translation does not change the measure, the conditional volume follows from the two collinear ranges and this nonsingular constant Jacobian:

$$d(\delta\alpha_A) d(\delta\alpha_B) = \frac{|(P_1 \cdot P_5)(P_2 \cdot P_5)|}{2|P_1 \cdot P_2|^2} d\Delta_5(q_C) d\Sigma_5(q_C) \sim \lambda \lambda^2 = \lambda^3.$$

The remaining transverse components are supplied by the residual momenta already counted in the two collinear measures. As required by the parameter-space vector, all six propagator virtualities are then $q_e^2 = O(\lambda^2)$. Therefore

$$I_{\text{mom}}^{\text{scalar}} \sim \underbrace{(\lambda^D)^2}_{\text{collinear measures}} \underbrace{\lambda^3}_{\text{restricted support}} \underbrace{(\lambda^{-2})^6}_{\text{propagators}} = \lambda^{2D-9} = \lambda^{-1-4\epsilon}. \quad (3.98)$$

The agreement of Eqs. (3.83), (3.86) and (3.98) checks the cancellation depth first in the original LP-parameter domain, then in local dissected coordinates, and finally as restricted loop-momentum support. In the (q_A, q_B) basis, the separate collinear envelopes have the same component powers as their average momenta, but the additional factor λ^3 is a correlation in their joint support. It cannot be recovered by comparing the average-momentum scaling and marginal width of each loop separately.

The algebraic comparison with the Crown also has a direct momentum-space interpretation. At the vertex joining the two edges of each collinear path, the attached external momentum is shared between the two propagator momenta. This relation follows from the momentum-space Landau equations before the Gaussian loop integrations are performed. Using the independent loop momenta q_A and q_B introduced above Eq. (3.88), differentiate the Schwinger exponent in Eq. (2.3) with respect to each. The difference of the two equations cancels the common contribution from the dependent momentum q_C and gives

$$0 = \frac{1}{2} \left(\frac{\partial \mathcal{S}}{\partial q_A} - \frac{\partial \mathcal{S}}{\partial q_B} \right) = \tilde{x}_0 q_A + \tilde{x}_1 (q_A + p_1) - \tilde{x}_2 q_B - \tilde{x}_3 (q_B + p_2). \quad (3.99)$$

At the pinch, Eq. (3.99) becomes

$$0 = [(\tilde{x}_0 + \tilde{x}_1)\alpha_A^* + \tilde{x}_1] P_1 - [(\tilde{x}_2 + \tilde{x}_3)\alpha_B^* + \tilde{x}_3] P_2.$$

Since P_1 and P_2 are linearly independent at generic wide angle, their coefficients vanish separately. In particular,

$$\alpha_A^* = -\frac{\tilde{x}_1}{\tilde{x}_0 + \tilde{x}_1}, \quad q_A^* = -\frac{\tilde{x}_1}{\tilde{x}_0 + \tilde{x}_1} P_1, \quad q_A^* + P_1 = \frac{\tilde{x}_0}{\tilde{x}_0 + \tilde{x}_1} P_1.$$

Thus the proper-time ratio $\tilde{r}_A = \tilde{x}_0/\tilde{x}_1$ fixes the two longitudinal shares. The second coefficient above fixes the sharing on path B ; substituting these results into either unsubtracted Landau equation fixes the analogous sharing on path C .

Equation (3.99) thereby supplies a momentum-space interpretation of the three collinear-path sharings. Physically, each external lightlike momentum splits at the vertex between the two propagators of its path, and the two propagator momenta remain in the same collinear sector. This is the Coleman–Norton picture of the positive pinch: the active massless propagators describe on-shell classical trajectories, and the two null momenta on each path differ by the attached null external momentum, which places them on the same collinear ray [42, 55]. Accommodating the three distinct collinear sectors simultaneously produces the momentum-space constraints on their longitudinal shares. The comparison with HRF is structural: the momentum-space Landau equations fix these three collinear-splitting fractions, while the LP analysis independently fixes the three ratios through $K\boldsymbol{\rho} = -\mathbf{b}$ in Eq. (3.56).

In the Crown there are four such collinear two-edge paths: each external lightlike particle splits into two propagator momenta in its own collinear sector, and no external momentum is inserted at either common hard vertex. Momentum conservation among the four collinear sectors leaves one common sharing parameter free, which is the momentum-space counterpart of the zero mode of K_C . In the Fish there are only three collinear two-edge paths, carrying the p_1 -, p_2 - and p_5 -collinear sectors, while p_3 and p_4 enter at their two common hard vertices. These hard insertions supply the inhomogeneous terms $\mathbf{b}$; since K is invertible, the balance equations fix the unique ratio vector $\boldsymbol{\rho} = -K^{-1}\mathbf{b}$. The remaining path-magnitude Landau equation is $\Phi(\boldsymbol{\rho}) = 0$, which by Eq. (3.60) is equivalent to $\Gamma_5 = 0$; it ensures that these uniquely fixed collinear splittings are compatible with the full five-point kinematics. The contrast is therefore explicit: the Crown condition $r_A = r_B = r_C = r_D$ supplies only three independent relative constraints and leaves their common value free, whereas the Fish fixes all three ratios ρ_A, ρ_B, ρ_C . In momentum space this complete fixing restricts the longitudinal integration domain; its measure is the λ^3 factor labelled “restricted support” in Eq. (3.98).

3.2.5 Simultaneous fixed-transverse MRK and near-planarity

An interesting limit, natural to consider here, combines the approach to planarity with MRK. The general kinematic chart in Eq. (3.45) provides a straightforward parametrisation of this limit. To separate its two ingredients, introduce two independent positive dimensionless parameters. The parameter x controls the longitudinal hierarchy, and hence the rapidity separations, through $X_{34}, X_{45} \propto x^{-1}$, while λ controls the departure from planarity through the imaginary parts of w and $\bar{w}$:

$$\begin{aligned} X_{34} &= \frac{\hat{X}_{34}}{x}, & X_{45} &= \frac{\hat{X}_{45}}{x}, \\ w &= -\frac{1}{\xi} + \frac{i}{2}\lambda, & \bar{w} &= -\frac{1}{\xi} - \frac{i}{2}\lambda. \end{aligned} \tag{3.100}$$

with $\widehat{X}_{34}, \widehat{X}_{45}, \xi > 0$ and $|\mathbf{p}_4|^2 > 0$ fixed. Thus $x \rightarrow 0$ produces the fixed-transverse MRK hierarchy, whereas $\lambda \rightarrow 0$ approaches planarity. We correlate the two limits into a one-parameter family by setting

$$\lambda = \delta^a, \quad x = \delta^b, \quad a > 0, \quad b \geq 0. \quad (3.101)$$

For $b > 0$, both longitudinal ratios grow while all transverse scales remain fixed, giving standard fixed-transverse MRK. In this hierarchy p_2, p_3 carry large p^+ , while p_1, p_5 carry large p^- . These rapidity groupings do not identify either pair as a single collinear mode. At $b = 0$ the longitudinal hierarchy and the corresponding grouping are absent, and the kinematics returns to the near-planar wide-angle case analysed in Sec. 3.2.4.

The transverse approach to planarity is controlled independently by λ . To display it directly in momentum variables, the $w, \bar{w}$ relations in Eq. (3.100) may be rewritten as

$$|\mathbf{p}_3|^2 = |\mathbf{p}_4|^2 \left(\frac{1}{\xi^2} + \frac{\lambda^2}{4} \right), \quad 2\mathbf{p}_3 \cdot \mathbf{p}_4 = \frac{2|\mathbf{p}_4|^2}{\xi}. \quad (3.102)$$

It follows that

$$\frac{\Gamma_5}{s_{12}^2} = -|\mathbf{p}_4|^4 \lambda^2 = -|\mathbf{p}_4|^4 \delta^{2a}. \quad (3.103)$$

The Gram invariant Γ_5 encodes the departure from planarity. In the normalisation of Eq. (3.103), and since $|\mathbf{p}_4|^2$ is fixed in the present limit, this departure is governed directly by $\lambda = \delta^a$.

We now apply this composite limit to the Fish seed. For the fixed incoming-leg convention, the attachment audit finds an HR precisely for the two permutations in which p_1, p_2, p_5 are attached to the three trivalent path vertices and p_3, p_4 to the two four-valent vertices. These two HR-supporting permutations, which differ only by exchanging p_3 and p_4 between the four-valent vertices, are shown in Fig. 10(a,b). Panel (c) displays the distinct $p_3 \leftrightarrow p_5$ interchange used below to illustrate why putting p_4 at a four-valent vertex is not by itself sufficient.

To see how the MRK hierarchy changes the wide-angle near-planar stationary locus, recall for the representative permutation in Fig. 10(a) the exact stationary identity in Eq. (3.61):

$$\mathcal{F} = \underbrace{\frac{\Gamma_5}{4q_{AB}q_{BC}q_{CA}} x_1 x_3 x_5}_{\text{Gram-normal term}} + q_{AB} x_5 g_{AB} + q_{BC} x_1 g_{BC} + q_{CA} x_3 g_{CA},$$

$$g_{AB} = f_A f_B, \quad g_{BC} = f_B f_C, \quad g_{CA} = f_C f_A. \quad (3.104)$$

Here $f_A = x_0 - \rho_A x_1$, $f_B = x_2 - \rho_B x_3$ and $f_C = x_4 - \rho_C x_5$. The preceding subsection showed how the saturated derivative ideal recovers these factors from the leading polynomial and pairs them into the generator ideal $\langle f_A f_B, f_B f_C, f_C f_A \rangle$; here we retain that result without repeating the construction.

Equation (3.104) is not the representation supplied to HRF. After inserting the composite scaling, HRF starts from the unaligned leading polynomial $\mathcal{F}_0$ in its ordinary Mandelstam form. The Gram-normal term is subleading in δ and is therefore absent from $\mathcal{F}_0$. We use

the exact identity above only to track how the previously established stationary structure and generator sectors degenerate in the MRK hierarchy. As established in Eqs. (3.51) and (3.56), $\rho = (\rho_A, \rho_B, \rho_C)^T = -K^{-1}\mathbf{b}$ gives the stationary values of the three path ratios (r_A, r_B, r_C) . Hence the generator locus lies in the positive LP domain precisely when all three ρ_I are positive.

For this diagram, the composite limit defined by Eqs. (3.100) and (3.101) gives

$$(\rho_A, \rho_B, \rho_C) \sim \left(\xi, \frac{x}{\widehat{X}_{34}}, \xi \right). \quad (3.105)$$

For every nonzero x these ratios are positive, so the candidate locus remains in the physical domain and passes the HRF positivity test. The trajectory approaches the boundary of the positive planar chamber through $\rho_B \rightarrow 0$ as $x \rightarrow 0$. Exchanging p_3 and p_4 between the two four-valent vertices gives the hard-vertex-reflected configuration in Fig. 10(b), which passes the same test; both candidates are certified below. The corresponding edge-momentum configuration for the representative permutation is displayed below in Fig. 11.

The situation changes for Fig. 10(c), obtained from panel (b) by interchanging p_3 and p_5 while keeping p_1, p_2, p_4 fixed. At $b = 0$ this configuration still supports the near-planar wide-angle HR of Sec. 3.2.4. For $b > 0$, however, p_3 and p_5 approach the p_2 - and p_1 -directed ends of the rapidity ordering, respectively, so their interchange alone is not a symmetry of the MRK kinematics. Its stationary ratios behave as

$$(\rho_A, \rho_B, \rho_C)_{p_3 \leftrightarrow p_5} \sim \left(\widehat{X}_{45} \frac{(1 + \xi)^2}{\xi^2 x}, -\frac{1 + \xi}{\xi}, -\frac{1 + \xi}{\xi} \right). \quad (3.106)$$

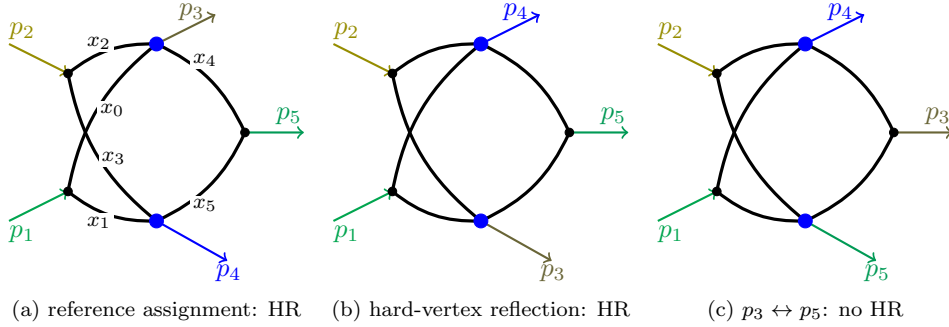


Figure 10. External-leg permutations relevant to the simultaneous MRK and near-planar analysis. The internal Fish topology and the positions of its three trivalent vertices and two blue four-valent vertices are fixed in all panels. The LP labels $x_0, \dots, x_5$ are displayed in panel (a) and apply to the corresponding edges in all three panels. Panels (a) and (b) are the only two assignments among the twelve inequivalent permutations for which HRF finds an HR; they place p_1, p_2, p_5 at the trivalent vertices and p_3, p_4 at the four-valent vertices. Panel (c) interchanges p_3 and p_5 relative to panel (b), leaving p_4 at a four-valent vertex, but has no HR for $b > 0$. The external-leg colours retain a distinct shade for each resolved direction. Related green shades group p_1, p_5 at the large- p^- end of the rapidity ordering, and related olive shades group p_2, p_3 at the large- p^+ end; this grouping does not identify either pair as a single collinear mode. The mid-rapidity outgoing line p_4 is blue.

Two ratios are negative throughout the positive physical parameter domain, so the stationary locus leaves $\mathfrak{C}_{\text{Fish}}^+$ and the HR is lost. A complete beam reflection would also interchange $p_1 \leftrightarrow p_2$; it is distinct from the single outgoing-leg interchange in panel (c).

For either permutation with positive stationary ratios, solving the homogeneity and hierarchy conditions of Eq. (2.36) for the unaligned expansion of the complete $\mathcal{P} = \mathcal{U} + \mathcal{F}$ determines, for each fixed pair of rates a, b , the candidate total vector

$$\vec{v}_{\text{HR}}^{\text{MRK+pl}} = (-2a, -2a, b - 2a, -2a, -2a, -2a; 1). \quad (3.107)$$

At $b = 0$, this reduces to the uniform near-planar wide-angle vector in Eq. (3.80), with $\lambda = \delta^a$. For the hard-vertex-reflected HR-supporting permutation, the corresponding vector follows from the graph automorphism $x_0 \leftrightarrow x_1$, $x_2 \leftrightarrow x_3$ and $x_4 \leftrightarrow x_5$; its layer weights are unchanged.

The hierarchy solution can then be read directly from the stationary form in Eq. (3.104). For $b > 0$, the relevant source terms of the rescaled polynomial are

$$\begin{aligned} \mathcal{F}^{(\text{HR})} = & \underbrace{(q_{AB}x_5g_{AB} + q_{BC}x_1g_{BC})}_{\substack{W_{\text{SL}} = -6a - b \\ \in \mathcal{I}_C^2}} + \underbrace{q_{CA}x_3g_{CA}}_{\substack{W_k = -6a \\ \in \mathcal{I}_C^2}} \\ & + \frac{\Gamma_5}{\underbrace{4q_{AB}q_{BC}q_{CA}}_{\substack{W_{\text{HR}} = -4a \\ \notin \mathcal{I}_C}}} x_1x_3x_5, \quad \mathcal{I}_C = \langle f_A, f_B, f_C \rangle. \end{aligned} \quad (3.108)$$

The first term defines $W_{\text{SL}} = -6a - b$. The second has the intervening nominal weight $W_k = -6a$, strictly between W_{SL} and $W_{\text{HR}} = -4a$, yet its complete coefficient also vanishes on the same pinch component.¹⁵ The weights of the individual monomials in the first line do not change. Rather, both underbraced contributions vanish on the common pinch component $C = \{f_A = f_B = f_C = 0\}$. Resolving the cancellation in the transverse normal directions suppresses their first nonzero contributions, relative to their nominal layers, by δ^{2a+b} and δ^{2a} , respectively. After cancellation, both therefore contribute effectively at δ^{-4a} , alongside the nonvanishing Gram-normal term in the second line of Eq. (3.108) and the leading term of $\mathcal{U}^{(\text{HR})}$, which also has weight $-4a$. This is the first explicit application of the intervening ideal-layer test in Eq. (2.39). At $b = 0$ the first two weights in Eq. (3.108) coincide, so no intervening layer remains.

What is inherited from the wide-angle Fish is the cancellation locus, not its scaling: the relative MRK and Gram-normal rates determine both the total vector and this layer structure. In projective LP-parameter space the locus itself moves towards a boundary, with only the middle path ratio tending to zero. Dissection is now applied as the final certification step.

¹⁵Additional occupied layers need not be intervening. In the spacelike-collinear Fish, for example, the further layers satisfy $W_k > W_{\text{HR}}$; none lies between W_{SL} and W_{HR} .

This moving locus is most transparent in local coordinates

$$\begin{aligned}
x_0 &= \rho_A u_A + y_A, & x_1 &= u_A, \\
x_2 &= \rho_B u_B + y_B, & x_3 &= u_B, \\
x_4 &= \rho_C u_C + y_C, & x_5 &= u_C.
\end{aligned} \tag{3.109}$$

Here (u_A, u_B, u_C) are tangential path-magnitude coordinates along the cancellation locus, whereas $(y_A, y_B, y_C) = (f_A, f_B, f_C)$ are signed normal displacements from it, in the same notation as Eq. (3.84). The local dissection and lower-facet test give the weights

$$(u_A, u_B, u_C, y_A, y_B, y_C) \sim (\delta^{-2a}, \delta^{-2a}, \delta^{-2a}, \delta^{-a}, \delta^{-a+2b}, \delta^{-a}). \tag{3.110}$$

Their pullback reproduces Eq. (3.107), independently checking the scaling already determined by the hierarchy equations. The exact leading local Lee–Pomeransky polynomial contains seven monomials of affine rank six in the six local variables. It therefore defines a full lower facet and upgrades the candidate to a certified scaleful HR.

Having certified the region, we now determine its power counting in parameter space. For unit propagator powers, the local-coordinate weights in Eq. (3.110) give the measure weight $-9a + 2b$. The physical layer in Eq. (3.108) has weight $-4a$, so $\mathcal{P}^{-D/2}$ contributes $2aD$. The resulting count is

$$I_{\text{MRK+pl}}^{\text{scalar}} \sim \delta^{-9a+2b+2aD}. \tag{3.111}$$

Normalising the equal-rate point by a redefinition of δ gives

$$\begin{aligned}
\vec{v}_{\text{HR}}^{\text{MRK+pl}} &= (-2, -2, -1, -2, -2, -2; 1), & W_{\text{SL}} &= -7, & W_{\text{HR}} &= -4, \\
\text{intermediate layer: } & W = -6, & I^{\text{scalar}} &\sim \delta^{1-4\epsilon}.
\end{aligned} \tag{3.112}$$

Taken together, the hierarchy solution, local certification and power count define a one-parameter family of certified HR scalings, since only the ratio b/a is invariant under a redefinition of δ . At $b = 0$ the uniform near-planar wide-angle result is recovered and the intermediate layer merges with the superleading layer; for $b > 0$ the MRK hierarchy makes the vector non-uniform and separates these layers. The boundary $a = 0$ removes the approach to the Gram surface, so generic fixed-transverse MRK alone is not certified by this construction.

Momentum-space interpretation. Because $b = 0$ recovers the near-planar wide-angle HR, the natural question is how the MRK hierarchy deforms the average edge-momentum configuration and the local integration widths within the same three-path organisation. We keep a, b general in the external kinematics, the average configuration and the independent-loop power count below. For the representative HR-supporting permutation choose

$$q_0 = r, \quad q_1 = p_1 - r, \quad q_2 = p_3 - \ell, \quad q_3 = \ell + p_2 - p_3, \quad q_4 = r - \ell, \quad q_5 = p_5 - r + \ell. \tag{3.113}$$

The variables (r, ℓ) provide a compact exact routing, but do not isolate the asymptotic modes in the most transparent loop basis. For the width analysis we instead use $q_A \equiv q_1 = p_1 - r$

and $q_C \equiv q_4 = r - \ell$. This affine change has unit absolute Jacobian and aligns the two independent integrations with the pinched propagator pairs used in the power count. Choose the otherwise irrelevant longitudinal boost so that the nonzero components of p_4 are $O(1)$. Since the transverse scales remain fixed, and writing each triple in the global frame as $(p^+, p^-, |\mathbf{p}_\perp|)_{(p_2, p_1)}$, the external momenta then have the schematic global-frame scalings

$$\begin{aligned} p_1 &\sim (0, \delta^{-b}, \mathbf{0}_\perp), & p_2 &\sim (\delta^{-b}, 0, \mathbf{0}_\perp), \\ p_3 &\sim (\delta^{-b}, \delta^b, 1), & p_4 &\sim (1, 1, 1), & p_5 &\sim (\delta^b, \delta^{-b}, 1), \end{aligned} \quad (+, -, \perp)_{(p_2, p_1)}, \quad (3.114)$$

up to signs and nonzero $O(1)$ coefficients. Thus p_2, p_3 share the large- p^+ end of the rapidity ordering, while p_1, p_5 share the large- p^- end. Their transverse momenta remain fixed while the corresponding large lightcone components grow as δ^{-b} , so the normalised opening angles within both pairs scale as δ^b . Equivalently, $-s_{23}/s_{12} \sim -s_{15}/s_{12} \sim \delta^{2b}$, although s_{23} and s_{15} themselves remain of fixed order. The two pairs therefore occupy corresponding ends of the MRK rapidity ordering. They must not be identified as single collinear modes: their relative transverse momenta remain of fixed order even while their normalised opening angles shrink against the growing beam scale.

We first determine the average edge momenta q_e^* , in the mixed-representation sense defined in Sec. 4. Writing their global-frame component scaling as $((q_e^*)^+, (q_e^*)^-, |q_{e\perp}^*|) \sim (\delta^{a_e}, \delta^{b_e}, \delta^{c_e})_{(p_2, p_1)}$, the mixed-representation stationary equations give Table 4 for general $a > 0$ and $b \geq 0$; Fig. 11 displays the corresponding momentum configuration. The average-component and saddle-virtuality columns follow from the exact mixed-representation stationary solution, whereas the region-scale column follows from the LP-parameter vector. Neither virtuality column is an integration width. At the stationary point, the leading $O(\delta^b)$ minus and $O(1)$ in-plane transverse components cancel in $q_2^* = p_3 - \ell^*$. Thus $q_2^* \sim (\delta^{-b}, \delta^{2a+b}, \delta^a)_{(p_2, p_1)}$ and $(q_2^*)^2 = O(\delta^{2a})$, although the region vector assigns $q_2^2 = O(\delta^{2a-b})$ across the local integration region. The routing variables in Eq. (3.113) themselves have average scalings

$$r^* \sim (\delta^{2a+b}, \delta^{-b}, \delta^a)_{(p_2, p_1)}, \quad \ell^* \sim (\delta^b, \delta^b, 1)_{(p_2, p_1)}. \quad (3.115)$$

Here $r^* = q_0^*$, while $\ell^* = r^* - q_4^* = p_3 - q_2^*$. The $O(\delta^{-b})$ minus components cancel in the first difference and the $O(\delta^{-b})$ plus components cancel in the second. For determining integration widths we therefore use the mode-adapted variables (q_A, q_C) defined above.

With the average momentum directions established, we next determine the integration widths and use them for the momentum-space power count. We begin with the LP-parameter vector in Eq. (3.107), which fixes the region-scale propagator virtualities according to $q_e^2 \sim \delta^{-v_e}$; for example,

$$q_1^2 \sim \delta^{-v_1} = \delta^{2a}, \quad q_2^2 \sim \delta^{-v_2} = \delta^{2a-b}. \quad (3.116)$$

edge	average $(a_e, b_e, c_e)_{(p_2, p_1)}$	saddle $(q_e^*)^2$	region-scale q_e^2	interpretation
q_0	$(2a + b, -b, a)$	δ^{2a}	δ^{2a}	p_1 -directed, path A
q_1	$(2a + b, -b, a)$	δ^{2a}	δ^{2a}	p_1 -directed, path A
q_2	$(-b, 2a + b, a)$	δ^{2a}	δ^{2a-b}	p_2 -directed; deeper saddle value
q_3	$(0, 2a + b, a)$	δ^{2a}	δ^{2a}	vanishing fraction on path B
q_4	$(b, -b, 0)$	δ^{2a}	δ^{2a}	p_5 -directed, fixed $p_\perp$
q_5	$(b, -b, 0)$	δ^{2a}	δ^{2a}	p_5 -directed, fixed $p_\perp$

Table 4. Average edge-momentum modes, saddle virtualities and region-scale virtualities for the simultaneous MRK and near-planar HR at general rates $a > 0$ and $b \geq 0$.

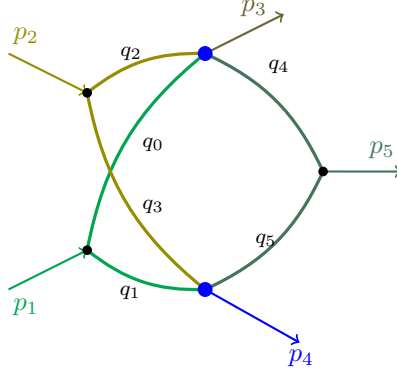


Figure 11. Average edge-momentum configuration at the positive Landau saddle of the MRK-planar HR for the representative permutation in Fig. 10(a). The edge labels correspond to Table 4; green, olive and dark green show paths A, B, C, whose average flows approach the p_1, p_2, p_5 directions. For $b > 0$, q_3 carries a fraction $O(\delta^b)$ relative to q_2 ; at $b = 0$ they are of the same order. This reconstruction has no independent Glauber loop.

Momentum conservation is built into the routing in Eq. (3.113). For general $a > 0$ and $b > 0$, use the independent loop momenta $q_A = q_1$ on path A and $q_C = q_4$ on path C, introduced above. The q_0, q_1 and q_4, q_5 pole pairs pinch, respectively, the small lightcone components of q_A and q_C . In the corresponding mode-adapted frames, the separation of each pole pair constrains the integration width of that component to δ^{2a+b} . Variations along the separate p_1 - and p_5 -collinear rays scale as δ^{-b} , while the region virtualities fix the transverse widths at δ^a . For these independent-loop widths we use a frame adapted to each mode. Let $\bar{p}_i$ denote a conjugate null reference direction. We may choose $\bar{p}_1 \parallel p_2$, so $(\bar{p}_1, p_1)$ agrees with the global frame up to normalisation. Since p_5 has nonzero transverse momentum in the global frame, $(\bar{p}_5, p_5)$ is instead a genuinely different local basis. The marginal widths are therefore

$$\Delta q_A \sim (\delta^{2a+b}, \delta^{-b}, \delta^a)_{(\bar{p}_1, p_1)}, \quad \Delta q_C \sim (\delta^{2a+b}, \delta^{-b}, \delta^a)_{(\bar{p}_5, p_5)}. \quad (3.117)$$

Equation (3.117) shows that the average $q_A = q_1$ triple in Table 4 agrees component by component with its marginal width for general a, b . The average $q_C = q_4$ triple $(2a + b, -b, a)_{(\bar{p}_5, p_5)}$ in its mode-adapted frame corresponds to the global entry $(b, -b, 0)_{(p_2, p_1)}$

in the table and likewise agrees with its marginal width. Thus the deeper saddle virtuality of the dependent edge q_2 does not contradict the average-width correspondence for the chosen independent loop basis.

We can now perform the momentum-space power count. Each balanced on-shell marginal width contributes δ^{aD} . The wide-angle correlation factor is $\lambda^3 = \delta^{3a}$: it is the conditional longitudinal volume obtained from the two invariant ranges and the nonsingular Jacobian displayed below Eq. (3.97). The MRK hierarchy supplies one further restriction. The path- B component of the stationary equation (3.99) gives

$$1 + \alpha_B^* = \frac{\tilde{x}_2}{\tilde{x}_2 + \tilde{x}_3} = O(\rho_B) = O(\delta^b),$$

where the Schwinger and LP ratios have the same regional scaling by Eq. (2.13). Thus the two edges on the middle path carry highly unequal longitudinal shares: in the routing of Eq. (3.113), $q_3 = p_2 - q_2$ is smaller than q_2 by $O(\delta^b)$. Producing $q_3^+ = O(1)$ from $p_2^+, q_2^+ = O(\delta^{-b})$ requires their leading components to cancel to relative accuracy δ^b . Since q_2 is a dependent combination of the two mode-adapted loop variables, this narrows their joint longitudinal support, rather than either marginal width, by one factor δ^b . The resulting correlation factor is therefore δ^{3a+b} . The region vector fixes $-\sum_e v_e = 12a - b$, so the six propagators contribute in total $\delta^{-(12a-b)}$. Momentum space therefore gives

$$I_{\text{MRK+pl}}^{\text{scalar}} \sim \delta^{aD} \delta^{aD} \delta^{3a+b} \delta^{-(12a-b)} = \delta^{-9a+2b+2aD}, \quad (3.118)$$

in exact agreement with Eq. (3.111). Thus the extra restriction in the count is a correlation of the collinear loop variables.

The qualitative path-flow asymmetry illustrated in Table 4 and Fig. 11 persists for $b > 0$ and explains the external-leg condition. In the reference orientation, q_2 is the only internal edge carrying the leading p^+ flow; q_3 has a vanishing longitudinal fraction, while paths A and C are directed towards the large- p^- end. Momentum conservation at the hard vertices therefore requires q_2 to meet the other large- p^+ momentum, p_3 , at the upper four-valent vertex, leaving the fixed-transverse momentum p_4 at the lower one. Interchanging q_2 and q_3 alone does not preserve this matching. The second HR-supporting assignment is instead the complete hard-vertex reflection: it exchanges $q_2 \leftrightarrow q_3$ together with $p_3 \leftrightarrow p_4$, $q_0 \leftrightarrow q_1$, and $q_4 \leftrightarrow q_5$.

This example separates three logically distinct operations in HRF. The decomposition of $\mathcal{F}$ identifies the cancellation factors and generator ideal; positivity of the stationary ratios selects the external-leg permutations for which the candidate locus lies in the positive LP domain; and the local hierarchy together with the lower-facet test of the complete $\mathcal{P} = \mathcal{U} + \mathcal{F}$ determines the region vector and certifies scalefulness. Thus the three-generator cancellation ideal found for the wide-angle near-planar Fish is inherited under the MRK deformation, while the MRK hierarchy moves its stationary locus towards a boundary or, for a different external-leg permutation, outside the positive domain.

3.2.6 The central-soft rapidity-ordered limit

This limit provides an alternative way to approach the normalised planar configuration. Rather than suppressing the parity-odd invariant at fixed transverse scales, it combines

strong rapidity ordering with one central momentum becoming transversely soft. It thereby separates features tied to the planar singular locus from those that depend on the particular asymptotic path used to reach it.

The soft limit. Starting again from the generic chart (3.45)–(3.46), introduce two independent dimensionless parameters. The parameter x controls the two large rapidity gaps, while τ measures the transverse softness of the mid-rapidity momentum relative to the fixed scale

$$Q_\perp^2 \equiv |\mathbf{p}_3|^2 = \frac{|\mathbf{p}_4|^2}{\zeta \bar{\zeta}}. \quad (3.119)$$

With $\kappa, \bar{\kappa}, \hat{X}_{34}, \hat{X}_{45}$ fixed and nonzero, the transverse part of the exact path is

$$\begin{aligned} \zeta &= \kappa\tau, & \bar{\zeta} &= \bar{\kappa}\tau, \\ |\mathbf{p}_4|^2 &= Q_\perp^2 \kappa \bar{\kappa} \tau^2, & |\mathbf{p}_5|^2 &= Q_\perp^2 (1 + \kappa\tau)(1 + \bar{\kappa}\tau). \end{aligned} \quad (3.120)$$

The two transverse-magnitude relations follow respectively from Eq. (3.119) and transverse momentum conservation, so they are not independent specifications. In the physical region $\bar{\kappa} = \kappa^*$. The longitudinal part of the same parametrisation gives directly

$$X_{34} \equiv \frac{p_3^+}{p_4^+} = \frac{\hat{X}_{34}}{x\tau}, \quad X_{45} \equiv \frac{p_4^+}{p_5^+} = \frac{\hat{X}_{45}\tau}{x(1 + \kappa\tau)(1 + \bar{\kappa}\tau)}. \quad (3.121)$$

In the representative frame $p_4^+ \propto \tau$, these invariant relations imply $p_3^+ \propto x^{-1}$ and $p_5^+ \propto x(1 + \kappa\tau)(1 + \bar{\kappa}\tau)$. Using Eq. (3.120) and on-shellness then gives $p_5^- = |\mathbf{p}_5|^2/p_5^+ \propto x^{-1}$. Thus p_3^+ and p_5^- both become large.

The corresponding exponentials of rapidity differences, written directly in terms of lightcone components, are

$$\begin{aligned} Y_{34} &\equiv e^{y_3 - y_4} = \sqrt{\frac{p_3^+ p_4^-}{p_3^- p_4^+}} = X_{34} \sqrt{\zeta \bar{\zeta}} = \frac{\hat{X}_{34} \sqrt{\kappa \bar{\kappa}}}{x}, \\ Y_{45} &\equiv e^{y_4 - y_5} = \sqrt{\frac{p_4^+ p_5^-}{p_4^- p_5^+}} = X_{45} \sqrt{\frac{(1 + \zeta)(1 + \bar{\zeta})}{\zeta \bar{\zeta}}} = \frac{\hat{X}_{45}}{x \sqrt{\kappa \bar{\kappa} (1 + \kappa\tau)(1 + \bar{\kappa}\tau)}}. \end{aligned} \quad (3.122)$$

The final equalities follow by substituting Eqs. (3.120) and (3.121) into the component-ratio forms. Thus the exponentials of the two rapidity gaps both scale as x^{-1} , or equivalently both gaps grow as $\log x^{-1}$, whereas the transverse ratio $|\mathbf{p}_4|^2/|\mathbf{p}_3|^2 = \kappa \bar{\kappa} \tau^2$ vanishes. At fixed nonzero τ , the limit $x \rightarrow 0$ has the ordinary MRK hierarchy. The joint limit $x, \tau \rightarrow 0$, however, departs from the standard MRK scaling in which all transverse ratios remain fixed. We therefore call it the central-soft rapidity-ordered limit, not a central-soft MRK sublimit.

For the one-parameter asymptotic expansion used by HRF we correlate the two parameters as

$$\tau = \delta^a, \quad x = \delta^b, \quad 0 < a < b. \quad (3.123)$$

The inequality ensures that both X_{34} and X_{45} diverge, but at different rates:

$$X_{34} = O(\delta^{-(a+b)}), \quad X_{45} = O(\delta^{-(b-a)}). \quad (3.124)$$

The unequal powers in Eq. (3.124) arise because the transversely soft momentum p_4 lies between the two rapidity gaps: although Y_{34} and Y_{45} both scale as x^{-1} , their conversion to plus-momentum ratios introduces opposite powers of τ .

The remaining hard momenta approach a $2 \rightarrow 2$ Regge configuration with an additional soft emission. Indeed, identify the hard subprocess as $p_1 + p_2 \rightarrow p_3 + p_5$, and set $s = s_{12}$ and $t = s_{23} < 0$. In the global $(+, -, \perp)_{(p_2, p_1)}$ lightcone frame, $s = p_1^- p_2^+$ and $-t = p_3^- p_2^+$. Momentum conservation and on-shellness therefore give

$$\begin{aligned} -\frac{t}{s} &= \frac{p_3^-}{p_1^-} = \frac{1}{1 + X_{34}\zeta\bar{\zeta} + X_{34}X_{45}(1 + \zeta)(1 + \bar{\zeta})} \\ &= \frac{x^2}{\widehat{X}_{34}\widehat{X}_{45}} \left[1 + \frac{\kappa\bar{\kappa}}{\widehat{X}_{45}}\tau x + \frac{x^2}{\widehat{X}_{34}\widehat{X}_{45}} \right]^{-1} = \frac{\delta^{2b}}{\widehat{X}_{34}\widehat{X}_{45}} \left[1 + O(\delta^{a+b}) \right]. \end{aligned} \quad (3.125)$$

Thus $-t/s \rightarrow 0$: p_3 becomes collinear to p_2 , while p_5 becomes collinear to p_1 . The leading Regge ratio is independent of the rate at which the additional momentum p_4 becomes soft.

A sharp interpretation of the present limit is obtained by considering the parity-odd invariant ϵ_5 . Combining its exact expression in Eq. (3.47) with Eqs. (3.119) and (3.120) gives

$$\frac{\epsilon_5}{s_{12}Q_\perp^2} = \zeta - \bar{\zeta} = \tau(\kappa - \bar{\kappa}). \quad (3.126)$$

Near-planar wide-angle scattering sends $\zeta - \bar{\zeta} \rightarrow 0$ at fixed $|\mathbf{p}_4|^2$, whereas the present limit sends $\zeta, \bar{\zeta} \rightarrow 0$ with their phase ratio generic. Both approach the normalised planar limit, but by distinct angular and soft degenerations; this fact alone does not identify their HRs.

The Fish in the soft limit: order alignment and the HRF. Let us now study this limit in the context of the Fish topology, which is the simplest topology where an HR arises. The three external-leg assignments needed below are collected in Fig. 12. Their internal orientation and LP-parameter labels are fixed; only the external legs are permuted. We denote the upper and lower four-point vertices by V_u and V_d , respectively. Panel (a) defines the representative HR assignment used in the construction below, with the soft leg attached to V_d . Panels (b) and (c) are the two negative controls analysed subsequently.

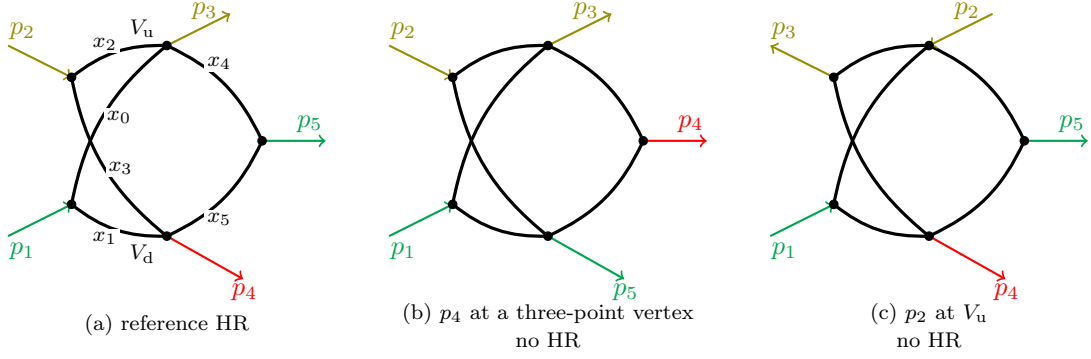


Figure 12. External-leg assignments used in the central-soft analysis. The internal Fish topology and its orientation are identical in all three panels; the LP labels shown in panel (a) are held fixed in panels (b) and (c). The external colours encode only the kinematic roles: Green for the p_1, p_5 directions, olive for the p_2, p_3 directions and red for the soft momentum p_4 . Panel (a) has an HR. Panel (b), used in Eq. (3.134), moves p_4 to the remaining three-point vertex. Panel (c), used in Eq. (3.136), keeps p_4 at V_d but moves the incoming particle p_2 to V_u .

For the representative central-soft assignment in Fig. 12(a) and $(a, b) = (1, 2)$, substitution of the kinematic expansion into the complete $\mathcal{F}$ gives, through native order δ^{-1} ,

$$\begin{aligned} \mathcal{F} = & -\delta^{-4} \hat{X}_{34} \hat{X}_{45} Q_{\perp}^2 x_2 (x_1 x_4 - x_0 x_5) \\ & + \delta^{-1} Q_{\perp}^2 \left[\hat{X}_{45} x_3 (x_1 x_4 - x_0 x_5) + \hat{X}_{45} x_2 (x_1 x_4 - x_0 x_5) \right. \\ & \quad \left. - \hat{X}_{34} \kappa \bar{\kappa} x_2 x_5 (x_0 + x_1) \right] + O(\delta^0). \end{aligned}$$

At fixed LP parameters the first line is therefore the entire leading polynomial. Its possible positive zero $x_1 x_4 - x_0 x_5 = 0$ is not stationary, since $\partial_{x_1} \mathcal{F}_{-4} = -\hat{X}_{34} \hat{X}_{45} Q_{\perp}^2 x_2 x_4 \neq 0$ in the positive orthant.

This is the role of the asymptotic-order alignment wrapper of Sec. 2.3: it brings terms from different native powers of δ to a common effective leading order before the core HRF search. Under $x_e \mapsto \delta^{\phi_e} x_e$, the choice below promotes the red δ^{-1} contribution to weight -10 , equal to that of the δ^{-4} term. The two uncoloured structures at order δ^{-1} instead have weight -7 , and every term in $O(\delta^0)$ has higher weight. The two weight- -10 contributions therefore give the selected face

$$\begin{aligned} \phi &= (-3, -3, 0, -3, -3, -3), \\ \mathcal{F}_{-10}^{\text{align}} &= -\hat{X}_{45} Q_{\perp}^2 (\hat{X}_{34} x_2 - x_3) (x_1 x_4 - x_0 x_5). \end{aligned} \quad (3.127)$$

The factorised structure in Eq. (3.127) is the central-soft specialisation of the cancellation sector $sf_1 f_2$ in the spacelike-collinear polynomial (3.21): it describes the same two-factor Fish pinch locus, but here that locus becomes visible only after order alignment. The simultaneous zero of the two displayed factors is stationary, so this aligned face supplies the starting polynomial for the core construction.

To express the locus in the common ratio notation, recall from Eq. (3.51) that $r_A = x_0/x_1$, $r_B = x_2/x_3$ and $r_C = x_4/x_5$, and let ρ_I denote the value of r_I at the pinch. Setting

the two factors in Eq. (3.127) to zero gives

$$\rho_B = \frac{1}{\widehat{X}_{34}} > 0, \quad \rho_A = \rho_C > 0. \quad (3.128)$$

To relate this result to the generic wide-angle locus, use the normal polynomials f_A, f_B, f_C defined in Eq. (3.57), whose pairwise products appear in the exact decomposition (3.61). On Eq. (3.128), the two factors of the aligned face become

$$x_1 x_4 - x_0 x_5 = x_1 f_C - x_5 f_A, \quad \widehat{X}_{34} x_2 - x_3 = \widehat{X}_{34} f_B. \quad (3.129)$$

As the kinematics enter the central-soft regime, one of the three independent normal conditions of the generic planar Fish locus is lost on the aligned leading face. Consequently, the generic locus $f_A = f_B = f_C = 0$ is contained in the larger locus defined by $f_B = 0$ and $x_1 f_C - x_5 f_A = 0$. Equivalently, r_B is fixed as in Eq. (3.128), while the common positive value $r_A = r_C$ remains free at this order. Thus it is the kinematics that approaches a boundary of the wide-angle domain; the corresponding leading LP-parameter locus enlarges. This singular-locus relation does not determine the region vector.

Returning now to the HRF construction, the order-aligned face in Eq. (3.127) is passed to the core algorithm. In the notation of the composition law (2.45), it gives the uniform second-stage scaling and the composed edge vector. We label quantities associated with this soft limit by the superscript *soft*:

$$\mathbf{v}_{\text{core}}^{\text{soft}} = (-1, -1, -1, -1, -1, -1), \quad \tilde{\mathbf{v}}^{\text{soft}} = \phi + \mathbf{v}_{\text{core}}^{\text{soft}} = (-4, -4, -1, -4, -4, -4). \quad (3.130)$$

Equation (3.130) completes the determination of the LP-parameter scaling. To compare the layers of $\mathcal{F}$ directly with those of $\mathcal{U}$, we make the explicit scale choice $\mu_{\text{LP}}^2 = s_{12}$ in Eq. (2.4). Thus the dimensionless LP polynomial is $\mathcal{P}_{s_{12}} = \mathcal{U} + \mathcal{F}/s_{12}$. Unlike the default fixed-scale convention, this reference scale varies with δ , and its scaling is included in the layers below. In the central-soft limit,

$$s_{12} = \widehat{X}_{34} \widehat{X}_{45} Q_{\perp}^2 \delta^{-4} + O(\delta^{-1}), \quad \frac{Q_{\perp}^2}{s_{12}} = \frac{\delta^4}{\widehat{X}_{34} \widehat{X}_{45}} [1 + O(\delta^3)].$$

This kinematic scaling has already been inserted in Eq. (3.127), which is why s_{12} does not appear there explicitly. Applying the composed vector in Eq. (3.130), the first two nonempty layers of $\mathcal{F}$, expressed in units of s_{12} , are

$$\begin{aligned} \left. \frac{\mathcal{F}}{s_{12}} \right|_{x_e \mapsto \delta \tilde{\mathbf{v}}_e^{\text{soft}} x_e} &= - \frac{\delta^{-9}}{\widehat{X}_{34}} (\widehat{X}_{34} x_2 - x_3) (x_1 x_4 - x_0 x_5) \\ &\quad - \frac{\delta^{-8}}{\widehat{X}_{34} \widehat{X}_{45}} x_0 x_3 x_4 + O(\delta^{-7}). \end{aligned} \quad (3.131)$$

The first term is the hard-scale-normalised form of the aligned cancellation sector in Eq. (3.127). In the $\mu_{\text{LP}}^2 = s_{12}$ normalisation the augmented vector and resolved weights are

$$\tilde{\mathbf{v}}_{\text{HR}, s_{12}}^{\text{soft}} = (-4, -4, -1, -4, -4, -4; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-9, -8). \quad (3.132)$$

The second term in Eq. (3.131) is the entire next nonvanishing $\mathcal{F}/s_{12}$ layer. It remains nonzero on the two-factor locus and has weight -8 , exactly the leading $\mathcal{U}$ weight. In the representative labelling,

$$\mathcal{U}_{-8} = x_0x_3 + x_1x_3 + x_0x_4 + x_1x_4 + x_3x_4 + x_0x_5 + x_1x_5 + x_3x_5. \quad (3.133)$$

The physical layer of $\mathcal{U} + \mathcal{F}/s_{12}$ therefore contains both this $\mathcal{F}/s_{12}$ monomial and $\mathcal{U}_{-8}$; because the former survives on the cancellation locus, this layer is scaleful. Together with the promoted factorised terms, its resolved support has affine rank six and the unique inward normal is precisely Eq. (3.132). The hierarchy gap is one.

This layered structure is the soft-limit counterpart of the spacelike-collinear HR in Eqs. (3.21)–(3.26). After translating the edge labels and specialising the kinematic coefficients, the factorised superleading sector is the same. In the spacelike-collinear expansion the core HRF acts directly on $\mathcal{F}_0$ and isolates the single-monomial obstruction $\mathcal{F}_{\text{Obs}}$; here alignment first exposes the same cancellation sector with zero obstruction, while restoration of the next native layer supplies the different monomial $x_0x_3x_4$. These different monomials are tied to the different region vectors selected by the two kinematic limits. Thus, while found through different HRF discovery routes, the two regions share the same cancellation locus and layered structure: in each case an uncanceled monomial lies one weight above the common cancellation sector and joins $\mathcal{U}$ in the physical layer.

External-leg conditions. The attachment scan yields a simple characterisation of the Fish seed in this limit. An HR occurs precisely when the soft particle p_4 is attached to one of the two four-point vertices and the two incoming particles p_1, p_2 are attached to two distinct three-point vertices. The remaining outgoing particles occupy the unused three- and four-point vertices. This criterion accounts for all six inequivalent attachments with an HR and is unchanged for the tested choices $(a, b) = (1, 2), (1, 3), (2, 3)$ of the exponents defined in Eq. (3.123).

The reason for requiring the soft particle to be emitted from a four-point vertex is already visible on the leading face exposed by the natural alignment vector $\phi = (-3, -3, 0, -3, -3, -3)$. For the complete external-leg assignment in Fig. 12(b), in which p_4 is emitted from the three-point vertex, the aligned face is

$$\mathcal{F}_{\text{mid}}^{\text{align}} = -\hat{X}_{45}Q_{\perp}^2x_1[(\hat{X}_{34}x_2 - x_3)x_4 + \hat{X}_{34}x_2x_5]. \quad (3.134)$$

The bracket can vanish for positive parameters, but the zero is not stationary:

$$\frac{\partial}{\partial x_5}[(\hat{X}_{34}x_2 - x_3)x_4 + \hat{X}_{34}x_2x_5] = \hat{X}_{34}x_2 > 0. \quad (3.135)$$

Attaching p_4 to a four-point vertex is not by itself sufficient. In the complete assignment of Fig. 12(c), the incoming particle p_2 is attached to the other four-point vertex. The aligned face has, up to a nonzero overall coefficient and graph reflection, the form

$$\mathcal{F}_{\text{in,end}}^{\text{align}} \propto (\hat{X}_{34}x_2 + x_3)(x_1x_4 - x_0x_5). \quad (3.136)$$

The first factor is strictly positive in the positive orthant. Setting only the second factor to zero does not make the polynomial stationary, so this attachment has no positive Landau pinch.

By contrast, the reference assignment in Fig. 12(a), in which both incoming particles are attached to three-point vertices, gives the factorised aligned face already displayed in Eq. (3.127), and hence the HR certified above. Here every three-point vertex carries an energetic external particle, so all three two-edge paths can support the collinear splittings required by the pinch. Moving either incoming particle to the second four-point vertex replaces the difference $\hat{X}_{34}x_2 - x_3$ by the strictly positive sum $\hat{X}_{34}x_2 + x_3$ in Eq. (3.136), eliminating the pinch.

For the hard-vertex-reflected configuration, the fixed-edge presentation is, up to a nonzero overall coefficient,

$$\begin{aligned}\mathcal{F}_{*,\text{opp}}^{\text{soft}} &\propto -(x_2 - \hat{X}_{34}x_3)(x_1x_4 - x_0x_5), \\ \phi_{\text{opp}} &= (-3, -3, -3, 0, -3, -3), & \mathbf{v}_{\text{core,opp}}^{\text{soft}} &= (-1, -1, -1, -1, -1, -1), \\ \bar{v}_{\text{HR,opp},s_{12}}^{\text{soft}} &= (-4, -4, -4, -1, -4, -4; 1).\end{aligned}\tag{3.137}$$

The reflected external-leg assignment is analysed independently with the same fixed edge labels. The graph automorphism $V_u \leftrightarrow V_d$ relates its certificate to the representative one.

Power counting in parameter space. We continue in the $\mu_{\text{LP}}^2 = s_{12}$ normalisation used in Eqs. (3.131) and (3.132). Normalising the first factor by the nonzero kinematic ratio X_{34} , define the local cancellation factors

$$f_1^{\text{soft}} = x_2 - X_{34}^{-1}x_3 = x_2 - \hat{X}_{34}^{-1}\delta^3x_3, \quad f_2^{\text{soft}} = x_1x_4 - x_0x_5.\tag{3.138}$$

The second form of f_1^{soft} follows from Eqs. (3.121) and (3.123) with $(a, b) = (1, 2)$, for which $X_{34} = \hat{X}_{34}\delta^{-3}$. The two factors have generic weights -1 and -8 , respectively. The HRF gap fixes one additional power of suppression for their product. We use the endpoint chart that assigns this power to f_1^{soft} , so that its resolved weight is 0, while f_2^{soft} retains weight -8 . The resolved polynomial $\mathcal{P}_{s_{12}}$ therefore has weight -8 , and its integrand factor scales as

$$\mathcal{P}_{s_{12}}^{-D/2} \sim (\delta^{-8})^{-D/2} = \delta^{4D}.\tag{3.139}$$

This asymmetric assignment of the cancellation depth is a choice of dissection chart, not a statement that the physical pinch has only codimension one. The invariant component is still $f_1^{\text{soft}} = f_2^{\text{soft}} = 0$. The radial-ratio dissection of Sec. 3.2.1 can likewise represent the unit gap by a radial variable controlling the product and a ratio variable distributing the suppression between f_1^{soft} and f_2^{soft} . Its pullback gives the vector in Eq. (3.132); the endpoint chart above is the dissection used for the certified power count.

For unit propagator powers, the parameter-space measure is counted separately in the same chart. Use $(f_1^{\text{soft}}, x_0, x_1, x_3, x_4, x_5)$ as local coordinates. Since $x_2 = f_1^{\text{soft}} + \hat{X}_{34}^{-1}\delta^3x_3$, replacing dx_2 by df_1^{soft} at fixed x_3 has unit Jacobian. The resolved weight of df_1^{soft} is zero, while the other five differentials each have weight -4 . Thus the full parameter-space measure

scales as δ^{-20} . Combining it with the cancellation-aware integrand in Eq. (3.139) gives

$$\frac{I_{\text{soft}}^{\text{scalar}}}{(s_{12})^{D-6}} \sim \delta^{4D-20} = \delta^{-4-8\epsilon}, \quad D = 4 - 2\epsilon. \quad (3.140)$$

Momentum-space comparison. The momentum-space reconstruction uses the same s_{12} normalisation. For the representative external assignment choose from the outset the mode-adapted independent variables $r = q_0$ and $\ell_G = q_0 - q_4 = p_3 - q_2$. The exact routing is then

$$(q_0, q_1, q_2, q_3, q_4, q_5) = (r, p_1 - r, p_3 - \ell_G, \ell_G + p_2 - p_3, r - \ell_G, p_5 - r + \ell_G). \quad (3.141)$$

The region vector in Eq. (3.132) gives

$$\frac{q_e^2}{s_{12}} \sim (\delta^4, \delta^4, \delta, \delta^4, \delta^4, \delta^4).$$

The component-level reconstruction contains more information than these virtualities. The consequences of the parameter-space cancellations enter through the certified region vector, which fixes the six virtuality powers above. To obtain a consistent momentum-space realisation, we substitute the external scaling in Eqs. (3.120)–(3.123), with $(a, b) = (1, 2)$, into the routing (3.141) and impose momentum conservation on the branch realising those virtualities. The resulting compatible edge-momentum scalings are given in Table 5. In the global $(+, -, \perp)_{(p_2, p_1)}$ lightcone frame, where p_2 carries only plus momentum and p_1 only minus momentum, an entry $(a_e, b_e, c_e)_{(p_2, p_1)}$ means

$$\left(\frac{q_e^+}{\sqrt{s_{12}}}, \frac{q_e^-}{\sqrt{s_{12}}}, \frac{|q_{e\perp}|}{\sqrt{s_{12}}} \right)_{(p_2, p_1)} \sim (\delta^{a_e}, \delta^{b_e}, \delta^{c_e})_{(p_2, p_1)}.$$

These are valuations of a compatible edge-momentum routing at the pinch, not the local widths of independent loop-integration variables.

edge	$(a_e, b_e, c_e)_{(p_2, p_1)}$	q_e^2/s_{12}	momentum scaling
q_0	$(4, 0, 2)$	δ^4	$(p_1\text{-collinear})^2$
q_1	$(4, 0, 2)$	δ^4	$(p_1\text{-collinear})^2$
q_2	$(0, 1, 2)$	δ	longitudinally dominated, p_2 -directed
q_3	$(3, 1, 2)$	δ^4	soft $\times$ $(p_1\text{-collinear})$
q_4	$(4, 0, 2)$	δ^4	$(p_1\text{-collinear})^2$
q_5	$(4, 0, 2)$	δ^4	$(p_1\text{-collinear})^2$

Table 5. Complete edge-momentum scalings of a compatible routing for the representative hidden-region assignment. The component exponents and virtualities are normalised to the hard scale s_{12} . They complement, and should not be confused with, the local loop-integration widths in Eq. (3.143).

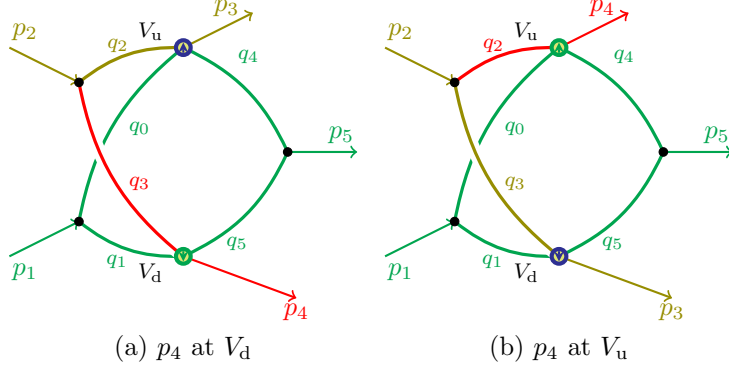


Figure 13. Momentum-space realisation of the two hard-vertex assignments in the central-soft rapidity-ordered limit. Olive marks the edge carrying the leading p_2, p_3 lightcone direction, green the $(p_1\text{-collinear})^2$ edges, and red the internal soft $\times$ $(p_1\text{-collinear})$ edge adjacent to the soft external momentum p_4 . The complete component scalings for panel (a) are given in Table 5; panel (b) follows by the reflection described in the text. The GreenYellow discs follow the oriented Glauber-flow convention defined in Fig. 4 and used in Ref. [49]. Thus the upper and lower arrows point in opposite directions. In panel (a) the upper blue rim records the hard complementary channel and the lower green rim the p_1 -collinear one; these rim colours exchange in panel (b). In panel (a), the symbols at V_u and V_d correspond to $q_0 - q_4 = \ell_G$ and $q_1 - q_5 = p_1 - p_5 - \ell_G$, respectively. Panel (b) is the hard-vertex-reflected assignment.

Panel (a) of Fig. 13 corresponds directly to the compatible edge modes in Table 5. The colours encode their leading kinematic association, while the table records the complete component scaling. Panel (b) is obtained by the hard-vertex-reflected permutation $q_2 \leftrightarrow q_3$, together with $q_0 \leftrightarrow q_1$ and $q_4 \leftrightarrow q_5$, and therefore requires no second table. The olive and red scalings exchange positions, while all green edges retain $(p_1\text{-collinear})^2$.

Here $(p_1\text{-collinear})^2$ denotes the standard p_1 -collinear scaling with small parameter δ^2 , so its exponent vector is $2(2, 0, 1) = (4, 0, 2)$. Similarly, the soft $\times$ $(p_1\text{-collinear})$ entry obeys $(3, 1, 2) = (1, 1, 1) + (2, 0, 1)$. The origin of the remaining entries is transparent in Eq. (3.141). The momenta $q_0 = r$ and $q_1 = p_1 - r$, as well as $q_4 = r - \ell_G$ and $q_5 = p_5 - r + \ell_G$, share the $(p_1\text{-collinear})^2$ scaling. The momentum $q_2 = p_3 - \ell_G$ retains the leading p_2, p_3 lightcone direction but has a larger minus component and is therefore longitudinally dominated. In $q_3 = \ell_G + p_2 - p_3$, cancellation of the leading p_2, p_3 plus components gives the soft $\times$ $(p_1\text{-collinear})$ scaling. All six propagators are infrared relative to the hard scale, with q_2 the least suppressed.

The Glauber momentum flow shown in Fig. 13 is $\ell_G = q_0 - q_4$ at V_u . The exact routing in Eq. (3.141) carries the same loop through V_d , where the transfer is $q_1 - q_5 = p_1 - p_5 - \ell_G$. Thus the two four-point vertices lie on one Glauber exchange, rather than defining two independent loops.

Having identified a compatible momentum-mode assignment and its routing, we now turn to the widths of the loop integrations around the pinch. These are not given by Table 5. To determine a width, hold the other loop components fixed and regard each

active propagator as a pole in the chosen integration component u . Locally,

$$\mathcal{D}_e = A_e u + B_e + i0, \quad u_e = -\frac{B_e}{A_e} - \frac{i0}{A_e}.$$

Two propagators form a pinching pole pair when the signs of their coefficients A_e are opposite: their poles then approach the real contour from opposite sides. The scaling of the separation between their real parts fixes the local integration width Δu . This is a momentum-space pole analysis performed after the parameter-space cancellation analysis has fixed the propagator virtualities, not an additional cancellation condition. The general longitudinal pole rule and the transverse homogeneity rule used below are collected in Sec. 4.3.

The following estimates use both kinds of information collected in Table 5. The component scalings fix the pole coefficients A_e , while the virtuality scalings fix the size of the corresponding denominators; their ratio determines each pole displacement and hence the local width.

In the routing (3.141), the denominators associated with (q_0, q_1) are both of order δ^4 , while their coefficients of r^+ are of order one. Their opposite-side poles therefore fix $\Delta r^+ \sim \delta^4$. The unpinched minus component has $\Delta r^- \sim \delta^0$, set by the large minus component of p_1 , while the transverse width is δ^2 . The pair (q_4, q_5) forms the p_5 -collinear path, since $q_4 + q_5 = p_5$. Because p_5 becomes collinear to p_1 in this limit, the same global lightcone frame applies: the two denominators are of order δ^4 , while their coefficients of ℓ_G^+ are of order one and have opposite signs. Hence $\Delta \ell_G^+ \sim \delta^4$.

To determine the remaining longitudinal width of ℓ_G , use the q_2 and q_3 relations in Eq. (3.141),

$$q_2 = p_3 - \ell_G, \quad q_3 = \ell_G + p_2 - p_3.$$

Regarded as functions of ℓ_G^- , these two denominators supply the pole pair in the global ℓ_G^- contour. Their coefficients are $-q_2^+ \sim -\delta^0$ and $+q_3^+ \sim \delta^3$, which have opposite signs on the physical branch. Combining these coefficients with the corresponding virtualities, the two pole displacements have the same scaling,

$$\frac{q_2^2}{q_2^+} \sim \delta, \quad \frac{q_3^2}{q_3^+} \sim \frac{\delta^4}{\delta^3} = \delta.$$

They therefore pinch the contour over the common width $\Delta \ell_G^- \sim \delta$.

The transverse width is not determined by another pole pair. A transverse displacement changes the three relevant denominators according to

$$\Delta \mathcal{D}_e|_{\perp} \sim q_{e\perp} \cdot \Delta \ell_{G\perp} + |\Delta \ell_{G\perp}|^2, \quad e = 3, 4, 5. \quad (3.142)$$

The propagator q_2 is excluded here because it is longitudinally dominated: $q_2^+ q_2^- \sim \delta$, whereas $q_{2\perp}^2 \sim \delta^4$. Transverse changes of order δ^4 are therefore subleading to its virtuality and do not determine the width. For the three propagators obeying the balanced on-shell condition in Eq. (3.142), Table 5 gives $|q_{e\perp}| \sim \delta^2$ and $q_e^2 \sim \delta^4$. Requiring the transverse displacement to preserve these virtuality orders, and taking the widest allowed support,

gives $|\Delta\ell_{G\perp}| \sim \delta^2$. This is the explicit example of the general transverse homogeneity condition in Eq. (4.9). In units of $\sqrt{s_{12}}$, the complete result is

$$\Delta r \sim (\delta^4, 1, \delta^2)_{(p_2, p_1)}, \quad \Delta\ell_G \sim (\delta^4, \delta, \delta^2)_{(p_2, p_1)}. \quad (3.143)$$

Here the triples again denote plus, minus and transverse components in the global (p_2, p_1) frame. At the pinch, $r = q_0 \sim (\delta^4, 1, \delta^2)_{(p_2, p_1)}$, as shown in Table 5; its fluctuation therefore has the same scaling as its value in this routing. This is the balanced on-shell $(p_1\text{-collinear})^2$ mode. The exact routing similarly gives $\ell_G|_{\text{pinch}} \sim (\delta^4, \delta, \delta^2)_{(p_2, p_1)}$, again matching its fluctuation width. In this case, however, $\ell_G^+ \ell_G^- \sim \delta^5$, whereas $\ell_{G\perp}^2 \sim \delta^4$, so the mode is transverse dominated and hence Glauber. Thus, in this mode-adapted independent-loop basis, the compatible routing and fluctuation scalings agree for both loops even though their mode types differ.

It is the integration widths in Eq. (3.143), rather than the central edge modes themselves, that determine the loop-measure factors in the power count. The two loop measures scale as δ^{2D} and δ^{2D+1} , respectively. The six inverse propagators contribute δ^{-21} . Therefore

$$\frac{I_{\text{soft}}^{\text{scalar}}}{(s_{12})^{D-6}} \sim \underbrace{\delta^{2D}}_{r \text{ measure}} \underbrace{\delta^{2D+1}}_{\ell_G \text{ measure}} \underbrace{\delta^{-21}}_{\text{six propagators}} = \delta^{4D-20}, \quad (3.144)$$

in agreement with the parameter-space result Eq. (3.140). The restricted support identified by HRF is already encoded in the narrower ℓ_G integration width; it must not be included as an additional factor.

Restoring the dimensionful normalisation and using $s_{12}/Q_\perp^2 \sim \delta^{-4}$, this common parameter- and momentum-space power count becomes

$$I_{\text{soft}}^{\text{scalar}} \sim (s_{12})^{D-6} \delta^{4D-20} \sim (Q_\perp^2)^{D-6} \delta^4. \quad (3.145)$$

In summary, this soft-limit example requires order alignment before the core HRF construction exposes the two-factor Fish pinch. Within the Fish topology, the external-leg analysis requires both incoming particles to be attached to three-point vertices and the soft particle to a four-point vertex. The composed vector in Eq. (3.132) gives the parameter-space power count in Eq. (3.140); its momentum-space realisation contains one balanced on-shell $(p_1\text{-collinear})^2$ loop and one transverse-dominated Glauber loop, and leads to the matching power count in Eqs. (3.144) and (3.145). The comparison with the spacelike-collinear Fish is complementary: after translating the edge labels, the factorised superleading sector and cancellation locus coincide, but the spacelike-collinear HR is exposed by obstruction removal whereas the present soft-limit HR first requires order alignment. The two kinematic limits select different resolved monomials and hence different region vectors. The relation to the generic wide-angle near-planar locus was established above in Eqs. (3.128) and (3.129): the selected rapidity-ordered face retains two combinations of the three normal conditions and therefore has an enlarged leading locus. Thus the same Fish seed and related cancellation geometry underlie both expansions, while their region vectors, momentum modes and scalar powers remain distinct.

3.2.7 Comparison of the certified five-point regions

Before turning to the six-point examples, we compare the four certified HRs found for the Fish seed. Their organising relation is the singular locus: each belongs to the positive planar Fish family of Eq. (3.69), or to a controlled boundary stratum of its closure. Table 6 separates this inherited geometry from the expansion-dependent total vectors, scalar powers, momentum-space integration widths and correlated support selected by the asymptotic trajectory and the permutation of external legs.

Expansion	total edge vector	scalar power	momentum-space widths and support
Spacelike collinear	$(-2, -1, -2, -2, -2, -1; 1)$ Eq. (3.23)	$\delta^{-1-4\epsilon}$	$\Delta\ell_s \sim (\delta, \delta, \delta)_{(p_2, p_1)},$ $\Delta\ell_G \sim (\delta, \delta^2, \delta)_{(p_2, p_1)}$
Near-planar wide angle	$(-2, -2, -2, -2, -2, -2; 1)$ Eq. (3.80)	$\lambda^{-1-4\epsilon}$	$\Delta q_I \sim (1, \lambda^2, \lambda)_{(P_I, \bar{P}_I)}, I = A, B,$ $\mathcal{V}_{\parallel}^{\text{corr}} \sim \lambda^3$
MRK plus near-planarity, $a > 0, b > 0$	$(-2a, -2a, b - 2a, -2a, -2a, -2a; 1)$ Eq. (3.107)	$\delta^{-a+2b-4a\epsilon}$	$\Delta q_A \sim (\delta^{2a+b}, \delta^{-b}, \delta^a)_{(\bar{p}_1, p_1)},$ $\Delta q_C \sim (\delta^{2a+b}, \delta^{-b}, \delta^a)_{(\bar{p}_5, p_5)},$ $\mathcal{V}_{\parallel}^{\text{corr}} \sim \delta^{3a+b}$
Central-soft rapidity ordered	$(-4, -4, -1, -4, -4, -4; 1)$ Eq. (3.132)	δ^{4D-20} $= \delta^{-4-8\epsilon}$	$\Delta r \sim (\delta^4, 1, \delta^2)_{(p_2, p_1)},$ $\Delta\ell_G \sim (\delta^4, \delta, \delta^2)_{(p_2, p_1)}$

Table 6. Comparison of the certified five-point hidden regions. Each row uses the LP normalisation of its defining subsection, and the equation below each vector fixes the edge convention and representative permutation of external legs. The last column gives loop-integration widths in the mode-adapted lightcone frame identified by its subscript. The factor $\mathcal{V}_{\parallel}^{\text{corr}}$ denotes the additional longitudinal restriction generated by cross-loop correlations; where it is absent, there is no such cross-loop correlation.

Two distinctions are important when reading the last column. First, the preceding momentum-space tables record compatible edge-momentum scalings, whereas Table 6 records independent-loop integration widths; the frame subscript in $(+, -, \perp)_{(n_+, n_-)}$ names the plus and minus reference directions. Second, the component widths determine the D -dependent part of the measure power, whereas $\mathcal{V}_{\parallel}^{\text{corr}}$ contributes only to its integer part. For the central-soft row, $\mu_{\text{LP}}^2 = s_{12}$, the widths are in units of $\sqrt{s_{12}}$, and the listed scalar power multiplies $(s_{12})^{D-6}$; in fixed $Q_{\perp}$ units it becomes δ^4 , as in Eq. (3.145).

The four cases in Table 6 are distinct regions, with different kinematic expansions, active cancellation ideals, vectors and momentum-space realisations. Their stronger common feature is that their singular loci belong to the positive planar Fish family defined by Eq. (3.69), or to a controlled boundary stratum of its closure. The near-planar wide-angle example is the nondegenerate interior case. The spacelike-collinear limit reaches the $s_{23} = 0$ boundary with the surviving ratio positive, whereas the timelike-collinear range fails precisely this positivity test. The simultaneous MRK-planar family approaches $\rho_B = 0$ from positive values as in Eq. (3.105), whereas the central-soft limit retains Eq. (3.128) on an aligned face. Within the simultaneous MRK-planar family, only the ratio b/a distinguishes inequivalent asymptotic trajectories. The equal-rate representative has $a = b$, while the boundary $b = 0$

removes the MRK hierarchy and recovers the near-planar wide-angle row with $\lambda = \delta^a$. For $b > 0$, only the representative of Eq. (3.107) and its hard-vertex-reflected partner pass the positivity test. The interchange $p_4 \leftrightarrow p_5$ is immaterial in the spacelike-collinear and near-planar wide-angle limits. In the central-soft limit, by contrast, the HR requires the soft leg p_4 at a four-point Fish vertex and the two incoming legs at distinct three-point vertices; the directly reflected vector is given in Eq. (3.137).

Two complementary relations organise the four rows. The first pair comprises the spacelike-collinear and central-soft limits, both factorisation limits in which an underlying $2 \rightarrow 2$ hard process emerges from the $2 \rightarrow 3$ kinematics. After translating the edge labels, their factorised superleading sectors have the same two-factor form and define the same single-generator pinch. Their HRF discovery routes nevertheless differ: the spacelike-collinear construction isolates a nontrivial obstruction, whereas the central-soft construction first requires order alignment. Their momentum-space realisations accordingly combine a Glauber loop with, respectively, a soft or a balanced on-shell collinear loop.

The near-planar wide-angle and simultaneous MRK-planar limits form the second pair. Both retain the three-normal, three-generator planar locus. The MRK hierarchy drives $\rho_B \rightarrow 0$ and narrows the correlated longitudinal support, while a path-adapted independent-loop basis retains matching average and marginal scalings and generates no independent Glauber loop. Together, these two pairs of limits distinguish inherited cancellation geometry from the expansion-dependent region vector and momentum modes.

This positive-chamber organisation explains why the same Fish seed supports HRs in all four limits without identifying the resulting regions: the singular locus is inherited from the planar family, whereas the certified region vector belongs to the trajectory approaching that family or one of its boundary strata. In the central-soft example, order alignment selects the active part of the inherited ideal before the core HRF construction. The scalar powers in Table 6 are reproduced in both parameter and momentum space.

3.3 Six-point integrals: the twisted-box seed and its hexagon realisation

At six points, the first hidden regions in our survey appear already at one loop. The relevant graph is a massless hexagon with the non-planar permutation of external legs displayed in Fig. 14(a), rather than the usual cyclic planar ordering. We refer to this graph as the *twisted hexagon*. Contracting x_0 and x_3 gives the twisted-box boundary in Fig. 14(b). The contraction merges the (p_3, p_6) and (p_4, p_5) pairs into two four-point vertices without reducing the loop order. The twisted box in panel (b) is the minimal six-point HR seed identified here. The twisted hexagon in panel (a) is its simplest six-propagator realisation, and the same externally labelled seed can occur as a contraction minor of more complicated graphs.

The figure fixes both the permutation of external legs and the LP-parameter convention. Throughout this six-point section, p_1, p_2 are future-directed incoming momenta and $p_3, \dots, p_6$ are future-directed outgoing momenta, so $p_1 + p_2 = p_3 + p_4 + p_5 + p_6$. For any

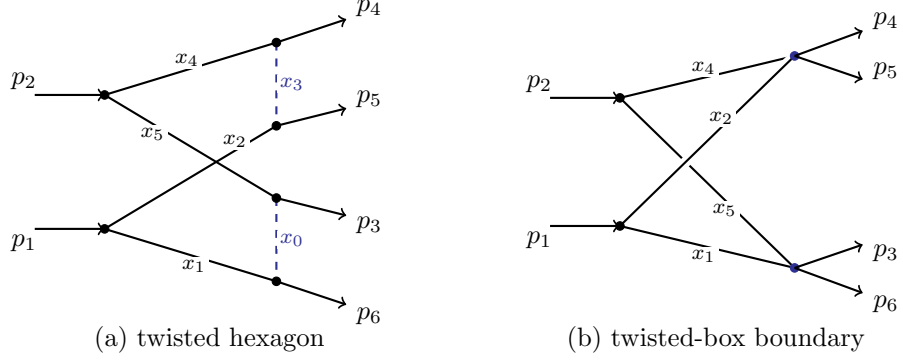


Figure 14. The six-point topology and its self-crossing boundary. Panel (a) defines the LP parameters $x_0, \dots, x_5$ on the twisted hexagon, whose vertices are all three-point. Contracting the dashed-blue edges x_0 and x_3 gives panel (b), with active parameters x_1, x_2, x_4, x_5 and two four-point hard vertices shown in blue. The crossings of drawn lines are not graph vertices.

index set I , define

$$s_I \equiv \left(- \sum_{i \in I \cap \{1,2\}} p_i + \sum_{j \in I \cap \{3,4,5,6\}} p_j \right)^2. \quad (3.146)$$

Thus, for example, $s_{145} = (-p_1 + p_4 + p_5)^2$, whereas $s_{36} = (p_3 + p_6)^2$. In these variables the graph polynomials are

$$\begin{aligned} \mathcal{F} = & \textcolor{brown}{s}_{16}x_0x_2 + \textcolor{brown}{s}_{156}x_0x_3 + \textcolor{brown}{s}_{23}x_0x_4 + \textcolor{brown}{s}_{15}x_1x_3 + \textcolor{brown}{s}_{145}x_1x_4 + \textcolor{green}{s}_{36}x_1x_5 \\ & + \textcolor{green}{s}_{45}x_2x_4 + \textcolor{brown}{s}_{245}x_2x_5 + \textcolor{brown}{s}_{24}x_3x_5, \end{aligned} \quad (3.147)$$

$$\mathcal{U} = x_0 + x_1 + x_2 + x_3 + x_4 + x_5. \quad (3.148)$$

Here and below, invariant labels are colour-coded by their physical-sheet sign: **green** denotes positive and **burgundy** negative. All coefficients in Eq. (3.147) have a fixed sign throughout the physical $2 \rightarrow 4$ sheet. In particular, the three-particle invariants shown there each contain one incoming and two outgoing momenta and are negative. Indeed, in the lightcone frame defined by $(+, -, \perp)_{(p_2, p_1)}$, any sum K of outgoing momenta obeys $0 \leq K^+ \leq p_2^+$ and $0 \leq K^- \leq p_1^-$. It follows that $(-p_i + K)^2 \leq 0$ for $i = 1, 2$, with strict inequality at wide angle.

3.3.1 The wide-angle self-crossing hidden region

The twisted hexagon has no positive pinch at generic massless wide-angle kinematics. A special wide-angle surface is nevertheless sufficient. In the physical self-crossing configuration of Ref. [64], each incoming particle splits into two collinear lines which undergo two separate hard scatterings. Let the two pairs of collinear daughters of p_1 and p_2 carry the longitudinal fractions $x, 1-x$ and $y, 1-y$, with $0 < x, y < 1$. For the external assignment in Fig. 14, the exact double-scattering configuration obeys

$$xp_1 + yp_2 = p_4 + p_5, \quad (1-x)p_1 + (1-y)p_2 = p_3 + p_6. \quad (3.149)$$

The collinear assignments follow from the same Coleman–Norton argument used for the previous wide-angle examples, the four-point Crown and the five-point Fish. At a positive physical Landau pinch, the four active internal propagators represent on-shell classical trajectories [42, 55]. Momentum conservation at either incoming three-point vertex decomposes the future-directed null momentum p_i into two future-directed null momenta; both daughters must therefore lie on the p_i ray. The two hard vertices then join one daughter from each incoming direction, realising the two subprocesses in Eq. (3.149). Consequently, each outgoing pair has zero net transverse momentum in the incoming centre-of-mass frame. The same geometry has a dual Wilson-loop description: the two non-adjacent null edges associated with the incoming momenta intersect, and their intersection divides the first into fractions x and $1 - x$, and the second into y and $1 - y$.

Momentum conservation in the two $2 \rightarrow 2$ subprocesses fixes the invariant values at the pinch:

$$\begin{aligned} s_{36}^{(0)} &= (1-x)(1-y)s_{12}, & s_{45}^{(0)} &= xy s_{12}, \\ s_{145}^{(0)} &= -y(1-x)s_{12}, & s_{245}^{(0)} &= -x(1-y)s_{12}, \\ x s_{15}^{(0)} &= y s_{24}^{(0)}, & (1-x)s_{16}^{(0)} &= (1-y)s_{23}^{(0)}. \end{aligned} \quad (3.150)$$

For the present cyclic assignment, define the three six-point cross ratios

$$U_{\times} \equiv \frac{s_{36}s_{45}}{s_{145}s_{245}}, \quad V_{\times} \equiv \frac{s_{16}s_{24}}{s_{156}s_{245}}, \quad W_{\times} \equiv \frac{s_{15}s_{23}}{s_{145}s_{156}}. \quad (3.151)$$

These are the cross ratios conventionally denoted by u, v, w in Ref. [64]; capital letters and the subscript $\times$ distinguish them from the lightcone transverse coordinate w introduced below. For generic six-point kinematics these three cross ratios are independent. Using the invariant relations in x and y above, one may verify explicitly that $U_{\times} = 1$ and $V_{\times}/W_{\times} = 1$; in the latter ratio, s_{156} cancels. The physical self-crossing surface is therefore

$$U_{\times} = 1, \quad V_{\times} = W_{\times}. \quad (3.152)$$

These conditions are the invariant image of the double-scattering configuration rather than an additional kinematic assumption. The first is the singular condition that makes the twisted-box polynomial factorise (see Eq. (3.156) below); the second selects its physical six-point self-crossing branch. Correspondingly, the standard six-point discriminant¹⁶

$$\Delta_6 = (1 - U_{\times} - V_{\times} - W_{\times})^2 - 4U_{\times}V_{\times}W_{\times} \quad (3.153)$$

reduces to $(V_{\times} - W_{\times})^2$ at $U_{\times} = 1$. A convenient physical regulator separates the two scattering points by a small spacelike transverse recoil $\Delta_{\perp}$. Following the general self-crossing parametrisation of Ref. [64], set

$$\Lambda_{\times} \equiv s_{12}xy(1-x)(1-y), \quad \sigma_{\times} \equiv \frac{|\Delta_{\perp}|^2}{\Lambda_{\times}} \rightarrow 0^+. \quad (3.154)$$

¹⁶Writing the six dual cusps as null rays X_i in the six-dimensional conformal embedding space, Δ_6 is proportional, after projective normalisation, to $-\det(X_i \cdot X_j)$. Thus $\sqrt{\Delta_6}$ measures the normalised embedding-space pseudo-volume of the six rays, and $\Delta_6 = 0$ means that they are linearly dependent; see Ref. [65].

The part of the departure from the pinch that survives on the boundary $x_0 = x_3 = 0$ relevant below is

$$\begin{aligned} s_{36}(\sigma_\times) &= s_{36}^{(0)} - \sigma_\times \Lambda_\times + \mathcal{O}(\sigma_\times^2), & s_{45}(\sigma_\times) &= s_{45}^{(0)} - \sigma_\times \Lambda_\times + \mathcal{O}(\sigma_\times^2), \\ s_{145}(\sigma_\times) &= s_{145}^{(0)} - \sigma_\times \Lambda_\times + \mathcal{O}(\sigma_\times^2), & s_{245}(\sigma_\times) &= s_{245}^{(0)} - \sigma_\times \Lambda_\times + \mathcal{O}(\sigma_\times^2). \end{aligned} \quad (3.155)$$

The two relations in the final line of Eq. (3.150) are exact at the pinch. Under a generic transverse recoil their deviations begin at relative order $|\Delta_\perp|/\sqrt{\Lambda_\times} = \mathcal{O}(\sqrt{\sigma_\times})$. The affected invariants multiply x_0 or x_3 in Eq. (3.147), so these earlier corrections vanish identically on the boundary and do not enter its HRF analysis.

Equation (3.155) gives $U_\times = 1 - \sigma_\times + \mathcal{O}(\sigma_\times^2)$. Along the one-parameter path used here we hold the nonsingular cross ratio fixed, $V_\times = W_\times = V_\times^{(0)}$. Expansion of Eq. (3.153) then gives

$$\Delta_6 = -4\sigma_\times V_\times^{(0)}(1 - V_\times^{(0)}) + \mathcal{O}(\sigma_\times^2), \quad 0 < V_\times^{(0)} < 1.$$

Thus $\Lambda_\times$ sets the normalisation of $\sigma_\times$, while the finite coefficient is fixed by the nonsingular kinematics on the self-crossing surface. All cyclic two-particle invariants $s_{i,i+1}$ remain finite and nonzero in this wide-angle limit.

The input to HRF is the complete unfactorised LP polynomial of Eq. (3.147); the self-crossing parametrisation enters through its Mandelstam coefficients. The boundary scan identifies $x_0 = x_3 = 0$, and on this stratum Part I extracts the restricted leading polynomial shown in the first line below. The derivative harvest then finds the two factors displayed in the second line:

$$\begin{aligned} \mathcal{F}_{\text{SL,WA}} &= s_{145}^{(0)} x_1 x_4 + s_{36}^{(0)} x_1 x_5 + s_{45}^{(0)} x_2 x_4 + s_{245}^{(0)} x_2 x_5 \\ &= \frac{1}{s_{36}^{(0)}} \underbrace{(s_{145}^{(0)} x_4 + s_{36}^{(0)} x_5)}_{f_{1,\text{WA}}} \underbrace{(s_{36}^{(0)} x_1 + s_{245}^{(0)} x_2)}_{f_{2,\text{WA}}}. \end{aligned} \quad (3.156)$$

Here the second equality follows from the first condition in Eq. (3.152). In particular, expanding the factorised form replaces its only apparent rational coefficient according to $s_{145}^{(0)} s_{245}^{(0)} / s_{36}^{(0)} = s_{45}^{(0)}$, so the result is precisely the polynomial in the first line. Each displayed factor combines a positive and a negative term and therefore has a positive zero.

Pairing these harvested factors retains the generator $g_{1,\text{WA}} = f_{1,\text{WA}} f_{2,\text{WA}}$. The certified active scaling is $(-1, -1, -1, -1)$, and restoring the contracted coordinates gives

$$\vec{v}_{\text{HR,WA}} = (\mathbf{v}_{\text{HR,WA}}; 1) = (0, -1, -1, 0, -1, -1; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-2, -1). \quad (3.157)$$

The zero entries for x_0 and x_3 have the usual hard-subgraph interpretation: the corresponding momenta have $q_0^2, q_3^2 \sim \sigma_\times^0$ and are absorbed into the two four-point hard vertices of Fig. 14(b). Thus neither edge is a Glauber propagator in the wide-angle region. The status of these two resolved edges changes in the NMRK and DSC specialisations below. The two contributions to the resolved-leading layer on this stratum are

$$\begin{aligned} \mathcal{U}|_{x_0=x_3=0} &= x_1 + x_2 + x_4 + x_5, \\ \mathcal{F}_1|_{x_0=x_3=0} &= -\Lambda_\times (x_1 x_4 + x_1 x_5 + x_2 x_4 + x_2 x_5) \\ &= -\Lambda_\times (x_1 + x_2)(x_4 + x_5). \end{aligned} \quad (3.158)$$

Here $\mathcal{F}_1$ is the coefficient of $\sigma_\times$ in the physical expansion of $\mathcal{F}$. Applying the vector in Eq. (3.157) after restricting the complete LP polynomial to $x_0 = x_3 = 0$ gives

$$\begin{aligned}\mathcal{P}_{\text{WA}}^{(\text{HR})}(\mathbf{x}; \sigma_\times, \mathbf{s}) &\equiv \mathcal{P}(\sigma_\times^{v_{\text{HR}, \text{WA}}} \mathbf{x}; \sigma_\times, \mathbf{s})|_{x_0=x_3=0} \\ &= \sigma_\times^{-2} \mathcal{F}_{\text{SL}, \text{WA}} + \sigma_\times^{-1} (\mathcal{U} + \mathcal{F}_1)|_{x_0=x_3=0} + \mathcal{O}(\sigma_\times^0).\end{aligned}\quad (3.159)$$

Thus the restricted $\mathcal{F}_1$ contribution is part of the resolved-leading layer and supplies the expansion-parameter direction needed by the lower-facet certificate. Dissecting in the two local normals defined by $f_{1, \text{WA}}$ and $f_{2, \text{WA}}$ produces two endpoint charts; the complete restricted LP support has the required codimension-one lower facet in each. For unit propagator powers the resulting scalar contribution scales as

$$I_\times^{\text{scalar}} \sim \sigma_\times^{D/2-3} = \sigma_\times^{-1-\epsilon}, \quad D = 4 - 2\epsilon. \quad (3.160)$$

The momentum-space interpretation is especially transparent in a routing adapted to the double scattering. Momentum conservation at the vertices gives

$$\ell_\times \equiv q_1 - (1-x)p_1 = xp_1 - q_2 = q_4 - yp_2 = (1-y)p_2 - q_5. \quad (3.161)$$

On the self-crossing pinch, the double-scattering assignments in Eq. (3.149) imply separately that $\ell_\times^\star = 0$. The two contracted propagators have hard virtualities there:

$$(q_0^\star)^2 = (1-x)s_{16}^{(0)} = (1-y)s_{23}^{(0)} = \mathcal{O}(\Lambda_\times), \quad (q_3^\star)^2 = xs_{15}^{(0)} = ys_{24}^{(0)} = \mathcal{O}(\Lambda_\times). \quad (3.162)$$

This hard scaling shows that, in this region, the twisted-hexagon realisation can be contracted to the twisted box considered here. The four active entries $v_e = -1$ in Eq. (3.157) fix their virtualities to

$$\frac{q_e^2}{\Lambda_\times} \sim \sigma_\times, \quad e \in \{1, 2, 4, 5\}. \quad (3.163)$$

Expanding these denominators in the shifted loop momentum of Eq. (3.161) gives

$$\begin{aligned}q_1^2 &= 2(1-x)p_1^- \ell_\times^+ + \ell_\times^2, & q_2^2 &= -2xp_1^- \ell_\times^+ + \ell_\times^2, \\ q_4^2 &= 2yp_2^+ \ell_\times^- + \ell_\times^2, & q_5^2 &= -2(1-y)p_2^+ \ell_\times^- + \ell_\times^2,\end{aligned}\quad (3.164)$$

The opposite longitudinal coefficients exhibit the q_1, q_2 pinch in $\ell_\times^+$ and the q_4, q_5 pinch in $\ell_\times^-$, while $\ell_\times^2 = 2\ell_\times^+ \ell_\times^- - |\ell_{\times\perp}|^2$. Since $p_1^-, p_2^+ = \mathcal{O}(\sqrt{\Lambda_\times})$, the virtualities in Eq. (3.163) require $\ell_\times^\pm = \mathcal{O}(\sigma_\times \sqrt{\Lambda_\times})$ and $|\ell_{\times\perp}| = \mathcal{O}(\sqrt{\sigma_\times \Lambda_\times})$. The latter is precisely the physical recoil scale in Eq. (3.154). This is the same local-denominator analysis used for the fluctuation widths in the discussion leading to Eq. (3.143). In the present one-loop problem it is the complete momentum-space reconstruction: since $\ell_\times^\star = 0$, the loop-momentum scaling and its local integration width coincide,

$$\frac{(\Delta\ell_\times^+, \Delta\ell_\times^-, |\Delta\ell_{\times\perp}|)}{\sqrt{\Lambda_\times}} \sim (\sigma_\times, \sigma_\times, \sqrt{\sigma_\times})_{(p_2, p_1)}. \quad (3.165)$$

Thus the single integration loop is transverse dominated and hence Glauber, although none of the individual propagators is Glauber: the four active propagators are collinear and q_0, q_3 are hard. Finally,

$$d^D \ell_\times \sim \sigma_\times^{D/2+1}, \quad \prod_{e \in \{1,2,4,5\}} \frac{1}{q_e^2} \sim \sigma_\times^{-4}, \quad I_\times^{\text{scalar}} \sim \sigma_\times^{D/2-3}, \quad (3.166)$$

which confirms Eq. (3.160) directly in momentum space.

We shall now approach this surface along two further kinematic limits: zero-recoil NMRK and the double spacelike-collinear (DSC) limit. In each limit, order alignment selects a leading face whose two cancellation factors inherit the zero sets of $f_{1,\text{WA}}$ and $f_{2,\text{WA}}$; exact normalised identities on the common pinch are displayed below. Thus both composite-limit HRs inherit the wide-angle cancellation locus, although their total region vectors and occupied layers depend on the expansion.

We first define the six-point variables and the two limiting paths. We then analyse the inherited one-loop cancellation mechanism in each limit, displaying the alignment vector, the factorised aligned face, the relative and combined scaling vectors, and the final validation against the complete graph polynomial. We then turn to two-loop graphs which contain the twisted-box seed as an externally labelled contraction minor.

3.3.2 Six-point lightcone variables and the zero-recoil NMRK surface

We use the minimal lightcone variables of Ref. [60]. Take legs 1 and 2 incoming, set $P = p_4^+$, and define the longitudinal ratios by

$$p_3^+ = P X_{34}, \quad p_4^+ = P, \quad p_5^+ = \frac{P}{X_{45}}, \quad p_6^+ = \frac{P}{X_{45} X_{56}}. \quad (3.167)$$

Writing $Q = q_1^\perp \bar{q}_1^\perp = |q_1^\perp|^2$, the transverse coordinates are

$$\begin{aligned} p_4^\perp &= -\frac{q_1^\perp}{z-1}, & p_5^\perp &= \frac{z q_1^\perp}{w(z-1)}, \\ p_6^\perp &= \frac{z(w-1) q_1^\perp}{w(z-1)}, & p_3^\perp &= -p_4^\perp - p_5^\perp - p_6^\perp, \end{aligned} \quad (3.168)$$

with barred equations for the conjugate transverse components. Thus the four real coordinates $Q, X_{34}, X_{45}, X_{56}$, together with the complex transverse coordinates z, w and their barred partners, parametrise generic massless six-point kinematics up to an irrelevant longitudinal boost. No asymptotic limit has yet been taken.

Central NMRK is the hierarchy

$$X_{34} = \frac{\hat{X}_{34}}{\eta}, \quad X_{56} = \frac{\hat{X}_{56}}{\eta}, \quad X_{45}, Q, z, \bar{z}, w, \bar{w} = O(1), \quad \eta \rightarrow 0^+. \quad (3.169)$$

On its physical sheet, $\bar{z} = z^*$ and $\bar{w} = w^*$. The limit studied in Sec. 3.3.3 combines the NMRK hierarchy in Eq. (3.169) with the zero-recoil condition

$$w = z, \quad \bar{w} = \bar{z} \quad (3.170)$$

which is not part of generic NMRK. It imposes $p_4^\perp + p_5^\perp = 0$. The self-crossing cross ratio obeys¹⁷

$$1 - U_\times = \frac{X_{45}(w - z)(\bar{w} - \bar{z})}{(1 + X_{45})(w\bar{w} + X_{45}z\bar{z})}. \quad (3.171)$$

Substituting the Mandelstam invariants obtained from the parametrisation in Eqs. (3.167) and (3.168) into Eq. (3.151) also gives $V_\times = W_\times$ when $w = z$ and $\bar{w} = \bar{z}$. Thus Eq. (3.170) places the kinematics on the complete self-crossing surface Eq. (3.152). The zero-recoil NMRK trajectory therefore lies on the same self-crossing singular surface as the wide-angle HR, while $\eta \rightarrow 0$ generates a rapidity hierarchy along that surface rather than measuring the distance from it. For the positive HRF chart below we use

$$Q, \hat{X}_{34}, X_{45}, \hat{X}_{56}, (1 - z)(1 - \bar{z}) > 0. \quad (3.172)$$

3.3.3 Twisted-hexagon HR in zero-recoil NMRK

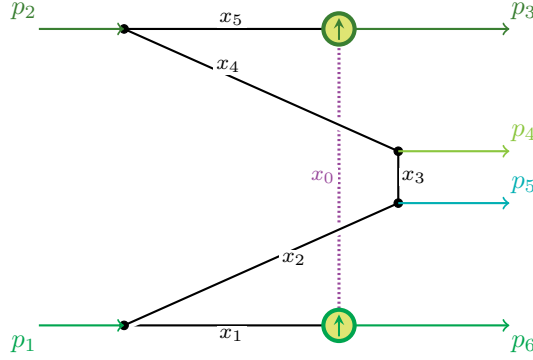


Figure 15. Momentum-space interpretation of the twisted-hexagon HR in NMRK. The embedding displays the rapidity ordering $p_3 \gg (p_4, p_5) \gg p_6$. The upper pair (p_2, p_3) and lower pair (p_1, p_6) have matching colours; the two line crossings are not additional vertices. The purple dotted edge labelled x_0 carries q_0 and is the explicit Glauber propagator in both zero-recoil NMRK and generic DSC. The oriented discs at the p_3 and p_6 vertices mark its two incident vertices and follow the convention defined in Fig. 4. Their olive and green rims record the two complementary collinear channels. The incoming three-point vertices carrying p_1 and p_2 are the factor-localisation vertices: they localise the two factor equations on the adjacent pairs (x_1, x_2) and (x_4, x_5) , respectively.

Substitution of Eq. (3.169) at fixed LP parameters gives a strict leading term

$$\mathcal{F}_{0,\text{NMRK}} = \eta^{-2} Q \hat{X}_{34} X_{45} \hat{X}_{56} x_1 x_5 \quad (\text{zero recoil}), \quad (3.173)$$

which is a single monomial and cannot reveal the pinch. The relevant monomials occupy different native η -orders. Following Sec. 2.3, HRF therefore searches the exposed lower faces of the augmented η -dependent support. This search selects the mixed-order face whose normal may be represented by

$$\phi_{\text{NMRK}} = (-1, -1, -2, -1, -2, -1). \quad (3.174)$$

¹⁷This cross-ratio parametrisation arises in the analysis of the two-gluon central-emission vertex; see App. B.1, in particular Eq. (B.9), of Ref. [66].

To display the structure of the selected face, strip off its leading η -power and a nonzero overall kinematic factor. Denoting equality up to this factor by $\doteq$, collect the LP parameters adjacent to the two incoming vertices into $\mathbf{x}_{p_1}$ and $\mathbf{x}_{p_2}$, and write

$$\mathcal{F}^{[\phi_{\text{NMRK}}]} \doteq \mathbf{x}_{p_1}^T M \mathbf{x}_{p_2}, \quad \mathbf{x}_{p_1} = \begin{pmatrix} x_2 \\ x_1 \end{pmatrix}, \quad \mathbf{x}_{p_2} = \begin{pmatrix} x_4 \\ x_5 \end{pmatrix}, \quad (3.175)$$

where, abbreviating $X = X_{45}$,

$$M = \begin{pmatrix} \frac{(w + Xz)(\bar{w} + X\bar{z})}{z\bar{z}} & -\hat{X}_{34}X \frac{w\bar{w} + Xz\bar{z}}{z\bar{z}} \\ -(w-1)(\bar{w}-1)X(1+X)\hat{X}_{56} & (w-1)(\bar{w}-1)\hat{X}_{34}X^2\hat{X}_{56} \end{pmatrix}. \quad (3.176)$$

The nontrivial stationary conditions are therefore

$$M\mathbf{x}_{p_2} = 0, \quad M^T\mathbf{x}_{p_1} = 0, \quad (3.177)$$

while

$$\det M = -\frac{(w-1)(\bar{w}-1)}{z\bar{z}} \hat{X}_{34}X^3\hat{X}_{56}(w-z)(\bar{w}-\bar{z}). \quad (3.178)$$

For generic NMRK this determinant is nonzero, so Eq. (3.177) admits only $\mathbf{x}_{p_1} = \mathbf{x}_{p_2} = 0$, not a solution in the positive orthant. On the additional surface Eq. (3.170), the matrix has rank one and the bilinear polynomial factorises. This matrix form merely makes the coefficient condition explicit. In the actual run the selected face is passed to the core HRF algorithm, whose derivative harvest recovers the two cancellation factors in the usual way. Define

$$f_{1,\text{NMRK}} = (1 + X_{45})x_4 - \hat{X}_{34}X_{45}x_5, \quad f_{2,\text{NMRK}} = (1 + X_{45})x_2 - (1 - z)(1 - \bar{z})X_{45}\hat{X}_{56}x_1. \quad (3.179)$$

The aligned face is then

$$\mathcal{F}_{\star,\text{NMRK}} \equiv \mathcal{F}^{[\phi_{\text{NMRK}}]} = \mathcal{F}_{\text{SL},\text{NMRK}} = Q\hat{X}_{34}\hat{X}_{56}f_{1,\text{NMRK}}f_{2,\text{NMRK}}, \quad \mathcal{F}_{\text{Obs},\text{NMRK}} = 0. \quad (3.180)$$

Both linear factors have positive zeros in the domain Eq. (3.172). Homogeneity of the aligned face alone leaves a uniform freedom and therefore does not yet fix the core scaling. The Part II hierarchy inequalities, applied to the complete aligned LP polynomial in the original LP variables, fix the remaining core scaling quoted below. We then pass to local cancellation-resolving coordinates to give an independent lower-facet certificate and to make the composition of the core and alignment scalings transparent.

Here the local certificate is compact enough to display explicitly. In one physical sign sector take

$$t_1 = f_{1,\text{NMRK}}, \quad t_2 = f_{2,\text{NMRK}}, \quad t_1, t_2 \geq 0, \quad (3.181)$$

and solve

$$x_4 = \frac{\hat{X}_{34}X_{45}x_5 + t_1}{1 + X_{45}}, \quad x_1 = \frac{(1 + X_{45})x_2 - t_2}{(1 - z)(1 - \bar{z})X_{45}\hat{X}_{56}}. \quad (3.182)$$

The tangential coordinates are (x_0, x_2, x_3, x_5) . Here the x_e denote the aligned LP parameters, after applying Eq. (3.174). Define the local scaling exponents by

$$x_e \mapsto \eta^{v_{x_e}^{\text{loc}}} x_e \quad (e \in \{0, 2, 3, 5\}), \quad t_i \mapsto \eta^{v_{t_i}^{\text{loc}}} t_i \quad (i \in \{1, 2\}). \quad (3.183)$$

In the edge-compatible ordered local coordinate system $(x_0, t_2, x_2, x_3, t_1, x_5)$, where t_2 occupies the slot of x_1 and t_1 the slot of x_4 , the transformed complete LP polynomial has the exact lower-facet normal

$$\vec{v}_{\text{loc,NMRK}} \equiv (v_{x_0}^{\text{loc}}, v_{t_2}^{\text{loc}}, v_{x_2}^{\text{loc}}, v_{x_3}^{\text{loc}}, v_{t_1}^{\text{loc}}, v_{x_5}^{\text{loc}}; 1) = (-1, -2, -4, -1, -2, -4; 1). \quad (3.184)$$

The last entry is the normalised η -component of the augmented normal. The Jacobian is nonzero and, up to sign, proportional to $[(1-z)(1-\bar{z})X_{45}(1+X_{45})\hat{X}_{56}]^{-1}$. Pulling this normal back to the aligned LP parameters reproduces the core scaling fixed by Part II. Composing it with the alignment vector in Eq. (3.174) then gives the certified vector in the original coordinates,

$$\begin{aligned} \mathbf{v}_{\text{core}} &= (-1, -4, -4, -1, -4, -4), \\ \mathbf{v}_{\text{HR}} &= \phi_{\text{NMRK}} + \mathbf{v}_{\text{core}} = (-2, -5, -6, -2, -6, -5), \\ \vec{v}_{\text{HR}} &= (\mathbf{v}_{\text{HR}}; 1), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-10, -6). \end{aligned} \quad (3.185)$$

The edge-compatible ordering in Eq. (3.184) makes the cancellation depth directly visible:

$$\mathbf{v}_{\text{loc,NMRK}} = \mathbf{v}_{\text{core}} + (0, 2, 0, 0, 2, 0).$$

Only the two normal entries differ. In those slots t_2 replaces x_1 and t_1 replaces x_4 ; each local normal width is suppressed by two powers of η relative to the corresponding edge-parameter scaling. Each transverse factor is suppressed by two powers, so their product $f_{1,\text{NMRK}}f_{2,\text{NMRK}}$ first contributes four powers above the individual superleading monomials, precisely at W_{HR} . The resolved lower facet contains this first nonzero contribution from $\mathcal{F}_{\text{SL,NMRK}}$, together with two $\mathcal{U}$ monomials and seven monomials from the remaining $\mathcal{F}$ polynomial. Its ten exponent points have affine rank six, and the inverse-map and pullback checks are exact. Thus the equality of the resolved $\mathcal{F}_{\text{SL,NMRK}}$ weight and W_{HR} is observed in the dissected polynomial rather than supplied as an assumption. Generic NMRK, for which the determinant above is nonzero, has no corresponding certificate for this seed.

Dissection enters once more, now for power counting rather than for determining the vector: the local normal widths determine the measure near the cancellation locus. For unit propagator powers the unrestricted edge measure has weight $\sum_e (v_{\text{HR}})_e = -26$. The actual local measure instead uses the normal widths displayed in the comparison above. The alignment Jacobian has weight $\sum_e (\phi_{\text{NMRK}})_e = -8$, while the six local variables in Eq. (3.184) have total weight -14 . The exact local measure therefore has weight $-8 - 14 = -22$. Equivalently, the two changes $-4 \rightarrow -2$ in that comparison restore four powers relative to the unrestricted edge measure. Since $\mathcal{P}$ has resolved weight -6 ,

$$I_{\text{NMRK}}^{\text{scalar}} \sim \eta^{-22} (\eta^{-6})^{-D/2} = \eta^{3D-22}. \quad (3.186)$$

Turning to momentum space, the inverse edge scalings give propagator virtuality powers $(2, 5, 6, 2, 6, 5)$, whose sum is 26. A momentum-space realisation of the complete power in Eq. (3.186) must therefore have local one-loop measure $d^D\ell \sim \eta^{3D+4}$. In the incoming-momentum lightcone frame $(+, -, \perp)_{(p_2, p_1)}$, where p_2 defines the plus direction and p_1 the minus direction, write

$$(\Delta\ell^+, \Delta\ell^-, |\Delta\ell_\perp|)_{(p_2, p_1)} \sim (\eta^a, \eta^b, \eta^c)_{(p_2, p_1)}.$$

The corresponding measure scales as $\eta^{a+b+(D-2)c}$. Matching this with η^{3D+4} gives

$$c = 3, \quad a + b = 10, \quad a + b - 2c = 4 > 0. \quad (3.187)$$

Here the D -dependent part fixes $c = 3$, while the remaining power fixes $a + b = 10$.

We now reconstruct the component widths directly from the pinching propagator pairs, thereby confirming Eq. (3.187) and fixing the individual longitudinal exponents a and b . Normalise momenta to the growing beam scale, so that p_2^+ and p_1^- are order one. The two central momenta and the corresponding self-crossing splitting fractions are then of order η . Let q_e denote the momentum through the edge labelled by x_e . Orienting (q_1, q_2) away from the p_1 vertex and (q_4, q_5) away from the p_2 vertex gives the exact routing

$$p_1 = q_1 + q_2, \quad p_2 = q_4 + q_5, \quad q_0 = q_1 - p_6 = p_3 - q_5, \quad q_3 = q_2 - p_5 = p_4 - q_4.$$

The edge values at the pinch obey

$$q_1^-, q_5^+ \sim \eta^0, \quad q_2^-, q_4^+ \sim \eta. \quad (3.188)$$

In the same hard-normalised frame, the two resolved t -channel propagators consequently have the average momentum scalings

$$q_0 \sim (\eta^2, \eta^2, \eta)_{(p_2, p_1)}, \quad q_3 \sim (\eta, \eta, \eta)_{(p_2, p_1)}. \quad (3.189)$$

Thus q_0 is Glauber whereas q_3 is soft, even though both have virtuality of order η^2 . Component scaling, rather than virtuality alone, distinguishes the two modes. Combining these coefficients with the virtuality powers $(2, 5, 6, 2, 6, 5)$, the two members of each pole pair give the same width,

$$\Delta\ell^+ \sim \frac{|q_1^2|}{|q_1^-|} \sim \frac{|q_2^2|}{|q_2^-|} \sim \eta^5, \quad \Delta\ell^- \sim \frac{|q_5^2|}{|q_5^+|} \sim \frac{|q_4^2|}{|q_4^+|} \sim \eta^5. \quad (3.190)$$

Positivity of the LP parameters at the pinch places the two poles in each pair on opposite sides of the contour. The active transverse momenta vanish at the pinch, so their first fluctuation is quadratic; the order- η^6 virtualities of q_2 and q_4 require $|\Delta\ell_\perp| \sim \eta^3$. Thus the symmetric hard-normalised frame gives

$$\Delta\ell \sim (\eta^5, \eta^5, \eta^3)_{(p_2, p_1)}. \quad (3.191)$$

A longitudinal boost shifts the first two entries oppositely but preserves their sum.

The pole analysis consequently gives $d^D \ell \sim \eta^{5+5+3(D-2)} = \eta^{3D+4}$. Including the six propagators completes the momentum-space power count,

$$I_{\text{NMRK}}^{\text{scalar}} \sim \frac{d^D \ell}{\prod_{e=0}^5 q_e^2} \sim \eta^{3D+4-26} = \eta^{3D-22},$$

in direct agreement with Eq. (3.186). Independently of the frame choice, the inequality $a + b - 2c = 4 > 0$ in Eq. (3.187) identifies the unique loop width as Glauber. Its average q_0 momentum is also Glauber, so x_0 is an explicit Glauber propagator. The two oriented markers in Fig. 15 are at its incident p_3 and p_6 vertices. They are distinct from the two incoming factor-localisation vertices.

3.3.4 Twisted-hexagon HR in generic DSC

The DSC limit uses the momentum-twistor chart of Ref. [67], not the lightcone transverse variables z, w above. Here a is the hard seed modulus, $\tau_i = (1 - \xi_i)/\xi_i$ encode the two longitudinal momentum fractions, and $z_D, \bar{z}_D$ are the two twistor roots that remain after the double-collinear deformation. The expansion parameter is $\varepsilon \rightarrow 0^+$. On the physical DSC sheet the symbols $z_D, \bar{z}_D$ are independent real variables rather than a conjugate pair. A convenient connected real slice is

$$a < 0, \quad -1 < \tau_1 < 0, \quad \tau_2 < -1, \quad z_D < 0, \quad \bar{z}_D < 0. \quad (3.192)$$

With p_1, p_2 incoming, $p_3, \dots, p_6$ outgoing and the invariants defined as in Eq. (3.146), the adjacent invariants have the following small- ε expansions:

$$\begin{aligned} s_{12} &= -\frac{\tau_2}{a(1+\tau_1)(1+\tau_2)} + O(\varepsilon), & s_{23} &= \frac{\varepsilon^2 \tau_1}{a \bar{z}_D} + O(\varepsilon^3), \\ s_{34} &= \frac{1}{(a-1)(1+\tau_2)} + O(\varepsilon), & s_{45} &= -\frac{1}{a}, \\ s_{56} &= \frac{\tau_1}{(a-1)(1+\tau_1)} + O(\varepsilon), & s_{16} &= \frac{\varepsilon^2 \tau_2 z_D}{a} + O(\varepsilon^3). \end{aligned} \quad (3.193)$$

Thus $s_{23}, s_{16} < 0$, while $s_{12}, s_{34}, s_{45}, s_{56} > 0$, as required for $2 \rightarrow 4$ scattering with legs 1 and 2 incoming. The two limits therefore use the same graph and physical scattering channel, but they are distinct paths in six-point kinematic space. In terms of the invariant cross ratios Eq. (3.151), the DSC chart obeys

$$1 - U_\times = O(\varepsilon^2), \quad V_\times = O(\varepsilon^2), \quad W_\times = O(\varepsilon^2), \quad V_\times - W_\times = O(\varepsilon^2). \quad (3.194)$$

It therefore approaches the endpoint $(U_\times, V_\times, W_\times) = (1, 0, 0)$ of the self-crossing surface, rather than remaining on its interior at finite ε . The DSC roots $z_D, \bar{z}_D$ consequently have no notational or algebraic identification with the NMRK transverse coordinates z, w .

As in the NMRK example, the cancelling monomials of Eq. (3.147) lie at different fixed-parameter orders. HRF therefore selects the alignment vector

$$\phi_{\text{DSC}} = (-2, -2, -2, -1, -2, -2). \quad (3.195)$$

Part I of the core HRF algorithm, applied to this selected face, identifies the cancellation factors

$$f_{1,\text{DSC}} = (1 + \tau_2)x_4 + x_5, \quad f_{2,\text{DSC}} = \tau_1 x_1 + (1 + \tau_1)x_2. \quad (3.196)$$

In terms of these factors, the aligned face is

$$\mathcal{F}_{\star,\text{DSC}} \equiv \mathcal{F}^{[\phi_{\text{DSC}}]} = \mathcal{F}_{\text{SL},\text{DSC}} = -\frac{f_{1,\text{DSC}}f_{2,\text{DSC}}}{a(1 + \tau_1)(1 + \tau_2)}, \quad \mathcal{F}_{\text{Obs},\text{DSC}} = 0. \quad (3.197)$$

The physical domain in Eq. (3.192) makes the two positive zeros explicit:

$$x_5 = -(1 + \tau_2)x_4 > 0, \quad x_2 = -\frac{\tau_1}{1 + \tau_1}x_1 > 0. \quad (3.198)$$

For crossed double-timelike collinear splittings $\tau_i > 0$, both factors are strictly positive for positive x_i , so they have no positive zeros and the pinch disappears.

Part II of the core HRF algorithm, applied to the aligned face alone, gives a unique candidate core vector, which is uniform:

$$\mathbf{v}_{\text{core},\text{DSC}}^{(0)} = (-1, -1, -1, -1, -1, -1) \quad (3.199)$$

This face-only result is provisional. Returning to the full native polynomial, Part II rescales both the LP parameters and the kinematics and tests every occupied layer. With $\mathcal{I}_D = \langle f_{1,\text{DSC}}, f_{2,\text{DSC}} \rangle$, the first three occupied layers of the accepted rescaled polynomial are

$$\mathcal{F}^{(\text{HR})} = \underbrace{\varepsilon^{-4}(\mathcal{F}_{-4} \in \mathcal{I}_D^2)}_{\text{ord}_{\mathcal{I}_D}=2} + \underbrace{\varepsilon^{-3}(\mathcal{F}_{-3} \in \mathcal{I}_D \setminus \mathcal{I}_D^2)}_{\text{ord}_{\mathcal{I}_D}=1} + \underbrace{\varepsilon^{-2}(\mathcal{F}_{-2} \notin \mathcal{I}_D)}_{\text{ord}_{\mathcal{I}_D}=0}. \quad (3.200)$$

The intervening ε^{-3} layer contains 52 nonzero terms. Since $\mathcal{F}_{-3} \in \mathcal{I}_D \setminus \mathcal{I}_D^2$, it vanishes to first order on the common pinch. At the same scaling,

$$\mathcal{U}^{(\text{HR})} = \varepsilon^{-2}\mathcal{U}_{-2} + x_3, \quad \mathcal{U}_{-2} = x_0 + x_1 + x_2 + x_4 + x_5, \quad (3.201)$$

Together with Eq. (3.201), the common resolved weight

$$-4 + 2 = -3 + 1 = -2 + 0 = -2$$

of the three occupied ideal layers fixes the corrected relative vector and the final original-coordinate answer:

$$\begin{aligned} \mathbf{v}_{\text{core},\text{DSC}} &= (0, 0, 0, 1, 0, 0), & \mathbf{v}_{\text{HR},\text{DSC}} &= \phi_{\text{DSC}} + \mathbf{v}_{\text{core},\text{DSC}} = (-2, -2, -2, 0, -2, -2), \\ \vec{v}_{\text{HR},\text{DSC}} &= (\mathbf{v}_{\text{HR},\text{DSC}}; 1), & (W_{\text{SL}}, W_{\text{HR}}) &= (-4, -2). \end{aligned} \quad (3.202)$$

The gap is two, and a local dissection pulls back to the same vector. This is why asymptotic-order alignment cannot be treated as a cosmetic preprocessing step: adding $\phi_{\text{DSC}} + \mathbf{v}_{\text{core},\text{DSC}}^{(0)}$ would give both the wrong vector and the wrong cancellation depth.

The corresponding momentum audit is even more direct. The unrestricted edge measure has weight -10 ; the two first-order normal restrictions restore two powers, and the resolved LP weight is -2 :

$$I_{\text{DSC}}^{\text{scalar}} \sim \varepsilon^{-10+2}(\varepsilon^{-2})^{-D/2} = \varepsilon^{D-8}. \quad (3.203)$$

The propagator virtuality powers are $(2, 2, 2, 0, 2, 2)$, again with sum 10. Thus the loop measure is ε^{D+2} . In the incoming-momentum lightcone frame $(+, -, \perp)_{(p_2, p_1)}$, where p_2 defines the plus direction and p_1 the minus direction, write

$$(\Delta\ell^+, \Delta\ell^-, |\Delta\ell_\perp|)_{(p_2, p_1)} \sim (\varepsilon^{a_{\text{DSC}}}, \varepsilon^{b_{\text{DSC}}}, \varepsilon^{c_{\text{DSC}}})_{(p_2, p_1)}$$

The exponents then satisfy

$$c_{\text{DSC}} = 1, \quad a_{\text{DSC}} + b_{\text{DSC}} = 4, \quad a_{\text{DSC}} + b_{\text{DSC}} - 2c_{\text{DSC}} = 2 > 0. \quad (3.204)$$

The individual component widths follow directly from the pinching propagator pairs. Normalise momenta to the hard beam scale, as in the NMRK analysis. Because the DSC splitting fractions τ_1 and τ_2 are fixed, the two lines in each pinching pair carry finite nonzero longitudinal fractions: $q_1^-, q_2^-, q_4^+, q_5^+ \sim \varepsilon^0$. Combining these coefficients with the order- ε^2 virtualities gives

$$\Delta\ell^+ \sim \frac{|q_1^2|}{|q_1^-|} \sim \frac{|q_2^2|}{|q_2^-|} \sim \varepsilon^2, \quad \Delta\ell^- \sim \frac{|q_4^2|}{|q_4^+|} \sim \frac{|q_5^2|}{|q_5^+|} \sim \varepsilon^2. \quad (3.205)$$

Positivity of the LP parameters at the pinch places the two poles in each pair on opposite sides of the contour. The transverse virtuality is also of order ε^2 , so $|\Delta\ell_\perp| \sim \varepsilon$. The symmetric hard-normalised frame therefore gives

$$(\Delta\ell^+, \Delta\ell^-, |\Delta\ell_\perp|)_{(p_2, p_1)} \sim (\varepsilon^2, \varepsilon^2, \varepsilon)_{(p_2, p_1)}. \quad (3.206)$$

For the average propagator momenta, the routing introduced in the NMRK analysis instead gives

$$q_0 \sim (\varepsilon^2, \varepsilon^2, \varepsilon)_{(p_2, p_1)}, \quad q_3 \sim (1, 1, 1)_{(p_2, p_1)}. \quad (3.207)$$

Accordingly, x_0 is again an explicit Glauber propagator, while x_3 is hard and is absorbed into the four-point hard vertex associated with the (p_4, p_5) subprocess. In DSC the Glauber scaling of the average q_0 momentum coincides with its local integration width. The resulting measure $\varepsilon^{2+2+(D-2)} = \varepsilon^{D+2}$ confirms Eq. (3.204) and fixes $a_{\text{DSC}} = b_{\text{DSC}} = 2$. A longitudinal boost shifts these two exponents oppositely while preserving their sum. Thus the DSC loop is Glauber, as in the NMRK example, and the marked Glauber vertices in Fig. 15 are incident to x_0 . The incoming three-point vertices carrying p_1 and p_2 are the separate factor-localisation vertices. The transverse width and longitudinal-to-transverse imbalance nevertheless differ from NMRK.

3.3.5 Inheritance from the wide-angle self-crossing region

We label the wide-angle factors in Eq. (3.156) so that $f_{1,\text{WA}}$ uses (x_4, x_5) and $f_{2,\text{WA}}$ uses (x_1, x_2) , matching the NMRK and DSC labels. Only on the positive double-scattering pinch of Eqs. (3.149) and (3.150), after taking the leading aligned restriction and identifying the same fractions x, y , do their monic forms obey

$$\begin{aligned} \frac{f_{1,\text{WA}}}{s_{145}^{(0)}} &= \frac{f_{1,\text{NMRK}}}{1 + X_{45}} = \frac{f_{1,\text{DSC}}}{1 + \tau_2} = x_4 - \frac{1 - y}{y} x_5, \\ \frac{f_{2,\text{WA}}}{s_{245}^{(0)}} &= \frac{f_{2,\text{NMRK}}}{1 + X_{45}} = \frac{f_{2,\text{DSC}}}{1 + \tau_1} = x_2 - \frac{1 - x}{x} x_1, \end{aligned} \quad \text{on the common positive pinch.} \quad (3.208)$$

The coefficient identifications responsible for these equalities are

$$\frac{\widehat{X}_{34}X_{45}}{1+X_{45}} = -\frac{1}{1+\tau_2} = \frac{1-y}{y}, \quad \frac{(1-z)(1-\bar{z})X_{45}\widehat{X}_{56}}{1+X_{45}} = -\frac{\tau_1}{1+\tau_1} = \frac{1-x}{x}. \quad (3.209)$$

Equation (3.149) identifies $(1-y)/y$ as the ratio of the p_2 -collinear fractions feeding (p_3, p_6) and (p_4, p_5) , respectively; $(1-x)/x$ is the analogous p_1 ratio. Thus Eq. (3.209) expresses the same two pinch momentum-sharing ratios in NMRK and DSC variables. All these equalities are pinch-restricted: there Eq. (3.208) is an identity in the LP parameters, not merely an equality of zero sets, but neither equation maps the full NMRK and DSC parametrisations. Their occupied layers, and hence their physical vectors, are different:

Expansion	certified edge vector	$(W_{\text{SL}}, W_{\text{HR}})$	cancellation depth
wide-angle self-crossing	$(0, -1, -1, 0, -1, -1)$	$(-2, -1)$	1
zero-recoil NMRK	$(-2, -5, -6, -2, -6, -5)$	$(-10, -6)$	4
generic DSC	$(-2, -2, -2, 0, -2, -2)$	$(-4, -2)$	2

Table 7. Comparison of the three certified one-loop twisted-hexagon regions.

In every row the displayed edge vector becomes the homogeneous LP vector by appending the final entry 1. Equation (3.208) shows that the two composite limits inherit the same seed cancellation geometry; the distinct resolved layers show why the three HRs must not be identified.

Finally, the positive Landau equations make the loop diagnosis a genuine pinch statement. For a one-loop graph they include $\sum_e x_e q_e^+ = \sum_e x_e q_e^- = 0$. Since every active $x_e > 0$, the nonzero q_e^+ and q_e^- cannot each have a common sign. The corresponding ℓ^- and ℓ^+ poles therefore approach from opposite half-planes. The purple dotted x_0 edge in Fig. 15 displays the corresponding explicit Glauber propagator. Its two incident vertices carry the oriented markers. The incoming p_1 and p_2 three-point vertices are the two factor-localisation vertices.

3.3.6 Two-loop topology test

Having established the one-loop twisted-hexagon realisation of the seed, we now ask which two-loop graphs retain its cancellation mechanism. The nearly horizontal chains in Fig. 15 are the (23) and (16) collinear sectors. A t -channel exchange is an internal path connecting these two chains. The central emissions p_4, p_5 may lie on the same such path or on two separate paths. Figure 16 defines the three tested graph configurations and all their edge parameters.

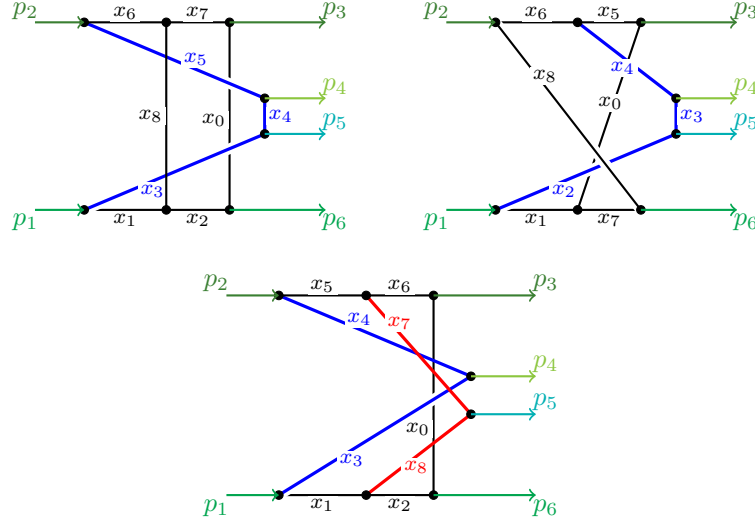


Figure 16. Two-loop graphs in the same rapidity-ordered embedding as Fig. 15. The upper-left graph is the planar hexagon–box. The upper-right and lower graphs have the same hexagon–pentagon internal topology but different external-leg attachments. In the upper two configurations both central emissions lie on the same t -channel path, shown in blue, and contraction of the additional edges recovers the externally labelled twisted-box seed. In the lower configuration the emissions lie on separate paths, shown in blue and red, and the externally labelled twisted-box seed is not a contraction minor. Black dots denote vertices.

Table 8 summarises the completed audits.

Graph	emission paths	labelled seed minor?	zero-recoil NMRK	generic DSC
twisted hexagon	same	yes	1 interior [1]	1 interior
planar hexagon–box	same	yes	13 interior + 12 boundary [7]	1 interior
hexagon–pentagon	same	yes	3 interior + 6 boundary [4]	none
hexagon–pentagon	separate	no	none [0]	none

Table 8. Certified HR counts in the one- and two-loop six-point topology audit. The square-bracketed NMRK entry counts distinct Part-I generators that supply at least one certified HR. “Labelled seed minor” uses the externally labelled sense defined in Sec. 2.3.2.

The entries count inequivalent certified HRs for the displayed permutation of external legs. Two certificates are identified when they have the same boundary stratum, exact cancellation ideal, normalised pullback vector and W_{HR} . The value of W_{SL} , and hence the cancellation depth assigned to a particular Part-I presentation, does not split an HR. “Interior” means that every edge parameter is active; “boundary” refers in this case to a certified codimension-one (one-edge contraction) stratum. Only codimension-one boundary strata were included in this two-loop audit; deeper boundary strata were not tested. The generator count is a coarser classification: it measures how many distinct Part-I cancellation structures supply HRs, but one generator can resolve into several HRs.

A staged candidate is a structural Part-I class specified by its restricted face polynomial and generator set. It is not counted as an HR unless the source-aware Part-II construction is applied to the full source decomposition, with every native kinematic layer of the original $\mathcal{F}$, together with $\mathcal{U}$, retained, and a scaleful lower-facet certificate is produced. The thirteen planar hexagon–box candidates produce 41 such certificates. After the above identification they give 25 HRs, 13 interior and 12 boundary, supplied by seven distinct generators. Two of these HRs each admit two different W_{SL} and cancellation-depth presentations, giving 27 strict certificate-presentation classes but no additional HRs. For the same-path hexagon–pentagon, three of the eight staged candidates have no common pinch in the positive domain, one has no certified facet after the complete dissection, and the remaining four produce nine HRs, three interior and six boundary, supplied by four distinct generators. The separate-path graph has no staged candidate.

Within this sample, placing both emissions on one exchange is necessary: the separate-exchange graph has no staged candidate in either limit and no externally labelled twisted-box contraction minor. For the same-path graphs, contracting the additional edges preserves the external momentum labels and their attachment to the seed vertices, so the seed HR is inherited on the corresponding contraction stratum. Since this is a deeper boundary, it lies outside the interior and codimension-one strata tested in the two-loop audit; the absence of a generic-DSC entry for the same-path hexagon–pentagon therefore does not exclude this inherited boundary HR. Its certified NMRK regions show in addition that the seed mechanism extends away from the contraction stratum in that limit.

3.4 Algorithmic capabilities demonstrated by the examples

The examples above demonstrate the successive capabilities collected in the algorithm summary of Sec. 2.5. The first is to recover cancellation factors from derivatives, as described in Sec. 2.2.1. In the Crown, individual derivatives mix terms weighted by the independent invariants s_{12} and s_{23} . Separating the corresponding kinematic components exposes the four factors in Eq. (3.2) and permits their pairing into two generators. The Crown descendants in Sec. 3.1.2 show that the same derivative harvest also identifies general multi-term cancellation factors and the generators constructed from them.

A second capability is demonstrated by the nonzero obstruction in the spacelike-collinear five-point Fish. At the common zero of f_1 and f_2 , the term $\mathcal{F}_{\text{Obs}} = sz(1 + \chi)x_2x_3x_6$ in Eq. (3.21) remains nonzero throughout the physical domain (3.20). Thus the strict leading polynomial $\mathcal{F}_0$ does not itself admit the stationary Landau solution; the stationary conditions (2.16) apply only to the selected $\mathcal{F}_{\text{SL}} = sf_1f_2$. Part I must therefore determine the decomposition $\mathcal{F}_\star = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{Obs}}$ of Eqs. (2.23)–(2.24) before Part II can determine the non-uniform vector in Eq. (3.23), under which the monomials of $\mathcal{F}_{\text{SL}}$ become individually superleading. The five-point seed and its vertex correction also introduce kinematic variables inside the cancellation factors; compare Eqs. (3.21) and (3.41). Thus a factorisation is physically admissible only if the factors have a common zero at positive LP parameters in the stated kinematic domain, as required by Eq. (2.26).

The near-planar wide-angle example demonstrates how factors can be recovered when no individual derivative displays them. Here the saturation procedure of Eq. (2.28), which

permits division only by quantities known to be nonzero in the chosen sector, becomes the explicit calculation in Eqs. (3.78) and (3.79). It recovers the three factors in Eq. (3.73), and hence the three pair-product generators appearing in the decomposition Eq. (3.77).

Correlating two kinematic limits can make the stationary ratios rate dependent. For the reference assignment in the simultaneous MRK and near-planar limit of Sec. 3.2.5, all three ratios in Eq. (3.105) remain positive at every finite $x > 0$, while $\rho_B \rightarrow 0$ as the MRK limit is approached. Thus this particular positive solution approaches the boundary of the positive LP orthant but does not leave it. This behaviour does not determine the result for another external-leg assignment: each inequivalent permutation must be tested separately unless it is related to the first by a symmetry. In the example, the two assignments in Fig. 10(a,b) retain positive stationary ratios, whereas the $p_3 \leftrightarrow p_5$ assignment in panel (c) gives the negative ratios in Eq. (3.106); its stationary solution therefore lies outside the positive LP orthant and it has no HR.

The same correlated limit presents a separate hierarchy issue. In Eq. (3.108), generator terms occur both at W_{SL} and at an intervening nominal weight W_k , and both contributions vanish on the same pinch. The Part-II hierarchy must therefore include both occupied cancellation layers.

Asymptotic-order alignment is needed when the relevant cancelling monomials start at different powers of the expansion parameter. Its general construction and the composition of the alignment and core vectors are given in Sec. 2.3, especially Eq. (2.45). The central-soft example in Sec. 3.2.6 and the six-point NMRK example in Sec. 3.3.3 illustrate this operation. The DSC example adds a further lesson: after alignment, the omitted layers of the full LP polynomial must be restored. The intervening layer in Eq. (3.200) vanishes on the same pinch and changes the provisional face-only scaling to the final vector in Eq. (3.202), following the general test in Eq. (2.39).

Across these examples, the leading nonzero dependence normal to the pinch and every restored contribution that cancels on the same locus enter at the same final scaling, even when they originate at different orders of the kinematic expansion. This saturation is an empirical regularity of the asymptotic hierarchy, not an assumption of the construction: Eq. (2.39) requires a resolved contribution only to enter at W_{HR} or above. The distinct polynomial sources must nevertheless be retained separately until their combined scaling has been checked.

Finally, the boundary wrapper acts around rather than inside the core construction. As described in Sec. 2.3.2 and illustrated by the Crown descendants and the two-loop audit in Table 8, it scans contraction strata and compares their restricted ideals with the parent ideal.

Independently, every candidate that passes the algebraic stages, whether interior or boundary, undergoes the final dissection certificate described in Sec. 2.2.2. This maps each relevant local lower-facet normal back to the original LP variables and thereby certifies the scaling; it is not part of the construction of the cancellation generators. The capabilities are therefore cumulative: the algebraic steps reconstruct information absent from the original Newton polytope, while the local lower-facet test supplies the final geometric certificate.

Allowing massive internal propagators would relax the standing massless multiaffinity

assumption. The resulting quadratic edge-parameter dependence changes the possible orders of cancellation and the local facet problem. It is a genuine extension of this framework, not another example in the present tour.

4 From parameter regions to momentum modes

The output of HRF is a parameter-space region. Its translation into momentum space is essential for going beyond individual integrals to amplitudes and observables: physical intuition about scale separation, and the formulation and proof of factorisation, are naturally expressed in terms of momentum modes. Momentum-space reconstruction is therefore the bridge from parameter-space region finding to amplitude-level physics. It is a logically subsequent problem, rather than an additional stage of the region-finding algorithm. We address that problem here, after the examples, by distinguishing three questions that are easily conflated: the virtuality of each propagator, the average loop-momentum configuration at the pinch and its associated edge momenta, and the local integration widths of the independent loop fluctuations. The second identifies the momentum modes in the usual EFT sense; the third determines the integration measure and hence enters power counting.

In the Schwinger representation, before performing the loop integrations, the loop-momentum integral is Gaussian at fixed proper times. For a region $\vec{v}$, we implement the scaling in Eq. (2.13) by writing $Q^2 \tilde{x}_e = \delta^{v_e} \hat{x}_e$, with $\hat{x}_e = O(1)$. At fixed values of the projective ratios of the $\hat{x}_e$, the centre of the resulting Gaussian defines the conditional average $\langle \ell \rangle_{\vec{v}}$. Its componentwise powers define the *average-momentum scaling*. The standard deviations obtained from the same Gaussian covariance define the marginal integration widths. Agreement between the componentwise average-momentum scaling and the marginal-width scaling is not assumed; it is tested separately in the examples.¹⁸ Throughout this section, q_e and $\mathcal{D}_e$ denote the oriented edge momenta and propagator denominators introduced in Eqs. (2.1) and (2.2).

4.1 Inherited virtualities and component constraints

Let the certified original-coordinate HR vector be $\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1) = (v_1, \dots, v_N; 1)$. Inverse LP-parameter scaling fixes the relative virtuality powers of the denominators in Eq. (2.2), as derived in Ref. [43, Sec. 2.3, especially Eqs. (46)–(48), and App. B],

$$\frac{\mathcal{D}_e}{\mu_{\text{LP}}^2} \sim \delta^{-v_e} \equiv \delta^{\kappa_e}, \quad \kappa_e = -v_e, \quad (4.1)$$

in the LP normalisation of Eq. (2.4). By default μ_{LP} is fixed. If instead the chosen reference scale varies in the limit — as in the central-soft rapidity-ordered limit of Sec. 3.2.6, where $\mu_{\text{LP}}^2 = s_{12}$ grows as δ^{-4} — virtuality powers relative to a fixed scale are obtained by adding

¹⁸A systematic mixed-representation definition of these averages, standard deviations and cross-loop covariances, together with a general momentum-reconstruction algorithm for facet and hidden regions, will be presented in forthcoming work [68]. That work will also place the faithful-routing and average-width correspondence observed below on a general footing and distinguish the role of the balanced on-shell condition for facet and hidden regions.

the common power carried by μ_{LP}^2 to every κ_e . HRF also supplies the cancellation ideal and the gap $W_{\text{HR}} - W_{\text{SL}}$. In momentum space this gap reappears as restricted support normal to the pinch. Thus HRF determines the virtualities and the amount of local restriction, but not a unique set of loop momenta or their separate lightcone components. For each mode-complete example presented in Sec. 3, we obtain a faithful loop routing by combining the virtuality constraints with componentwise momentum conservation in an exact kinematic chart. The pinching poles and local denominator homogeneity then determine the widths; any residual cross-loop restriction is determined from the joint momentum-space support and independently checked against the LP power count.

Choose null reference directions n_+ and n_- . We record this choice on every potentially ambiguous component triple by the subscript $(+, -, \perp)_{(n_+, n_-)}$, where the entries name the reference directions defining the plus and minus components, respectively. An edge value is then written as

$$q_e \sim (\delta^{a_e}, \delta^{b_e}, \delta^{c_e})_{(n_+, n_-)} \equiv (q_e^+, q_e^-, |q_{e\perp}|)_{(n_+, n_-)}. \quad (4.2)$$

In the absence of a cancellation between the leading longitudinal and transverse terms,

$$\kappa_e = \min(a_e + b_e, 2c_e). \quad (4.3)$$

When

$$a_e + b_e = 2c_e, \quad (4.4)$$

the longitudinal and transverse contributions have homogeneous scaling. We refer to Eq. (4.4) as the *balanced on-shell condition* for a momentum mode. The qualifier “balanced” emphasises this homogeneity and does not imply a small virtuality; the condition applies equally to UV modes. The complementary Glauber hierarchy, $q_e^+ q_e^- \ll q_{e\perp}^2$, has long been recognised in the classic factorisation analyses [8, 9]; see in particular Ref. [10, Sec. 8.4, especially Eq. (137)]. The standard hard, jet and soft scalings used here are reviewed in Ref. [43, Sec. 2.3, especially Eqs. (2.42)–(2.45)], while Ref. [6, Sec. 4, especially Eqs. (4.15)–(4.17)] gives explicit facet-region reconstructions based on the balanced on-shell condition. If the common power in Eq. (4.4) is smaller than κ_e , the longitudinal and transverse terms in $q_e^2 = q_e^+ q_e^- - q_{e\perp}^2$ must cancel at leading order. Such an enhancement can occur for individual edges even in facet regions, and does so in our examples. It cannot be inferred from the local triple (a_e, b_e, c_e) alone; whether and where it occurs is determined by the global solution of the scaling constraints and componentwise momentum conservation.

At each vertex, momentum conservation is imposed component by component. At valuation level this requires the smallest exponent in every signed sum to occur at least twice. We then verify the signs and leading coefficients in an exact kinematic chart. Equations (4.3) and (4.4), together with componentwise momentum conservation, constrain the component powers but do not by themselves provide a general existence or uniqueness theorem. In the examples we use the balanced on-shell condition for independent loop variables whenever it is compatible with the certified HR data; we do not impose it on every loop, since a transverse-dominated Glauber mode necessarily violates it. Leading cancellations between longitudinal and transverse terms can occur in either facet or hidden regions and can be obscured by a lightcone frame not aligned with the relevant collinear direction. For HRs,

moreover, the cancellation equations can correlate leading coefficients between momentum components or between loops. Such information is lost when only the powers of δ are retained.

4.2 Loop bases, routing freedom and correlated support

The edge virtualities inherited from HRF do not by themselves define a momentum region. Reconstructing the corresponding momentum region is a global problem and requires a consistent set of independent loop variables. Choose these variables as chords of a spanning tree, or use any affine rerouting by external momenta, and express every edge momentum in that basis. Two such bases describe the same region when their local fluctuations are related by a nonsingular transformation. In particular, a combination such as $q_i - q_j$ may be the natural loop variable even when both individual edge momenta look soft or collinear.

It is useful to write the distinction between the centre and the local support explicitly. Collecting the independent loop components into ℓ , a local region may be written as

$$\ell = \langle \ell \rangle_{\vec{v}} + T(\delta) \mathbf{u}, \quad d^{LD} \ell = |\det T(\delta)| d^{LD} \mathbf{u}, \quad (4.5)$$

where $\mathbf{u} = O(1)$. The matrix T records both the local component standard deviations and correlations between loop variables; equivalently, the covariance scales as TT^T . A product of separately listed component widths gives the measure only after a mode-adapted change of coordinates has made those correlations explicit, for example by putting T in triangular form. The component powers of $\langle \ell \rangle_{\vec{v}}$ do not determine T , and the marginal widths alone do not determine its determinant. The examples exhibit three distinct possibilities: average and marginal scalings can agree in a suitable loop basis; an individual dependent edge need not represent either member of that basis; and, even when they agree loop by loop, the joint support can acquire an additional suppression through correlations between several loops. The last mechanism occurs in the wide-angle Crown and in the near-planar five-point examples.

There is also a harmless longitudinal-frame freedom. A boost changes

$$a \longrightarrow a + \chi, \quad b \longrightarrow b - \chi, \quad c \longrightarrow c, \quad (4.6)$$

and therefore preserves $a + b$, all virtualities and the distinction between longitudinal and transverse dominance, whether the triple describes an average momentum or an integration width. A result which determines only $a + b$ and c can consequently establish this invariant class without fixing the separate plus and minus powers. We distinguish this diagnosis from a *mode-complete* solution, for which an exact chart and pole analysis determine a faithful component-level momentum routing and its local integration widths.

No constraint beyond Eqs. (4.1)–(4.3), vertex conservation, the already certified cancellation equations and the physical kinematic chart is imposed. The choice of a convenient loop basis and the use of the balanced on-shell condition where it is compatible with the certified HR organise the solutions but do not change their invariant content.

4.3 Pole pinches and local integration widths

The scaling of a momentum combination at the pinch and that of its local integration width are a priori distinct and must be determined independently. In the faithful routings of our examples they agree loop by loop, although the joint support may still contain additional cross-loop correlations. To determine the widths, express every denominator in a mode-adapted loop basis. If a denominator is linear in the selected component,

$$\mathcal{D}_e = A_e \ell^+ + B_e + i0, \quad \ell_e^+ = -\frac{B_e}{A_e} - \frac{i0}{A_e}, \quad (4.7)$$

then active denominators with opposite signs of A_e place the poles ℓ_e^+ on opposite sides of the real ℓ^+ axis. If $A_e \sim \delta^{\alpha_e}$ and $\mathcal{D}_e \sim \delta^{\kappa_e}$, their approach restricts the fluctuation to

$$\Delta \ell^+ \sim \delta^{\kappa_e - \alpha_e}. \quad (4.8)$$

A denominator may depend algebraically on ℓ^+ and nevertheless be subleading in this local fluctuation. Such a denominator does not generate the leading pinch.

The central-soft Fish analysis leading to Eq. (3.143) illustrates this rule explicitly: the (q_0, q_1) and (q_4, q_5) pairs determine the two plus widths, while (q_2, q_3) determines the remaining minus width.

Transverse widths are determined instead by homogeneity of the local denominator expansion. Suppose $q_{e\perp} \sim \delta^{c_e}$, $\mathcal{D}_e \sim \delta^{\kappa_e}$, and a transverse loop fluctuation obeys $\Delta \ell_\perp \sim \delta^\gamma$. If $\Delta q_{e\perp} = C_e \Delta \ell_\perp$, with an order-one routing coefficient C_e , then generically

$$\Delta \mathcal{D}_e|_\perp \sim q_{e\perp} \cdot C_e \Delta \ell_\perp + |C_e \Delta \ell_\perp|^2 \sim \delta^{\min(c_e + \gamma, 2\gamma)}, \quad \min(c_e + \gamma, 2\gamma) \geq \kappa_e. \quad (4.9)$$

The inequality must hold for every active denominator. The transverse width is the widest support, equivalently the smallest γ , satisfying all these conditions, with equality for at least one denominator. If the linear term vanishes, its first nonzero replacement in the Taylor expansion is used. For several transverse loop variables the conditions are solved jointly, including mixed terms and correlated directions. In Eq. (3.142), for example, $c_e = 2$ and $\kappa_e = 4$ give $\gamma = 2$.

For an independent loop fluctuation in a specified lightcone frame,

$$(\Delta \ell^+, \Delta \ell^-, |\Delta \ell_\perp|)_{(n_+, n_-)} \sim (\delta^a, \delta^b, \delta^c)_{(n_+, n_-)}, \quad (4.10)$$

we use the invariant classification of its integration width

$$\begin{aligned} a + b = 2c &: \text{balanced on-shell mode,} \\ a + b < 2c &: \text{longitudinal dominated,} \\ a + b > 2c &: \text{transverse dominated (Glauber).} \end{aligned} \quad (4.11)$$

These labels apply here to the width. When the average momentum has the same component powers, they also classify its mode; otherwise the average mode and the width must be stated separately. The last line concerns an independent integration variable, not an individual propagator: a Glauber loop may or may not be carried by an explicit Glauber propagator. The drawing convention is defined at its first use in Fig. 4. A marker denotes a vertex through which the loop flows, and its arrow follows the local t -channel orientation.

4.4 What the examples establish

The examples of Sec. 3 now form a useful hierarchy. Their momentum-space content is summarised in Table 9.

region	independent loop modes	Glauber structure and correlations
wide-angle Crown [45] and its audited boundary descendants	Crown: three collinear loops; descendants add soft or collinear loops	no Glauber loop; correlated longitudinal path sharing
Regge Crown [45]	two collinear loops; one Glauber loop	two marked Glauber vertices; opposite local orientations when opened in the t channel
two-loop spacelike-collinear Fish family	one soft loop; one Glauber loop	one marked Glauber vertex; all six-, seven- and eight-propagator UT integrals of Ref. [49] inherit the Fish seed modes
vertex-corrected spacelike-collinear Fish	one soft loop; one collinear loop; one Glauber loop	one marked Glauber vertex; the correction adds the collinear loop
five-point near-planar Fish	two balanced on-shell collinear loops	no Glauber loop; the dependent third collinear path restricts their joint support
five-point simultaneous MRK and near-planar Fish	two balanced on-shell, path-adapted collinear loops	no Glauber loop; correlated, MRK-narrowed longitudinal support
five-point central-soft rapidity-ordered Fish	one balanced on-shell (p_1 -collinear) ² loop; one Glauber loop	two marked Glauber vertices; restricted support is encoded in the Glauber width
six-point wide-angle self-crossing twisted box	one Glauber loop	no Glauber propagator; the four active propagators are collinear and the two four-point vertices are hard
six-point twisted hexagon in generic DSC or in zero-recoil NMRK	one Glauber loop	one Glauber propagator x_0 with its two incident vertices marked

Table 9. Independent loop modes and Glauber structure of the certified HRs.

Table 9 records the independent loop modes of the examples and the diagrammatic realisation of each Glauber loop. In every listed HR that exhibits Glauber scaling, the Glauber mode forms an independent integration loop rather than merely a flow through the diagram. The marked vertices identify where this loop flows; their arrows record its local t -channel orientation. The presence of an explicit Glauber propagator is a separate property. The wide-angle twisted box illustrates the alternative: its Glauber loop is realised by the common fluctuation of four collinear propagators around two hard vertices, with no explicit Glauber propagator.

The five-point spacelike-collinear example demonstrates mode inheritance. Ref. [49] computed the HR for every member of a complete basis of two-loop UT integrals, containing six, seven or eight propagators. Each inherits the mode-adapted Fish-seed loop momenta, reproduced in Eq. (3.38),

$$\ell_s = q_5 \sim (\delta, \delta, \delta), \quad \ell_G = q_1 - q_5 \sim (\delta, \delta^2, \delta). \quad (4.12)$$

Thus ℓ_s is soft and ℓ_G is Glauber. Inspecting the propagators individually would miss the latter mode. The Glauber loop has one marked collinear-to-soft vertex, as shown in Fig. 6. The vertex-corrected descendant preserves this soft–Glauber pair and adds one collinear loop, so the two cases are listed separately in Table 9.

In the central-soft rapidity-ordered example, using s_{12} as the hard scale, the realisation has one balanced on-shell loop and one Glauber loop,

$$\Delta r \sim (\delta^4, 1, \delta^2), \quad \Delta \ell_G \sim (\delta^4, \delta, \delta^2), \quad \ell_G = q_0 - q_4. \quad (4.13)$$

The first satisfies the balanced on-shell condition in Eq. (4.4), while the second is transverse dominated. The momentum-transfer fluctuations at the two four-point vertices differ only by a sign, and the corresponding markers in Fig. 13 have opposite local orientations. The longitudinal components of the Glauber loop’s integration width are unequal: $\Delta \ell_G^+ \sim \delta^4$ is narrower than $\Delta \ell_G^- \sim \delta$, realising in momentum space the restricted support found by HRF.

In the displayed mode-adapted frames, Eqs. (4.12) and (4.13) therefore provide two forms of unequal longitudinal Glauber width, respectively (δ, δ^2) and (δ^4, δ) ; the vertex-corrected Fish inherits the former. By contrast, the Regge Crown in Fig. 4, the wide-angle twisted box in Eq. (3.165), and the zero-recoil NMRK and DSC regions in Eqs. (3.191) and (3.206) have equal longitudinal widths in their symmetric frames.

Faithful routing, average-width correspondence and cross-mode correlations.

The mode-complete examples support a two-part conclusion. First, in every such example we find a faithful basis of independent loop momenta, with appropriate local lightcone directions, in which the conditional average-momentum scaling and the marginal integration width agree componentwise. This correspondence is found for balanced on-shell modes and also for the transverse-dominated Glauber modes. It is a statement about an adapted loop basis, not about an arbitrary edge momentum; the deeper saddle value of the dependent edge q_2 in the simultaneous MRK–planar Fish is an explicit illustration of this distinction.

Second, several HR examples show that the marginal modes can be mutually correlated. Momentum conservation among several collinear sectors may restrict their joint longitudinal support without changing any one marginal width. If the width of loop ℓ is $(\delta^{a_\ell}, \delta^{b_\ell}, \delta^{c_\ell})$, the integration-volume power separates as

$$\Omega_{\text{meas}}(D) = \sum_{\ell=1}^L [a_\ell + b_\ell + (D-2)c_\ell] + \gamma_{\text{corr}}, \quad \mathcal{V}_{\parallel}^{\text{corr}} \sim \delta^{\gamma_{\text{corr}}}. \quad (4.14)$$

Thus the marginal widths determine the D -dependent part of the measure, whereas the cross-mode correlation changes only its integer power. The wide-angle Crown and near-planar Fish exhibit this second effect; for the near-planar Fish $\gamma_{\text{corr}} = 3$ in powers of λ ,

while the simultaneous MRK–near-planar family has $\gamma_{\text{corr}} = 3a + b$. By contrast, in the local soft–Glauber examples the restriction is already absorbed into the narrower Glauber width and no additional joint-support factor is required.

For the NMRK and DSC one-loop six-point examples, we reconstructed the component widths explicitly from the pinching poles. The NMRK reconstruction is given in Eqs. (3.188)–(3.191), and the DSC reconstruction in Eqs. (3.205)–(3.206). In the symmetric hard-normalised frame their widths give

$$\begin{aligned} \text{zero-recoil NMRK : } (a_{\text{NMRK}}, b_{\text{NMRK}}, c_{\text{NMRK}}) &= (5, 5, 3), \\ a_{\text{NMRK}} + b_{\text{NMRK}} - 2c_{\text{NMRK}} &= 4 > 0, \\ \text{DSC : } (a_{\text{DSC}}, b_{\text{DSC}}, c_{\text{DSC}}) &= (2, 2, 1), \\ a_{\text{DSC}} + b_{\text{DSC}} - 2c_{\text{DSC}} &= 2 > 0. \end{aligned} \tag{4.15}$$

For a single loop with no additional support correlation, its measure exponent may be written as

$$a + b + (D - 2)c = Dc + (a + b - 2c).$$

Thus $a + b - 2c$ is the extra suppression relative to a balanced on-shell width with the same transverse exponent. In NMRK and DSC it equals the corresponding cancellation depth $W_{\text{HR}} - W_{\text{SL}}$, namely 4 and 2 in Table 7. The wide-angle member obeys the same relation: Eq. (3.165) gives $(a, b, c) = (1, 1, \frac{1}{2})$ in powers of $\sigma_{\times}$, so both the transverse-dominance excess and the cancellation depth are 1. The momentum-space width therefore already incorporates the restriction measured by the cancellation depth; the latter must not be included as a separate factor. A longitudinal boost acts as in Eq. (4.6), shifting the first two exponents oppositely but preserving their sum. Each region contains one Glauber loop and one explicit Glauber propagator, x_0 , whose incident p_3 and p_6 vertices carry the two oriented markers in Fig. 15. The incoming three-point vertices carrying p_1 and p_2 , selected by the cancellation factors, are instead the factor-localisation vertices. The other resolved transfer, q_3 , is soft in NMRK and hard in DSC.

4.5 Power counting as a consistency certificate

Once a loop basis and all local widths are known, momentum-space power counting follows directly from Eq. (2.2):

$$I_{\text{mom}} \sim \frac{\prod_{r=1}^L d^D \ell_r}{\prod_e \mathcal{D}_e^{\nu_e}}. \tag{4.16}$$

The measure uses the fluctuation widths of the independent loop variables, not the values at the pinch of a convenient set of edge momenta. The propagator virtualities are inherited from the parameter-space vector, while momentum conservation and the pole analysis determine the component widths and their correlations. The resulting momentum-space count must agree with the LP count, including the reduced support associated with the cancellation depth. Detailed calculations are given alongside the corresponding examples in Sec. 3.

This comparison does more than label a mode. It checks that the loop basis is complete, that no cancellation-normal width has been counted twice, and that a putative Glauber value is genuinely an integration direction. It is therefore a strong optional certificate for a newly discovered HR, while the existence and scaling of the HR itself remain parameter-space conclusions of the core algorithm.

5 Topological organisation of Landshoff hidden regions

The examples above reveal a common origin: every HR studied here can be associated with a wide-angle multiple-hard-scattering, or Landshoff, seed. In the graph-theoretic direction, the seed is exposed by a label-preserving contraction of a larger topology; in kinematic space, a specialised region descends from the relevant wide-angle limit through further kinematic specialisation. This motivates a conjectural organising principle: the presence of a topology supporting such a Landshoff ancestor is necessary for an HR. It is not sufficient. Whether the region is realised in a particular diagram and limit depends on additional data, including the external-leg assignment, positivity in the physical domain and the kinematic path by which the singular surface is approached. Together these determine whether the pinch is reached and which region vector results.

The six-point twisted box illustrates this dependence. Its positive wide-angle pinch occurs on the physical self-crossing surface rather than at generic wide-angle kinematics, and its NMRK and DSC regions are specialisations of that locus. More generally, the same topology may realise different HRs on collinear or high-energy faces of kinematic space. Topology therefore organises candidate mechanisms; it does not by itself determine the physical sheet, the required kinematic surface or the region vector.

The minimal member of the complete-bipartite family relevant here is the one-loop $K_{2,2}$ twisted-box seed in Fig. 14(b). Figure 17 displays four higher complete-bipartite incidence cores. We denote the vertices in the two parts by

$$\{A_1, \dots, A_m\} \quad \text{and} \quad \{B_1, \dots, B_n\}, \quad (5.1)$$

and assign the LP parameter x_{ia} to the edge joining A_i to B_a . Thus the pair of vertex labels, rather than a separate edge number, names every edge, and we use this x_{ia} notation throughout the present section. The external half-edges shown in the figure do not carry an x_{ia} label.

Although the two parts of a complete-bipartite graph are combinatorially interchangeable, its physical Landshoff realisation selects a preferred orientation. In all certified configurations considered here, the A_i are external collinear or multi-collinear factor-localisation vertices: the row variables x_{ia} encode how the attached momentum is shared among the incident on-shell lines. The cancellation ideal constrains these sharing fractions and their compatibility between different A_i ; an individual normal may therefore compare several A_i rather than belong to a single vertex. The B_a , by contrast, represent the hard-scattering components and may be internal or carry additional external legs. In specialised limits they may carry oriented Glauber flow, or resolve into vertices joined by an explicit Glauber

propagator. This physical interpretation explains why all A_i carry external legs in the configurations studied here, whereas external attachments to the B_a are optional.

The external half-edges in Fig. 17 display several realisations of this division. Every vertex of $K_{3,2}$ and $K_{3,3}$ carries an external leg, so these are respectively five- and six-point problems. By contrast, in the displayed $K_{4,2}$ and $K_{4,3}$ configurations only the four A_i vertices carry external legs, while the B_a vertices are internal; both are therefore four-point problems. The $K_{4,2}$ core also admits the five- and six-point configurations analysed in Sec. 5.2. Increasing the second partition in the right-hand column raises the loop order without enlarging the external kinematic space. In the left-hand column it also introduces additional external directions and the associated momentum-conservation and Gram constraints. The complete-bipartite language thus organises the internal incidence structure, but it does not make the four-, five- and six-point kinematics equivalent.

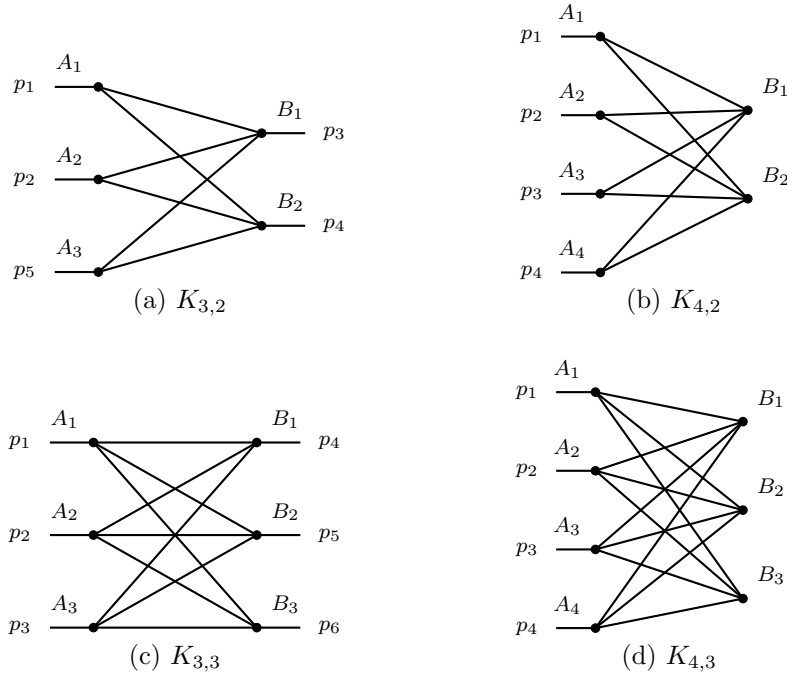


Figure 17. Four higher complete-bipartite incidence cores used in this section; the minimal $K_{2,2}$ member is the twisted-box seed in Fig. 14(b). The edge joining the explicitly named vertices A_i and B_a carries the LP parameter x_{ia} . The external half-edges make the kinematic distinction explicit: panels (a) and (c) are five- and six-point graphs, whereas panels (b) and (d) are both four-point graphs. Panel (a) uses the near-planar Fish permutation of external legs in Fig. 9; panel (b) is the Crown of Fig. 1. Panel (d) has the incidence topology of the baryon–baryon independent-scattering diagram in Fig. 14 of Ref. [48].

5.1 Complete-bipartite incidence cores

The simplest established wide-angle seeds have a common incidence description. After the external half-edges are suppressed, the six-point twisted box is $K_{2,2}$, the five-point Fish is $K_{3,2}$, and the Crown is $K_{4,2}$. Here “wide angle” includes special physical kinematic surfaces:

the twisted-box pinch requires Eq. (3.152), and the Fish requires the near-planar condition $\Gamma_5 = 0$, whereas the Crown pinch exists in an open wide-angle domain. With the vertex convention introduced above, the three established seeds all have $n = 2$, so their path ratios provide the most direct introduction. The two edges from A_i to B_1 and B_2 carry parameters x_{i1} and x_{i2} . Define

$$r_i = \frac{x_{i1}}{x_{i2}}, \quad R_i = \frac{r_i}{r_m} = \frac{x_{i1}x_{m2}}{x_{i2}x_{m1}}, \quad i = 1, \dots, m-1. \quad (5.2)$$

At a physical Landau pinch, the normalised pair $z_i = x_{i2}/(x_{i1} + x_{i2})$ and $1 - z_i$ encodes the momentum fractions carried by the two collinear partons emerging from A_i towards B_1 and B_2 .

We call the subset of LP-parameter space on which these path splittings are mutually compatible the *incidence locus*. It is the locus associated with the bipartite incidence pattern before the complete stationary and vanishing Landau equations are imposed. In its undeformed form, mutual compatibility means

$$R_i = 1, \quad i < m \quad \Longleftrightarrow \quad (r_1, \dots, r_m) = c(1, \dots, 1). \quad (5.3)$$

The $m - 1$ quantities $R_i - 1$ are normal coordinates associated with a basis of independent graph cycles. The collective ratio c , by contrast, is tangent to the incidence locus: varying it changes the common momentum sharing while preserving every relative compatibility condition.

The Crown is the simplest case. Its four path ratios in Eq. (3.3) obey three independent relative conditions. The four binomial cancellation factors in Eq. (3.2) form a redundant cyclic set of equations for the same codimension-three locus. Their common value $c = \rho$ remains free, so the incidence locus is already the LP-parameter part of the physical pinch.

The other two seeds separate the incidence count from the codimension of the full physical pinch. On the near-planar surface $\Gamma_5 = 0$, the Fish factors in Eq. (3.73) impose $r_I = \rho_I$, $I = A, B, C$. Two independent relative equations,

$$\frac{r_A}{r_C} = \frac{\rho_A}{\rho_C}, \quad \frac{r_B}{r_C} = \frac{\rho_B}{\rho_C}, \quad (5.4)$$

define its kinematically deformed incidence locus $\mathbf{r} = c\boldsymbol{\rho}$. The remaining stationary equation fixes the tangent coordinate to $c = 1$. Thus the Fish has two incidence normals but three independent LP-parameter normals f_A, f_B, f_C at the physical pinch. The separate kinematic condition $\Gamma_5 = 0$ is required because the graph polynomial evaluated at the stationary point is proportional to Γ_5 , as shown in Eq. (3.60).

For the twisted box, the single relative condition between its two path ratios defines a codimension-one incidence locus. The two hard-vertex momentum-balance factors $f_{1,\text{WA}}$ and $f_{2,\text{WA}}$ in Eq. (3.156) also fix the collective ratio, giving a codimension-two pinch within the active $K_{2,2}$ parameter space on the self-crossing surface (3.152). The three counts within the corresponding active incidence cores may therefore be summarised as

	codim C_{inc}	codim _{core} C_{pinch}
$K_{4,2}$ (Crown)	3	3
$K_{3,2}$ (Fish)	2	3
$K_{2,2}$ (twisted box)	1	2

(5.5)

We now generalise this construction to $K_{m,n}$. Its edge parameters form the $m \times n$ matrix

$$X = (x_{ia}), \quad i = 1, \dots, m, \quad a = 1, \dots, n. \quad (5.6)$$

The undeformed incidence locus is parametrised by

$$C_{\text{inc}} : \quad x_{ia} = u_i w_a. \quad (5.7)$$

The variables u_i and w_a , modulo the redundancy $u_i \mapsto cu_i$, $w_a \mapsto c^{-1}w_a$, parametrise motion tangent to C_{inc} . A set of local normal coordinates is provided by

$$R_{ia} = \frac{x_{ia}x_{mn}}{x_{in}x_{ma}}, \quad i < m, \quad a < n. \quad (5.8)$$

The index ranges give $(m-1)(n-1)$ such ratios. They are independent on the positive chart, so $R_{ia} - 1$ provide the same number of local normal coordinates to C_{inc} . Each R_{ia} is associated with the four-cycle $A_i - B_a - A_m - B_n - A_i$, is invariant under row and column rescalings $x_{ia} \mapsto \alpha_i \beta_a x_{ia}$. The incidence locus is the common level set $R_{ia} = 1$; equivalently, on the positive chart,

$$R_{ia} = 1 \iff x_{ia}x_{mn} - x_{in}x_{ma} = 0. \quad (5.9)$$

More invariantly, let

$$I_2(X) \equiv \langle x_{ia}x_{jb} - x_{ib}x_{ja} \mid i < j, \ a < b \rangle \quad (5.10)$$

denote the ideal generated by all 2×2 minors of X . Equation (5.7) guarantees that every such minor vanishes. Conversely, because $x_{mn} > 0$, the vanishing of the reference minors in Eq. (5.9) gives $x_{ia} = x_{in}x_{ma}/x_{mn}$. Defining $u_i = x_{in}/x_{mn}$ and $w_a = x_{ma}$ then reconstructs Eq. (5.7). Thus the factorised parametrisation and the simultaneous cancellation equations describe the same incidence locus in the positive domain.

Although $I_2(X)$ generally has more generators, its zero locus in the positive domain is C_{inc} . The parametrisation in Eq. (5.7) uses $m+n$ variables with one rescaling redundancy, so $\dim C_{\text{inc}} = m+n-1$. Its codimension in the mn -dimensional space of matrices X is therefore $(m-1)(n-1)$, in agreement with the count following Eq. (5.8). Since $K_{m,n}$ is connected and has $E = mn$ edges and $V = m+n$ vertices, the usual graph relation $L = E - V + 1$ gives

$$L(K_{m,n}) = E - V + 1 = mn - (m+n) + 1 = (m-1)(n-1) = \text{codim } C_{\text{inc}}. \quad (5.11)$$

The equality says that the number of normal coordinates of the incidence locus agrees with the loop order. It does not count all independent cancellation factors of the physical pinch, as the Fish and twisted-box examples demonstrate. The complete physical pinch must additionally satisfy the stationary and vanishing equations and positivity in the physical domain. In particular, kinematics can deform the incidence normals to $R_{ia} = \rho_{ia}(\mathbf{s})$, with polynomial representatives

$$f_{ia} = x_{ia}x_{mn} - \rho_{ia}x_{in}x_{ma}. \quad (5.12)$$

5.2 External-leg configurations of $K_{4,2}$

The external-leg configuration is part of the definition of a Landshoff seed. The same $K_{4,2}$ incidence core gives three sharply different Landau problems, shown in Fig. 18. The familiar Crown has external legs only at the four A_i vertices. Attaching a leg also at one or both B_a vertices instead gives five- and six-point kinematics, without changing the loop order.

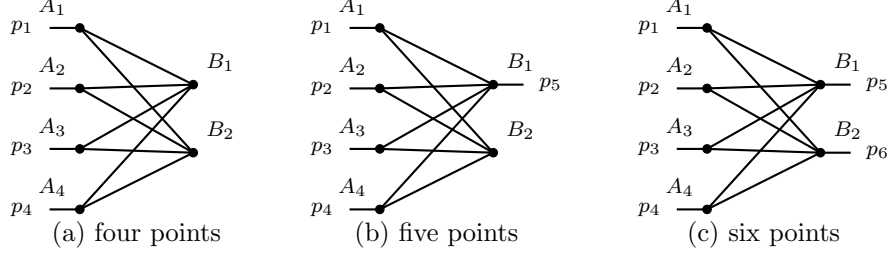


Figure 18. Three external-leg configurations of the same three-loop $K_{4,2}$ incidence core. The four-point configuration (a) is the Crown. In (b), only B_1 carries an additional external momentum; in (c), both B_1 and B_2 do. The change in external kinematics changes the existence and location of the positive Landau pinch even though the internal graph is unchanged.

For all three configurations, introduce the path coordinates

$$h_i = x_{i1} + x_{i2}, \quad z_i = \frac{x_{i2}}{h_i}, \quad 0 < z_i < 1, \quad i = 1, \dots, 4. \quad (5.13)$$

These coordinates are equivalent to the Crown path-ratio representation in Eq. (3.3), since $r_i = x_{i1}/x_{i2} = (1 - z_i)/z_i$. For the six-point configuration we retain the paper-wide convention that p_1, p_2 are incoming and $p_3, \dots, p_6$ are outgoing. To keep the following algebra symmetric, define the corresponding all-outgoing representatives $\hat{p}_r = \eta_r p_r$, where $\eta_1 = \eta_2 = -1$ and $\eta_3 = \dots = \eta_6 = 1$, so that $\sum_{r=1}^6 \hat{p}_r = 0$. For massless internal propagators, the exact Symanzik polynomials are

$$\mathcal{U} = e_3(\mathbf{h}) = \sum_{i < j < k} h_i h_j h_k, \quad \mathcal{F} = \left(\prod_{i=1}^4 h_i \right) \left[Q(\mathbf{z})^2 + \sum_{i=1}^4 z_i (1 - z_i) \hat{p}_i^2 \right], \quad Q(\mathbf{z}) = \hat{p}_5 + \sum_{i=1}^4 z_i \hat{p}_i. \quad (5.14)$$

Here $Q(\mathbf{z})$ measures the residual four-momentum imbalance at B_1 for the prospective pinch assignment in which the propagators joining A_i to B_1 and B_2 carry the fractions $z_i \hat{p}_i$ and $(1 - z_i) \hat{p}_i$, respectively. Overall external momentum conservation implies that the corresponding imbalance at B_2 is $\hat{p}_6 + \sum_{i=1}^4 (1 - z_i) \hat{p}_i = -Q(\mathbf{z})$. For general $\mathbf{z}$, neither imbalance need vanish; the ratio Landau equations below select the values for which they do. In particular, at massless on-shell kinematics, $\mathcal{F}_0 = (\prod_i h_i) Q(\mathbf{z})^2$. Let

$$G_{ij} = \hat{p}_i \cdot \hat{p}_j, \quad b_i = \hat{p}_i \cdot \hat{p}_5. \quad (5.15)$$

The ratio Landau equations are $\hat{p}_i \cdot Q(\mathbf{z}) = 0$, $i = 1, \dots, 4$. When $\det G \neq 0$, the four momenta $\hat{p}_1, \dots, \hat{p}_4$ form a basis of four-dimensional momentum space. Writing $\hat{p}_5 = \sum_j c_j \hat{p}_j$ and contracting with this basis gives $G\mathbf{c} = \mathbf{b}$. Substitution into Eq. (5.14) gives $Q(\mathbf{z}) = \sum_j (c_j + z_j) \hat{p}_j$, so the stationary equations are $G(\mathbf{c} + \mathbf{z}) = 0$. Inverting G therefore gives the unique solution

$$\mathbf{z}_\star = -G^{-1}\mathbf{b}, \quad Q(\mathbf{z}_\star) = 0, \quad (5.16)$$

At this point the residual imbalances vanish at both B_1 and B_2 , so momentum is conserved separately at the two vertices. The quantities $z_{\star i}$ and $1 - z_{\star i}$ are therefore the momentum fractions carried by the two propagators at the pinch. Equation (5.16) is also simply the expansion of $-\hat{p}_5$ in that basis. The corresponding invariant compatibility condition is the five-vector Gram relation

$$0 = \det \begin{pmatrix} G & \mathbf{b} \\ \mathbf{b}^T & \hat{p}_5^2 \end{pmatrix} = \det G (\hat{p}_5^2 - \mathbf{b}^T G^{-1} \mathbf{b}).$$

Thus, in addition to $G\mathbf{c} = \mathbf{b}$, it states $\hat{p}_5^2 = \mathbf{b}^T G^{-1} \mathbf{b} = \mathbf{c}^T G \mathbf{c}$. This Gram relation is automatic for physical six-point kinematics in four dimensions (with $\hat{p}_6$ fixed by momentum conservation); it is not an additional special kinematic hypersurface analogous to $\Gamma_5 = 0$ in the Fish example. The condition $\det G \neq 0$ merely selects the open chart in which the first four momenta provide a basis. The first-sheet condition is the transparent set of inequalities

$$0 < -(G^{-1}\mathbf{b})_i < 1, \quad i = 1, \dots, 4. \quad (5.17)$$

These inequalities define a nonempty open domain of physical $2 \rightarrow 4$ kinematics.

The Hessian determines the local normal form rather than the existence of the stationary solution. Setting $\mathbf{z} = \mathbf{z}_\star + \Delta\mathbf{z}$, Eq. (5.16) gives

$$\begin{aligned} Q(\mathbf{z}_\star + \Delta\mathbf{z}) &= \sum_{i=1}^4 \Delta z_i \hat{p}_i, \\ \mathcal{F}_0(\mathbf{z}_\star + \Delta\mathbf{z}) &= \left(\prod_{k=1}^4 h_k \right) \sum_{i,j=1}^4 G_{ij} \Delta z_i \Delta z_j, \\ \left. \frac{\partial^2 \mathcal{F}_0}{\partial z_i \partial z_j} \right|_{\mathbf{z}_\star} &= 2 \left(\prod_{k=1}^4 h_k \right) G_{ij}. \end{aligned} \quad (5.18)$$

Since G is non-singular, all four ratio deviations are genuine normal coordinates. To translate this statement into the cancellation-factor language of HRF, define the local factors

$$f_i^{(6)} \equiv h_i \Delta z_i = x_{i2} + \frac{(\text{adj}(G)\mathbf{b})_i}{\det G} h_i = x_{i2} - z_{\star i} (x_{i1} + x_{i2}).$$

The denominator is harmless on the chart $\det G \neq 0$; it may be cleared to obtain an equivalent polynomial representative without changing the local cancellation ideal. Since, in the on-shell limit considered, $G_{ii} = \hat{p}_i^2 = 0$, the polynomial can be written exactly as

$$\mathcal{F}_0 = 2 \sum_{1 \leq i < j \leq 4} G_{ij} \left(\prod_{k \neq i,j} h_k \right) f_i^{(6)} f_j^{(6)}.$$

The natural graph-dependent HRF presentation therefore contains the six generators $g_{ij}^{(6)} = f_i^{(6)} f_j^{(6)}$, with $i < j$. These are six generator polynomials, not six independent normal constraints: their common pinch component is defined by the four independent factors $f_i^{(6)}$. For the Crown, by contrast, the four factors in Eq. (3.2) describe only three independent ratio normals. A common shift of all four path ratios is tangent to its pinch locus, so the fourth factor vanishes once three independent ones do, and only the two disjoint-support products shown in Eq. (3.2) occur. Here the stationary point is isolated in the four ratio variables, so no ratio-tangent direction remains; the four path scales h_i are the tangent coordinates.

The scaling is fixed before introducing a local chart. The local LP analysis uses the same external-virtuality expansion as for the Crown. At finite regulator, take the four legs attached to A_i slightly off shell, $\hat{p}_i^2 = \delta\mu_i$, with fixed nonzero μ_i ; the massless on-shell point is recovered as $\delta \rightarrow 0$. The legs attached to B_1 and B_2 are kept on shell in the explicit certificate below. An order- δ continuation of their virtualities would add further terms at resolved-leading weight without changing the scaling. The Part-II homogeneity equations force a common edge weight v . Since $\mathcal{F}_0$ and $\mathcal{F}_1$ have degree four whereas $\mathcal{U}$ has degree three, equality at resolved-leading weight gives $1 + 4v = 3v$, and hence $v = -1$. Together with the Part-I decomposition above, Part II therefore determines the HR vector and both weights simultaneously:

$$\begin{aligned} \mathcal{F}_\star &= \mathcal{F}_0 = \mathcal{F}_{\text{SL}}, & \mathcal{F}_{\text{Obs}} &= 0, \\ \vec{v}_{\text{HR}} &= (-1, -1, -1, -1, -1, -1, -1, -1; 1), & (W_{\text{SL}}, W_{\text{HR}}) &= (-4, -3). \end{aligned} \quad (5.19)$$

In particular, every edge parameter scales as $x_{ia} \sim \delta^{-1}$.

The normal weights now follow directly from the second line of Eq. (5.18). Write $\Delta z_i \sim \delta^{\alpha_i}$. The core scaling just found gives $h_k \sim \delta^{-1}$, so the prefactor $\prod_k h_k$ has weight -4 , and its term proportional to $G_{ij} \Delta z_i \Delta z_j$ has weight $-4 + \alpha_i + \alpha_j$. Requiring this term at the resolved-leading weight $W_{\text{HR}} = -3$ gives $\alpha_i + \alpha_j = 1$. In the generic domain all six off-diagonal terms occur; equivalently, all six generator products $g_{ij}^{(6)}$ must be resolved. The resulting equations for every $i < j$ have the unique solution $\alpha_i = 1/2$ for all four factors. Writing the path sums of Eq. (5.13) as $h_i = \delta^{-1} H_i$, with H_i of order one, therefore gives the chart

$$\begin{aligned} x_{i1} &= \delta^{-1} H_i (1 - z_{\star i} - \delta^{1/2} y_i), \\ x_{i2} &= \delta^{-1} H_i (z_{\star i} + \delta^{1/2} y_i) \end{aligned} \quad (5.20)$$

with $H_i, y_i = \mathcal{O}(1)$. The same powers can be reconstructed from the collinear momentum scaling, but they have been derived here from the parameter-space HRF conditions. Substituting the chart into the exact polynomials in Eq. (5.14) gives the resolved-leading layer

$$\mathcal{P}_{\text{loc}} = \delta^{-3} \left[e_3(\mathbf{H}) + \left(\prod_{k=1}^4 H_k \right) \left(\sum_{i,j=1}^4 G_{ij} y_i y_j + \sum_{i=1}^4 \mu_i z_{\star i} (1 - z_{\star i}) \right) + \mathcal{O}(\delta^{1/2}) \right]. \quad (5.21)$$

Here the quadratic term in the y_i comes from $\mathcal{F}_0$. The final sum in parentheses is the contribution of the explicit p_i^2 term in Eq. (5.14), with $\hat{p}_i^2 = \delta\mu_i$, evaluated at the pinch. If this final, y_i -independent sum were absent, the remaining two terms would be homogeneous under $H_i \mapsto \lambda H_i$, $y_i \mapsto \lambda^{-1/2} y_i$: both $e_3(\mathbf{H})$ and $(\prod_i H_i) \sum_{i,j} G_{ij} y_i y_j$ scale as λ^3 . The off-shell contribution instead scales as λ^4 , and therefore breaks this residual homogeneity and makes the local region scaleful. The leading exponent support in Eq. (5.21) has full affine rank in the eight local coordinates, thereby certifying the Part-II solution in Eq. (5.19). The four cancellation-normal widths supply the factor $\delta^{4/2}$ in the measure. For unit propagator powers and $D = 4 - 2\epsilon$, the scalar power count is

$$I_{K_{4,2}}^{\text{scalar}} \sim \delta^{-8} \delta^{4/2} (\delta^{-3})^{-D/2} = \delta^{-3\epsilon}. \quad (5.22)$$

The six-point $K_{4,2}$ configuration is therefore a certified three-loop six-point wide-angle Landshoff seed.

The five-point configuration behaves differently. Removing the p_6 leg sets $\hat{p}_6 = 0$, and momentum conservation gives $\hat{p}_5 = -\sum_i \hat{p}_i$. Equation (5.16) then reduces to

$$\sum_{i=1}^4 (z_i - 1) \hat{p}_i = 0. \quad (5.23)$$

At generic five-point kinematics the four $\hat{p}_i$ are independent, so the only solution is $z_i = 1$. Equivalently, all x_{i1} vanish. The pinch has reached an LP-parameter boundary and the corresponding contraction is a massless vacuum banana, hence scaleless. There is therefore no generic wide-angle five-point HR for this configuration. Removing instead the leg attached to the other B vertex gives the equivalent boundary $z_i = 0$.

On the coplanar surface, however, $\det G = 0$. If $\sum_i c_i \hat{p}_i = 0$, Eq. (5.23) admits

$$z_i = 1 + t c_i. \quad (5.24)$$

Physical coplanar $2 \rightarrow 3$ configurations exist for which an interval of t places every z_i strictly between zero and one. The rank-three normal quadratic form then gives a near-coplanar five-point HR. Thus the same topology has no interior HR at generic five-point kinematics, but does have one after expanding about the coplanar Gram surface.

This mechanism has a close parallel in the wide-angle Fish of Sec. 3.2.4. Both problems require the Gram-degenerate surface $\Gamma_5 = 0$, but for different algebraic reasons. For the Fish, the ratio-stationary point exists at generic kinematics, while the polynomial evaluated there is proportional to Γ_5 , as in Eq. (3.60); planarity removes this nonzero obstruction. Here generic stationarity instead confines the pinch to the $z_i = 1$ boundary, while the rank loss at $\Gamma_5 = 0$ supplies the null direction that carries it into the positive interior. In both cases the physical five-point pinch has three independent ratio-normal directions, although in the present $K_{4,2}$ problem the fourth ratio direction is tangent to the coplanar family in Eq. (5.24).

This comparison suggests that massless five-point wide-angle HRs may require Gram-degenerate kinematics, but the examples do not establish such a general theorem. All certified five-point Fish regions in Sec. 3.2 approach its positive planar singular family or

a boundary stratum of its closure. Near-planarity need not, however, be imposed as a separate expansion: the spacelike-collinear and central-soft limits reach that locus or its boundary through their collinear or soft degeneration.

The near-planar five-point $K_{4,2}$ region may likewise serve as the ancestor of regions in further composite kinematic limits. Establishing such descendants would require analysing each specialised expansion and its positive cancellation locus separately; we do not pursue that question here.

Taken together, these configurations show that the external-leg attachment is part of the Landau classification: the same $K_{4,2}$ incidence core leads to distinct stationary loci and positivity properties. Each attachment must therefore be tested separately against the positivity, scaling, facet and scalefulness conditions of Sec. 2.

5.3 Further complete-bipartite cases

The $K_{4,3}$ graph in Fig. 17(d) is the natural next complete-bipartite continuation of the Crown. This incidence topology already appears in Fig. 14 of Botts and Sterman [48] as the independent-scattering diagram for baryon–baryon elastic scattering: its four external baryonic vertices are connected to three hard quark–quark scattering subgraphs. The corresponding scalar topology does support a generic wide-angle Landshoff HR. To see this directly, attach the four external legs to the four-vertex partition and take the on-shell expansion $p_i^2 = \delta P_i^2$, with fixed $P_i^2 > 0$. The leading two-forest polynomial can be described without displaying its 72 monomials. First make the relation between the twelve edge variables and the ratio coordinates explicit. Retaining a symmetric row–column normalisation, an exact chart of the positive orthant is

$$X = (x_{ia}) = \begin{pmatrix} u_1 w_1 R_{11} & u_1 w_2 R_{12} & u_1 w_3 \\ u_2 w_1 R_{21} & u_2 w_2 R_{22} & u_2 w_3 \\ u_3 w_1 R_{31} & u_3 w_2 R_{32} & u_3 w_3 \\ u_4 w_1 & u_4 w_2 & u_4 w_3 \end{pmatrix}. \quad (5.25)$$

Here the u_i and w_a are the tangent variables introduced in Eq. (5.7); setting all $R_{ia} = 1$ reduces Eq. (5.25) to that rank-one parametrisation. The chart has the single redundancy $u_i \mapsto \alpha u_i$, $w_a \mapsto w_a/\alpha$. If it is fixed by $w_3 = 1$, then $u_i = x_{i3}$, $w_a = x_{4a}/x_{43}$ for $a = 1, 2$, and the six R_{ia} are precisely the ratios in Eq. (5.8). Thus Eq. (5.25) is an exact change of coordinates on the positive orthant: the six tangent coordinates $(u_1, \dots, u_4, w_1, w_2)$ and the six normal ratios R_{ia} account for the twelve original variables x_{ia} . On C_{inc} , the six conditions $R_{ia} = 1$ leave the tangent coordinates free, and hence

$$\text{codim } C_{\text{inc}} = 6 = (4 - 1)(3 - 1) = 12 - 7 + 1 = L,$$

in agreement with the general count following Eq. (5.8). The last equality is the connected-graph loop formula $L = E - V + 1$, with twelve edges and seven vertices.

For a spanning two-forest $\mathcal{T}_{ij|kl}$ which separates external vertices i, j from k, l , define

$$\Phi_{ij|kl} = \sum_{\mathcal{T}_{ij|kl}} \prod_{(r,a) \notin \mathcal{T}_{ij|kl}} x_{ra}. \quad (5.26)$$

Each of the three channel polynomials contains 24 monomials. Using $s_{13} = -(s_{12} + s_{23})$, the superleading polynomial is

$$\begin{aligned}\mathcal{F}_{\text{SL}} = \mathcal{F}_0 &= s_{12}\Phi_{12|34} + s_{23}\Phi_{14|23} + s_{13}\Phi_{13|24} \\ &= s_{12}(\Phi_{12|34} - \Phi_{13|24}) + s_{23}(\Phi_{14|23} - \Phi_{13|24}), \quad \mathcal{F}_{\text{Obs}} = 0.\end{aligned}\quad (5.27)$$

This is the compact expression for $\mathcal{F}_{\text{SL}}$ used below. On the rank-one incidence locus C parametrised in Eq. (5.7), substituting $x_{ia} = u_i w_a$ into the 24 monomials of each channel polynomial gives the same positive polynomial,

$$\Phi_{12|34}|_C = \Phi_{14|23}|_C = \Phi_{13|24}|_C = 2 \left(\prod_{i=1}^4 u_i \right) e_3(\mathbf{u}) (w_1 w_2 w_3)^2 (w_1 + w_2 + w_3), \quad (5.28)$$

where $e_3(\mathbf{u}) = \sum_{i < j < k} u_i u_j u_k$. The common result is symmetric in the four u_i and independent of the 2|2 channel partition. Through Eq. (5.27), Eq. (5.28) gives $\mathcal{F}_0|_C = 0$. Direct differentiation of the full channel polynomials, followed by restriction to C , shows that their first normal derivatives agree there as well. Equivalently, after localising to the positive chart,

$$\Phi_{12|34} - \Phi_{13|24}, \quad \Phi_{14|23} - \Phi_{13|24} \in I_2(X)^2. \quad (5.29)$$

Membership in $I_2(X)^2$ means that both channel differences and all their first derivatives vanish on C . Equation (5.27) therefore gives $\mathcal{F}_0|_C = d\mathcal{F}_0|_C = 0$, so C is the positive Landau stationary locus

$$x_{ia} = u_i w_a > 0, \quad R_{ia} = \frac{x_{ia} x_{43}}{x_{i3} x_{4a}} = 1, \quad i = 1, 2, 3, \quad a = 1, 2. \quad (5.30)$$

The six displayed ratios are independent normal coordinates. Their quadratic layer is nondegenerate throughout the physical domain. With the tangent coordinates u_i, w_a held fixed, define the normal Hessian by

$$(H_R)_{ia,jb} := \left. \frac{\partial^2 \mathcal{F}_0}{\partial R_{ia} \partial R_{jb}} \right|_{R_{kc}=1}, \quad i, j = 1, 2, 3, \quad a, b = 1, 2.$$

Its determinant is

$$\begin{aligned}\det H_R &= 4s_{12}^2 s_{23}^2 (s_{12} + s_{23})^2 (w_1 w_2 w_3)^{12} (w_1 w_2 + w_1 w_3 + w_2 w_3)^3 \prod_{i=1}^4 u_i^6 \\ &\quad \times (u_1 u_2 u_3 + u_1 u_2 u_4 + u_1 u_3 u_4 + u_2 u_3 u_4)^6,\end{aligned}\quad (5.31)$$

which cannot vanish for $u_i, w_a > 0$ and $s_{12} > -s_{23} > 0$. The six normal stationarity equations are consequently independent, and C is locally the full positive Landau component rather than a sublocus of a larger stationary component.

The Part-II homogeneity equations give a uniform edge weight v . Since $\mathcal{U}$ has weight $6v$, whereas $\delta\mathcal{F}_1$ has weight $1 + 7v$, their balance fixes $v = -1$ and the resolved-leading weight to -6 . Individual monomials of $\mathcal{F}_0 = \mathcal{F}_{\text{SL}}$ then have weight -7 . Its first nonzero normal

layer is quadratic, so a normal width $R_{ia} - 1 \sim \delta^\alpha$ enters at weight $-7 + 2\alpha$. Equating this to -6 fixes $\alpha = 1/2$. The resulting local scaling is

$$x_{ia} = \delta^{-1} u_i w_a R_{ia}, \quad R_{ia} = 1 + \delta^{1/2} y_{ia}, \quad \vec{v}_{\text{HR}} = (\underbrace{-1, \dots, -1}_{12 \text{ edge components}}; 1), \quad (5.32)$$

where ratios not listed in Eq. (5.30) are fixed to one. The three contributions at weight -6 are therefore $\mathcal{U}$, $\delta\mathcal{F}_1$, and the quadratic normal layer of $\mathcal{F}_0$:

$$\mathcal{P} = \delta^{-6} \left[\mathcal{U}|_C + \mathcal{F}_1|_C + \frac{1}{2} \mathbf{y}^T H_R \mathbf{y} + \mathcal{O}(\delta^{1/2}) \right], \quad (W_{\text{SL}}, W_{\text{HR}}) = (-7, -6). \quad (5.33)$$

Here C denotes the rank-one locus. The leading polynomial has affine exponent rank twelve in the thirteen local variables (u_i, w_a, y_{ia}) ; the sole null direction is the coordinate redundancy $u_i \mapsto \alpha u_i$, $w_a \mapsto w_a/\alpha$. There is therefore no additional rescaling which could make the region scaleless. Finally, for unit propagator powers and $D = 4 - 2\epsilon$, the twelve edge measures, the six cancellation widths and $\mathcal{P}^{-D/2}$ give

$$I_{K_{4,3}}^{\text{scalar}} \sim \delta^{-12} \delta^{6/2} (\delta^{-6})^{-D/2} = \delta^{3-6\epsilon}. \quad (5.34)$$

Thus the positive Landau locus, the local lower facet and the absence of an extra scaling null direction all agree: $K_{4,3}$ is a certified six-loop wide-angle Landshoff seed.

The $K_{3,3}$ calculation gives a complementary warning. To test whether its two-forest polynomial can exhibit cancellation beyond quadratic order, it is enough to find one consistent kinematic slice. We choose the slice invariant under simultaneous permutations of the two three-vertex partitions,

$$s_{i,3+a} = \begin{cases} p, & i = a, \\ q, & i \neq a, \end{cases} \quad \text{Gram}_5 = -\frac{3}{16} p^2 (p - 4q)^2 (p + 2q), \quad (5.35)$$

which is realised at the symmetric point of a planar cyclic $3 \rightarrow 3$ kinematic family. This restriction is used only to establish the existence of an algebraic higher-order stationary locus, not to classify general $K_{3,3}$ kinematics.

On the nondegenerate Gram branch $p = 4q$, write the edge-parameter matrix $X = (x_{ia})$ in the same symmetry class as $x_{ii} = A$ and $x_{ia} = B$ for $i \neq a$. The exact polynomial and a representative of each of the two derivative orbits reduce to

$$\mathcal{F}|_{\text{sym}} = 3qB^2(A + 2B)^3, \quad \left. \frac{\partial \mathcal{F}}{\partial x_{ii}} \right|_{\text{sym}} = 3qB^2(A + 2B)^2, \quad \left. \frac{\partial \mathcal{F}}{\partial x_{ia}} \right|_{\text{sym}} = qB(A + 2B)^2(A + 5B), \quad i \neq a.$$

For $qB \neq 0$, the stationary branch is $A = -2B$. Removing the projective scale gives

$$X_\star \propto \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}. \quad (5.36)$$

There $\mathcal{F}$, its gradient and its Hessian vanish, while the third derivative in a transverse direction does not. Hence the polynomial vanishes to cubic order. The rows and columns of $X_\star$ sum to zero, however, so no nonzero point on this ray lies in the positive orthant. The two-forest structure therefore permits a higher-order cancellation algebraically, but positivity still excludes it from the first sheet.

Table 10 summarises the certified cases and separates them from algebraic stationary loci which fail the positive-orthant requirement.

core and loop order	locus and domain	present status
$K_{2,2}$ $L = 1$	hard-vertex momentum-balance locus; two linear defining factors; positive on the twisted-box boundary at the physical self-crossing surface (3.152)	certified six-point wide-angle HR; no positive pinch at generic wide-angle kinematics. Its NMRK and DSC specialisations inherit the same two factors
$K_{3,2}$ $L = 2$	hard-vertex momentum-balance locus; kinematically deformed path ratios; positive on the near-planar surface $\Gamma_5 = 0$	certified five-point Fish Landshoff HR (Figs. 17(a) and 9); no generic wide-angle HR
$K_{4,2}$, 4 pt $L = 3$	hard-vertex momentum-balance locus; common path ratio; positive at generic wide-angle kinematics	certified four-point Crown HR (Figs. 17(b) and 1); the same core organises its audited boundary descendants and Regge HRs
$K_{4,2}$, 5 pt $L = 3$	hard-vertex momentum-balance locus $Q(z) = 0$; no positive interior solution at generic wide-angle kinematics; positive on the coplanar Gram surface	the generic stationary point lies on a scaleless LP-parameter boundary; the coplanar solution is a certified HR, Eq. (5.24)
$K_{4,2}$, 6 pt $L = 3$	hard-vertex momentum-balance locus $Q(z) = 0$; positive at generic wide-angle kinematics	certified six-point Landshoff HR in the open physical domain (5.17)
$K_{3,3}$ $L = 4$	higher-order algebraic stationary locus; special planar slice; mixed-sign stationary parameter matrix	not a first-sheet Landau singularity and cannot generate an HR; see Fig. 17(c)
$K_{4,3}$ $L = 6$	rank-one incidence locus $x_{ia} = u_i w_a$; positive at generic wide-angle kinematics	certified four-point Landshoff HR with six independent quadratic cancellation normals; this is the Botts–Sterman independent-scattering topology; see Fig. 17(d)

Table 10. Status of the complete-bipartite Landshoff organisation.

5.4 Incidence cores and physical hidden regions

The topological analysis separates three levels intertwined in the examples. The incidence locus C_{inc} records the path-sharing compatibility intrinsic to a complete-bipartite core. Once the collinear lines are removed, the B_a components are mutually disconnected hard subdiagrams — the defining Landshoff structure. The paths from each factor-localisation vertex A_i distribute its collinear momentum among them, and the incidence conditions enforce compatible momentum fractions across the hard components. The rank-one form in Eq. (5.7) has one normal coordinate for each independent graph cycle, and hence codimension equal to the loop order, as shown in Eq. (5.11). The full physical pinch additionally obeys the remaining stationary and vanishing Landau equations and must lie in the positive physical domain. Only after a kinematic expansion is specified do the aligned LP polynomial and the HRF hierarchy determine the active cancellation ideal, region vector and cancellation depth. Topology therefore organises the possible mechanism, but does not by itself determine the HR.

The three established $K_{m,2}$ seeds exhibit this distinction quantitatively in Eq. (5.5). For the Crown, the incidence locus is already the LP-parameter part of the physical pinch, and both have codimension three. The Fish has two incidence normals, whereas a third stationary equation fixes the collective path-sharing coordinate; the separate condition $\Gamma_5 = 0$ is also needed to remove the kinematic obstruction. The twisted box has one incidence normal, while its two hard-vertex momentum-balance factors also fix the collective ratio and hence define a codimension-two pinch on the self-crossing surface. The two factors illustrate that topology carries the incidence path-compatibility, not necessarily the complete factor set or its scaling.

The external-leg analysis of $K_{4,2}$ gives a direct control on the scope of this organisation. The same incidence core supports the generic four-point Crown HR and a generic six-point HR, but its five-point configuration has no positive interior solution at generic wide-angle kinematics and acquires one only on the coplanar Gram surface. The $K_{3,3}$ example provides a complementary control: its two-forest polynomial admits a cubic stationary locus algebraically, but the stationary matrix has mixed signs and therefore lies outside the first sheet. Thus the incidence topology, the external-leg assignment, the kinematic surface and positivity all enter the Landau classification.

New wide-angle $K_{4,2}$ and $K_{4,3}$ seeds are certified; their specialised descendants remain to be analysed. Contraction minors instead relate different graphs. When the contracted graph reproduces the seed with compatible external attachments, the seed loop-momentum routing lifts directly and its HR survives on the corresponding, possibly deeper, contraction stratum. This label compatibility is automatic for the symmetric Crown but is a genuine restriction for the Fish and twisted-box descendants.

The Crown gives the strongest quantitative evidence: a Crown minor occurs in all 1081 four-loop topologies selected by the mixed-sign screen of Ref. [45]. The present audit certifies representative descendants and excludes all 16 topologies without it; the remaining Crown graphs inherit the boundary HR from the Crown minor, even though those strata were not enumerated topology by topology. The Regge audit supports extending this

Crown-minor criterion beyond the audited loop orders.

These results motivate two complementary conjectural principles. The first is topological: the minimal massless wide-angle Landshoff seeds that support HRs possess a complete-bipartite incidence core, with path-sharing or hard-vertex momentum-balance coordinates providing the natural normals. This includes seeds whose HRs are confined to special physical kinematic surfaces, such as the Fish on a near-planar surface and the twisted box on a self-crossing surface; it does not assert that their pinches persist at generic kinematics. Within the audited four-point class, the more specific Crown-minor conjecture is an inheritance criterion of this topological type: a Crown contraction minor controls both wide-angle and Regge HRs.

The second principle concerns the local analytic type of the pinch. Every genuine interior HR encountered so far arises from an ordinary transverse first-sheet pinch: after the stationary conditions remove the constant and linear terms, the graph polynomial retains a nonzero quadratic dependence on displacements normal to the pinch surface. If $I = \langle f_1, \dots, f_r \rangle$ is the saturated ideal generated by local normal coordinates, the invariant conjecture is

$$\mathcal{F}_{\text{SL}} \in I^2 \setminus I^3. \quad (5.37)$$

In words, $\mathcal{F}_{\text{SL}}$ vanishes exactly to quadratic order normal to the pinch; cubic and higher-order terms may be present as well. The two parts have different status. For a smooth pinch component, the Landau equations prove membership in I^2 , as shown in Appendix A. The conjectural statement is that the class in I^2/I^3 is nonzero, or equivalently that the transverse quadratic form does not vanish completely. At the level of the presentations used by HRF, every generator needed in the examples contains precisely two distinct cancellation factors with non-overlapping LP-parameter support. This is compatible with quadratic transverse vanishing, but is not its invariant content: generator presentations are not unique, and terms in I^3 may coexist with the nonzero quadratic class in I^2/I^3 .

The $K_{3,3}$ stationary locus shows why the physical qualification matters: its two-forest polynomial reaches cubic transverse order only outside the positive orthant. Thus any proof of the nonvanishing quadratic class must use more than multiaffinity alone; it must also incorporate graph-polynomial support, connectivity and the positive physical domain.

6 Conclusions and outlook

We have formulated the Hidden Region Finder as a procedure for addressing a central problem in applying the Method of Regions to non-Euclidean Feynman integrals: how to find and certify the complete set of regions when cancellations can conceal relevant scalings. The geometric Method of Regions retains the exponent support of the graph polynomial but is insensitive to the signs and kinematic dependence of its coefficients. HRF supplies this missing information by extracting candidate cancellation factors together with the scaling that enhances their constituent monomials relative to the remaining terms. On the pinch, these enhanced monomials cancel, so that their sum contributes at the same order as any non-cancelling obstruction and the other dominant terms of the complete Lee–Pomeransky (LP) polynomial. A candidate is retained only when the cancellation factors have a common zero in the positive LP domain and the resulting integral is scaleful. A local dissection around the cancellation locus identifies the relevant lower-facet normals. Their consistent pullback to the original LP parameters provides the final geometric certificate.

The examples show why each part of this construction is needed. Hidden regions may have a nonzero obstruction, a non-uniform scaling, multi-term polynomial cancellation factors or several generators. The five-point sector already exhibits almost this entire range. The Fish (the non-planar double box) was identified in the recent Letter [49] as the unique two-loop source of kinematics-dependent factorisation-violating contributions in the spacelike-collinear limit. In that limit it features a nonzero obstruction and a non-uniform vector, and both the seed and its vertex-corrected descendant involve kinematics-dependent cancellation factors that must be tested in the physical domain. For the same seed in the near-planar wide-angle limit, suitable combinations of polynomial derivatives are required to reveal all cancellation factors, yielding several generators. In its simultaneous MRK and near-planar limit, contributions from several orders of the kinematic expansion cancel on the same pinch and all affect the final scaling. Its central-soft limit separately requires asymptotic orders to be aligned before the cancellation structure can be determined. Alignment makes the cancelling terms comparable, while restoring the remaining contributing orders ensures that the full kinematic expansion determines the final scaling. In all such examples the active cancelling contributions saturate the hierarchy bound and enter at the same resolved weight. This is not a requirement of HRF: Eq. (2.39) allows a resolved contribution to enter above W_{HR} . The observed saturation is instead an empirical finding and supplies the dominant balance through which the combined leading support becomes scaleful.

Independent of this hierarchy problem, a distinct completeness issue is whether an HR appears only on a graph boundary, after one or more propagators have been contracted. The absence of an interior candidate therefore does not imply the absence of an HR: a complete analysis must also scan these contraction boundaries. An HR of a larger graph can originate in a seed subgraph exposed by contracting additional propagators. When inequivalent external attachments are not related by a symmetry, they must also be preserved; this defines an externally labelled contraction minor. For the Crown this qualification is automatic, and Ref. [45] demonstrated this mechanism for enlarged three-loop graphs with additional propagators, at the same loop order as the seed; here we extend it to representative four-

and five-loop descendants. The same mechanism exposes the twisted-box seed inside the six-point twisted hexagon and compatible descendants. Its wide-angle HR lies on the self-crossing surface; in the six-point NMRK and DSC specialisations, order alignment and restoration of all contributing orders determine its realisation.

Structural conclusions and conjectures.

Taken together, the examples suggest two complementary organising principles. First, massless wide-angle HRs originate from Landshoff seeds with complete-bipartite incidence cores. The evidence is not confined to the Crown: the mechanism recurs in the five-point Fish and six-point twisted-box families, in the additional complete-bipartite seeds, and in examples extending through six loops. Its appearance across external multiplicities, loop orders and kinematic limits suggests that it is a widespread source of HRs, with potential relevance to many scattering processes. Within the audited four-point class, the principle is realised concretely through inheritance from a Crown contraction minor. Second, every genuine interior HR found here is associated with an ordinary transverse first-sheet pinch: after the stationary conditions remove the constant and linear terms, a nonzero quadratic dependence normal to the pinch remains. In the particular generator presentations produced by HRF, every generator required in the examples has the form $f_i f_j$, with two distinct cancellation factors of non-overlapping LP-parameter support. Although this statement is not invariant under a change of generators, its persistence across all examples is a valuable empirical regularity. These principles constrain, but do not alone determine, the existence of an HR, which also depends on the external-leg assignment, the kinematic limit and positivity. Both principles are formulated and supported in Sec. 5.4. The local ideal-theoretic statement underlying the second is explained in Appendix A.

The Fish topology in the spacelike-collinear limit also shows that one HR can admit several local coordinate descriptions. A fixed cancellation depth may be distributed continuously among several directions normal to the pinch. A centred chart treats their simultaneous approach, whereas endpoint charts isolate limiting directions; all describe the same HR after translation back to the original LP parameters. Endpoint charts can be particularly efficient for evaluating integrals, but may overlap ordinary facet regions and display rapidity divergences absent from a centred presentation. Thus the separation into regional contributions, including its overlap prescription, must be made after the local coordinates are chosen. The individual regional terms and their regulators may therefore depend on the dissection chart, while their properly combined expansion does not.

Seed inheritance and kinematic specialisation.

The HRs of the Crown, Fish and twisted-box seeds in wide-angle kinematics have a common Coleman–Norton backbone. At the positive physical pinch, the active massless propagators describe on-shell classical trajectories [42, 55]. At each external splitting vertex the two null propagator momenta are related by momentum conservation to the attached null external momentum; on the physical branch this places the internal lines on its collinear ray. The resulting collinear splittings must then be compatible at two hard vertices. In parameter space, that compatibility is encoded by the path-sharing or hard-vertex momentum-balance cancellation factors. Thus the collinear-mode assignments

of the three seed families follow from their physical Landau pinches rather than constituting independent mode assumptions.

The six-point family gives the clearest chain of kinematic inheritance. The full one-loop twisted hexagon has no positive pinch at generic wide-angle kinematics, but its contraction stratum $x_0 = x_3 = 0$ exposes the $K_{2,2}$ twisted-box seed, which has a certified HR on the physical self-crossing surface. The HRs of this seed in zero-recoil NMRK and DSC are aligned degenerations of its wide-angle cancellation locus. Their construction requires asymptotic-order alignment before the cancellation structure and scaling can be determined. In the DSC presentation, contributions from more than one order of the kinematic expansion must also be retained. Thus the cancellation geometry is inherited after specialisation, but the final region vector and cancellation depth must be determined anew in the specialised expansion.

The momentum-space reconstruction sharpens the comparison among three limits of the twisted-box seed. Its wide-angle, zero-recoil NMRK and DSC HRs all contain one independent Glauber loop. The two hard-vertex momentum-balance factors constrain its two longitudinal directions. At wide angle the Glauber loop is the common fluctuation of four collinear propagators and no individual propagator is Glauber, whereas in NMRK and DSC x_0 is an explicit Glauber propagator. The degeneration therefore changes the support and diagrammatic realisation of the loop rather than creating it. Across the three regions the transverse-dominance excess $a + b - 2c$ equals the cancellation depth, with respective values 1, 4 and 2, as shown in Sec. 4.

The Crown topology supplies a complementary pattern. Its HR in generic wide-angle kinematics has no Glauber loop, whereas the HR in its Regge specialisation develops one. The Crown contraction minor organises both the wide-angle and three-channel Regge audits, but the Regge singular locus is defined by fewer equations and its region vector is fixed anew by the specialised kinematics. This is a second example in which the wide-angle seed organises the limiting region without making the two HRs equivalent.

For the Fish topology, the HR in near-planar wide-angle kinematics provides the origin of HRs in a variety of specialised limits that expose near-planar configurations. These include the spacelike-collinear, central-soft and simultaneous MRK–near-planar limits. The corresponding HRs approach the positive planar singular family or strata of its closure, but their cancellation factors, region vectors and momentum modes differ. A controlled derivation of these rapidity-ordered regions from the wide-angle limit of the Fish remains open, because matching the singular loci is not enough: one must also match the scalings and every contributing order of the specialised expansion.

These examples separate two inheritance mechanisms. Graph-theoretic inheritance can expose a seed through a compatible contraction minor, both in graphs with additional propagators at the same loop order and in higher-loop topologies. Kinematic specialisation and order alignment can instead carry a physical pinch into a Regge, collinear or correlated limit. The mechanisms can occur together, as in the twisted hexagon, but they are logically distinct operations. In either case common ancestry does not identify the resulting HRs; positivity, the external-leg assignment and the remaining edges decide whether the inherited mechanism is realised. Both mechanisms are encoded in the two-forest polynomial, but the

relevant cancellations and scaling are selected by the particular kinematic expansion.

Momentum-space conclusions.

The translation from a certified parameter region to momentum space is a separate step.¹⁹ The parameter-space vector directly supplies the propagator virtuality scalings, but it does not determine a basis of loop momenta or their separate lightcone components. Momentum-space reconstruction uses these virtualities, together with momentum conservation and the pole structure, to determine the conditional average loop momenta and the local widths of their independent fluctuations. The averages and widths are obtained separately, so their componentwise agreement is not assumed. The first conclusion is an existence statement: in every reconstructed example we find a faithful routing — an adapted basis of independent loop momenta with appropriate local lightcone directions — in which the componentwise scaling of the average momentum agrees with the corresponding integration width. Linear independence alone is not sufficient, and an arbitrary loop routing need not display this correspondence componentwise. This includes Glauber loops, which may be carried by a difference of edge momenta even when no individual propagator is Glauber, or by an explicit Glauber propagator, as in the NMRK and DSC examples. The examples therefore provide no counterexample to this correspondence, although they do not establish it in general. These widths determine the D -dependent part of the momentum-space measure, as made explicit in Eq. (4.14).

The second conclusion concerns longitudinal correlations imposed by Landshoff multiple scattering. Momentum conservation correlates the fractions carried along distinct collinear paths. In the explicit momentum-space reconstructions, this restricts the joint longitudinal support of the loop modes for the Crown in wide-angle kinematics and for the Fish both in near-planar kinematics and in the simultaneous MRK–near-planar limit. The incidence analysis of Sec. 5 exhibits the same path-compatibility structure in every positive complete-bipartite Landshoff example: its defining conditions enforce compatible momentum fractions across the disconnected hard subdiagrams. With several independent loop modes this becomes a cross-loop restriction. In the one-loop twisted-box family, the same path compatibility is encoded within the component widths of its single loop.

The correlated support enters through a Jacobian among longitudinal components. It can therefore change only the integer part of the power count, not its D -dependent coefficient, which is fixed by the transverse integration dimensions. When a Glauber loop is present, this same longitudinal restriction is already incorporated in the integration width discussed above. Both equal and unequal longitudinal-width scalings occur. The spacelike-collinear Fish family, its vertex-corrected descendant and the central-soft Fish have unequal plus and minus widths. The Regge Crown and the wide-angle, zero-recoil NMRK and DSC twisted-box regions instead have equal longitudinal widths in their symmetric frames. In either pattern the restriction is already contained in the Glauber integration width, so no additional cross-mode factor is required.

Returning to parameter space, the correlated longitudinal support reconstructed in momentum space is already visible before dissection as the cancellation encoded by the

¹⁹A systematic reconstruction for facet and hidden regions is being developed in forthcoming work [68].

incidence conditions. The parameter-space cancellation and the momentum-space support restriction have a common origin in the global compatibility among distinct collinear paths. This interpretation is an empirical conclusion from the examples, not a general theorem, and both the average-width correspondence and the separation of any residual correlation require an appropriate independent-loop basis. Whenever a complete momentum-space reconstruction was obtained, direct power counting agreed with the parameter-space count, including the restricted support associated with the cancellation depth. Because the virtuality scaling is supplied by the parameter-space region and the component reconstruction also uses physical mode and pinch information, this agreement is a strong consistency test rather than a wholly independent derivation or part of the existence test for an HR.

Scope and outlook.

The present construction assumes massless propagators and scalar unit numerators. Numerators can alter powers but not denominator-defined regions. Once those regions have been identified, incorporating numerator power counting is relatively straightforward, but is essential for determining their consequences in gauge-theory amplitudes. Internal masses instead introduce quadratic dependence on individual parameters and permit new patterns of cancellation and scaling. Treating masses is therefore a principal extension of HRF and of the restricted ASY2 treatment, with direct applications to heavy-quark pair production near threshold, where potential modes play a central role.

Two complementary momentum-space programmes, both drawing on the parameter-space method in different ways, are under way. The first will systematically convert the parameter-space region vectors obtained from the geometric Method of Regions and HRF into faithful loop-momentum routings, widths and correlations for facet and hidden regions [68]. In parallel, building upon the wide-angle graphical algorithms of Refs. [43, 44], a direct all-order graphical description of both classes in the massless $2 \rightarrow 2$ Regge limit is being developed [69].

The construction of asymptotic expansions is one of several applications of Landau analysis; another is the study of the discontinuity and cut structure of Feynman integrals. These applications are related, and we envisage using the first-sheet analysis developed here to explore their connections in future work.

On the theoretical side, priorities are to understand the two inheritance mechanisms more systematically: the transmission of seed HRs through compatible contraction minors, which is most transparent in momentum space, and the relation between HRs obtained from different kinematic specialisations of the same wide-angle seed. A separate structural question is whether the observed $f_i f_j$ generator form and the absence of a purely cubic leading transverse cancellation on the positive first sheet are universal, and how the order of transverse vanishing controls the region scaling. Broader interior and boundary surveys provide a practical means of testing these inheritance patterns and cancellation structures across further topologies and kinematic limits.

On the applications side, this work is relevant to kinematic expansions of massless scattering processes more generally, opening a broad range of applications. The Regge limit is one important example, raising the general question of how QCD amplitudes behave at

high energy. Its classical foundations are gluon Reggeisation and BFKL evolution [70–74]. Modern analyses have approached this problem both through Glauber-operator formulations in SCET [75–77] and through Reggeisation and fixed-order amplitudes [78–83]. Within this broader programme, one theoretical question is the appearance of Regge cuts, to which HRs are expected to be relevant. In the planar theory Regge-cut contributions first arise at six points and have been studied extensively in the multi-Regge limit [84–86]; beyond the planar limit, Regge-cut contributions occur already at four and five points [58, 59, 80, 82].

A second direction concerns multi-collinear and soft limits and spacelike-collinear splitting amplitudes at higher perturbative orders. Spacelike-collinear limits are particularly interesting since, in contrast to their timelike counterparts, they do not admit strict collinear factorisation [57, 87–89]. The recent Letter [49] showed that the entire kinematics-dependent factorisation-violating contribution at two loops originates in a single HR. Whether HRs continue to control such contributions at higher orders, and how this mechanism extends to multi-collinear and soft limits, are important questions for future work.

For limits involving soft or collinear final-state particles, the expansion acts directly on external-particle degrees of freedom describing real radiation. In observables, this expansion must be combined with phase-space integration and measurement restrictions, which determine which configurations contribute. The spacelike-collinear HR identified using HRF in the Letter [49] is the virtual counterpart of the region that restores PDF factorisation in gap-between-jets observables [11, 90, 91]. This real–virtual connection may help to expose the region structure underlying Glauber-sensitive logarithms in jet observables more generally [92, 93].

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A Landau conditions and the squared cancellation ideal

This appendix isolates the local statement behind the product-generator construction in Sec. 2. Fix the kinematics and, for a boundary region, first restrict to the corresponding contraction stratum. Let A denote the space of active LP parameters and let $C \subset A$ be a smooth component of a pinch locus contained in the positive orthant. Suppose that the selected cancellation factors

$$\mathcal{I}_C = \langle f_1, \dots, f_r \rangle \tag{A.1}$$

generate the radical ideal of C locally and have independent gradients there. We show that the Landau equations on C are equivalent to quadratic vanishing in the square of its local defining ideal.

The transverse-coordinate argument. The implicit-function theorem allows the f_i to be completed to local coordinates $(f_1, \dots, f_r, y_1, \dots, y_{N-r})$, where the y_a are tangent to C . In these coordinates a function regular near C has the expansion

$$\mathcal{F}_{\text{SL}}(f, y) = A(y) + \sum_{i=1}^r f_i B_i(y) + \sum_{1 \leq i < j \leq r} f_i f_j C_{ij}(f, y). \quad (\text{A.2})$$

The condition $\mathcal{F}_{\text{SL}}|_C = 0$ sets $A(y) = 0$. The stationary Landau conditions set every derivative in a transverse direction to zero on C , and hence $B_i(y) = 0$. The tangential derivatives already vanish because the restriction of $\mathcal{F}_{\text{SL}}$ to C is identically zero. Consequently

$$\mathcal{F}_{\text{SL}} = \sum_{i < j} f_i f_j C_{ij}(f, y) \in \mathcal{I}_C^2. \quad (\text{A.3})$$

This gives an intuitive interpretation of the result: a stationary zero has neither a constant displacement from the zero locus nor a linear displacement transverse to it. Its first possible nonzero term is quadratic in the cancellation factors.

The individual transverse coordinates are not canonical. They may be changed by an invertible local transformation, and a factorised quadratic term can be approached anisotropically. Consequently, the weights assigned to separate f_i , and the facets exposed in a particular dissection chart, are not invariants. What is invariant is the common locus C , the lowest nonzero degree in its transverse Taylor expansion, and the scaling obtained after pulling the resolved normal data back to the original LP parameters. Ratio coordinates parameterise the corresponding projectivised normal directions; their endpoint charts may expose the bounding facets of a centred common face, as in Sec. 3.2.1.

The ideal-theoretic statement. For completeness, let $\mathcal{O}_{A,p}$ be the local ring of the ambient active-parameter space at a point $p \in C$, and use the same symbol $\mathcal{I}_C$ for the corresponding local ideal. Working in this local ring means restricting attention to a neighbourhood of p and allowing division by functions that are nonzero there. Because C is smoothly embedded in the smooth space A , the conormal sequence is short exact²⁰ [94]:

$$0 \longrightarrow \mathcal{I}_C / \mathcal{I}_C^2 \xrightarrow{d} \Omega_A|_C \longrightarrow \Omega_C \longrightarrow 0. \quad (\text{A.4})$$

This sequence separates ambient differentials normal to C , represented by $\mathcal{I}_C / \mathcal{I}_C^2$, from the intrinsic differentials along C , represented by Ω_C . If $\mathcal{F}_{\text{SL}}|_C = 0$, then $\mathcal{F}_{\text{SL}} \in \mathcal{I}_C$. Its class in $\mathcal{I}_C / \mathcal{I}_C^2$ is mapped by the first arrow to $d\mathcal{F}_{\text{SL}}|_C$. The stationary Landau equations make this image zero, and injectivity therefore gives

$$[\mathcal{F}_{\text{SL}}] = 0 \quad \text{in } \mathcal{I}_C / \mathcal{I}_C^2, \quad \Longleftrightarrow \quad \mathcal{F}_{\text{SL}} \in \mathcal{I}_C^2. \quad (\text{A.5})$$

²⁰A sequence $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ is short exact when the first map is injective, the second is surjective, and the image of the first equals the kernel of the second.

Conversely, the product rule shows immediately that every element of $\mathcal{I}_C^2$ and all of its first derivatives vanish on C . Thus, at a smooth component, the Landau conditions and membership in the square of the local defining ideal are locally equivalent. Since all active parameters are nonzero, the same statement follows if the Landau equations are written with logarithmic derivatives $x_e \partial_{x_e}$.

Localisation and limitations. Equation (A.5) is a local result, which is exactly what is required for a specified physical component of the pinch. In a global polynomial ring it may take the form

$$h\mathcal{F}_{\text{SL}} \in \mathcal{I}_C^2, \quad h|_C \neq 0, \quad (\text{A.6})$$

or, equivalently, membership in the saturation of $\mathcal{I}_C^2$ by factors known not to vanish on that component. The regular multipliers in Eq. (2.25) are the local coefficients produced in this way.

Smoothness and radicality are essential to this formulation. On a singular or nonreduced component the conormal map in Eq. (A.4) need not be injective, so vanishing of a function and its first derivatives does not by itself prove membership in the ordinary square of an arbitrarily chosen defining ideal. In such a case one must first resolve the relevant smooth positive component, for example in local dissection coordinates, or use the appropriate differential or symbolic-power formulation. For HRF this makes the Jacobian-rank test on the candidate cancellation factors part of the certificate for applying the two-factor construction.

Finally, the Landau equations establish only $\mathcal{F}_{\text{SL}} \in \mathcal{I}_C^2$. They do not imply that the quadratic class is nonzero. The stronger condition

$$[\mathcal{F}_{\text{SL}}] \neq 0 \quad \text{in } \mathcal{I}_C^2/\mathcal{I}_C^3 \quad (\text{A.7})$$

is the statement that the transverse Hessian is nonvanishing. It distinguishes an ordinary quadratic pinch from a higher-order degeneracy and is the genuinely conjectural part of Eq. (5.37).

B Notation summary

Table 11: Principal notation used throughout the paper.

Symbol	Meaning
δ	kinematic expansion parameter
$\mathbf{s}$	real kinematic coordinates; their physical domain $\mathcal{K}$ is specified by explicit inequalities
$x_e, \quad \mathbf{x} \equiv (x_e)_{e \in E}$	independent LP edge parameter and the vector of all such parameters; each x_e is integrated over $(0, \infty)$

Continued on next page

Table 11 (continued)

Symbol	Meaning
$q_e, \mathcal{D}_e = q_e^2 + i0$	oriented edge momentum and massless propagator denominator
μ_{LP}	auxiliary reference mass in the dimensionless LP polynomial; fixed as $\delta \rightarrow 0$ unless stated otherwise
$\mathcal{P}_{\mu_{\text{LP}}} = \mathcal{U} + \mathcal{F}/\mu_{\text{LP}}^2$	dimensionless Lee–Pomeransky polynomial; the scale label is suppressed for the default fixed choice
$X[\mathcal{G}; \alpha]$	graph-dependent quantity; the optional second entry distinguishes inequivalent regions or branches of the same graph
$\mathcal{P}^{(R)}[\mathcal{G}]$	LP polynomial of $\mathcal{G}$ after the rescaling associated with region R ; parenthesised superscripts denote rescaled expansions, not graph identity
$\deg_x \mathcal{U} = L, \deg_x \mathcal{F} = L + 1$	massless graph-polynomial degrees; both polynomials are linear in each individual x_e
$m_i = c_i(\mathbf{s})\delta^{a_i}\mathbf{x}^{\mathbf{r}_i}$	monomial of $\mathcal{P}$
$\mathbf{r}_i$	length- N LP-parameter exponent vector of m_i
$\vec{r}_i = (\mathbf{r}_i; a_i)$	augmented exponent point
$\vec{v}_R = (\mathbf{v}_R; 1)$	normalised augmented region vector; $\mathbf{v}_R$ gives the LP-parameter exponents
$\mathcal{F}_0$	strict fixed- $\mathbf{x}$ leading polynomial
$\mathcal{F}^{[\phi]}$	face selected by the preliminary alignment vector ϕ
$\vec{\phi} = (\phi; 1), \quad x_e = \delta^{\phi_e} y_e$	augmented alignment normal and the associated first transformation
$\mathcal{F}_\star$	HRF starting polynomial, either $\mathcal{F}_0$ or $\mathcal{F}^{[\phi]}$
$\mathcal{M}_\star$	monomial support of the HRF starting polynomial
$\mathcal{M}_{\text{Obs}}$	obstruction monomial support selected for removal
$\mathcal{M}_{\text{SL}} = \mathcal{M}_\star \setminus \mathcal{M}_{\text{Obs}}$	candidate superleading monomial support
$\mathcal{F}_{\text{SL}}$	superleading cancellation sector
$\mathcal{F}_{\text{Obs}}$	obstruction polynomial
f_j	non-monomial cancellation factor
C	selected smooth component of the common positive cancellation locus
$\mathcal{I}_C$	radical local ideal of the selected smooth pinch component
$\mathcal{I}_C^2$	square of the local defining ideal, generated locally by all pairwise products $f_i f_j$

Continued on next page

Table 11 (continued)

Symbol	Meaning
$\text{supp}_x(f_j)$	LP parameters on which f_j depends
$g_k = f_{i_k} f_{j_k}$	massless pair generator with distinct factors and disjoint LP-parameter support
$G = \{g_1, \dots, g_{n_{\text{gen}}}\}$	candidate generator set, retained when it yields an admissible decomposition
$n_{\text{gen}} = G $	number of generators in the selected presentation
$\mathcal{I}_{\text{gen}}(G)$	candidate ideal generated by G , retained only if the obstruction search finds an admissible decomposition
$\mathbf{v}_{\text{core}}^{(0)}$	provisional face-only HRF scaling in the aligned variables
$y_e = \delta^{(v_{\text{core}})_e} z_e$	corrected second-stage HRF scaling after all LP layers are restored
$\mathbf{v}_{\text{HR}} = \boldsymbol{\phi} + \mathbf{v}_{\text{core}}$	composition of the two edge-parameter transformations
$\vec{v}_{\text{HR}} = (\mathbf{v}_{\text{HR}}; 1)$	certified total hidden-region vector; the last component is appended once and is not added stage by stage
$\mathcal{P}_{W_k}$	sum of monomials in the occupied LP layer of nominal weight W_k
W_{SL}	common individual weight of SL monomials
W_{HR}	first resolved nonvanishing LP weight after cancellation
$W_{\text{HR}} - W_{\text{SL}}$	cancellation depth

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